Liu, Chapter 5: Coherent sheaves and Cech cohomology

Solutions to the exercises

Conventions. Rings are commutative with 1. Results of the text are cited by their numbers in Liu ("Prop. 1.12", "Thm. 2.19", ...), and results of earlier chapters as "Prop. 3.3.6", "Ex. 2.4.8" etc. Exercise labels are those of the book, numbered within each section of the chapter: for instance Exercise 2.10 is the tenth exercise at the end of Section 5.2. Each exercise is restated before its solution; sub-items that the book displays as nested lists are written as running prose.

Honest notes. Several printed statements are false as printed or need a hypothesis, and each solution says what is proved instead; in particular Ex. 1.8 (false for subsheaves of an arbitrary module), 1.24 (counterexample in characteristic different from $2, 3$), 1.29(b),(c) (counterexample with a disconnected X'), 3.5(a),(b) (false when the special fibre is 0-dimensional), and 3.12 (the geometrically integral case is proved only for Y integral and f flat). **Partly proved:** in Ex. 2.17(c) associativity of the cup product is proved but the general anti-commutativity sign rule only for $p = 0$ or $q = 0$, and one statement of Section 5.2 (the commutation with direct limits) is proved only for finite coverings. Each section below begins with its own list of notes. Some solutions cite standard results which are named where used.

5.1 Coherent sheaves on a scheme

Conventions for this section. Results of the text are cited by their numbers in Liu ("Prop. 1.5", "Thm. 1.7", ..., all in Section 5.1). For a morphism $f : X \to Y$ of ringed spaces, $f^{\#} : \mathcal{O}_Y \to f_*\mathcal{O}_X$ makes $f_*\mathcal{F}$ an $\mathcal{O}_Y$-module, and $\rho : f^{-1}\mathcal{O}_Y \to \mathcal{O}_X$ denotes the adjoint homomorphism. Recall that $f^{-1}\mathcal{G}$ is the sheaf associated to the presheaf $U \mapsto \varinjlim_{V \supseteq f(U)} \mathcal{G}(V)$, that $(f^{-1}\mathcal{G})_x = \mathcal{G}_{f(x)}$, and that $f^*\mathcal{G} = f^{-1}\mathcal{G} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X$. For an affine scheme $\operatorname{Spec} A$ and an A-module M, $\widetilde{M}$ is the associated sheaf; $\widetilde{M}(D(f)) = M_f$. A coherent sheaf is quasi-coherent: by Def. 1.10, near any point there is a surjection $\mathcal{O}^n \to \mathcal{F}$ whose kernel is again finitely generated, so (shrinking the neighborhood) there is a surjection $\mathcal{O}^m \to \mathrm{Ker}$, i.e. locally an exact sequence $\mathcal{O}^m \to \mathcal{O}^n \to \mathcal{F} \to 0$ (Def. 1.4); this is what allows us to apply Prop. 1.11 and Thm. 1.7 to coherent sheaves. We use freely that on a locally Noetherian scheme every affine open subset has a Noetherian ring, and that the principal open sets form a base of an affine scheme, so that a morphism of sheaves is determined by its values on principal opens, and compatible families of maps on principal opens glue to a morphism of sheaves.

Honest notes. Ex. 1.8 is *false as printed* (for subsheaves of an arbitrary $\mathcal{O}_X$-module $\mathcal{H}$): a counterexample is given and the statement is proved when $\mathcal{H}$ is quasi-coherent. In Ex. 1.1(a) the word "immersion" is made precise (homeomorphism onto its image with surjective maps on

stalks, as for immersions of schemes). In Ex. 1.10(c) the sheaf is described by its sections over affine opens, as the statement says.

Exercise 1.1

> **Problem.** Let $f : X \to Y$ be a continuous map of ringed topological spaces. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module and $\mathcal{G}$ an $\mathcal{O}_Y$-module. (a) Show that there exists a canonical homomorphism $f^*f_*\mathcal{F} \to \mathcal{F}$ (resp. $\mathcal{G} \to f_*f^*\mathcal{G}$) of $\mathcal{O}_X$-modules (resp. of $\mathcal{O}_Y$-modules) (use Exercise 2.2.13(a)). Show that the first homomorphism is an isomorphism if f is an immersion. (b) Show that there exists a canonical isomorphism $\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$. (c) Show that there exists a canonical homomorphism $\mathcal{G} \otimes_{\mathcal{O}_Y} f_*\mathcal{O}_X \to f_*f^*\mathcal{G}$ and that it is an isomorphism if f is an affine morphism of schemes and if $\mathcal{G}$ is quasi-coherent (note that $f_*\mathcal{O}_X$ is quasi-coherent).

Solution. *Two adjunctions.* We first prove two facts, which give (b) and which produce the maps of (a). Regard $\mathcal{F}$ as an $f^{-1}\mathcal{O}_Y$-module through ρ.

(A) For an $f^{-1}\mathcal{O}_Y$-module $\mathcal{M}$ and an $\mathcal{O}_X$-module $\mathcal{F}$, restriction along $m \mapsto m \otimes 1$ gives a bijection

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathcal{M} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X, \mathcal{F}) \xrightarrow{\sim} \mathrm{Hom}_{f^{-1}\mathcal{O}_Y}(\mathcal{M}, \mathcal{F}).$$

Indeed, given an $f^{-1}\mathcal{O}_Y$-linear $\lambda : \mathcal{M} \to \mathcal{F}$, the maps $\mathcal{M}(U) \otimes_{f^{-1}\mathcal{O}_Y(U)} \mathcal{O}_X(U) \to \mathcal{F}(U)$, $m \otimes a \mapsto a\lambda(m)$ are well defined (for $r \in f^{-1}\mathcal{O}_Y(U)$ we have $\lambda(rm) = \rho(r)\lambda(m)$), $\mathcal{O}_X(U)$-linear and compatible with restrictions. By the universal property of the sheaf associated to a presheaf they induce an $\mathcal{O}_X$-linear map $\mathcal{M} \otimes \mathcal{O}_X \to \mathcal{F}$ extending λ. It is the only one: an $\mathcal{O}_X$-linear map is determined on the presheaf $U \mapsto \mathcal{M}(U) \otimes \mathcal{O}_X(U)$ by its values on the elements $m \otimes 1$ (which generate), and a map out of the associated sheaf is determined by its restriction to the presheaf.

(B) For an $\mathcal{O}_Y$-module $\mathcal{G}$ and an $\mathcal{O}_X$-module $\mathcal{F}$ there is a bijection

$$\mathrm{Hom}_{f^{-1}\mathcal{O}_Y}(f^{-1}\mathcal{G}, \mathcal{F}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F}).$$

Let $\iota : \mathcal{G}(V) \to (f^{-1}\mathcal{G})(f^{-1}V)$ be the canonical map (class of s in the presheaf, then sheafify). To θ associate $\psi_V = \theta_{f^{-1}V} \circ \iota : \mathcal{G}(V) \to \mathcal{F}(f^{-1}V)$. For $a \in \mathcal{O}_Y(V)$, $\iota(as) = \rho(a) \cdot \iota(s)$ (where we write $\rho(a)$ for the image of a in $f^{-1}\mathcal{O}_Y(f^{-1}V)$ and then in $\mathcal{O}_X(f^{-1}V)$, which is $f^{\#}(a)$), so $\psi_V(as) = f^{\#}(a)\psi_V(s)$ and ψ is $\mathcal{O}_Y$-linear. Conversely let $\psi : \mathcal{G} \to f_*\mathcal{F}$ be $\mathcal{O}_Y$-linear. On the presheaf $P(U) = \varinjlim_{V \supseteq f(U)} \mathcal{G}(V)$ define $P(U) \to \mathcal{F}(U)$ by sending the class of $s \in \mathcal{G}(V)$ to $\psi_V(s)|_U$ (this makes sense since $U \subseteq f^{-1}V$, and it is compatible with the transition maps because ψ commutes with restrictions). It induces $\theta : f^{-1}\mathcal{G} \to \mathcal{F}$. It is $f^{-1}\mathcal{O}_Y$-linear: for $b \in \mathcal{O}_Y(V')$, $s \in \mathcal{G}(V'')$ representing sections over $U \subseteq f^{-1}(V' \cap V'')$, we get $\theta(bs) = \psi(bs)|_U = f^{\#}(b)\psi(s)|_U = \rho(b)\theta(s)$. The two constructions are inverse to each other, since both are checked on the presheaf level where they are obvious.

Composing, $\mathrm{Hom}_{\mathcal{O}_X}(f^*\mathcal{G}, \mathcal{F}) \simeq \mathrm{Hom}_{f^{-1}\mathcal{O}_Y}(f^{-1}\mathcal{G}, \mathcal{F}) \simeq \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{G}, f_*\mathcal{F})$. This is the canonical isomorphism asked for in (b). Both bijections are visibly additive (indeed natural in $\mathcal{F}$ and $\mathcal{G}$), which gives (b).

(a) Define $\eta : \mathcal{G} \to f_*f^*\mathcal{G}$ as the $\mathcal{O}_Y$-linear map corresponding to $\mathrm{id}_{f^*\mathcal{G}}$ under (b) (with $\mathcal{F} = f^*\mathcal{G}$): explicitly $\eta_V(s) = \iota(s) \otimes 1 \in (f^*\mathcal{G})(f^{-1}V)$. Define $\varepsilon : f^*f_*\mathcal{F} \to \mathcal{F}$ as the $\mathcal{O}_X$-linear

map corresponding under (b) to $\mathrm{id}_{f_*\mathcal{F}}$: explicitly it sends $t \otimes a$, with t the class of a section $t' \in \mathcal{F}(f^{-1}V)$, to $a \cdot t'|_U$. These are the canonical homomorphisms.

Now let f be an immersion, i.e. a homeomorphism onto its image such that $\mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$ is surjective for all x (this holds for immersions of schemes: a closed immersion followed by an open immersion). We check that ε is an isomorphism on stalks at $x \in X$; put $y = f(x)$. By Prop. 1.14(a), $(f^* f_* \mathcal{F})_x = (f_*\mathcal{F})_y \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x}$, and ε_x is $u \otimes a \mapsto a\,\bar{u}$ where $\bar{u}$ is the image of u under the canonical map $(f_*\mathcal{F})_y = \varinjlim_{V \ni y} \mathcal{F}(f^{-1}V) \to \mathcal{F}_x$. Since f is a homeomorphism onto its image, every open neighborhood of x is of the form $f^{-1}V$ with $V \ni y$ open, and one can shrink V at will; hence this canonical map is an isomorphism. It remains to see that $\mathcal{F}_x \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{X,x} \to \mathcal{F}_x$, $u \otimes a \mapsto au$ is an isomorphism. Write $R = \mathcal{O}_{Y,y}$ and $S = \mathcal{O}_{X,x} = R/I$. Then $\mathcal{F}_x \otimes_R S = \mathcal{F}_x / I\mathcal{F}_x = \mathcal{F}_x$ because I acts as 0 on the S-module $\mathcal{F}_x$, and under this identification the map is the identity. So ε is an isomorphism.

(c) For $V \subseteq Y$ open, define

$$\mathcal{G}(V) \otimes_{\mathcal{O}_Y(V)} (f_*\mathcal{O}_X)(V) = \mathcal{G}(V) \otimes_{\mathcal{O}_Y(V)} \mathcal{O}_X(f^{-1}V) \longrightarrow (f^*\mathcal{G})(f^{-1}V) = (f_* f^* \mathcal{G})(V), \quad s \otimes a \mapsto \eta_V(s) \cdot a.$$

This is well defined: for $b \in \mathcal{O}_Y(V)$, $\eta_V(bs)\, a = \eta_V(s)\, f^{\#}(b) a$ because η_V is $\mathcal{O}_Y$-linear and the structure of $\mathcal{O}_Y$-module on $f_* f^* \mathcal{G}$ goes through $f^{\#}$. It is compatible with restrictions, so it is a morphism from the presheaf $V \mapsto \mathcal{G}(V) \otimes \mathcal{O}_X(f^{-1}V)$ to the sheaf $f_* f^* \mathcal{G}$, and it factors uniquely through the associated sheaf $\mathcal{G} \otimes_{\mathcal{O}_Y} f_*\mathcal{O}_X$. Call the resulting $\mathcal{O}_Y$-linear map ψ.

Now assume f affine and $\mathcal{G}$ quasi-coherent. The statement is local on Y and the construction of ψ is compatible with restriction to open subsets of Y (replace X by $f^{-1}V$), so we may assume $Y = \operatorname{Spec} A$, hence $X = \operatorname{Spec} B$ is affine, and $\mathcal{G} = \widetilde{M}$ with $M = \mathcal{G}(Y)$ (Thm. 1.7). The sheaf $f_*\mathcal{O}_X$ is $\widetilde{B}$ (B viewed as an A-module) since $(f_*\mathcal{O}_X)(D(a)) = B_a$ for $a \in A$ (here $D(a)$ has preimage $D(f^{\#}a)$ and $B_{f^{\#}a} = B \otimes_A A_a$). By Prop. 1.5(d), $\mathcal{G} \otimes f_*\mathcal{O}_X = (M \otimes_A B)^{\sim}$, with global sections $M \otimes_A B$. The target $f_* f^* \mathcal{G}$ is quasi-coherent by Prop. 1.14(b),(c). So ψ is a morphism of quasi-coherent sheaves on the affine scheme Y, and it is an isomorphism as soon as $\psi(Y)$ is: indeed, both sheaves are associated to modules and ψ is $\mathcal{O}_Y$-linear, so for $a \in A$ and a section $a^{-n}u|_{D(a)}$ (u a global section) of the source over $D(a)$ we get $\psi_{D(a)}(a^{-n}u|_{D(a)}) = a^{-n}\psi(Y)(u)|_{D(a)}$; thus ψ is the map induced by $\psi(Y)$ on every $D(a)$, namely its localization at a, and localization preserves isomorphisms. Now $\psi(Y) : M \otimes_A B \to (f^*\mathcal{G})(X)$ is $m \otimes b \mapsto \eta_Y(m)\, b$. This is the map Φ_M which, by Prop. 1.14(b), is an isomorphism: indeed in the proof of Prop. 1.14(b) one takes a presentation $K \to L \to M \to 0$ with K, L free, uses that f^* of the corresponding sequence of sheaves is right exact, and compares with $K \otimes B \to L \otimes B \to M \otimes B \to 0$; for free modules $A^{(I)}$ the map Φ is the identity $B^{(I)} \to B^{(I)}$ (since $f^*\mathcal{O}_Y^{(I)} = \mathcal{O}_X^{(I)}$ and X is quasi-compact, so global sections commute with direct sums), Φ is natural in M, and both functors $M \mapsto M \otimes_A B$ and $M \mapsto (f^*\widetilde{M})(X)$ are right exact (the latter by Prop. 1.8 applied to the quasi-coherent sheaves $f^*\widetilde{K}, f^*\widetilde{L}, f^*\widetilde{M}$ and the quasi-coherent images/kernels of Prop. 1.12(c)). The five lemma gives that Φ_M is an isomorphism. Hence ψ is an isomorphism. □

Exercise 1.2

> **Problem.** Let A be an integral domain with field of fractions K, and let $x_0 \in X = \operatorname{Spec} A$. We construct a sheaf $\mathcal{F}$ on X by setting $\mathcal{F}(U) = K$ if $x_0 \in U$ and $\mathcal{F}(U) = 0$ otherwise. The restriction maps are the obvious ones. Show that $\mathcal{F}$ is an $\mathcal{O}_X$-module. Under what condition is $\mathcal{F}$ quasi-coherent?

Solution. The restriction maps are id_K if $x_0 \in V \subseteq U$ and 0 otherwise. This is a presheaf. It is a sheaf: let $U = \bigcup U_i$ and $s_i \in \mathcal{F}(U_i)$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$. If $x_0 \notin U$ then all $\mathcal{F}(U_i) = 0$ and $\mathcal{F}(U) = 0$, and there is nothing to prove. If $x_0 \in U$, the U_i containing x_0 have $s_i \in K$, and for two of them the overlap contains x_0, so $s_i = s_j =: \lambda$. The element $\lambda \in \mathcal{F}(U)$ restricts to s_i for $x_0 \in U_i$, and to $0 = s_i$ for $x_0 \notin U_i$. It is unique, since $\mathcal{F}(U) = K \to \mathcal{F}(U_i) = K$ is the identity for some $U_i \ni x_0$.

Module structure: let $\mathfrak{p}$ be the prime corresponding to x_0. For $x_0 \in U$ let $\mathcal{O}_X(U)$ act on $\mathcal{F}(U) = K$ through $\mathcal{O}_X(U) \to \mathcal{O}_{X,x_0} = A_{\mathfrak{p}} \subseteq K$ (germ at x_0 followed by the inclusion into K, A being a domain); for $x_0 \notin U$ the module $\mathcal{F}(U) = 0$ is trivial. Then $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module. For $V \subseteq U$ and $a \in \mathcal{O}_X(U)$, $s \in \mathcal{F}(U)$: if $x_0 \in V$ then $a|_V$ and a have the same germ at x_0, so $(as)|_V = a|_V s|_V$; if $x_0 \notin V$ both sides are 0. So $\mathcal{F}$ is an $\mathcal{O}_X$-module.

Claim: $\mathcal{F}$ is quasi-coherent if and only if x_0 is the generic point of X, i.e. $\mathfrak{p} = (0)$.

Suppose $\mathfrak{p} = 0$. Then $x_0 \in U$ iff $U \neq \emptyset$ (the generic point lies in every nonempty open set). Now $D(f) \neq \emptyset$ iff $f \neq 0$, and for $f \neq 0$ we have $K_f = K$ (f is already invertible in K), while $D(0) = \emptyset$. Hence $\mathcal{F}(D(f)) = K = \widetilde{K}(D(f))$ for $f \neq 0$ and $\mathcal{F}(D(0)) = 0$, with identity restriction maps (and the A_f-module structures agree: $A_f \subseteq K$ acts by multiplication). Since the principal opens form a base, $\mathcal{F} \simeq \widetilde{K}$, which is quasi-coherent by Prop. 1.5(c).

Suppose $\mathfrak{p} \neq 0$ and, for contradiction, that $\mathcal{F}$ is quasi-coherent. By Thm. 1.7, $\mathcal{F} \simeq \widetilde{M}$ with $M = \mathcal{F}(X) = K$ (as an A-module, $A \subseteq K$). Pick $0 \neq f \in \mathfrak{p}$. Then $x_0 \notin D(f)$, so $\mathcal{F}(D(f)) = 0$, but $\widetilde{M}(D(f)) = K_f = K \neq 0$. An isomorphism of sheaves cannot exist; contradiction. (If A is a field, X is one point, x_0 is the generic point, and $\mathcal{F} = \widetilde{K}$.) □

Exercise 1.3

> **Problem.** Let X be a scheme and $f \in \mathcal{O}_X(X)$. Show that the sheaf $f\mathcal{O}_X$ (Exercise 2.3.4) is finitely generated quasi-coherent. Show that in general it is not an invertible sheaf.

Solution. Recall that $f\mathcal{O}_X$ is the sheaf of ideals with $(f\mathcal{O}_X)(U) = f|_U \mathcal{O}_X(U)$ for U affine. Let $\mu : \mathcal{O}_X \to \mathcal{O}_X$ be multiplication by f. We show $f\mathcal{O}_X = \operatorname{Im}\mu$. Let $U = \operatorname{Spec} A$ be affine and $a = f|_U \in A$. By Prop. 1.5(b), applied to the exact sequence $A \xrightarrow{a} A \to A/aA \to 0$ and to $0 \to aA \to A$, the sheaf $\operatorname{Im}(\mu|_U)$ is $(aA)^\sim \subseteq \widetilde{A} = \mathcal{O}_U$, whose sections over U are $aA = f|_U \mathcal{O}_X(U)$. So $\operatorname{Im}\mu$ and $f\mathcal{O}_X$ are subsheaves of $\mathcal{O}_X$ with the same sections over each affine open, hence equal.

$\mathcal{O}_X$ is quasi-coherent, so $\operatorname{Im}\mu$ is quasi-coherent by Prop. 1.12(c). It is finitely generated because $\mu : \mathcal{O}_X \to f\mathcal{O}_X$ is a surjective homomorphism (Def. 1.10 with $n = 1$, $U = X$).

Not invertible in general. Stalks of $f\mathcal{O}_X$ are $(f\mathcal{O}_X)_x = f_x \mathcal{O}_{X,x}$ (image of the stalk map). If $f\mathcal{O}_X$ were invertible, then for each x the module $f_x\mathcal{O}_{X,x}$ would be isomorphic to $\mathcal{O}_{X,x}$, in particular have zero annihilator in $\mathcal{O}_{X,x}$. Two examples where this fails: (i) $f = 0$ on a nonempty scheme: $f\mathcal{O}_X = 0$, whose stalks are $0 \neq \mathcal{O}_{X,x}$. (ii) $X = \operatorname{Spec} k[T]/(T^2)$ (k a field), $f = T$: the unique stalk is $T \cdot k[T]/(T^2)$, which is annihilated by T, whereas $T \neq 0$ in $\mathcal{O}_{X,x} = k[T]/(T^2)$. So $f\mathcal{O}_X$ is not invertible, although X is Noetherian and f is neither a unit nor zero. (If X is integral and $f \neq 0$, then multiplication by f is injective on stalks and $f\mathcal{O}_X \simeq \mathcal{O}_X$, so one needs zero divisors, or $f = 0$, for a counterexample.) □

Exercise 1.4

Problem. Let $X = \operatorname{Spec} A$ be an affine scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Show that for any affine open subset U of X, we have a canonical isomorphism $\mathcal{F}(X) \otimes_A \mathcal{O}_X(U) \simeq \mathcal{F}(U)$. Show that this is false if U is not affine (consider Example 2.3.6 with $M = \mathbf{Z}[T]/(T, p)$).

Solution. Put $M = \mathcal{F}(X)$, so that $\mathcal{F} \simeq \widetilde{M}$ (Thm. 1.7). Let $U = \operatorname{Spec} B$ be an affine open subset of X; then $B = \mathcal{O}_X(U)$ is an A-algebra. The canonical map is

$$\theta : M \otimes_A B \to \mathcal{F}(U), \qquad m \otimes b \mapsto m|_U \cdot b.$$

θ is an isomorphism. The sets $D(g) \subseteq U$, $g \in A$, cover U; since U is quasi-compact, $U = D(g_1) \cup \cdots \cup D(g_r)$. Each $D(g_i)$ is a principal open set of U, namely $D_U(g_i|_U)$, and $B_{g_i} = \mathcal{O}_X(D(g_i)) = A_{g_i}$. The g_i generate the unit ideal of B (as $\operatorname{Spec} B = \bigcup D_U(g_i)$). A homomorphism θ of B-modules is an isomorphism as soon as every θ_{g_i} is: kernel and cokernel have all localizations 0 at the g_i, and if $\sum b_i g_i^n = 1$ (possible since the g_i generate the unit ideal, hence so do their powers), every element killed by some power of each g_i is 0. So it suffices that θ_{g_i} is an isomorphism. We have $(M \otimes_A B)_{g_i} = M_{g_i} \otimes_{A_{g_i}} B_{g_i} = M_{g_i}$ (using $B_{g_i} = A_{g_i}$). Applying Prop. 1.6 to $\mathcal{F}|_U$ on the affine scheme U and $g_i|_U$ gives an isomorphism $\mathcal{F}(U)_{g_i} \to \mathcal{F}(D(g_i))$, and Prop. 1.6 applied to $\mathcal{F}$ on X gives $M_{g_i} \simeq \mathcal{F}(D(g_i))$. The composite

$$M_{g_i} = (M \otimes_A B)_{g_i} \xrightarrow{\theta_{g_i}} \mathcal{F}(U)_{g_i} \to \mathcal{F}(D(g_i))$$

sends $m/g_i^n \otimes 1$ to $g_i^{-n} m|_{D(g_i)}$, so it is the isomorphism $M_{g_i} \simeq \mathcal{F}(D(g_i))$. As the second arrow is an isomorphism, so is θ_{g_i}. Hence θ is an isomorphism.

Failure for non-affine U. Let $A = \mathbf{Z}[T]$, p a prime number, $\mathfrak{m} = (T, p)$, a maximal ideal ($A/\mathfrak{m} = \mathbf{F}_p$), x the corresponding closed point, $M = A/\mathfrak{m} = \mathbf{F}_p$ and $\mathcal{F} = \widetilde{M}$. Let $U = X \setminus \{x\} = D(T) \cup D(p)$. (It is not affine, as the computation below shows; we do not need this.) Since $\mathfrak{m}M = 0$, for every prime $\mathfrak{q} \neq \mathfrak{m}$ there is an element of $\mathfrak{m} \setminus \mathfrak{q}$, invertible in $A_\mathfrak{q}$ and killing $M_\mathfrak{q}$, so $M_\mathfrak{q} = \mathcal{F}_\mathfrak{q} = 0$. A section of a sheaf all of whose stalks over U vanish is 0, so $\mathcal{F}(U) = 0$. On the other hand, $\mathcal{O}_X(U) = \{(a, b) \in A_T \times A_p : a = b \text{ in } A_{Tp}\} = A_T \cap A_p$ inside $\operatorname{Frac} A$ (A is a domain). Since $A = \mathbf{Z}[T]$ is a UFD and T, p are non-associate primes, $A_T \cap A_p = A$: if $a/T^n = b/p^m$ with $a, b \in A$, then $ap^m = bT^n$, so $T^n \mid a$ and $a/T^n \in A$. So $\mathcal{F}(X) \otimes_A \mathcal{O}_X(U) = \mathbf{F}_p \otimes_A A = \mathbf{F}_p \neq 0 = \mathcal{F}(U)$, and the canonical map is not an isomorphism. □

Exercise 1.5

Problem. Let X be a scheme, and let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$-modules. (a) Show that $U \mapsto \operatorname{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$ defines an $\mathcal{O}_X$-module. This sheaf will be denoted $\mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$. (b) Let us suppose $X = \operatorname{Spec} A$. Show that the canonical map $\mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G}) \to \operatorname{Hom}_A(\mathcal{F}(X), \mathcal{G}(X))$ is bijective if $\mathcal{F}$ is quasi-coherent.

Solution. (a) Write $H(U) = \operatorname{Hom}_{\mathcal{O}_X|_U}(\mathcal{F}|_U, \mathcal{G}|_U)$ (an abelian group). For $V \subseteq U$ the restriction $\varphi \mapsto \varphi|_V$ is defined, and H is a presheaf. $H(U)$ is an $\mathcal{O}_X(U)$-module via $(a\varphi)_W(s) = a|_W \, \varphi_W(s)$

for $W \subseteq U$ open, $s \in \mathcal{F}(W)$; one checks that $a\varphi$ is again $\mathcal{O}_X$-linear (since $\mathcal{O}_X(W)$ is commutative) and the module axioms. Clearly $(a\varphi)|_V = a|_V \varphi|_V$. It remains to check the sheaf property. Let $U = \bigcup U_i$.

Separation: if $\varphi|_{U_i} = 0$ for all i, then for $W \subseteq U$ and $s \in \mathcal{F}(W)$ the section $\varphi_W(s) \in \mathcal{G}(W)$ restricts to $\varphi_{W\cap U_i}(s|_{W\cap U_i}) = 0$ on each $W \cap U_i$, so $\varphi_W(s) = 0$.

Gluing: given $\varphi^{(i)} \in H(U_i)$ with $\varphi^{(i)}|_{U_i\cap U_j} = \varphi^{(j)}|_{U_i\cap U_j}$, define for $W \subseteq U$ open and $s \in \mathcal{F}(W)$ the section $\varphi_W(s) \in \mathcal{G}(W)$ by gluing the sections $\varphi^{(i)}_{W\cap U_i}(s|_{W\cap U_i})$ of $\mathcal{G}$ over the cover $\{W \cap U_i\}$; they agree on the overlaps by hypothesis. The family φ_W is compatible with restrictions and is $\mathcal{O}_X(W)$-linear (check after restriction to each $W \cap U_i$), and restricts to $\varphi^{(i)}$ on U_i. So H is a sheaf and an $\mathcal{O}_X$-module.

(b) Here the map is $\varphi \mapsto \varphi_X$, from $\mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})(X) = \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$ to $\mathrm{Hom}_A(M,N)$, where $M = \mathcal{F}(X)$, $N = \mathcal{G}(X)$. By Thm. 1.7, $\mathcal{F} \simeq \widetilde{M}$, so every section of $\mathcal{F}$ over $D(f)$ is of the form $f^{-n}\, m|_{D(f)}$ with $m \in M$.

Injective. Let $\varphi_X = 0$. For $s = f^{-n}m|_{D(f)}$ we get $\varphi_{D(f)}(s) = f^{-n}\varphi_{D(f)}(m|_{D(f)}) = f^{-n}\varphi_X(m)|_{D(f)} = 0$. If U is any open set and $s \in \mathcal{F}(U)$, cover U by principal opens $D(f)$; $\varphi_U(s)$ vanishes on each, so $\varphi_U(s) = 0$. Thus $\varphi = 0$.

Surjective. Let $u \in \mathrm{Hom}_A(M,N)$. For $f \in A$ define $u_f : M_f = \mathcal{F}(D(f)) \to \mathcal{G}(D(f))$ by $u_f(m/f^n) = f^{-n}u(m)|_{D(f)}$. It is well defined: if $m/f^n = m'/f^{n'}$ in M_f, then $f^k(f^{n'}m - f^n m') = 0$ in M for some k, hence $f^k(f^{n'}u(m) - f^n u(m')) = 0$ in N; restricting to $D(f)$, where f is a unit, gives $f^{-n}u(m)| = f^{-n'}u(m')|$. It is A_f-linear, and compatible with the restriction maps for $D(g) \subseteq D(f)$ (all maps are induced by u and the localization maps). Since the principal opens form a base of the topology, the u_f glue to a morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$, which is $\mathcal{O}_X$-linear (it is A_f-linear on each $D(f)$ and linearity can be checked locally on sections). By construction $\varphi_X = u_1 = u$. □

Exercise 1.6

> **Problem.** Let X be a locally Noetherian scheme, and let $\mathcal{F},\mathcal{G}$ be $\mathcal{O}_X$-modules. (a) Show that if $\mathcal{F}$ is coherent and $\mathcal{G}$ quasi-coherent, then $\mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$ is quasi-coherent (use Exercise 1.2.8). (b) Show that if $\mathcal{F}$ and $\mathcal{G}$ are coherent, then so is $\mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F},\mathcal{G})$.

Solution. (a) Quasi-coherence is local, so let $X = \operatorname{Spec} A$ with A Noetherian. Then $\mathcal{F} = \widetilde{M}$ with $M = \mathcal{F}(X)$ finitely generated (Prop. 1.11), hence finitely presented since A is Noetherian, and $\mathcal{G} = \widetilde{N}$, $N = \mathcal{G}(X)$.

Algebraic fact (the content of Exercise 1.2.8). For $f \in A$ the natural map $\lambda_f : \mathrm{Hom}_A(M,N)_f \to \mathrm{Hom}_{A_f}(M_f,N_f)$, $(u/f^n) \mapsto (m/f^k \mapsto u(m)/f^{n+k})$ is an isomorphism when M is finitely presented. Indeed, take $A^m \to A^n \to M \to 0$ exact. Then $0 \to \mathrm{Hom}_A(M,N) \to N^n \to N^m$ is exact; localizing (exact) gives $0 \to \mathrm{Hom}_A(M,N)_f \to N_f^n \to N_f^m$ exact. Also $A_f^m \to A_f^n \to M_f \to 0$ is exact, so $0 \to \mathrm{Hom}_{A_f}(M_f,N_f) \to N_f^n \to N_f^m$ is exact, and λ_f is compatible with the maps to N_f^n and N_f^m (these being identities). Hence λ_f is an isomorphism (both are the kernel of the same map).

By Ex. 1.5(b) applied to the affine scheme $D(f) = \operatorname{Spec} A_f$ and the quasi-coherent sheaf $\mathcal{F}|_{D(f)} = (M_f)^{\sim}$, we have $\mathcal{H}\mathrm{om}(\mathcal{F},\mathcal{G})(D(f)) \simeq \mathrm{Hom}_{A_f}(M_f,N_f)$ via $\varphi \mapsto \varphi_{D(f)}$, compatibly

with restriction to $D(g) \subseteq D(f)$. Combining with λ_f, we get isomorphisms $\mathrm{Hom}_A(M,N)_f \simeq \mathcal{H}\mathrm{om}(\mathcal{F},\mathcal{G})(D(f))$ compatible with restrictions, so $\mathcal{H}\mathrm{om}(\mathcal{F},\mathcal{G}) \simeq \mathrm{Hom}_A(M,N)^\sim$, which is quasi-coherent (Prop. 1.5(c)).

(b) By (a), $\mathcal{H} = \mathcal{H}\mathrm{om}(\mathcal{F},\mathcal{G})$ is quasi-coherent. Let $U = \operatorname{Spec} A$ be an affine open subset with A Noetherian. Then $\mathcal{H}(U) \simeq \mathrm{Hom}_A(M,N)$ with $M = \mathcal{F}(U)$, $N = \mathcal{G}(U)$ finitely generated. If M is generated by n elements, $\mathrm{Hom}_A(M,N)$ embeds in N^n and is a finitely generated module, A being Noetherian. The same argument applies to every affine open U of X (its ring is Noetherian), so $\mathcal{H}$ is quasi-coherent with $\mathcal{H}(U)$ finitely generated for all affine U, i.e. (iii) of Prop. 1.11 holds; since X is locally Noetherian, $\mathcal{H}$ is coherent by that proposition. □

Exercise 1.7

> **Problem.** Let X be a scheme. Show that any finitely generated sheaf of ideals $\mathcal{J}$ is quasi-coherent. If X is locally Noetherian, show that any quasi-coherent sheaf of ideals is coherent.

Solution. Let $\mathcal{J} \subseteq \mathcal{O}_X$ be a finitely generated sheaf of ideals and $x \in X$. By Def. 1.10 there are an open neighborhood U of x and a surjection $\pi : \mathcal{O}_U^n \to \mathcal{J}|_U$. Let $\iota : \mathcal{J}|_U \to \mathcal{O}_U$ be the inclusion. Then $\iota \circ \pi : \mathcal{O}_U^n \to \mathcal{O}_U$ is a morphism of quasi-coherent sheaves whose image is $\mathcal{J}|_U$ (the image sheaf of $\iota\pi$ is the image of ι since π is surjective, i.e. $\mathcal{J}|_U$). By Prop. 1.12(c), $\mathcal{J}|_U$ is quasi-coherent. Quasi-coherence being local (Def. 1.4), $\mathcal{J}$ is quasi-coherent.

Now let X be locally Noetherian and $\mathcal{J}$ a quasi-coherent sheaf of ideals. For an affine open U, $\mathcal{O}_X(U)$ is Noetherian and $\mathcal{J}(U)$ is an ideal of it, hence finitely generated. So $\mathcal{J}$ is quasi-coherent with $\mathcal{J}(U)$ finitely generated for every affine open U, i.e. property (iii) of Prop. 1.11 holds, and by that proposition (X locally Noetherian) $\mathcal{J}$ is coherent. □

Exercise 1.8

> **Problem.** Let X be a scheme. Let $\mathcal{F},\mathcal{G}$ be two quasi-coherent subsheaves of an $\mathcal{O}_X$-module $\mathcal{H}$. Show that $\mathcal{F}\cap\mathcal{G}$ and $\mathcal{F}+\mathcal{G}$ are quasi-coherent, where $\mathcal{F}+\mathcal{G}$ is the sheaf associated to the presheaf $U \mapsto \mathcal{F}(U)+\mathcal{G}(U)$. Show that these properties are true for coherent sheaves if X is locally Noetherian.

Solution. *The statement is false as printed* when $\mathcal{H}$ is an arbitrary $\mathcal{O}_X$-module, and it is true when $\mathcal{H}$ is quasi-coherent (which is surely what is intended). We give the counterexample and then prove the corrected statement, including the coherent version.

Counterexample. Let $X = \operatorname{Spec} R$ for a discrete valuation ring R with fraction field K, so $X = \{\xi, s\}$ with generic point ξ and closed point s, and the opens are $\emptyset, \{\xi\}, X$. Let $\mathcal{J} \subseteq \mathcal{O}_X$ be given by $\mathcal{J}(X) = 0$, $\mathcal{J}(\{\xi\}) = K = \mathcal{O}_X(\{\xi\})$, $\mathcal{J}(\emptyset) = 0$. This is a sheaf (any cover of X contains X) and a sheaf of ideals. It is not quasi-coherent: otherwise $\mathcal{J} \simeq \mathcal{J}(X)^\sim = 0$ by Thm. 1.7, but $\mathcal{J}(\{\xi\}) = K \neq 0$. Let $\mathcal{H} = \mathcal{O}_X^2/\mathcal{R}$ where $\mathcal{R} = \{(a,-a) : a \in \mathcal{J}\} \simeq \mathcal{J}$, and let $\mathcal{F}$ (resp. $\mathcal{G}$) be the image in $\mathcal{H}$ of $\mathcal{O}_X \oplus 0$ (resp. $0 \oplus \mathcal{O}_X$). The map $\mathcal{O}_X \to \mathcal{H}$, $a \mapsto [(a,0)]$ is injective ($(a,0) \in \mathcal{R}$ forces $a = 0$), so $\mathcal{F} \simeq \mathcal{O}_X$, likewise $\mathcal{G} \simeq \mathcal{O}_X$; both are coherent, hence quasi-coherent. At a point x, $\mathcal{F}_x \cap \mathcal{G}_x$ consists of the classes $[(a,0)] = [(0,b)]$, i.e. $(a,-b) \in \mathcal{R}_x$, i.e. $a = b \in \mathcal{J}_x$. So $\mathcal{F}\cap\mathcal{G}$ is the image of $\mathcal{J}$ under the injection $\mathcal{J} \to \mathcal{F}$, $a \mapsto [(a,0)]$, hence $\simeq \mathcal{J}$, which is not

quasi-coherent. Also $\mathcal{F}+\mathcal{G}=\mathcal{H}$, and $\mathcal{H}$ is not quasi-coherent either: otherwise the kernel $\mathcal{R}$ of $\mathcal{O}_X^2 \to \mathcal{H}$ would be quasi-coherent by Prop. 1.12(c). (So both claims fail for non-quasi-coherent $\mathcal{H}$, and X is even Noetherian, so the coherent version fails as well.)

Corrected statement and proof. Let $\mathcal{H}$ be quasi-coherent and $\mathcal{F},\mathcal{G} \subseteq \mathcal{H}$ quasi-coherent subsheaves. Consider $u : \mathcal{F}\oplus\mathcal{G} \to \mathcal{H}$, $(s,t) \mapsto s-t$. The sheaf $\mathcal{F}\oplus\mathcal{G}$ is quasi-coherent by Prop. 1.12(a), $\mathcal{H}$ is quasi-coherent, so $\operatorname{Ker} u$ and $\operatorname{Im} u$ are quasi-coherent by Prop. 1.12(c). Now $\operatorname{Ker} u = \{(s,s)\} \simeq \mathcal{F}\cap\mathcal{G}$ via $(s,s)\mapsto s$, so $\mathcal{F}\cap\mathcal{G}$ is quasi-coherent. And for $v : \mathcal{F}\oplus\mathcal{G}\to\mathcal{H}$, $(s,t)\mapsto s+t$, $\operatorname{Im} v$ is the subsheaf of $\mathcal{H}$ whose sections are those that are locally of the form $s+t$, i.e. the sheaf associated to $U \mapsto \mathcal{F}(U)+\mathcal{G}(U)$; thus $\mathcal{F}+\mathcal{G}=\operatorname{Im} v$ is quasi-coherent by Prop. 1.12(c).

Coherent case. Let X be locally Noetherian and $\mathcal{F},\mathcal{G}$ coherent subsheaves of a quasi-coherent (in particular, of a coherent) $\mathcal{H}$. By the above, $\mathcal{F}\cap\mathcal{G}$ and $\mathcal{F}+\mathcal{G}$ are quasi-coherent. For U affine, $(\mathcal{F}\cap\mathcal{G})(U)=\mathcal{F}(U)\cap\mathcal{G}(U)$ is a submodule of the finitely generated module $\mathcal{F}(U)$ over the Noetherian ring $\mathcal{O}_X(U)$, hence finitely generated; and $\mathcal{F}+\mathcal{G}=\operatorname{Im} v$ with v as above. Since $\operatorname{Ker} v$ is quasi-coherent (Prop. 1.12(c)), Prop. 1.8 applied on the affine scheme U to $0\to\operatorname{Ker} v\to\mathcal{F}\oplus\mathcal{G}\to\operatorname{Im} v\to 0$ shows that $(\mathcal{F}+\mathcal{G})(U)$ is a quotient of $\mathcal{F}(U)\oplus\mathcal{G}(U)$, hence finitely generated. By Prop. 1.11 both are coherent. (If $\mathcal{H}$ is coherent this is also Prop. 1.12(e).) □

Exercise 1.9

> **Problem.** Let $\mathcal{F}$ be a coherent sheaf on a locally Noetherian scheme X. (a) Show that the support $\operatorname{Supp}\mathcal{F}$ of $\mathcal{F}$ (Exercise 2.2.5) is closed. (b) Show that the multiplication by scalars defines a canonical homomorphism of $\mathcal{O}_X$-modules $\alpha : \mathcal{O}_X \to \mathcal{H}\mathrm{om}_{\mathcal{O}_X}(\mathcal{F},\mathcal{F})$. Show that $\operatorname{Ker}\alpha$ is quasi-coherent. We denote it by $\operatorname{Ann}\mathcal{F}$. This sheaf is called the annihilator of $\mathcal{F}$. (c) Show that for any affine open subset U of X, we have $(\operatorname{Ann}\mathcal{F})(U)=\{a\in\mathcal{O}_X(U) \mid a\mathcal{F}(U)=0\}$. (d) Show that $\operatorname{Supp}\mathcal{F}$ coincides with the closed subset $V(\operatorname{Ann}\mathcal{F})$. Show that $\mathcal{F}$ is the direct image of a coherent sheaf on the closed subscheme $V(\operatorname{Ann}\mathcal{F})$ under the closed immersion $V(\operatorname{Ann}\mathcal{F})\to X$.

Solution. Here $\operatorname{Supp}\mathcal{F}=\{x\in X : \mathcal{F}_x\neq 0\}$ and, for a quasi-coherent ideal $\mathcal{J}$, $V(\mathcal{J})=\{x : \mathcal{J}_x\neq\mathcal{O}_{X,x}\}$.

A preliminary algebraic fact. *Let A be a ring, M a finitely generated A-module, $\mathfrak{p}$ a prime. Then $M_\mathfrak{p}\neq 0$ iff $\operatorname{Ann}_A(M)\subseteq\mathfrak{p}$; and $\operatorname{Ann}_{A_\mathfrak{p}}(M_\mathfrak{p})=\operatorname{Ann}_A(M)_\mathfrak{p}$.* Let $m_1,\dots,m_r$ generate M. If $a/s\in\operatorname{Ann}(M_\mathfrak{p})$, then $t_i a\, m_i=0$ for some $t_i\notin\mathfrak{p}$; with $t=\prod t_i$, $ta\in\operatorname{Ann}_A M$ and $a/s=ta/(ts)\in\operatorname{Ann}(M)_\mathfrak{p}$. The other inclusion is clear. Now $M_\mathfrak{p}=0$ iff $\operatorname{Ann}(M_\mathfrak{p})=A_\mathfrak{p}$ iff $\operatorname{Ann}(M)_\mathfrak{p}=A_\mathfrak{p}$ iff $\operatorname{Ann}(M)\not\subseteq\mathfrak{p}$.

(a) Let $U=\operatorname{Spec}A$ be an affine open subset with A Noetherian. Then $\mathcal{F}|_U=\widetilde{M}$ with $M=\mathcal{F}(U)$ finitely generated (Prop. 1.11). By the fact above, $\operatorname{Supp}\mathcal{F}\cap U=\{\mathfrak{p} : M_\mathfrak{p}\neq 0\}=V(\operatorname{Ann}_A M)$, closed in U. Cover X by such U; a subset whose intersection with each member of an open cover is closed is closed. So $\operatorname{Supp}\mathcal{F}$ is closed.

(b) For an open set U and $a\in\mathcal{O}_X(U)$ let $\alpha_U(a)\in\mathcal{H}\mathrm{om}(\mathcal{F},\mathcal{F})(U)$ be the endomorphism $s\mapsto as$ of $\mathcal{F}|_U$. It is an $\mathcal{O}_X|_U$-linear map by commutativity of $\mathcal{O}_X(W)$, and $a\mapsto\alpha_U(a)$ is $\mathcal{O}_X(U)$-linear and compatible with restrictions. So $\alpha:\mathcal{O}_X\to\mathcal{H}\mathrm{om}(\mathcal{F},\mathcal{F})$ is a homomorphism of $\mathcal{O}_X$-modules. By Ex. 1.6(b), $\mathcal{H}\mathrm{om}(\mathcal{F},\mathcal{F})$ is coherent, so α is a morphism of quasi-coherent sheaves and $\operatorname{Ker}\alpha$ is quasi-coherent (indeed coherent) by Prop. 1.12(c),(e).

(c) By definition $(\operatorname{Ann}\mathcal{F})(U) = \operatorname{Ker}\alpha_U = \{a \in \mathcal{O}_X(U) : a \cdot s = 0 \text{ for all } W \subseteq U \text{ and } s \in \mathcal{F}(W)\}$. If a is in this set then $a\mathcal{F}(U) = 0$. Conversely let $U = \operatorname{Spec} A$ be affine, $M = \mathcal{F}(U)$ and $aM = 0$. For $f \in A$, $\mathcal{F}(D(f)) = M_f$ and $aM_f = 0$. So a kills all sections over principal opens; since a section over any open set is locally a section over a principal open, $a \cdot s = 0$ for all sections, i.e. $\alpha_U(a) = 0$.

(d) Let $U = \operatorname{Spec} A$ be affine, $M = \mathcal{F}(U)$, $I = \operatorname{Ann}_A M$. By (c), applied on every $D(f)$, $(\operatorname{Ann}\mathcal{F})(D(f)) = \operatorname{Ann}_{A_f}(M_f)$, which equals I_f by the same argument as in the preliminary fact (M_f is finitely generated over A_f). Since $\operatorname{Ann}\mathcal{F}$ is quasi-coherent, $(\operatorname{Ann}\mathcal{F})|_U = \widetilde{I}$ (the two sheaves have the same sections on principal opens). Then for $\mathfrak{p} \in U$: $\mathfrak{p} \in V(\operatorname{Ann}\mathcal{F})$ iff $I_\mathfrak{p} \neq A_\mathfrak{p}$ iff $I \subseteq \mathfrak{p}$ iff $M_\mathfrak{p} \neq 0$ iff $\mathfrak{p} \in \operatorname{Supp}\mathcal{F}$ (the preliminary fact). As such U cover X, $\operatorname{Supp}\mathcal{F} = V(\operatorname{Ann}\mathcal{F})$.

Now let $\mathcal{J} = \operatorname{Ann}\mathcal{F}$, a quasi-coherent (in fact coherent) sheaf of ideals, $Z = V(\mathcal{J})$ with its closed subscheme structure (Prop. 1.15, Rem. 1.16), $i : Z \to X$ the closed immersion; recall $\mathcal{O}_Z = i^{-1}(\mathcal{O}_X/\mathcal{J})$, i.e. $i_*\mathcal{O}_Z = \mathcal{O}_X/\mathcal{J}$. For every open U, $\mathcal{J}(U) = \operatorname{Ker}\alpha_U$ annihilates $\mathcal{F}(U)$, so $\mathcal{F}$ is a module over the presheaf of rings $U \mapsto \mathcal{O}_X(U)/\mathcal{J}(U)$, hence over its sheafification $\mathcal{O}_X/\mathcal{J}$. Then $\mathcal{G} := i^{-1}\mathcal{F}$ is a module over $i^{-1}(\mathcal{O}_X/\mathcal{J}) = \mathcal{O}_Z$. We have $i_*\mathcal{G} = i_*i^{-1}\mathcal{F} \simeq \mathcal{F}$ through the canonical map $\mathcal{F} \to i_*i^{-1}\mathcal{F}$: on stalks, at $x \in Z$ both sides are $\mathcal{F}_x$ (as i is a homeomorphism onto the closed set Z, $(i_*\mathcal{G})_x = \mathcal{G}_x = \mathcal{F}_x$ as in Ex. 1.1(a)), and at $x \notin Z$ both are 0 ($\mathcal{F}_x = 0$ since $x \notin \operatorname{Supp}\mathcal{F} = Z$, and $(i_*\mathcal{G})_x = 0$ because x has a neighborhood missing the closed set Z). The $\mathcal{O}_X$-module structures agree by construction.

$\mathcal{G}$ is coherent on Z: this is local, so take $U = \operatorname{Spec} A \subseteq X$ affine with A Noetherian, $\mathcal{F}|_U = \widetilde{M}$, $I = \operatorname{Ann}_A M$. Then $Z \cap U = \operatorname{Spec}(A/I)$ and M is an A/I-module, finitely generated. Let $\mathcal{N} = \widetilde{M}$ on $\operatorname{Spec} A/I$ (coherent, A/I Noetherian, Prop. 1.11). We have $i_*\mathcal{N}(D(f)) = \mathcal{N}(D(\bar{f})) = M_{\bar{f}} = M_f$, so $i_*\mathcal{N} \simeq \mathcal{F}|_U$, and $\mathcal{G}|_{Z\cap U} = i^{-1}\mathcal{F}|_U \simeq i^{-1}i_*\mathcal{N} \simeq \mathcal{N}$ (again by stalks: $(i^{-1}i_*\mathcal{N})_z = (i_*\mathcal{N})_{i(z)} = \mathcal{N}_z$). Hence $\mathcal{G}$ is coherent and $\mathcal{F} = i_*\mathcal{G}$. □

Exercise 1.10

> **Problem.** Let X be a scheme. (a) Let $\mathcal{N}$ be the sheaf of nilpotent elements of $\mathcal{O}_X$ (defined in the proof of Proposition 2.4.2). Show that it is quasi-coherent. (b) Let us suppose X locally Noetherian. Show that the set of $x \in X$ such that $\mathcal{O}_{X,x}$ is reduced is open in X. (c) Let $\mathcal{I}$ be a quasi-coherent sheaf of ideals of $\mathcal{O}_X$. Show that $U \mapsto \sqrt{\mathcal{I}(U)}$, for every affine open U, defines a quasi-coherent sheaf of ideals of $\mathcal{O}_X$.

Solution. (a) Here $\mathcal{N}(V) = \{s \in \mathcal{O}_X(V) : s_x \text{ is nilpotent in } \mathcal{O}_{X,x} \text{ for all } x \in V\}$, a sheaf of ideals with stalk $\mathcal{N}_x = \operatorname{Nil}(\mathcal{O}_{X,x})$. Let $U = \operatorname{Spec} A$ be affine and $f \in A$. We claim $\mathcal{N}(D(f)) = \operatorname{Nil}(A_f) = \operatorname{Nil}(A)_f$. If $s \in \operatorname{Nil}(A_f)$ then every germ is nilpotent. Conversely if all germs $s_\mathfrak{p}$ ($\mathfrak{p} \in D(f)$, primes of A_f) are nilpotent, then $s_\mathfrak{p} \in \mathfrak{p}A_\mathfrak{p}$, so $s \in \mathfrak{p}$ for all primes $\mathfrak{p}$ of A_f, so $s \in \operatorname{Nil}(A_f)$. Finally $\operatorname{Nil}(A_f) = \operatorname{Nil}(A)_f$: if $(a/f^k)^n = 0$ then $f^m a^n = 0$ for some m, so for $N \geq n, m$ we get $(fa)^N = f^{N-m}a^{N-n} \cdot f^m a^n = 0$; hence $a/f^k = fa/f^{k+1} \in \operatorname{Nil}(A)_f$. The other inclusion is clear. These identifications are compatible with restriction (they are inclusions in $A_f \subseteq$ ambient localizations), so $\mathcal{N}|_U \simeq \operatorname{Nil}(A)^\sim$ (a morphism of sheaves given on principal opens), which is quasi-coherent by Prop. 1.5(c). Since affine opens cover X, $\mathcal{N}$ is quasi-coherent.

(b) By (a), $\mathcal{N} \subseteq \mathcal{O}_X$ is a quasi-coherent sheaf of ideals, hence coherent by Ex. 1.7 (X locally Noetherian). By Ex. 1.9(a), $\operatorname{Supp}\mathcal{N}$ is closed. Now $\mathcal{O}_{X,x}$ is reduced iff $\mathcal{N}_x = \operatorname{Nil}(\mathcal{O}_{X,x}) = 0$ iff $x \notin \operatorname{Supp}\mathcal{N}$. So the set of x with $\mathcal{O}_{X,x}$ reduced is $X \setminus \operatorname{Supp}\mathcal{N}$, which is open.

(c) Define $\mathcal{J} \subseteq \mathcal{O}_X$ by $\mathcal{J}(V) = \{s \in \mathcal{O}_X(V) : s_x \in \sqrt{\mathcal{I}_x} \text{ for all } x \in V\}$, a sheaf of ideals (the condition is local on V). Let $U = \operatorname{Spec} A$ be affine, $I_0 = \mathcal{I}(U) \subseteq A$. Since $\mathcal{I}$ is quasi-coherent, $\mathcal{I}|_U = \widetilde{I_0}$, so $\mathcal{I}(D(f)) = I_0 A_f$ and $\mathcal{I}_\mathfrak{p} = I_0 A_\mathfrak{p}$. Claim: $\mathcal{J}(D(f)) = \sqrt{I_0 A_f}$. If $s^n \in I_0 A_f$, then $s^n_\mathfrak{p} \in I_0 A_\mathfrak{p}$ for all $\mathfrak{p} \in D(f)$, so $s \in \mathcal{J}(D(f))$. Conversely if $s \notin \sqrt{I_0 A_f}$, there is a prime $\mathfrak{q}$ of A_f containing $I_0 A_f$ with $s \notin \mathfrak{q}$; then $\sqrt{I_0 A_\mathfrak{q}} \subseteq \mathfrak{q} A_\mathfrak{q} \not\ni s_\mathfrak{q}$, so $s \notin \mathcal{J}(D(f))$. Next, $\sqrt{I_0 A_f} = (\sqrt{I_0})_f$: if $(a/f^k)^n \in I_0 A_f$, then $f^m a^n \in I_0$ for some m, hence for $N \geq n, m$, $(fa)^N = f^{N-m} a^{N-n} \cdot f^m a^n \in I_0$, so $fa \in \sqrt{I_0}$ and $a/f^k = fa/f^{k+1} \in (\sqrt{I_0})_f$; the converse is clear. So $\mathcal{J}|_U \simeq (\sqrt{I_0})^\sim$ is quasi-coherent, and $\mathcal{J}(U) = \sqrt{I_0} = \sqrt{\mathcal{I}(U)}$. Thus $\mathcal{J}$ is a quasi-coherent sheaf of ideals with $\mathcal{J}(U) = \sqrt{\mathcal{I}(U)}$ for every affine open U. It is the only subsheaf of $\mathcal{O}_X$ with this property, because a subsheaf of $\mathcal{O}_X$ is determined by its sections over the affine opens (they form a base). □

Exercise 1.11

Problem. Let X be a locally Noetherian scheme, $\varphi : \mathcal{F} \to \mathcal{G}$ a homomorphism of coherent sheaves. Show that $\{x \in X \mid \varphi_x \text{ is an isomorphism}\}$ is an open subset of X.

Solution. By Prop. 1.12(e), $\mathcal{K} = \operatorname{Ker} \varphi$ and $\mathcal{C} = \operatorname{Coker} \varphi$ are coherent. Taking stalks is exact, so $(\operatorname{Ker} \varphi)_x = \operatorname{Ker} \varphi_x$ and $(\operatorname{Coker} \varphi)_x = \operatorname{Coker} \varphi_x$. Hence φ_x is an isomorphism iff $\mathcal{K}_x = 0$ and $\mathcal{C}_x = 0$, i.e. iff $x \notin \operatorname{Supp} \mathcal{K} \cup \operatorname{Supp} \mathcal{C}$. By Ex. 1.9(a) both supports are closed. So the set in question is the complement of a closed set, hence open. □

Conventions for Exercises 1.12–1.22. Chapter and section numbers of the book are cited as in the text ("Prop. 1.14" is Prop. 1.14 of Section 5.1; "Lemma 1.2.7" is Nakayama's lemma, Chapter 1, Section 2). Stalks are written $\mathcal{F}_x$, $k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$. Several solutions use the following standard facts. *Fact A.* If N is a finitely generated A-module, $\mathfrak{p}$ a prime and $N_\mathfrak{p} = 0$, then $N_f = 0$ for some $f \notin \mathfrak{p}$ (each of finitely many generators is killed by an element outside $\mathfrak{p}$; take f to be their product). *Fact B.* A coherent sheaf is quasi-coherent: if $\mathcal{F}$ is coherent and $x \in X$, there are a neighbourhood U and a surjection $\mathcal{O}_X^n|_U \to \mathcal{F}|_U$; its kernel is finitely generated, so after shrinking U there is a surjection $\mathcal{O}_X^m|_U \to \operatorname{Ker}$, and $\mathcal{O}_X^m|_U \to \mathcal{O}_X^n|_U \to \mathcal{F}|_U \to 0$ is exact (Def. 1.4). Hence, if X is locally Noetherian and $U = \operatorname{Spec} A$ is an affine open, a coherent $\mathcal{F}$ satisfies $\mathcal{F}|_U = \widetilde{M}$ with $M = \mathcal{F}(U)$ a finitely generated A-module (Thm. 1.7, Prop. 1.11), and A is Noetherian. Finally, a morphism of sheaves is an isomorphism (resp. surjective) iff it is so on all stalks.

Exercise 1.12

Problem. Let X be a scheme. A sheaf $\mathcal{F}$ on X is *free of rank n* if $\mathcal{F} \simeq \mathcal{O}_X^n$. It is *locally free of rank n* if there is a covering $\{X_i\}_i$ of X such that $\mathcal{F}|_{X_i}$ is free of rank n for all i, and *locally free of finite rank* if there are a covering $\{X_i\}$ and integers n_i with $\mathcal{F}|_{X_i}$ free of rank n_i. Suppose X locally Noetherian.

(a) Let $\mathcal{F}$ be a coherent sheaf on X. Show that if $\mathcal{F}_x$ is free of rank n over $\mathcal{O}_{X,x}$, then there is an open neighbourhood U of x such that $\mathcal{F}|_U$ is free of rank n. Deduce that $\mathcal{F}$ is locally free of rank n if and only if $\mathcal{F}_x$ is free of rank n over $\mathcal{O}_{X,x}$ for every $x \in X$. In the remainder of the exercise $\mathcal{F}$ satisfies this condition.

(b) Show that $\mathcal{F}^\vee := \operatorname{Hom}_{\mathcal{O}_X}(\mathcal{F}, \mathcal{O}_X)$ is locally free of rank n.

(c) Show that the canonical homomorphism $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{F}^\vee \to \mathcal{O}_X$ is an isomorphism if $n = 1$.

(d) Let $\mathrm{Pic}(X)$ be the set of isomorphism classes of invertible sheaves on X. Show that the tensor product makes $\mathrm{Pic}(X)$ a commutative group with unit element the class of $\mathcal{O}_X$, and that the inverse of the class of $\mathcal{L}$ is the class of $\mathcal{L}^\vee$.

Solution. (a) Let $x \in X$ and choose an affine open neighbourhood $V = \operatorname{Spec} A$ of x, with A Noetherian. By Fact B, $\mathcal{F}|_V = \widetilde{M}$ with M finitely generated; let $\mathfrak{p} \subset A$ be the prime of x. Since $M_\mathfrak{p} = \mathcal{F}_x$ is free of rank n, there are $m_1, \dots, m_n \in M$ whose images form a basis of $M_\mathfrak{p}$ (a basis consists of fractions m/a, $a \notin \mathfrak{p}$, and m/a and $m/1$ differ by a unit). Define $\varphi : A^n \to M$, $e_i \mapsto m_i$. Then $\varphi_\mathfrak{p}$ is an isomorphism. The kernel K and cokernel C of φ are finitely generated (A is Noetherian) and $K_\mathfrak{p} = C_\mathfrak{p} = 0$, so by Fact A there is $f \notin \mathfrak{p}$ with $K_f = C_f = 0$. Hence $\varphi_f : A_f^n \to M_f$ is an isomorphism, and by Prop. 1.5(b) the induced map $\mathcal{O}_X^n|_U \to \mathcal{F}|_U$ on $U = D(f) = \operatorname{Spec} A_f$ is an isomorphism. So $\mathcal{F}|_U$ is free of rank n.

If $\mathcal{F}$ is locally free of rank n, then $\mathcal{F}_x \simeq \mathcal{O}_{X,x}^n$ for every x (take the stalk of $\mathcal{F}|_{X_i} \simeq \mathcal{O}_{X_i}^n$ at $x \in X_i$). Conversely, if every $\mathcal{F}_x$ is free of rank n, the first part gives for every x an open $U_x \ni x$ with $\mathcal{F}|_{U_x}$ free of rank n; these U_x cover X.

(b) The question is local, and $\mathcal{F}^\vee|_U = \mathrm{Hom}_{\mathcal{O}_U}(\mathcal{F}|_U, \mathcal{O}_U)$ by definition of the sheaf Hom (Ex. 1.5(a)); it depends only on $\mathcal{F}|_U$ up to isomorphism. So let $\mathcal{F}|_U \simeq \mathcal{O}_U^n$. For every open $V \subseteq U$, a homomorphism $\mathcal{O}_V^n \to \mathcal{O}_V$ is determined by the images $a_1, \dots, a_n \in \mathcal{O}_X(V)$ of the basis vectors, and any (a_i) occurs. Thus $\mathrm{Hom}(\mathcal{O}_V^n, \mathcal{O}_V) \simeq \mathcal{O}_X(V)^n$, compatibly with restriction and $\mathcal{O}_X(V)$-module structure, i.e. $\mathcal{F}^\vee|_U \simeq \mathcal{O}_U^n$.

(c) The canonical map $\mathrm{ev} : \mathcal{F} \otimes \mathcal{F}^\vee \to \mathcal{O}_X$ is the one induced on the presheaf $V \mapsto \mathcal{F}(V) \otimes_{\mathcal{O}_X(V)} \mathcal{F}^\vee(V)$ by $u \otimes \psi \mapsto \psi(u)$, followed by sheafification. Being an isomorphism is local. Let $\mathcal{F}|_U = \mathcal{O}_U e$ with e a generator (so $\mathcal{F}|_U \simeq \mathcal{O}_U$). By (b), $\mathcal{F}^\vee|_U = \mathcal{O}_U e^*$, where $e^*(e) = 1$ and e^* is a generator. Then $\mathcal{F} \otimes \mathcal{F}^\vee|_U \simeq \mathcal{O}_U\,(e \otimes e^*)$ is free of rank 1 with generator $e \otimes e^*$, and $\mathrm{ev}(e \otimes e^*) = e^*(e) = 1$. Hence $\mathrm{ev}\,|_U$ sends a generator of a free rank-one module to a generator of $\mathcal{O}_U$, so it is an isomorphism.

(d) Here "invertible" means Def. 1.21: $\mathcal{L}|_U \simeq \mathcal{O}_U$ locally. This is exactly "locally free of rank 1" in the sense of the exercise. (For locally Noetherian X, Def. 1.21 also records the equivalent stalk condition: $\mathcal{L}$ coherent with $\mathcal{L}_x$ free of rank 1 for all x; this follows from (a) and from coherence of $\mathcal{O}_X$, Prop. 1.11.) The tensor product of two invertible sheaves is invertible: $(\mathcal{L} \otimes \mathcal{M})|_U \simeq \mathcal{O}_U \otimes \mathcal{O}_U \simeq \mathcal{O}_U$ on a common trivializing open. Hence $\otimes$ induces a product on isomorphism classes ($\mathcal{L} \simeq \mathcal{L}'$, $\mathcal{M} \simeq \mathcal{M}'$ implies $\mathcal{L} \otimes \mathcal{M} \simeq \mathcal{L}' \otimes \mathcal{M}'$). It is associative and commutative because of the canonical isomorphisms $(\mathcal{L} \otimes \mathcal{M}) \otimes \mathcal{N} \simeq \mathcal{L} \otimes (\mathcal{M} \otimes \mathcal{N})$ and $\mathcal{L} \otimes \mathcal{M} \simeq \mathcal{M} \otimes \mathcal{L}$ of the tensor product of sheaves of modules. The class of $\mathcal{O}_X$ is a unit because $\mathcal{O}_X \otimes \mathcal{L} \simeq \mathcal{L}$. For an invertible $\mathcal{L}$, $\mathcal{L}^\vee$ is invertible by (b) (with $n = 1$), and $\mathcal{L} \otimes \mathcal{L}^\vee \simeq \mathcal{O}_X$ by (c). So $[\mathcal{L}^\vee]$ is the inverse of $[\mathcal{L}]$ (and inverses are unique in a monoid). □

Exercise 1.13

Problem. Let X be a locally Noetherian scheme and $\varphi : \mathcal{F} \to \mathcal{G}$ a surjective morphism between two locally free sheaves of equal (finite) rank. Show that φ is an isomorphism (use Prop. 1.2.6).

Solution. "Equal rank" is read as: for every x, $\mathcal{F}_x \simeq \mathcal{O}_{X,x}^n \simeq \mathcal{G}_x$ with the same $n = n(x)$. (This is automatic if both sheaves are locally free of the same constant rank n.) The Noetherian hypothesis is not needed.

It suffices to show that every stalk map $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is injective, since φ is already surjective. Put $R = \mathcal{O}_{X,x}$, with maximal ideal $\mathfrak{m}$ and residue field k. Then $\varphi_x : R^n \to R^n$ is a surjective R-linear map; let K be its kernel. As the target R^n is free, the exact sequence $0 \to K \to R^n \to R^n \to 0$ splits, so $R^n \simeq K \oplus R^n$, and K, a quotient of R^n, is finitely generated. Tensor with k: since the sequence is exact with flat (free) third term, it stays exact by Prop. 1.2.6, giving $0 \to K \otimes k \to k^n \to k^n \to 0$. Comparing dimensions, $K \otimes_R k = 0$, i.e. $K = \mathfrak{m}K$. By Nakayama's lemma (Lemma 1.2.7) $K = 0$. Hence φ_x is bijective for all x, and φ is an isomorphism. □

Exercise 1.14

Problem. Let X be an integral scheme and $\mathcal{F}$ a quasi-coherent sheaf on X.

(a) Let $\mathcal{F}(U)_{\text{tors}}$ be the set of $x \in \mathcal{F}(U)$ such that $ax = 0$ for some non-zero $a \in \mathcal{O}_X(U)$. Show that it is a submodule of $\mathcal{F}(U)$.

(b) Show that there exists a unique subsheaf $\mathcal{F}_{\text{tors}}$ of $\mathcal{F}$ such that $\mathcal{F}_{\text{tors}}(U) = \mathcal{F}(U)_{\text{tors}}$ for every affine open subset U of X. Show that $\mathcal{F}_{\text{tors}}$ is quasi-coherent.

(c) $\mathcal{F}$ is *torsion-free* if $\mathcal{F}_{\text{tors}} = 0$. Show that $\mathcal{F}/\mathcal{F}_{\text{tors}}$ is always torsion-free.

(d) Suppose moreover that X is locally Noetherian and that $\mathcal{F}$ is coherent and non-zero. Show that $\operatorname{Supp}\mathcal{F} = X$ if and only if $\mathcal{F} \neq \mathcal{F}_{\text{tors}}$. Show that there exist $n \geq 0$ and a non-empty open subset U of X such that $\mathcal{O}_X^n|_U \simeq \mathcal{F}|_U$ (by convention $\mathcal{O}_X^n = \mathcal{O}_X$ if $n = 0$).

Solution. Recall that on an integral scheme $\mathcal{O}_X(U)$ is an integral domain for every non-empty open U, and for non-empty opens $V \subseteq U$ the restriction $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$ is injective. Hence restrictions of non-zero elements are non-zero. For $U = \emptyset$ we set $\mathcal{F}(\emptyset)_{\text{tors}} = 0 = \mathcal{F}(\emptyset)$.

(a) Let $U \neq \emptyset$. If $ax = 0$, $by = 0$ with $a, b \neq 0$, then $ab \neq 0$ (domain) and $ab(x + y) = 0$; and $a(cx) = c(ax) = 0$ for all $c \in \mathcal{O}_X(U)$. Also $0 \in \mathcal{F}(U)_{\text{tors}}$ (take $a = 1$). So $\mathcal{F}(U)_{\text{tors}}$ is a submodule. Moreover, if $x \in \mathcal{F}(U)_{\text{tors}}$ and $V \subseteq U$ is non-empty open, then $x|_V \in \mathcal{F}(V)_{\text{tors}}$ (use $a|_V \neq 0$).

(b) *A local-to-global remark.* Let $V = \operatorname{Spec} R$ be affine, $s \in \mathcal{F}(V)$, and suppose V is covered by opens V_l with $s|_{V_l}$ torsion. Then s is torsion. Indeed, refining we may take $V_l = D(g_l)$ principal, finitely many (as V is quasi-compact), non-empty. If $a_l s|_{D(g_l)} = 0$ with $0 \neq a_l \in R_{g_l}$, write $a_l = b_l/g_l^N$ with $b_l \in R$; $b_l \neq 0$ since $R \to R_{g_l}$ is injective for the domain R and $a_l \neq 0$. Then $b_l s|_{D(g_l)} = 0$, and $b = \prod_l b_l \neq 0$ satisfies $bs|_{D(g_l)} = 0$ for all l, so $bs = 0$.

Definition and uniqueness. For any open U set $\mathcal{F}_{\text{tors}}(U) = \{s \in \mathcal{F}(U) : s|_V \in \mathcal{F}(V)_{\text{tors}}$ for all affine open $V \subseteq U\}$. By (a) this is a submodule; it is stable under restriction, and it is a sheaf: if $s \in \mathcal{F}(U)$ and $s|_{U_i} \in \mathcal{F}_{\text{tors}}(U_i)$ for a cover $\{U_i\}$, then for an affine $V \subseteq U$ the sets $V \cap U_i$ are covered by affine opens V' with $s|_{V'}$ torsion, so $s|_V$ is torsion by the remark; and the sheaf axioms are inherited from $\mathcal{F}$. If U is affine, then $\mathcal{F}_{\text{tors}}(U) = \mathcal{F}(U)_{\text{tors}}$: "$\subseteq$" take $V = U$; "$\supseteq$" follows from the last sentence of (a). Conversely, if $\mathcal{H} \subseteq \mathcal{F}$ is any subsheaf with $\mathcal{H}(V) = \mathcal{F}(V)_{\text{tors}}$ for all affine V, then for any open U, $\mathcal{H}(U) = \{s \in \mathcal{F}(U) : s|_V \in \mathcal{H}(V)\ \forall$ affine $V \subseteq U\}$ (affines cover U and $\mathcal{H}$ is a subsheaf), which is $\mathcal{F}_{\text{tors}}(U)$. This proves existence and uniqueness.

Quasi-coherence. Let $U = \operatorname{Spec} A$ be affine, $M = \mathcal{F}(U)$, so $\mathcal{F}|_U = \widetilde{M}$ (Thm. 1.7), and $N = M_{\text{tors}}$. For $0 \neq g \in A$ we claim $(M_g)_{\text{tors}} = N_g$ inside M_g. The inclusion $\supseteq$ is clear ($am = 0$ gives $(a/1)(m/g^k) = 0$, $a/1 \neq 0$). For $\subseteq$: if $(a/g^j)(m/g^k) = 0$ with $a/g^j \neq 0$, then $a \neq 0$ and $g^N am = 0$ for some N; $g^N a \neq 0$ since A is a domain, so $m \in N$. Now $\widetilde{N} \subseteq \widetilde{M} = \mathcal{F}|_U$ is a subsheaf (Prop. 1.5(b)) with $\widetilde{N}(D(g)) = N_g$, and $\mathcal{F}_{\text{tors}}(D(g)) = (M_g)_{\text{tors}} = N_g$ (principal opens are affine). Two subsheaves of $\mathcal{F}|_U$ having the same sections on all principal opens coincide (principal opens form a base). Hence $\mathcal{F}_{\text{tors}}|_U = \widetilde{N}$, which is quasi-coherent (Prop. 1.5(c)); as quasi-coherence is local, $\mathcal{F}_{\text{tors}}$ is quasi-coherent.

(c) $\mathcal{Q} = \mathcal{F}/\mathcal{F}_{\text{tors}}$ is quasi-coherent by Prop. 1.12(c). For $U = \operatorname{Spec} A$ affine, with M, N as above, Prop. 1.5(b) gives $\mathcal{Q}|_U = (M/N)^{\sim}$ and $\mathcal{Q}(U) = M/N$. If $a\bar{m} = 0$ in M/N with $a \neq 0$, then $am \in N$, so $bam = 0$ for some $b \neq 0$; as $ba \neq 0$ we get $m \in N$. So $\mathcal{Q}(U)_{\text{tors}} = 0$ for every affine U, whence $\mathcal{Q}_{\text{tors}} = 0$ (a subsheaf vanishing on all affine opens vanishes).

(d) Let ξ be the generic point. *Claim 1:* $\operatorname{Supp} \mathcal{F} = X$ *iff* $\mathcal{F}_\xi \neq 0$. Only "$\Leftarrow$" needs proof. Let $y \in X$; ξ generizes y, so in an affine neighbourhood $\operatorname{Spec} A$ of y the point ξ is the prime (0) and $y = \mathfrak{q}$, and $M_{(0)}$ is a localization of $M_{\mathfrak{q}}$. If $M_{\mathfrak{q}} = 0$ then $M_{(0)} = 0$. Hence $\mathcal{F}_\xi \neq 0$ implies $\mathcal{F}_y \neq 0$ for all y.

Claim 2: $\mathcal{F}_\xi \neq 0$ *iff* $\mathcal{F} \neq \mathcal{F}_{\text{tors}}$. Let $U = \operatorname{Spec} A \neq \emptyset$ be affine, $M = \mathcal{F}(U)$, $N = M_{\text{tors}}$, $K = \operatorname{Frac} A$. Then $\mathcal{F}_\xi = M \otimes_A K$. Since $N \otimes K = 0$ (every element of N is killed by a non-zero element), exactness of $\otimes K$ gives $M \otimes K = (M/N) \otimes K$; and M/N is torsion-free (proof of (c)), so it embeds in $(M/N) \otimes K$. Hence $\mathcal{F}_\xi \neq 0$ iff $M \neq N$. Now $\mathcal{F} = \mathcal{F}_{\text{tors}}$ iff $M = N$ for every affine U, iff $\mathcal{F}_\xi = 0$ (the condition $M = N$ for one non-empty affine U is equivalent to $\mathcal{F}_\xi = 0$, which does not depend on U). This proves Claim 2, and with Claim 1 the first assertion.

Free on an open set. Let $U = \operatorname{Spec} A$ be a non-empty affine open, A a Noetherian domain, $M = \mathcal{F}(U)$ finitely generated (Fact B). The K-vector space $M \otimes K$ is finite-dimensional, say of dimension $n \geq 0$, and is spanned by the images of elements of M, so we can choose $m_1, \ldots, m_n \in M$ whose images form a basis. Let $\varphi : A^n \to M$, $e_i \mapsto m_i$. Then $\varphi \otimes K$ is an isomorphism, so $\operatorname{Ker} \varphi \otimes K = 0 = \operatorname{Coker} \varphi \otimes K$ (K is flat). These modules are finitely generated, so by Fact A (applied to the prime (0)) there is $0 \neq f \in A$ killing both. Then φ_f is an isomorphism, so $\mathcal{O}_X^n|_{D(f)} \simeq \mathcal{F}|_{D(f)}$ with $D(f) \neq \emptyset$ (as $f \neq 0$ in a domain).

Remark on the convention. If $\mathcal{F} \neq \mathcal{F}_{\text{tors}}$ then $\mathcal{F}_\xi \neq 0$, so $n \geq 1$ and the convention for $n = 0$ is irrelevant. If $\mathcal{F} = \mathcal{F}_{\text{tors}}$ then $n = 0$ and the proof gives $\mathcal{F}|_U = 0$. In that case the statement must be read with $\mathcal{O}_X^0 = 0$; with the book's convention $\mathcal{O}_X^0 = \mathcal{O}_X$ it would be false (e.g. $X = \mathbb{A}^1_k$, $\mathcal{F}$ the skyscraper sheaf $k(0)$ at the origin, which is coherent and torsion; $\mathcal{F}|_U = 0 \not\simeq \mathcal{O}_U$ for $U = X \setminus \{0\}$, and for any non-empty open U the restriction $\mathcal{F}|_U$ is 0 or a skyscraper, never $\simeq \mathcal{O}_U$). The assertion is proved as stated with $\mathcal{O}_X^0 = 0$. □

Exercise 1.15

Problem. Let $\mathcal{F}$ be a coherent sheaf on a locally Noetherian scheme X. For $x \in X$ set $\phi(x) = \dim_{k(x)}(\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} k(x))$.

(a) Let $n \in \mathbb{N}$. Using Nakayama's lemma, show that $\{x \in X \mid \phi(x) \leq n\}$ is open in X. In particular, if X is irreducible with generic point ξ, then $\phi(x) \geq \phi(\xi)$ for all $x \in X$.

(b) Let A be a ring and $\alpha : A^n \to M$ a homomorphism of A-modules. Suppose that for every $\mathfrak{p} \in \operatorname{Spec} A$ the homomorphism $\alpha \otimes \operatorname{Id}_{k(\mathfrak{p})} : k(\mathfrak{p})^n \to M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p})$ is injective. Show

that $\operatorname{Ker}\alpha \subseteq \mathcal{N}^n$, where $\mathcal{N}$ is the nilradical of A.

(c) Suppose ϕ is constant of value n on X and X is reduced. Show that $\mathcal{F}$ is locally free of rank n.

(d) Show by an example that (c) is false if X is not reduced.

Solution. (a) Let $x \in X$ with $\phi(x) = r \leq n$. Take an affine open $U = \operatorname{Spec} A \ni x$, A Noetherian, $\mathcal{F}|_U = \widetilde{M}$ with M finitely generated (Fact B), and let $\mathfrak{p}$ be the prime of x. The $k(\mathfrak{p})$-vector space $M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p}$ has dimension r; choose $m_1, \ldots, m_r \in M$ whose images form a basis. By Nakayama (Lemma 1.2.7, for the finitely generated $A_\mathfrak{p}$-module $M_\mathfrak{p}$ and the submodule generated by the m_i), $m_1, \ldots, m_r$ generate $M_\mathfrak{p}$. Let C be the cokernel of $A^r \to M$, $e_i \mapsto m_i$; it is finitely generated and $C_\mathfrak{p} = 0$, so $C_f = 0$ for some $f \notin \mathfrak{p}$ (Fact A). For $y = \mathfrak{q} \in D(f)$, $M_\mathfrak{q}$ is generated by the images of the m_i, so $\dim M_\mathfrak{q}/\mathfrak{q}M_\mathfrak{q} \leq r \leq n$. Hence $D(f) \subseteq \{\phi \leq n\}$ is a neighbourhood of x, and $\{\phi \leq n\}$ is open.

If X is irreducible with generic point ξ and $x \in X$, put $n = \phi(x)$. The open set $\{\phi \leq n\}$ contains x, hence is non-empty, hence contains ξ (the generic point lies in every non-empty open set). So $\phi(\xi) \leq n = \phi(x)$.

(b) Let $v = (a_1, \ldots, a_n) \in \operatorname{Ker}\alpha \subseteq A^n$ and $\mathfrak{p} \in \operatorname{Spec} A$. The image $\bar{v} \in k(\mathfrak{p})^n$ of v satisfies $(\alpha \otimes \mathrm{Id})(\bar{v}) = \overline{\alpha(v)} = 0$, so $\bar{v} = 0$ by injectivity, i.e. $a_i \in \mathfrak{p}A_\mathfrak{p} \cap A = \mathfrak{p}$ for all i. As $\mathfrak{p}$ was arbitrary, $a_i \in \bigcap_\mathfrak{p} \mathfrak{p} = \mathcal{N}$. (No finiteness hypothesis is needed.)

(c) Let $x \in X$ and $U = \operatorname{Spec} A$, M, $\mathfrak{p}$ as in (a), now with $r = \phi(x) = n$. As in (a) there are $m_1, \ldots, m_n \in M$ and $f \notin \mathfrak{p}$ such that $\alpha : A^n \to M$, $e_i \mapsto m_i$, becomes surjective after inverting f. Replace A, M by A_f, M_f; A_f is reduced because X is reduced. For every prime $\mathfrak{q}$ of A_f, the map $\alpha \otimes k(\mathfrak{q}) : k(\mathfrak{q})^n \to M_\mathfrak{q} \otimes k(\mathfrak{q})$ is surjective (right exactness of $\otimes$) between spaces of the same dimension $n = \phi(\mathfrak{q})$, hence injective. By (b), $\operatorname{Ker}\alpha \subseteq \mathcal{N}^n = 0$. So α is an isomorphism, i.e. $\mathcal{F}|_{D(f)} \simeq \mathcal{O}^n_{D(f)}$ (Prop. 1.5(b)). Thus $\mathcal{F}$ is locally free of rank n.

(d) Let k be a field, $A = k[\varepsilon]/(\varepsilon^2)$, $X = \operatorname{Spec} A$ (one point, not reduced), $M = A/\varepsilon A \simeq k$ and $\mathcal{F} = \widetilde{M}$. It is coherent (Prop. 1.11), ϕ is the constant $\dim_k M/\varepsilon M = 1$ on the one point of X, but $\mathcal{F}$ is not locally free: the only non-empty open set is X, and $M \not\simeq A^n$ for any n because $\varepsilon M = 0$ whereas $\varepsilon A^n \neq 0$ for $n \geq 1$, and $M \neq 0$. □

Exercise 1.16

Problem. Let $f : X \to S$ be a morphism of schemes, and consider a base change Here $X_T = X \times_S T$, $f_T : X_T \to T$ and $p : X_T \to X$ are the projections, $\pi : T \to S$ is the given morphism, and $f \circ p = \pi \circ f_T$. (a) Let $\mathcal{F}$ be an arbitrary $\mathcal{O}_X$-module. Show that there is a canonical homomorphism $\pi^* f_* \mathcal{F} \to (f_T)_* p^* \mathcal{F}$ (use Exercise 1.1).

(b) Suppose $T \to S$ is flat and $\mathcal{F}$ is quasi-coherent. Show that $\pi^* f_* \mathcal{F} \to (f_T)_* p^* \mathcal{F}$ is an isomorphism if X is Noetherian or if f is separated and quasi-compact (see also Prop. 3.1.24).

(c) Let X and f be as in (b), and let $\mathcal{G}$ be a locally free sheaf on S. Show that there is a canonical isomorphism $f_* \mathcal{F} \otimes_{\mathcal{O}_S} \mathcal{G} \simeq f_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^* \mathcal{G})$ (see also Prop. 2.32).

Solution. *The adjunction of Exercise 1.1(b).* For a morphism $g : Y \to Z$ of schemes, an $\mathcal{O}_Z$-module $\mathcal{G}$ and an $\mathcal{O}_Y$-module $\mathcal{H}$ there is a natural bijection $\operatorname{Hom}_{\mathcal{O}_Y}(g^* \mathcal{G}, \mathcal{H}) \simeq \operatorname{Hom}_{\mathcal{O}_Z}(\mathcal{G}, g_* \mathcal{H})$.

Indeed, $g^*\mathcal{G} = g^{-1}\mathcal{G} \otimes_{g^{-1}\mathcal{O}_Z} \mathcal{O}_Y$, so $\mathcal{O}_Y$-linear maps $g^*\mathcal{G} \to \mathcal{H}$ correspond to $g^{-1}\mathcal{O}_Z$-linear maps $g^{-1}\mathcal{G} \to \mathcal{H}$ (extension of scalars), and these correspond to maps of sheaves $\mathcal{G} \to g_*\mathcal{H}$ by the adjunction $g^{-1} \dashv g_*$ of sheaves of sets; linearity over $g^{-1}\mathcal{O}_Z$ translates into $\mathcal{O}_Z$-linearity. In particular there are unit maps $\eta : \mathcal{G} \to g_*g^*\mathcal{G}$, sending a section t to the section $g^*t := t \otimes 1$.

(a) Since $f \circ p = \pi \circ f_T$, we have $\pi_*(f_T)_*p^*\mathcal{F} = f_*p_*p^*\mathcal{F}$. Let $\theta : \pi^*f_*\mathcal{F} \to (f_T)_*p^*\mathcal{F}$ be the map adjoint (under $\pi^* \dashv \pi_*$) to the composite

$$f_*\mathcal{F} \xrightarrow{f_*(\eta_\mathcal{F})} f_*p_*p^*\mathcal{F} = \pi_*(f_T)_*p^*\mathcal{F}.$$

Unwinding: if $V \subseteq S$, $W \subseteq T$ are open with $\pi(W) \subseteq V$, $s \in \mathcal{F}(f^{-1}V)$ and $a \in \mathcal{O}_T(W)$, then θ sends $a \cdot \pi^*s|_W$ (the section of $\pi^*f_*\mathcal{F}$ over W attached to s) to $a \cdot (p^*s)|_{f_T^{-1}(W)} \in (p^*\mathcal{F})(f_T^{-1}W)$; note $f_T^{-1}(W) \subseteq p^{-1}f^{-1}(V)$.

(b) The statement is local on T, so we may restrict to an affine open $W = \operatorname{Spec} A' \subseteq T$ whose image lies in an affine open $V = \operatorname{Spec} A \subseteq S$. Replace X by $X_V = f^{-1}(V)$ (still Noetherian, resp. with f separated quasi-compact); then $f_T^{-1}(W) = X_V \times_V W$, which we call X_W. A' is flat over A, and so is every localization $A'_{g'}$ ($g' \in A'$).

Step 1: X_V has a finite affine cover $\{U_i\}$ such that each $U_i \cap U_j$ is covered by finitely many affine opens U_{ijk}. If X_V is Noetherian, every open subset is quasi-compact. If f is separated and quasi-compact, X_V is quasi-compact and $U_i \cap U_j$ is affine for affine U_i, U_j (separatedness over the affine V).

Step 2. Put $M = \mathcal{F}(X_V)$ and let $B_i = \mathcal{O}_X(U_i)$ etc. For any flat A-algebra A'' (we shall use $A'' = A'_{g'}$, including $g' = 1$) let $X'' = X_V \times_V \operatorname{Spec} A''$ and $p'' : X'' \to X_V$. For an affine open $U = \operatorname{Spec} B \subseteq X_V$, $p''^{-1}(U) = \operatorname{Spec}(B \otimes_A A'')$ is affine and by Prop. 1.14(b) $p''^*\mathcal{F}|_{p''^{-1}U} = (\mathcal{F}(U) \otimes_B (B \otimes_A A''))^\sim$, so $(p''^*\mathcal{F})(p''^{-1}U) = \mathcal{F}(U) \otimes_A A''$, the identification being the map $s \otimes a \mapsto a \cdot p''^*s$. The sets $p''^{-1}(U_i)$ cover X'' and $p''^{-1}(U_i) \cap p''^{-1}(U_j) = \bigcup_k p''^{-1}(U_{ijk})$. The sheaf axiom for $p''^*\mathcal{F}$ and for $\mathcal{F}$ gives two exact sequences connected by natural maps

$$0 \to M \otimes_A A'' \to \bigoplus_i \mathcal{F}(U_i) \otimes_A A'' \to \bigoplus_{i,j,k} \mathcal{F}(U_{ijk}) \otimes_A A'',$$
$$0 \to (p''^*\mathcal{F})(X'') \to \bigoplus_i (p''^*\mathcal{F})(p''^{-1}U_i) \to \bigoplus_{i,j,k} (p''^*\mathcal{F})(p''^{-1}U_{ijk}).$$

The first is obtained from $0 \to M \to \bigoplus \mathcal{F}(U_i) \to \bigoplus \mathcal{F}(U_{ijk})$ (exact by the sheaf axiom, $(s_i) \mapsto (s_i|{-}s_j|)$) by tensoring with the flat A-module A''. The second and third vertical maps are isomorphisms by the above, so by the four lemma the first vertical map $M \otimes_A A'' \to (p''^*\mathcal{F})(X'')$ is an isomorphism.

Step 3. Now $f_*\mathcal{F}|_V$ is quasi-coherent (Prop. 1.14(c), applied to $X_V \to V$), hence $= \widetilde{M}$ with $M = (f_*\mathcal{F})(V) = \mathcal{F}(X_V)$ (Thm. 1.7). By Prop. 1.14(b), $(\pi^*f_*\mathcal{F})|_W = (M \otimes_A A')^\sim$. On the other hand, for a principal open $D(g') \subseteq W$, $((f_T)_*p^*\mathcal{F})(D(g')) = (p^*\mathcal{F})(X_{D(g')})$, which by Step 2 for $A'' = A'_{g'}$ equals $M \otimes_A A'_{g'} = (M \otimes_A A')_{g'}$; by the description of θ in (a), θ is on $D(g')$ the identification of Step 2. Thus $\theta|_W$ is an isomorphism on all principal opens, which form a base of W, hence is an isomorphism of sheaves. (We did not need that $(f_T)_*p^*\mathcal{F}$ is quasi-coherent a priori; it follows.)

(c) *The map.* For $V \subseteq S$ open, $s \in \mathcal{F}(f^{-1}V) = (f_*\mathcal{F})(V)$ and $g \in \mathcal{G}(V)$, put $f^*g := \eta(g)|_{f^{-1}V}$ and consider $s \otimes f^*g \in (\mathcal{F} \otimes f^*\mathcal{G})(f^{-1}V)$. For $a \in \mathcal{O}_S(V)$, $(as) \otimes f^*g = s \otimes f^*(ag)$ (both

equal $f^{\#}(a)\,(s \otimes f^*g)$). So this defines a map of presheaves $V \mapsto (f_*\mathcal{F})(V) \otimes_{\mathcal{O}_S(V)} \mathcal{G}(V) \to f_*(\mathcal{F} \otimes f^*\mathcal{G})(V)$; as the target is a sheaf, it factors through the sheafification and gives $\mu : f_*\mathcal{F} \otimes \mathcal{G} \to f_*(\mathcal{F} \otimes f^*\mathcal{G})$. It is functorial in $\mathcal{G}$ and compatible with finite direct sums in $\mathcal{G}$, and for $\mathcal{G} = \mathcal{O}_S$ it is the identity of $f_*\mathcal{F}$ (as $f^*\mathcal{O}_S = \mathcal{O}_X$).

It is an isomorphism. This is local on S; on an open V with $\mathcal{G}|_V \simeq \mathcal{O}_V^n$, functoriality and additivity reduce $\mu|_V$ to the case $\mathcal{O}_V^n$, where it is the identity of $(f_*\mathcal{F}|_V)^n$ (both sides equal $(f_*\mathcal{F}|_V)^n$, because f^* and $\otimes$ commute with finite direct sums and f_* also does). Here "locally free" means locally free of finite rank, as in Ex. 1.12. The hypotheses on X, f and the quasi-coherence of $\mathcal{F}$ are therefore not needed for this statement; they are only needed for $f_*\mathcal{F}$ to be quasi-coherent, so that the formula stays inside the world of quasi-coherent sheaves (Prop. 1.14(c)).

Remark (locally free of infinite rank). If $\mathcal{G}|_V \simeq \mathcal{O}_V^{(I)}$ with I infinite, the hypotheses of (b) do give the result. They imply that $f^{-1}(W)$ is quasi-compact for every affine open $W \subseteq S$ (X Noetherian, resp. f quasi-compact). The sections of a direct sum of sheaves over a quasi-compact open set are the direct sum of the sections (see Ex. 1.18(a) below). Hence for affine $W \subseteq V$ both $(f_*\mathcal{F})^{(I)}(W)$ and $f_*(\mathcal{F}^{(I)})(W)$ equal $\mathcal{F}(f^{-1}W)^{(I)}$, and $\mu|_V$ is an isomorphism on a base of the topology. □

Exercise 1.17

> **Problem.** Let X be a scheme and $\mathcal{A}$ a sheaf of (commutative) $\mathcal{O}_X$-algebras. We say that $\mathcal{A}$ is *quasi-coherent* if it is quasi-coherent as an $\mathcal{O}_X$-module.
>
> (a) Suppose $\mathcal{A}$ quasi-coherent. Show that there exists a unique X-scheme $\pi : Z \to X$ with π affine and $\pi_*\mathcal{O}_Z = \mathcal{A}$. We denote Z by $\operatorname{Spec}\mathcal{A}$.
>
> (b) Show that $\operatorname{Spec}\mathcal{A} \to X$ is finite if and only if $\mathcal{A}$ is quasi-coherent and finitely generated as an $\mathcal{O}_X$-module.
>
> (c) Let $f : W \to X$ be an affine morphism. Show that $f_*\mathcal{O}_W$ is a quasi-coherent $\mathcal{O}_X$-algebra and that $W \simeq \operatorname{Spec} f_*\mathcal{O}_W$ as X-schemes.

Solution. Here "affine morphism" means: preimages of affine opens are affine. $\pi_*\mathcal{O}_Z = \mathcal{A}$ is understood as an isomorphism of sheaves of $\mathcal{O}_X$-algebras (the $\mathcal{O}_X$-algebra structure of $\pi_*\mathcal{O}_Z$ being given by $\pi^{\#}$), and uniqueness is up to unique isomorphism of X-schemes.

A preliminary fact. Let $\mathcal{A}$ be quasi-coherent and $V \subseteq U$ affine opens of X, $R = \mathcal{O}_X(U)$, $R' = \mathcal{O}_X(V)$. Then $\mathcal{A}(V) = \mathcal{A}(U) \otimes_R R'$ (via the restriction map). Indeed, apply Prop. 1.14(b) to the open immersion $j : V \to U$ and the quasi-coherent sheaf $\mathcal{G} = \mathcal{A}|_U = \mathcal{A}(U)^{\sim}$: it gives $j^*\mathcal{G} = (\mathcal{A}(U) \otimes_R R')^{\sim}$, and $j^*\mathcal{G} = \mathcal{G}|_V$ for an open immersion; taking sections over V gives the claim.

(a) *Uniqueness.* If $\pi : Z \to X$ is affine with $\pi_*\mathcal{O}_Z = \mathcal{A}$, then for every affine open $U \subseteq X$, $\pi^{-1}(U)$ is an affine scheme with ring $\mathcal{O}_Z(\pi^{-1}U) = \mathcal{A}(U)$, and $\pi| : \pi^{-1}U \to U$ corresponds to the structure map $\mathcal{O}_X(U) \to \mathcal{A}(U)$. Hence $\pi^{-1}(U) \simeq \operatorname{Spec}\mathcal{A}(U)$ as U-schemes, canonically, and these isomorphisms are compatible with the inclusions $\pi^{-1}(V) \subseteq \pi^{-1}(U)$ (for $V \subseteq U$) because both correspond to the restriction map $\mathcal{A}(U) \to \mathcal{A}(V)$. A scheme is determined up to unique isomorphism by an open cover and the gluing data, so Z is unique.

Existence. For every affine open $U \subseteq X$ put $Z_U = \operatorname{Spec}\mathcal{A}(U)$ with its structure morphism $\pi_U : Z_U \to U$ given by $\mathcal{O}_X(U) \to \mathcal{A}(U)$. If $V \subseteq U$ are affine, the preliminary fact says

$\mathcal{A}(V) = \mathcal{A}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{O}_X(V)$, hence $Z_V = Z_U \times_U V = \pi_U^{-1}(V)$, an open subscheme of Z_U; call $\iota_{VU} : Z_V \to Z_U$ this open immersion over X. By construction $\iota_{UU} = \mathrm{id}$ and $\iota_{VU}\iota_{V'V} = \iota_{V'U}$ for $V' \subseteq V \subseteq U$ (all maps are induced by restriction maps of $\mathcal{A}$, which are transitive). For two affine opens U, U', the open subscheme $\pi_U^{-1}(U \cap U') \subseteq Z_U$ is the union of the images of the Z_V, $V \subseteq U \cap U'$ affine, and similarly for U'; the isomorphisms $\iota_{VU}(Z_V) \simeq Z_V \simeq \iota_{VU'}(Z_V)$ agree on $Z_{V''}$ for $V'' \subseteq V$ affine, hence glue (two affines $V_1, V_2 \subseteq U \cap U'$ have $V_1 \cap V_2$ covered by affines $V'' \subseteq V_1 \cap V_2$ on which both identifications are $\iota_{V''U}\iota_{V''U'}^{-1}$) to isomorphisms $\varphi_{UU'} : \pi_U^{-1}(U \cap U') \to \pi_{U'}^{-1}(U \cap U')$ satisfying the cocycle condition on triple overlaps (check on the Z_V with V affine inside $U \cap U' \cap U''$, where it is the transitivity above). Glue the Z_U along the $\varphi_{UU'}$ to a scheme Z with a morphism $\pi : Z \to X$ (the π_U agree on overlaps). Then $\pi^{-1}(U) = Z_U$ is affine for every affine open U of X, so π is affine, and $(\pi_*\mathcal{O}_Z)(U) = \mathcal{O}(Z_U) = \mathcal{A}(U)$ compatibly with restrictions and with the $\mathcal{O}_X$-algebra structures. Since the affine opens form a base of the topology, $\pi_*\mathcal{O}_Z \simeq \mathcal{A}$.

(b) Let $\pi : Z = \operatorname{Spec}\mathcal{A} \to X$. Recall that π is finite if for every affine open $U \subseteq X$, $\pi^{-1}(U)$ is affine and $\mathcal{O}_Z(\pi^{-1}U)$ is a finite $\mathcal{O}_X(U)$-module (Ex. 3.3.15 shows that it is equivalent to ask this for the members of one affine open cover of X; this is also visible from the argument below).

($\Leftarrow$) Suppose $\mathcal{A}$ is quasi-coherent and finitely generated. By Prop. 1.11, $\mathcal{A}(U)$ is finitely generated over $\mathcal{O}_X(U)$ for every affine open U ((ii)$\Rightarrow$(iii)). Since $\pi^{-1}(U) = \operatorname{Spec}\mathcal{A}(U)$ by (a), π is finite.

($\Rightarrow$) $\mathcal{A} = \pi_*\mathcal{O}_Z$ is quasi-coherent by hypothesis in the definition of $\operatorname{Spec}\mathcal{A}$. If π is finite, there is an affine open cover $\{U_i\}$ of X with $\mathcal{A}(U_i)$ finitely generated over $\mathcal{O}_X(U_i)$, say by n_i elements. Then $\mathcal{O}_X^{n_i}|_{U_i} \to \mathcal{A}|_{U_i} = \mathcal{A}(U_i)^\sim$ is surjective (Prop. 1.5(b)), so $\mathcal{A}$ is finitely generated (Def. 1.10).

(c) f is affine, hence quasi-compact; and it is separated since for affine open $V \subseteq X$, $f^{-1}(V) = \operatorname{Spec} B$ with B an $\mathcal{O}_X(V)$-algebra and the diagonal of $\operatorname{Spec} B \to \operatorname{Spec}\mathcal{O}_X(V)$ corresponds to the surjection $B \otimes B \to B$, a closed immersion. $\mathcal{O}_W$ is quasi-coherent on W, so $f_*\mathcal{O}_W$ is quasi-coherent on X by Prop. 1.14(c). It is a sheaf of commutative $\mathcal{O}_X$-algebras via $f^\#$. By (a), $Z = \operatorname{Spec} f_*\mathcal{O}_W \to X$ exists. For each affine open $U \subseteq X$, $f^{-1}(U)$ is an affine scheme, so the canonical morphism $\psi_U : f^{-1}(U) \to \operatorname{Spec}\mathcal{O}_W(f^{-1}U) = \operatorname{Spec}(f_*\mathcal{O}_W)(U) = Z_U = \pi^{-1}(U)$ is an isomorphism of U-schemes. For $V \subseteq U$ affine, the maps ψ_U, ψ_V are compatible with the inclusions $f^{-1}V \subseteq f^{-1}U$ and $\pi^{-1}V \subseteq \pi^{-1}U$ (functoriality of $T \mapsto \operatorname{Spec}\mathcal{O}(T)$ for the open immersion $f^{-1}V \to f^{-1}U$ of affine schemes). Hence the ψ_U agree on overlaps, and glue to an isomorphism $W \to Z$ of X-schemes. □

Exercise 1.18

Problem. Let X be a Noetherian or separated and quasi-compact scheme. Let $\mathcal{L}$ be an invertible sheaf on X and fix a section $s \in \mathcal{L}(X)$.

(a) For $n \geq 1$ write $\mathcal{L}^n = \mathcal{L}^{\otimes n}$ and $\mathcal{L}^{\otimes 0} = \mathcal{O}_X$. Let $\mathcal{A} = \bigoplus_{n \geq 0} \mathcal{L}^{\otimes n}$. Show that $\mathcal{A}$ has a natural structure of $\mathcal{O}_X$-algebra. Let $Z = \operatorname{Spec}\mathcal{A}$ and $\pi : Z \to X$ the canonical morphism. Show that for every $x \in X$ there is an open neighbourhood $U \ni x$ with $\pi^{-1}(U) \simeq \mathbb{A}^1_U$. Deduce that Z is Noetherian or separated and quasi-compact.

(b) Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module and $\mathcal{G} = \pi^*\mathcal{F}$. Show that $\mathcal{G}(\pi^{-1}(X_s)) = \bigoplus_{n \geq 0}(\mathcal{L}^n \otimes_{\mathcal{O}_X} \mathcal{F})(X_s)$ (use Exercise 1.1(c)).

(c) Regard s as an element of $\mathcal{O}_Z(Z)$. Show that $\mathcal{G}(Z_s) = \mathcal{G}(Z)_s$ and that $Z_s \subset \pi^{-1}(X_s)$.

(d) Give a new proof of Lemma 1.25.

Solution. Throughout, $X_s = \{x : \mathcal{L}_x = s_x\mathcal{O}_{X,x}\}$ and, for $h \in \mathcal{O}(Y)$, $Y_h = \{y : h_y \in \mathcal{O}^*_{Y,y}\}$. Fix a finite affine open cover $X = X_1 \cup \cdots \cup X_r$ with $\mathcal{L}|_{X_i} = e_i\mathcal{O}_{X_i}$ free (possible since X is quasi-compact and $\mathcal{L}$ is invertible), and write $s|_{X_i} = h_i e_i$ with $h_i \in \mathcal{O}_X(X_i)$. Then $X_s \cap X_i = D(h_i)$, which is affine; so X_s is a finite union of affine opens, hence quasi-compact.

(a) *Algebra structure.* The canonical isomorphisms $\mathcal{L}^n \otimes \mathcal{L}^m \simeq \mathcal{L}^{n+m}$ (associativity of $\otimes$, and $\mathcal{O}_X \otimes \mathcal{M} \simeq \mathcal{M}$ when n or m is 0) define a multiplication $\mathcal{A} \otimes \mathcal{A} \to \mathcal{A}$ with $\mathcal{L}^n \cdot \mathcal{L}^m \subseteq \mathcal{L}^{n+m}$, associative with unit $1 \in \mathcal{L}^{\otimes 0} = \mathcal{O}_X$. It is commutative: on X_i, $\mathcal{L}^n|_{X_i} = \mathcal{O}e_i^n$ where $e_i^n = e_i \otimes \cdots \otimes e_i$, and both $e_i^n \otimes e_i^m$ and $e_i^m \otimes e_i^n$ map to e_i^{n+m}. In particular, on X_i,

$$\mathcal{A}|_{X_i} = \bigoplus_{n\geq 0} \mathcal{O}_{X_i} e_i^n \simeq \mathcal{O}_{X_i}[T], \qquad e_i \leftrightarrow T,$$

the sheaf of polynomial algebras. Each $\mathcal{L}^n$ is invertible, hence quasi-coherent, so $\mathcal{A}$ is quasi-coherent by Prop. 1.12(a). Thus $Z = \operatorname{Spec}\mathcal{A}$ exists by Ex. 1.17(a).

Local structure. Let $x \in X$; take an affine open $U \ni x$ on which $\mathcal{L} = e\mathcal{O}_U$ is free. Sections of a direct sum of sheaves over a quasi-compact open set are the direct sum of the sections (a section of the sheaf $\bigoplus \mathcal{M}_n$ is a family (t_n) that locally has only finitely many non-zero components; on a quasi-compact open set finitely many neighbourhoods suffice, so only finitely many $t_n \neq 0$). Hence $\mathcal{A}(U) = \mathcal{O}_X(U)[T]$ and $\pi^{-1}(U) = \operatorname{Spec}\mathcal{A}(U) = \operatorname{Spec}\mathcal{O}_X(U)[T] = \mathbb{A}^1_U$ (Ex. 1.17(a)).

Noetherian or separated quasi-compact. If X is Noetherian, cover it by finitely many affine opens U_i as above, each $\mathcal{O}_X(U_i)$ Noetherian. Then Z is covered by the finitely many affine schemes $\operatorname{Spec}\mathcal{O}_X(U_i)[T]$, which are Noetherian by Hilbert's basis theorem; so Z is Noetherian. If X is separated and quasi-compact: π is affine, hence quasi-compact (by the same finite affine cover $\{V_j\}$ of X: $\pi^{-1}(V_j)$ is affine) and separated (as in Ex. 1.17(c)), so $Z \to X \to \operatorname{Spec}\mathbb{Z}$ is separated, and Z is quasi-compact because it is a finite union of the affine opens $\pi^{-1}(V_j)$.

(b) *The projection formula for an affine morphism (Exercise 1.1(c)).* Let $g : Z \to X$ be affine and $\mathcal{F}$ quasi-coherent on X. The map $\mathcal{F} \otimes_{\mathcal{O}_X} g_*\mathcal{O}_Z \to g_*g^*\mathcal{F}$ is defined on the presheaf $V \mapsto \mathcal{F}(V) \otimes_{\mathcal{O}_X(V)} \mathcal{O}_Z(g^{-1}V)$ by $t \otimes b \mapsto b \cdot g^*t$ ($g^*t = t \otimes 1$) and then sheafified; it is $g_*\mathcal{O}_Z$-linear. It is an isomorphism: this is local on X, so let $X = \operatorname{Spec}R$, $Z = \operatorname{Spec}C$, $\mathcal{F} = \widetilde{N}$. Then $g^*\mathcal{F} = (N \otimes_R C)^\sim$ (Prop. 1.14(b)) and $g_*((N \otimes_R C)^\sim) = (N \otimes_R C)^\sim$ viewed over R (on $D(r)$ both sides have sections $(N \otimes_R C)_r$); also $\widetilde{N} \otimes_{\mathcal{O}_X} \widetilde{C} = (N \otimes_R C)^\sim$ by Prop. 1.5(d), and the map is the identity of this sheaf.

Apply this to $g = \pi$: $\pi_*\mathcal{G} \simeq \mathcal{F} \otimes \pi_*\mathcal{O}_Z = \mathcal{F} \otimes \mathcal{A} = \bigoplus_{n\geq 0}(\mathcal{F} \otimes \mathcal{L}^n)$ ($\otimes$ commutes with direct sums), and this is an isomorphism of $\mathcal{A}$-modules. Since $\pi^{-1}(X_s)$ is the preimage of X_s,

$$\mathcal{G}(\pi^{-1}(X_s)) = (\pi_*\mathcal{G})(X_s) = \Big(\bigoplus_{n\geq 0} \mathcal{F} \otimes \mathcal{L}^n\Big)(X_s) = \bigoplus_{n\geq 0}(\mathcal{L}^n \otimes \mathcal{F})(X_s),$$

the last equality because X_s is quasi-compact (see the remark in (a)). The same computation with X_s replaced by X (quasi-compact) gives $\mathcal{G}(Z) = \bigoplus_{n\geq 0}(\mathcal{F} \otimes \mathcal{L}^n)(X)$. In these descriptions, the restriction maps are componentwise, and the element $s \in \mathcal{L}(X) \subset \mathcal{A}(X) = \mathcal{O}_Z(Z)$ acts by $t \mapsto t \otimes s$, sending the n-th component to the $(n+1)$-st.

(c) By (a), Z is Noetherian or separated quasi-compact, and $\mathcal{G} = \pi^*\mathcal{F}$ is quasi-coherent (Prop. 1.14(b)). Prop. 1.6 applied to Z, $\mathcal{G}$ and $s \in \mathcal{O}_Z(Z)$ gives $\mathcal{G}(Z)_s \simeq \mathcal{G}(Z_s)$, which is the first assertion.

For the inclusion, it suffices to work above each trivializing affine open $U = X_i$. There $\pi^{-1}(U) = \operatorname{Spec} \mathcal{O}_X(U)[T]$ and $s = hT$ with $h = h_i$ (here T corresponds to e_i). Hence $Z_s \cap \pi^{-1}(U) = D(hT) \subseteq D(h) = \pi^{-1}(D(h)) = \pi^{-1}(X_s \cap U)$. Since the X_i cover X, $Z_s \subseteq \pi^{-1}(X_s)$.

(d) *A new proof of Lemma 1.25.* Let $\mathcal{F}$, $\mathcal{L}$, s be as in the lemma ($f \in \mathcal{F}(X)$, $g \in \mathcal{F}(X_s)$ below), and use $\mathcal{G} = \pi^*\mathcal{F}$, $P_n = (\mathcal{F} \otimes \mathcal{L}^n)(X)$, $Q_n = (\mathcal{F} \otimes \mathcal{L}^n)(X_s)$, $P = \bigoplus P_n = \mathcal{G}(Z)$, $\mathcal{G}(W) = \bigoplus Q_n$ for $W = \pi^{-1}(X_s) \supseteq Z_s$. By (b), the restriction $\mathcal{G}(Z) \to \mathcal{G}(W)$ is $\bigoplus(P_n \to Q_n)$ and multiplication by s is the shift $t \mapsto t \otimes s$. By (c), $\mathcal{G}(Z_s) = P_s$.

(a) of Lemma 1.25. Let $f \in P_0 = \mathcal{F}(X)$ with $f|_{X_s} = 0$. The image of f in $\mathcal{G}(W) = \bigoplus Q_n$ is the element with only the 0-th component, $f|_{X_s}$; it is 0. Hence $f|_{Z_s} = 0$ in $\mathcal{G}(Z_s) = P_s$, i.e. $f/1 = 0$ in P_s. So $s^k f = 0$ in P for some $k \geq 0$, i.e. $f \otimes s^k = 0$ in $P_k = (\mathcal{F} \otimes \mathcal{L}^k)(X)$. (If $k = 0$ then $f = 0$ and every $n \geq 1$ works.) Hence $f \otimes s^n = 0$ for all $n \geq \max(k, 1)$.

(b) of Lemma 1.25. Claim: the restriction $\rho : \mathcal{G}(W) \to \mathcal{G}(Z_s)$ is injective. It suffices to prove injectivity above every $U = X_i \cap X_s = D(h) \subseteq X_i$ (the opens $\pi^{-1}(D(h_i))$ cover W). With $\mathcal{O} = \mathcal{O}_X(X_i)$, $N = \mathcal{F}(D(h))$, we have $\pi^{-1}(D(h)) = \operatorname{Spec} \mathcal{O}_h[T]$ and $\mathcal{G}$ on it is $(N[T])^\sim$ (Prop. 1.14(b)), while $Z_s \cap \pi^{-1}(D(h)) = D(hT) = \operatorname{Spec} \mathcal{O}_h[T, T^{-1}]$ with sections $N[T, T^{-1}]$. The restriction $N[T] \to N[T, T^{-1}]$ is injective, proving the claim locally; so if $\sigma \in \mathcal{G}(W)$ and $\rho(\sigma) = 0$ then σ vanishes on each $\pi^{-1}(D(h_i))$, hence $\sigma = 0$.

Now let $g \in \mathcal{F}(X_s) = Q_0 \subseteq \mathcal{G}(W)$. Its image in $\mathcal{G}(Z_s) = P_s$ has the form t/s^k with $t \in P$, $k \geq 0$, i.e. $s^k \rho(g) = \rho(t|_W)$ where $t = \sum_m t_m$, $t_m \in P_m$. As ρ is $\mathcal{O}_Z(Z)$-linear, $\rho(s^k g) = \rho(t|_W)$, so by the claim $s^k g = t|_W$ in $\mathcal{G}(W) = \bigoplus Q_n$. The left side is $g \otimes s^k|_{X_s} \in Q_k$; comparing k-th components, $t_k|_{X_s} = g \otimes s^k|_{X_s}$. So $g \otimes s^k|_{X_s}$ lifts to $t_k \in (\mathcal{F} \otimes \mathcal{L}^k)(X)$, and for $n \geq n_0 := \max(k, 1)$ the section $t_k \otimes s^{n-k} \in (\mathcal{F} \otimes \mathcal{L}^n)(X)$ lifts $g \otimes s^n|_{X_s}$. This is Lemma 1.25(b). □

Exercise 1.19

> **Problem.** By considering (if necessary infinite) direct sums of sheaves of the form $\mathcal{O}_X(-m)$, show that there does not exist an integer n_0 as in Theorem 1.27 which suits all coherent sheaves on X, and that this theorem is false for quasi-coherent sheaves on X.

Solution. *Setting.* Theorem 1.27 concerns a projective scheme X over a ring A with its sheaf $\mathcal{O}_X(1)$. The statement needs X to be genuinely "big"; it is false for, e.g., $X = \operatorname{Spec} A$ itself (a point of $\mathbb{P}^0_A$), where every $\mathcal{O}_X(n) = \mathcal{O}_X$ is generated by global sections. I prove it for $X = \mathbb{P}^d_A$ with $d \geq 1$ and $A \neq 0$ (in particular for projective spaces over fields); the same argument works for any X with $\mathcal{O}_X(-k)(X) = 0$ and $\mathcal{O}_X(-k) \neq 0$ for all $k \geq 1$. The sheaves below are finitely generated quasi-coherent (and coherent if A is Noetherian), as in Thm. 1.27.

Basic fact. For $k < 0$, $\mathcal{O}_X(k)(X) = 0$ (Lemma 1.22), whereas $\mathcal{O}_X(k)_x \simeq \mathcal{O}_{X,x} \neq 0$ for every x ($X \neq \emptyset$ as $A \neq 0$). Hence $\mathcal{O}_X(k)$ is not generated by global sections, at any point (Def. 1.2: the map $\mathcal{O}_X(k)(X) \otimes \mathcal{O}_{X,x} \to \mathcal{O}_X(k)_x$ has source 0 and non-zero target).

No uniform n_0. Suppose n_0 is such that $\mathcal{F}(n)$ is generated by global sections for all $n \geq n_0$ and for every finitely generated quasi-coherent (or coherent) $\mathcal{F}$. Take $\mathcal{F} = \mathcal{O}_X(-m)$ with $m = n_0 + 1$ (invertible, so finitely generated quasi-coherent, and coherent if A is Noetherian by Prop. 1.11). Then $\mathcal{F}(n_0) = \mathcal{O}_X(-m) \otimes \mathcal{O}_X(n_0) = \mathcal{O}_X(-1)$ (Example 1.19: $\mathcal{O}_X(a) \otimes \mathcal{O}_X(b) = \mathcal{O}_X(a + b)$), which is not generated by global sections. Contradiction. So n_0 depends on $\mathcal{F}$.

Failure for quasi-coherent sheaves. Let $\mathcal{F} = \bigoplus_{m\geq 1} \mathcal{O}_X(-m)$, quasi-coherent by Prop. 1.12(a) (not finitely generated). We claim that $\mathcal{F}(n)$ is not generated by global sections for *any* $n \in \mathbb{Z}$, so no n_0 exists at all. Fix n, take $m > \max(n, 0)$ and consider the projection $q : \mathcal{F}(n) = \bigoplus_{m'\geq 1} \mathcal{O}_X(n - m') \to \mathcal{O}_X(n - m)$ ($\otimes$ commutes with direct sums), a surjective homomorphism of $\mathcal{O}_X$-modules. If $\mathcal{F}(n)$ were generated by global sections, so would be its quotient $\mathcal{O}_X(n - m)$ (the images $q(t)$, $t \in \mathcal{F}(n)(X)$, generate every stalk of the quotient by surjectivity of q_x). But $n - m < 0$, contradicting the basic fact. Hence Theorem 1.27 fails for quasi-coherent sheaves. □

Exercise 1.20

> **Problem.** Let B be a graded algebra over a ring A and C an A-algebra. Let $f : \operatorname{Proj}(B \otimes_A C) \to \operatorname{Proj} B$ be the projection morphism (Prop. 3.1.9). For any graded B-module M, endow $M \otimes_A C$ with the grading $(M \otimes_A C)_n = M_n \otimes_A C$. Show that it is a graded $B \otimes_A C$-module and that $f^*(\widetilde{M}) = (M \otimes_A C)^{\sim}$.

Solution. Put $E = B \otimes_A C$ (graded by $E_d = B_d \otimes_A C$) and $N = M \otimes_A C$. Since $\otimes_A C$ commutes with direct sums, $N = \bigoplus_{n\in\mathbb{Z}}(M_n \otimes_A C)$, so the grading makes sense. For $b \in B_d$, $c \in C$, $m \in M_n$, $c' \in C$: $(b \otimes c)(m \otimes c') = bm \otimes cc' \in M_{n+d} \otimes C = N_{n+d}$. As the elements $b \otimes c$ (resp. $m \otimes c'$) generate E_d (resp. N_n) additively, $E_d N_n \subseteq N_{n+d}$. So N is a graded E-module.

Let $\phi : B \to E$, $b \mapsto b \otimes 1$, a graded homomorphism, and let $g = f : \operatorname{Proj} E \to \operatorname{Proj} B$ be the induced morphism; for homogeneous $u \in B_+$, $g^{-1}(D_+(u)) = D_+(\phi(u))$ (proof of Prop. 3.1.9, via Lemma 2.3.40), and $\operatorname{Proj} E = \bigcup_u g^{-1}D_+(u)$ because $\operatorname{Proj} B = \bigcup D_+(u)$, u homogeneous in B_+. Fix such u, of degree $r \geq 1$. We have shown in the proof of Prop. 3.1.9 that the canonical map $B_{(u)} \otimes_A C \to E_{(\phi(u))}$, $(b/u^k) \otimes c \mapsto (b \otimes c)/\phi(u)^k$ is an isomorphism.

Modules of degree 0. Localization commutes with $\otimes_A C$, so $N_{\phi(u)} = M_u \otimes_A C$, and the grading of $N_{\phi(u)}$ is $(N_{\phi(u)})_e = (M_u)_e \otimes_A C$ (again because $\otimes_A C$ commutes with direct sums). Taking $e = 0$ gives an isomorphism $M_{(u)} \otimes_A C \xrightarrow{\sim} N_{(\phi(u))}$, $(m/u^k) \otimes c \mapsto (m \otimes c)/\phi(u)^k$.

Construction of the map. Define a morphism of $\mathcal{O}_{\operatorname{Proj} B}$-modules $\beta : \widetilde{M} \to g_*\widetilde{N}$ on the base of principal opens as follows: for $D_+(u)$, $\widetilde{M}(D_+(u)) = M_{(u)}$ and $(g_*\widetilde{N})(D_+(u)) = \widetilde{N}(D_+(\phi u)) = N_{(\phi(u))}$ (Prop. 1.17(i)); let $\beta_u(m/u^k) = (m \otimes 1)/\phi(u)^k$. It is $B_{(u)}$-linear (where $B_{(u)}$ acts on the target through $B_{(u)} \to E_{(\phi u)}$) and compatible with restriction to $D_+(uv) \subseteq D_+(u)$ (both restrictions are given by the same formulas). By the adjunction $g^* \dashv g_*$ (see the discussion in Ex. 1.16), β corresponds to a morphism $\gamma : g^*\widetilde{M} \to \widetilde{N}$.

γ is an isomorphism. It suffices to check this on the open sets $D_+(\phi(u)) = g^{-1}D_+(u)$, which cover $\operatorname{Proj} E$. Here $D_+(u) = \operatorname{Spec} B_{(u)}$ and $D_+(\phi u) = \operatorname{Spec} E_{(\phi u)}$ are affine, so by Prop. 1.14(b),

$$g^*\widetilde{M}|_{D_+(\phi u)} = \big(M_{(u)} \otimes_{B_{(u)}} E_{(\phi u)}\big)^{\sim} = \big(M_{(u)} \otimes_{B_{(u)}} (B_{(u)} \otimes_A C)\big)^{\sim} = (M_{(u)} \otimes_A C)^{\sim},$$

and γ on this open set is the map induced by $M_{(u)} \otimes_A C \to N_{(\phi u)}$, $(m/u^k) \otimes c \mapsto (m \otimes c)/\phi(u)^k$ (this is what the adjunction produces: γ sends $a \cdot g^*(m/u^k)$ to $a \cdot \beta(m/u^k)$). That map is an isomorphism by the previous paragraph. Hence γ is an isomorphism, i.e. $f^*(\widetilde{M}) = (M \otimes_A C)^{\sim}$. □

Exercise 1.21

Problem. Let B be a graded Noetherian ring and M a finitely generated graded B-module. Show that $\widetilde{M}$ is coherent on $\operatorname{Proj} B$.

Solution. Here $\widetilde{M}$ is the quasi-coherent sheaf of Prop. 1.17. Coherence is local (Def. 1.10), and $\operatorname{Proj} B$ is covered by the $D_+(u) = \operatorname{Spec} B_{(u)}$ with $u \in B_+$ homogeneous (of degree $r \geq 1$), on which $\widetilde{M}|_{D_+(u)} = (M_{(u)})^\sim$ by Prop. 1.17(i). By Prop. 1.11, a quasi-coherent sheaf which is finitely generated on a locally Noetherian scheme is coherent. So it suffices to prove:

Claim. $B_{(u)}$ is a Noetherian ring and $M_{(u)}$ is a finitely generated $B_{(u)}$-module.

Given the claim, $\operatorname{Spec} B_{(u)}$ is a Noetherian scheme, $\widetilde{M_{(u)}}$ is quasi-coherent, and a surjection $B_{(u)}^q \to M_{(u)}$ gives, by Prop. 1.5(b), a surjection $\mathcal{O}^q \to \widetilde{M_{(u)}}$; so $\widetilde{M_{(u)}}$ is finitely generated, hence coherent by Prop. 1.11.

Proof of the claim (this is Ex. 2.3.12 for $B_{(u)}$; here is the argument). Since B is Noetherian, $B_0 = B/B_+$ is Noetherian and the ideal B_+ is finitely generated; taking the homogeneous components of finitely many generators, $B_+ = (x_1, \dots, x_s)$ with x_i homogeneous of degree $d_i \geq 1$. Then $B = B_0[x_1, \dots, x_s]$: by induction on n, every $b \in B_n$, $n \geq 1$, lies in B_+, so $b = \sum y_i x_i$ with y_i homogeneous of degrees $n - d_i < n$.

Let $B' = B_0[x_1^r, \dots, x_s^r, u] \subseteq B$, a finitely generated algebra over B_0, hence Noetherian. B is a finitely generated B'-module: it is generated by the monomials $x_1^{\alpha_1} \cdots x_s^{\alpha_s}$ with $\alpha_i < r$ (since $B = B_0[x_i]$ and $x_i^r \in B'$). Consider the Veronese subring $B^{(r)} = \bigoplus_{k \geq 0} B_{kr}$ and the $B^{(r)}$-module $M^{(r)} = \bigoplus_{k \in \mathbb{Z}} M_{kr}$. Since M is finitely generated over B, it is finitely generated over B', hence Noetherian over B'; $B^{(r)} \supseteq B'$ and $M^{(r)}$ are B'-submodules of B, resp. M (the elements x_i^r and u have degrees divisible by r), so both are finitely generated B'-modules, hence finitely generated over $B^{(r)} \supseteq B'$. Therefore $B^{(r)}$ is a finitely generated B_0-algebra (generated by the generators of B' and the finitely many module generators over B'), thus Noetherian.

The map $B^{(r)} \to B_{(u)}$, $\sum_k a_k \mapsto \sum_k a_k/u^k$ ($a_k \in B_{kr}$) is a surjective ring homomorphism; so $B_{(u)}$ is Noetherian. Similarly $M^{(r)} \to M_{(u)}$, $\sum m_k \mapsto \sum m_k/u^k$ ($m_k \in M_{kr}$), is surjective and semilinear over $B^{(r)} \to B_{(u)}$; since $M^{(r)}$ is finitely generated over $B^{(r)}$, the images of its generators generate $M_{(u)}$ over $B_{(u)}$. This proves the claim and the exercise. □

Exercise 1.22

Problem. Let B be a homogeneous algebra over a ring A, quotient of some $A[T_0, T_1, \dots, T_m]$. Let M be a graded B-module and fix $n \in \mathbb{Z}$. Let $\mathcal{O}_X(1)$ denote the sheaf on $X = \operatorname{Proj} B$ given by the closed immersion $X \to \mathbb{P}_A^m$. Consider the graded B-module $M(n)$ given by $M(n)_d = M_{d+n}$ for every $d \in \mathbb{Z}$. Show that $\widetilde{M(n)} = \widetilde{M}(n) := \widetilde{M} \otimes \mathcal{O}_X(n)$.

Solution. *Setup.* Let $\pi : A[T_0, \dots, T_m] \to B$ be the surjective graded homomorphism, $t_i = \pi(T_i) \in B_1$, $P = \mathbb{P}_A^m = \operatorname{Proj} A[T]$ and $\iota : X \to P$ the closed immersion induced by π (Lemma 2.3.40: $\iota^{-1} D_+(T_i) = D_+(t_i) =: U_i$). Since B_+ is generated by $t_0, \dots, t_m$ (as B is a homogeneous algebra), $X = \bigcup U_i$. As in Def. 1.26, $\mathcal{O}_X(n) = \iota^* \mathcal{O}_P(n)$. By Example 1.19, $\mathcal{O}_P(n)|_{D_+(T_i)} = T_i^n \mathcal{O}_P|_{D_+(T_i)}$ is free with basis $\varepsilon_i = T_i^n$, and on $D_+(T_i T_j)$ one has $\varepsilon_j = (T_j/T_i)^n \varepsilon_i$. Pulling back ($\iota^* \mathcal{O}_P = \mathcal{O}_X$): $\mathcal{O}_X(n)|_{U_i} = \mathcal{O}_{U_i} e_i$ is free with basis $e_i = \iota^* \varepsilon_i$, and $e_j = (t_j/t_i)^n e_i$ on $U_i \cap U_j$.

Local isomorphisms. On $U_i = \operatorname{Spec} B_{(t_i)}$, $\widetilde{M(n)}|_{U_i} = (M(n)_{(t_i)})^\sim$ (Prop. 1.17(i)). Define

$$\lambda_i : M(n)_{(t_i)} \to M_{(t_i)}, \qquad m/t_i^k \mapsto m/t_i^{n+k} \quad (m \in M(n)_k = M_{n+k}).$$

It is a well-defined $B_{(t_i)}$-linear bijection: $M(n)_{t_i} = M_{t_i}$ as B_{t_i}-modules but with the grading shifted, so $M(n)_{(t_i)} = (M(n)_{t_i})_0 = (M_{t_i})_n$, and λ_i is multiplication by the unit t_i^{-n} of B_{t_i} (of degree $-n$), which maps $(M_{t_i})_n$ bijectively onto $(M_{t_i})_0 = M_{(t_i)}$. Define

$$\Psi_i : \widetilde{M(n)}|_{U_i} \xrightarrow{\lambda_i^\sim} \widetilde{M}|_{U_i} \xrightarrow{x \mapsto x \otimes e_i} (\widetilde{M} \otimes \mathcal{O}_X(n))|_{U_i}.$$

The second map is an isomorphism because $\mathcal{O}_X(n)|_{U_i} = \mathcal{O}_{U_i} e_i$ is free of rank one with basis e_i.

Gluing. On the stalk at $\mathfrak{p} \in U_i$, $\widetilde{M(n)}_{\mathfrak{p}} = M(n)_{(\mathfrak{p})}$, whose elements are m/v with $m \in M_{n+k}$, $v \in B_k \setminus \mathfrak{p}$ homogeneous, and $\Psi_i(m/v) = \big(m/(vt_i^n)\big) \otimes e_i$ (take t_i^k replaced by v; this is the localization of the formula for λ_i, consistent because λ_i is multiplication by t_i^{-n}). If also $\mathfrak{p} \in U_j$, then

$$\Psi_j(m/v) = \Big(\frac{m}{vt_j^n}\Big) \otimes e_j = \Big(\frac{m}{vt_j^n}\Big)\Big(\frac{t_j}{t_i}\Big)^n \otimes e_i = \Big(\frac{m}{vt_i^n}\Big) \otimes e_i = \Psi_i(m/v).$$

Hence the Ψ_i agree on all stalks, thus on $U_i \cap U_j$, and glue to a morphism $\Psi : \widetilde{M(n)} \to \widetilde{M} \otimes \mathcal{O}_X(n)$, which is an isomorphism because it is so on each U_i. This is the canonical isomorphism $\widetilde{M(n)} = \widetilde{M}(n)$.

Remark. For $M = B$ this gives $\widetilde{B(n)} \simeq \widetilde{B} \otimes \mathcal{O}_X(n) = \mathcal{O}_X(n)$, so the two descriptions of $\mathcal{O}_X(n)$ (as $\widetilde{B(n)}$ in Example 1.19 and as $\iota^*\mathcal{O}_{\mathbb{P}^m_A}(n)$ in Def. 1.26) agree. □

Conventions for these exercises. Results of the text are cited by their numbers in Liu ("Prop. 1.31", "Lemma 1.25", ..., all in Section 5.1). For a scheme $Y = \operatorname{Proj} S[T_0, \dots, T_d]$ we write $\mathcal{O}_Y(m)$ for the twists, and for a morphism $g : Z \to Y$ the symbol g^*T_i denotes the global section of $g^*\mathcal{O}_Y(1)$ induced by T_i (Prop. 1.31(a)). For an invertible sheaf $\mathcal{L}$ and $s \in \mathcal{L}(X)$, X_s is the open set of Def. 1.24.

Honest notes. (1) In Ex. 1.23 "automorphism" must mean automorphism of k-schemes (a field automorphism of $\mathbb{C}$ induces an automorphism of $\mathbb{P}^n_{\mathbb{C}}$ that is not linear); the fact that $\operatorname{Pic} \mathbb{P}^n_k$ is generated by $\mathcal{O}(1)$ is admitted, as in the statement. (2) Ex. 1.24 is *false as printed*: a counterexample is given, and the statement is proved under the extra hypothesis that σ^* preserves the space of sections $f^*H^0(\mathbb{P}^n, \mathcal{O}(1))$ (this is automatic when that space is all of $H^0(X, f^*\mathcal{O}(1))$). (3) In Ex. 1.25(a) the equivalence with flatness is proved directly, without Ex. 1.15. (4) In Ex. 1.28 the "X'" of the first line is read as X. (5) In Ex. 1.27(c) one needs $A \neq 0$ (for $A = 0$ everything is empty). (6) Ex. 1.29(b) and (c) are *false as printed* for an arbitrary faithfully flat morphism (counterexample given); they are proved when $X' = X \otimes_A B$ for a faithfully flat ring map $A \to B$, which is the situation in which they are used in Ex. 1.30. The lemma on flat base change of global sections needed for this is proved in the solution of 1.29. (7) In Ex. 1.30(a) the sections $s_0, \dots, s_n$ have to be chosen suitably (an arbitrary generating family does not work; example given). (8) In Ex. 1.33(b) the map ρ is read as the projection $X \times_Y \operatorname{Spec} \mathcal{O}_{Y,y} \to X$ (otherwise $\rho^*\mathcal{F}$ makes no sense). Exercises are cited below only with the needed argument given. Everything else is proved completely.

Exercise 1.23

Problem. (Automorphisms of $\mathbb{P}^n_k$) Let k be a field, $B = k[T_0, \dots, T_n]$ with $n \geq 1$ and $X = \operatorname{Proj} B$. We are going to show that the automorphisms of X are linear; that is, that they come from k-linear transformations of B. We admit that the group $\operatorname{Pic}(X)$ (see Exercise 1.12) is generated by the class of $\mathcal{O}_X(1)$ (Proposition 7.2.9). Let σ be an automorphism of X. (a) Show that we have an isomorphism $\lambda : \sigma^*\mathcal{O}_X(1) \simeq \mathcal{O}_X(1)$. (b) Let $\tau : \mathcal{O}_X(1)(X) \to \sigma^*\mathcal{O}_X(1)(X) \to \mathcal{O}_X(1)(X)$ be the composition of the canonical homomorphism followed by $\lambda(X)$. Show that τ is an automorphism of the k-vector space $B_1 = \mathcal{O}_X(1)(X)$. (c) Show that τ canonically induces an automorphism $\tilde{\tau}$ of the graded k-algebra B and that σ is the automorphism of X induced by $\tilde{\tau}$.

Solution. Write $\mathcal{O}(m) = \mathcal{O}_X(m)$. Recall: $\mathcal{O}(m) \otimes \mathcal{O}(m') = \mathcal{O}(m+m')$ (Example 1.19 of the text), $\mathcal{O}(m)(X) = B_m$ for $m \geq 0$ and $= 0$ for $m < 0$ (Lemma 1.22); inverse images commute with tensor products and send invertible sheaves to invertible sheaves. As said, σ is an automorphism of the k-scheme X.

Two preliminary facts. (F1) For a linear form $t \in B_1 = \mathcal{O}(1)(X)$, the section t generates $\mathcal{O}(1)_x$ exactly when $x \in D_+(t)$. Indeed, for $x \in D_+(T_j)$ we have $\mathcal{O}(1)|_{D_+(T_j)} = T_j\mathcal{O}$, so $t = (t/T_j)T_j$ with $t/T_j \in \mathcal{O}(D_+(T_j))$, which is a unit at x iff $t \notin x$, i.e. $x \in D_+(t)$.

(F2) (Uniqueness, a strengthening of Rem. 1.32: the morphism is not assumed to be the one constructed in Prop. 1.31(b).) Let S be a ring, $Y = \mathbb{P}^d_S = \operatorname{Proj} S[T_0, \dots, T_d]$, Z an S-scheme, $\mathcal{M}$ an invertible sheaf on Z and $s_0, \dots, s_d \in \mathcal{M}(Z)$ generating $\mathcal{M}$. Let $g : Z \to Y$ be an S-morphism and $\mu : g^*\mathcal{O}_Y(1) \to \mathcal{M}$ an isomorphism with $\mu(g^*T_i) = s_i$ for all i. Then $g^{-1}(D_+(T_i)) = Z_{s_i}$ and g is the morphism attached to $(\mathcal{M}, s_i)$ in Prop. 1.31(b). In particular two such morphisms (with possibly different μ) coincide. *Proof of (F2).* Let $z \in Z$, $y = g(z)$. We have $(g^*\mathcal{O}(1))_z = \mathcal{O}(1)_y \otimes \mathcal{O}_{Z,z}$. If $y \in D_+(T_i)$ then T_i generates $\mathcal{O}(1)_y$ by (F1), so g^*T_i generates the stalk at z. If $y \notin D_+(T_i)$ choose j with $y \in D_+(T_j)$; then $T_i = (T_i/T_j)T_j$ with $T_i/T_j \in \mathfrak{m}_y$, so the germ of g^*T_i at z lies in $\mathfrak{m}_z(g^*\mathcal{O}(1))_z$ and is not a generator. Since μ is an isomorphism sending g^*T_i to s_i, $g^{-1}D_+(T_i) = Z_{s_i}$. On Z_{s_i} the morphism g maps to $D_+(T_i) = \operatorname{Spec} S[T_j/T_i]_j$, so it is determined by the ring homomorphism $T_j/T_i \mapsto g^{\#}(T_j/T_i) = g^*T_j/g^*T_i = s_j/s_i \in \mathcal{O}_Z(Z_{s_i})$. These data are exactly those defining the morphism of Prop. 1.31(b). $\square$

(a) Put $\mathcal{L} = \sigma^*\mathcal{O}(1)$, an invertible sheaf. Since $\operatorname{Pic}(X)$ is generated by $\mathcal{O}(1)$, $\mathcal{L} \simeq \mathcal{O}(m)$ for some $m \in \mathbb{Z}$. By Prop. 1.31(a) (its proof does not use that σ is over a base), $\mathcal{L}$ is generated by the global sections σ^*T_i. If $m < 0$, then $\mathcal{O}(m)(X) = 0$ would force $\mathcal{O}(m)_x = 0$ for all x, which is absurd as $\mathcal{O}(m)_x \simeq \mathcal{O}_{X,x} \neq 0$. So $m \geq 0$. The same argument for σ^{-1} gives $(\sigma^{-1})^*\mathcal{O}(1) \simeq \mathcal{O}(m')$ with $m' \geq 0$. Then

$$\mathcal{O}(1) \simeq \sigma^*(\sigma^{-1})^*\mathcal{O}(1) \simeq \sigma^*\mathcal{O}(m') \simeq (\sigma^*\mathcal{O}(1))^{\otimes m'} \simeq \mathcal{O}(mm').$$

Taking global sections, $n+1 = \dim_k B_1 = \dim_k B_{mm'} = \binom{n+mm'}{n}$ (here $mm' \geq 0$). Since $n \geq 1$, $j \mapsto \binom{n+j}{n}$ is strictly increasing on $j \geq 0$ (the increment is $\binom{n+j}{n-1} > 0$), and takes the value $n+1$ at $j = 1$. Hence $mm' = 1$, so $m = 1$, and λ exists.

(b) The canonical map $\mathcal{F}(X) \to (\sigma^*\mathcal{F})(X)$ is $s \mapsto \sigma^{-1}s \otimes 1$. Since σ is an isomorphism of ringed spaces, $\sigma^{-1}\mathcal{O}_X \to \mathcal{O}_X$ is an isomorphism and $\sigma^{-1}\mathcal{F}(U) = \mathcal{F}(\sigma(U))$, so $\sigma^*\mathcal{F} \simeq \sigma^{-1}\mathcal{F}$ and $\mathcal{F}(X) \to (\sigma^*\mathcal{F})(X)$ is bijective. It is k-linear because σ is a k-morphism. As $\lambda(X)$ is a k-linear bijection, τ is a k-linear bijection of B_1.

(c) Since $B = k[T_0, \dots, T_n]$ is the symmetric algebra of B_1, τ extends uniquely to a graded k-algebra endomorphism $\tilde{\tau}$ of B, and $\tilde{\tau}$ is an automorphism with inverse $\widetilde{\tau^{-1}}$. A graded auto-

morphism $\tilde\tau$ induces the automorphism $g_\tau = \operatorname{Proj}\tilde\tau$ of X with $g_\tau^{-1}(D_+(f)) = D_+(\tilde\tau f)$, given there by $\tilde\tau : B_{(f)} \to B_{(\tilde\tau f)}$.

Claim: for every $\tau \in GL(B_1)$ (with $t_i = \tau(T_i)$) the t_i generate $\mathcal{O}(1)$, and g_τ is the morphism of Prop. 1.31(b) attached to $(\mathcal{O}(1), t_i)$. The T_i are k-linear combinations of the t_j, so the t_j generate wherever the T_i do. By (F1), $X_{t_i} = D_+(t_i) = D_+(\tilde\tau T_i)$ and the morphism of Prop. 1.31(b) maps $D_+(t_i)$ to $D_+(T_i)$ via $T_j/T_i \mapsto t_j/t_i = \tilde\tau(T_j)/\tilde\tau(T_i) = \tilde\tau(T_j/T_i)$. This is the restriction of g_τ. The sets $D_+(t_i)$ cover X, which proves the claim.

Now by definition of τ, λ is an isomorphism $\sigma^*\mathcal{O}(1) \to \mathcal{O}(1)$ with $\lambda(\sigma^* T_i) = \tau(T_i) = t_i$. By (F2) (with $S = k$), σ is the morphism attached to $(\mathcal{O}(1), t_i)$, i.e. $\sigma = g_\tau = \operatorname{Proj}\tilde\tau$. So σ is induced by the automorphism $\tilde\tau$ of the graded algebra B. (The choice of λ is unique up to $\operatorname{Aut}\mathcal{O}(1) = \mathcal{O}(X)^* = k^*$, so τ and $\tilde\tau$ are determined by σ up to the scalar actions $c^{\deg}$, which induce the identity of X.) □

Exercise 1.24

> **Problem.** Let X be an algebraic variety over a field k. Let $f : X \to Y := \mathbb{P}^n_k$ be a morphism of algebraic k-varieties. Let $\mathcal{L} = f^*\mathcal{O}_Y(1)$. Let σ be an automorphism of X such that $\sigma^*\mathcal{L} \simeq \mathcal{L}$. Show that there exists an automorphism σ' of Y such that $\sigma' \circ f = f \circ \sigma$.

Solution. Automorphisms are k-automorphisms. Notation: $B = k[T_0, \dots, T_n]$, $s_i = f^*T_i \in \mathcal{L}(X)$, $V = \sum_i k\, s_i \subseteq \mathcal{L}(X)$, the image of $f^* : B_1 \to \mathcal{L}(X)$; the s_i generate $\mathcal{L}$ (Prop. 1.31(a)).

The statement as printed is false. Let $\operatorname{char} k \neq 2, 3$ (e.g. $k = \mathbb{C}$), $X = \operatorname{Proj} k[s,t] = \mathbb{P}^1_k$, $n = 3$, and $F_0 = s^4$, $F_1 = s^3t$, $F_2 = st^3$, $F_3 = t^4 \in \mathcal{O}_X(4)(X) = k[s,t]_4$. They generate $\mathcal{O}_X(4)$ (F_0 generates on $D_+(s)$, F_3 on $D_+(t)$), so Prop. 1.31(b) gives $f : X \to \mathbb{P}^3_k$ with $\mathcal{L} = f^*\mathcal{O}(1) \simeq \mathcal{O}_X(4)$ and $f^*T_i \leftrightarrow F_i$ (one checks that f is even a closed immersion; this is not needed). Let $\tilde\tau$ be the graded automorphism $s \mapsto s$, $t \mapsto s + t$ of $k[s,t]$ and $\sigma = \operatorname{Proj}\tilde\tau$. By (F2) of the solution of 1.23 (Claim in (c)), $\sigma^*\mathcal{O}_X(1) \simeq \mathcal{O}_X(1)$ with $\sigma^* s \leftrightarrow s$, $\sigma^* t \leftrightarrow s + t$; taking tensor powers, $\sigma^*\mathcal{O}_X(4) \simeq \mathcal{O}_X(4)$ with $\sigma^* F \leftrightarrow \tilde\tau(F) = F(s, s+t)$. So $\sigma^*\mathcal{L} \simeq \mathcal{L}$.

Suppose $\sigma' \in \operatorname{Aut}(\mathbb{P}^3_k)$ with $\sigma' f = f\sigma =: \varphi$. By 1.23(c), $\sigma'^*\mathcal{O}(1) \simeq \mathcal{O}(1)$ with $\sigma'^* T_i \leftrightarrow \sum_j a_{ij} T_j$ for some invertible matrix (a_{ij}). Then $\varphi^*\mathcal{O}(1) = (f\sigma)^*\mathcal{O}(1) \simeq \sigma^*\mathcal{O}_X(4) \simeq \mathcal{O}_X(4)$ with $\varphi^* T_i \leftrightarrow G_i := \tilde\tau(F_i)$, and also $\varphi^*\mathcal{O}(1) = (\sigma' f)^*\mathcal{O}(1) = f^*\sigma'^*\mathcal{O}(1) \simeq \mathcal{O}_X(4)$ with $\varphi^* T_i \leftrightarrow H_i := \sum_j a_{ij} F_j$. Comparing the two isomorphisms gives an automorphism of $\mathcal{O}_X(4)$ sending G_i to H_i for all i. Automorphisms of $\mathcal{O}_X(4)$ are multiplications by units of $\mathcal{O}_X(X) = k$, so $H_i = cG_i$ for some $c \in k^*$. In particular $(s+t)^4 = G_3 = c^{-1}H_3 \in \operatorname{span}(s^4, s^3t, st^3, t^4)$. But $(s+t)^4 = s^4 + 4s^3t + 6s^2t^2 + 4st^3 + t^4$ has the nonzero coefficient 6 at s^2t^2. Contradiction. So no such σ' exists.

Corrected statement. Assume, in addition, (i) the sections $s_0, \dots, s_n$ are linearly independent over k (i.e. $f^* : B_1 \to \mathcal{L}(X)$ is injective), and (ii) there is an isomorphism $\mu : \sigma^*\mathcal{L} \to \mathcal{L}$ such that the k-linear automorphism $\tilde\mu : \mathcal{L}(X) \to \mathcal{L}(X)$, $\tilde\mu(s) = \mu(X)(\sigma^* s)$, satisfies $\tilde\mu(V) \subseteq V$. (The map $\tilde\mu$ is bijective by 1.23(b). Condition (ii) holds automatically if $V = \mathcal{L}(X)$, i.e. if f is given by the complete linear system; in that case the printed statement is true as soon as (i) holds.) Then there is $\sigma' \in \operatorname{Aut}(\mathbb{P}^n_k)$ with $\sigma' f = f\sigma$.

Proof. $\tilde\mu|_V$ is injective with V finite dimensional and $\tilde\mu(V) \subseteq V$, so $\tilde\mu(V) = V$. Write $\tilde\mu(s_i) = \sum_j c_{ij} s_j$; by (i) the matrix (c_{ij}) is uniquely determined and invertible. Let $\tau \in GL(B_1)$ with

$\tau(T_i) = \sum_j c_{ij} T_j =: t_i$ and $\sigma' = \operatorname{Proj} \tilde{\tau}$. By the Claim in 1.23(c), $\sigma'^* \mathcal{O}(1) \simeq \mathcal{O}(1)$ with $\sigma'^* T_i \leftrightarrow t_i$. Hence $(\sigma' f)^* \mathcal{O}(1) = f^* \sigma'^* \mathcal{O}(1) \simeq f^* \mathcal{O}(1) = \mathcal{L}$ with $(\sigma' f)^* T_i \leftrightarrow f^* t_i = \sum_j c_{ij} s_j = \tilde{\mu}(s_i)$. On the other hand $(f\sigma)^* \mathcal{O}(1) = \sigma^* \mathcal{L} \xrightarrow{\mu} \mathcal{L}$ with $(f\sigma)^* T_i = \sigma^* s_i \mapsto \tilde{\mu}(s_i)$. The sections $\tilde{\mu}(s_i)$ generate $\mathcal{L}$ (they span the same space as the s_j). Both $\sigma' f$ and $f\sigma$ are k-morphisms $X \to \mathbb{P}^n_k$ satisfying the hypothesis of (F2) for the data $(\mathcal{L}, \tilde{\mu}(s_i))$, so they are equal. □

Exercise 1.25

> **Problem.** Let $f : X \to Y$ be a finite dominant morphism of integral locally Noetherian schemes. The integer $n = [K(X) : K(Y)]$ is called the degree of f. (a) Show that for any $y \in Y$, $\mathcal{O}(X_y)$ is a $k(y)$-vector space of finite dimension and that $\dim_{k(y)} \mathcal{O}(X_y) \geq n$. Show that f is flat if and only if there is equality for every $y \in f(X)$ (use Exercise 1.15). (b) Let us suppose f is flat above y and that X_y is made up of exactly n points. Show that f is etale at every point of $f^{-1}(y)$.

Solution. Local set-up. Let $y \in Y$ and let $\operatorname{Spec} A \subseteq Y$ be an affine open neighborhood, $\mathfrak{p} \subset A$ the prime of y. As f is finite, $f^{-1}(\operatorname{Spec} A) = \operatorname{Spec} B$ with B a finite A-algebra; it is nonempty as f is dominant. Since X, Y are integral, A, B are domains. The kernel of $A \to B$ is nilpotent because f is dominant, hence zero. So $A \subseteq B$, $K := K(Y) = \operatorname{Frac} A$, and $B \otimes_A K$ is a domain finite over the field K, hence a field, equal to $\operatorname{Frac} B = K(X)$. Thus $n = \dim_K(B \otimes_A K) < \infty$. Also f is closed (f is finite) and dominant, hence surjective: $f(X) = Y$. The fiber $X_y = X \times_Y \operatorname{Spec} k(y) = \operatorname{Spec}(B \otimes_A k(y))$, so $\mathcal{O}(X_y) = B \otimes_A k(y) = B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}$.

(a) Let $M = B_\mathfrak{p}$, a finite torsion-free module over the Noetherian local domain $A_\mathfrak{p}$; $M \otimes K = K(X)$ has dimension n. Then $\mathcal{O}(X_y) = M/\mathfrak{p}M$ is finite dimensional over $k(y)$ because M is finite. Let $r = \dim_{k(y)} M/\mathfrak{p}M$. By Nakayama's lemma M is generated by r elements, i.e. there is a surjection $\alpha : A^r_\mathfrak{p} \to M$. Tensoring with the flat module K gives a surjection $K^r \to K^n$, so $r \geq n$.

Flatness. f is flat iff $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$ for all x. For fixed $\mathfrak{p}$, the module $B_\mathfrak{p}$ is flat over $A_\mathfrak{p}$ iff $(B_\mathfrak{p})_\mathfrak{q} = B_\mathfrak{q}$ is flat over $A_\mathfrak{p}$ for all maximal ideals $\mathfrak{q}$ of $B_\mathfrak{p}$ (Tor commutes with localization of the $B_\mathfrak{p}$-module), and these are the primes of B above $\mathfrak{p}$ (the maximal ideals of the finite extension $A_\mathfrak{p} \subseteq B_\mathfrak{p}$ contract to the maximal ideal of $A_\mathfrak{p}$). Hence f is flat iff $B_\mathfrak{p}$ is flat over $A_\mathfrak{p}$ for all y (i.e. for all $y \in f(X) = Y$). If $M = B_\mathfrak{p}$ is flat, then, being finitely presented over a local ring, it is free; its rank is $\dim_K M \otimes K = n$. So $r = n$. Conversely assume $r = n$. Then $K \otimes \alpha : K^n \to K^n$ is a surjective endomorphism of a finite dimensional space, hence injective; by flatness of K, $K \otimes \operatorname{Ker} \alpha = 0$. But $\operatorname{Ker} \alpha \subseteq A^n_\mathfrak{p} \subseteq K^n$ and $\operatorname{Ker} \alpha \to K \otimes \operatorname{Ker} \alpha \subseteq K^n$ is the inclusion; hence $\operatorname{Ker} \alpha = 0$ and $M \simeq A^n_\mathfrak{p}$ is free, in particular flat. This proves (a). (The argument replaces Exercise 1.15.)

(b) Etale means unramified and flat (Section 4.3 of the text): $\mathfrak{m}_y \mathcal{O}_{X,x} = \mathfrak{m}_x$ and $k(x)/k(y)$ finite separable. Let y be as in (b) and $A, B, \mathfrak{p}$ as above. Flatness above y means that $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,y}$ for all $x \in f^{-1}(y)$, i.e. $B_\mathfrak{p}$ is flat over $A_\mathfrak{p}$ as in (a). So $B_\mathfrak{p}$ is free of rank n, and $F := B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p} = \mathcal{O}(X_y)$ has dimension exactly n over $k(y)$. F is a finite dimensional algebra over a field, hence artinian, hence the product of its localizations $F_x = \mathcal{O}_{X,x}/\mathfrak{m}_y \mathcal{O}_{X,x}$ at its finitely many points $x \in X_y = f^{-1}(y)$. By assumption there are exactly n points, each $F_x \neq 0$ has $k(y)$-dimension ≥ 1, and the dimensions add up to n. So $\dim F_x = 1$ for each x, i.e. $F_x = k(y) \cdot 1$ is a field. Then $F_x = \mathcal{O}_{X,x}/\mathfrak{m}_y \mathcal{O}_{X,x}$ being a field forces $\mathfrak{m}_y \mathcal{O}_{X,x} = \mathfrak{m}_x$ and $k(x) = F_x = k(y)$, which

is trivially separable over $k(y)$. Hence f is unramified and flat, i.e. etale, at every $x \in f^{-1}(y)$ (f is finite, hence of finite type, and X, Y are locally Noetherian as required). $\square$

Exercise 1.26

> **Problem.** Let $\mathcal{L}$ be an invertible sheaf on a scheme X. We call a point $x \in X$ such that $H^0(X, \mathcal{L}) \otimes \mathcal{O}_{X,x} \subseteq \mathfrak{m}_x \mathcal{L}_x$ a base point of $\mathcal{L}$. Let $x_1, \ldots, x_n \in X$ be points that are not base points. Let us suppose that X is an algebraic variety over a field k with k infinite or $\operatorname{Card} k \geq n$. Show that there exists a section $s \in H^0(X, \mathcal{L})$ such that $\mathcal{L}_{x_i} = s_{x_i} \mathcal{O}_{X,x_i}$ for every $i \leq n$.

Solution. (The inclusion in the definition means that the image of $H^0(X, \mathcal{L})$ in $\mathcal{L}_x$ lies in $\mathfrak{m}_x \mathcal{L}_x$.) Only the facts that X is a k-scheme and $|k| \geq n$ (or k infinite) are used. Let $W = H^0(X, \mathcal{L})$, a k-vector space. For each i let $\varepsilon_i : W \to \mathcal{L}_{x_i} / \mathfrak{m}_{x_i} \mathcal{L}_{x_i}$ be the germ map; it is k-linear, the target being a $k(x_i)$-vector space, hence a k-vector space. Put $W_i = \operatorname{Ker} \varepsilon_i = \{s : s_{x_i} \in \mathfrak{m}_{x_i} \mathcal{L}_{x_i}\}$. Since x_i is not a base point, some $s \in W$ has $s_{x_i} \notin \mathfrak{m}_{x_i} \mathcal{L}_{x_i}$, so $W_i \neq W$.

Lemma. A vector space W over k is not a union of m proper subspaces $W_1, \ldots, W_m$ if $m \leq |k|$ (in particular if k is infinite). Induction on m; $m = 1$ is clear. Let $m \geq 2$ and suppose $W = \bigcup W_i$. By the induction hypothesis $W \neq \bigcup_{i \geq 2} W_i$, so there is $b \in W \setminus \bigcup_{i \geq 2} W_i$; necessarily $b \in W_1$. Take $a \in W \setminus W_1$. For $\lambda \in k$, $a + \lambda b \notin W_1$ (else $a \in W_1$). For $i \geq 2$, W_i contains at most one of the elements $a + \lambda b$: if it contained $a + \lambda b$ and $a + \lambda' b$ with $\lambda \neq \lambda'$, then $b \in W_i$. So at most $m - 1$ of the $|k| \geq m$ elements $a + \lambda b$ lie in $\bigcup_{i \geq 2} W_i$, and some $a + \lambda b$ lies in no W_i, a contradiction. $\square$

By the lemma there is $s \in W$ with $s \notin W_i$ for all $i \leq n$, i.e. $s_{x_i} \notin \mathfrak{m}_{x_i} \mathcal{L}_{x_i}$. As $\mathcal{L}_{x_i} = \mathcal{O}_{X,x_i} e$ is free of rank 1 over the local ring $\mathcal{O}_{X,x_i}$, writing $s_{x_i} = ae$ we get $a \notin \mathfrak{m}_{x_i}$, so a is a unit and s_{x_i} generates $\mathcal{L}_{x_i}$. $\square$

Exercise 1.27

> **Problem.** (d-uple embedding) Let A be a ring, and $X = \operatorname{Proj} A[T_0, \ldots, T_n]$ with $n \geq 1$. Let us fix an integer $d \geq 1$. (a) Let $\{M_\alpha\}_\alpha$ denote the set of monomials of degree d in $A[T_0, \ldots, T_n]$. This set is of cardinal $N := \binom{n+d}{n}$. Let us consider the homomorphism $\varphi : A[S_\alpha]_\alpha \to A[T_0, \ldots, T_n]$ which sends S_α to $M_\alpha(T_0, \ldots, T_n)$. Show that φ induces a closed immersion $f : X \to \mathbb{P}_A^{N-1} = \operatorname{Proj} A[S_\alpha]_\alpha$ (Lemma 2.3.40). (b) Show that $\{M_\alpha\}_\alpha$ generates $H^0(X, \mathcal{O}_X(d))$, that $\mathcal{O}_X(d) \simeq f^* \mathcal{O}_{\mathbb{P}_A^{N-1}}(1)$, and that f is the morphism associated to $\mathcal{O}_X(d)$ and the global sections $\{M_\alpha\}_\alpha$ as in Proposition 1.31(b). (c) Show that $\mathcal{O}_X(d)$ is very ample if and only if $d \geq 1$.

Solution. Write $B = A[T_0, \ldots, T_n]$ and $P = \mathbb{P}_A^{N-1}$. Recall $\mathcal{O}_X(d)(X) = B_d$, a free A-module with basis $\{M_\alpha\}$ (Lemma 1.22).

(b), first part. For $x \in D_+(T_i)$ the sheaf $\mathcal{O}_X(d)$ is $T_i^d \mathcal{O}_X$ on $D_+(T_i)$ (Example 1.19 of the text), and T_i^d is one of the M_α. Hence the M_α generate $\mathcal{O}_X(d)$, and they span $H^0(X, \mathcal{O}_X(d)) = B_d$ over A.

Let $f : X \to P$ be the morphism of Prop. 1.31(b) attached to $(\mathcal{O}_X(d), \{M_\alpha\})$. Then $f^* \mathcal{O}_P(1) \simeq \mathcal{O}_X(d)$ with $f^* S_\alpha \leftrightarrow M_\alpha$. We describe f. For a monomial M_α the set X_{M_α} is $D_+(M_\alpha)$: on

$D_+(T_i)$ we have $M_\alpha = (M_\alpha/T_i^d)T_i^d$ and $M_\alpha/T_i^d \in B_{(T_i)}$ is a unit at the prime $\mathfrak{p}$ iff $M_\alpha \notin \mathfrak{p}$. By the proof of Prop. 1.31(b), (F2) of the solution of 1.23, $f^{-1}(D_+(S_\alpha)) = D_+(M_\alpha) = \operatorname{Spec} B_{(M_\alpha)}$, and f is on it given by the ring homomorphism

$$\varphi_\alpha : A[S_\beta/S_\alpha]_\beta \to B_{(M_\alpha)}, \qquad S_\beta/S_\alpha \mapsto M_\beta/M_\alpha.$$

This is exactly the ring homomorphism induced by φ on the degree-0 localizations, so f is the morphism of Proj's induced by φ (the graded ring map $A[S_\alpha] \to B^{(d)} = \bigoplus_k B_{kd}$, and $\operatorname{Proj} B^{(d)} = \operatorname{Proj} B$ via the covering by the $D_+(M_\alpha)$). This is the last assertion of (b), and the isomorphism $\mathcal{O}_X(d) \simeq f^*\mathcal{O}_P(1)$ is the one of Prop. 1.31(b).

(a) We show that each φ_α is surjective. An element of $B_{(M_\alpha)}$ is m/M_α^k with $m \in B_{kd}$, a combination of monomials T^c of degree kd. Splitting the kd variable factors of T^c into k groups of d, we write $T^c = M_{\beta_1} \cdots M_{\beta_k}$, so $T^c/M_\alpha^k = \prod_l (M_{\beta_l}/M_\alpha) = \varphi_\alpha(\prod_l S_{\beta_l}/S_\alpha)$. Therefore $f| : D_+(M_\alpha) = f^{-1}(D_+(S_\alpha)) \to D_+(S_\alpha)$ is the morphism of affine schemes attached to a surjective ring homomorphism, i.e. a closed immersion. The $D_+(S_\alpha)$ cover P and being a closed immersion is local on the target, so f is a closed immersion.

(c) Assume $A \neq 0$ in the "only if" part. If $d \geq 1$, f is a closed immersion with $f^*\mathcal{O}_P(1) \simeq \mathcal{O}_X(d)$, so $\mathcal{O}_X(d)$ is very ample (Def. 1.26). Conversely let $d \leq 0$ and suppose $\mathcal{O}_X(d) \simeq i^*\mathcal{O}_{\mathbb{P}^M_A}(1)$ for an immersion $i : X \to \mathbb{P}^M_A$. By Prop. 1.31(a) the sheaf $i^*\mathcal{O}(1)$ is generated by the global sections i^*T_j.

If $d < 0$, then $\mathcal{O}_X(d)(X) = 0$ (Lemma 1.22), so the stalks of $\mathcal{O}_X(d)$ would be 0; but they are isomorphic to $\mathcal{O}_{X,x} \neq 0$ ($X \neq \emptyset$ since $A \neq 0$). Contradiction.

If $d = 0$: let $\phi : i^*\mathcal{O}(1) \to \mathcal{O}_X$ be an isomorphism and $a_j = \phi(i^*T_j) \in \mathcal{O}_X(X) = B_0 = A$. The a_j generate $\mathcal{O}_X$. For each prime $\mathfrak{p}$ of A there is a point x over $\mathfrak{p}$ ($X \to \operatorname{Spec} A$ is surjective), and some a_j is a unit in $\mathcal{O}_{X,x}$, hence $a_j \notin \mathfrak{p}$ ($A_\mathfrak{p} \to \mathcal{O}_{X,x}$ is local). So $(a_0, \ldots, a_M) = A$, and the constant sections a_j generate $\mathcal{O}_{\operatorname{Spec} A}$; Prop. 1.31(b) gives $\varsigma : \operatorname{Spec} A \to \mathbb{P}^M_A$ attached to them. Let $\pi : X \to \operatorname{Spec} A$ be the structure morphism and $g = \varsigma\pi$. Then $g^*\mathcal{O}(1) = \pi^*\varsigma^*\mathcal{O}(1) \simeq \pi^*\mathcal{O}_{\operatorname{Spec} A} = \mathcal{O}_X$ with $g^*T_j \leftrightarrow a_j$. By (F2) of 1.23, $i = g = \varsigma\pi$. But i is an immersion, hence injective on points, while π is not injective: for any prime $\mathfrak{p}$ the fiber over $\mathfrak{p}$ is $\mathbb{P}^n_{k(\mathfrak{p})}$ with $n \geq 1$, which contains the two distinct points $[1 : 0 : \cdots : 0]$ and $[0 : 1 : 0 : \cdots : 0]$. Contradiction. □

Exercise 1.28

> **Problem.** Let X' be a projective scheme over a ring A, and let $\mathcal{L}, \mathcal{M}$ be two invertible sheaves generated by their global sections. Let $f : X \to \mathbb{P}^n_A$, $g : X \to \mathbb{P}^m_A$ be morphisms such that $\mathcal{L} \simeq f^*\mathcal{O}_{\mathbb{P}^n_A}(1)$ and $\mathcal{M} \simeq g^*\mathcal{O}_{\mathbb{P}^m_A}(1)$. Let $Y = \mathbb{P}^n_A \times_A \mathbb{P}^m_A$, and let p, q denote the respective projections of Y onto $\mathbb{P}^n_A$ and $\mathbb{P}^m_A$. (a) Show that $\mathcal{L} \otimes \mathcal{M} \simeq h^*\mathcal{N}$, where $h = (f, g)$ and $\mathcal{N} := p^*\mathcal{O}_{\mathbb{P}^n_A}(1) \otimes q^*\mathcal{O}_{\mathbb{P}^m_A}(1)$. (b) Let $i : Y \to \mathbb{P}^{nm+n+m}_A$ be the Segre embedding (Lemma 3.3.31). Show that $\mathcal{N} \simeq i^*\mathcal{O}_{\mathbb{P}^{nm+n+m}_A}(1)$. (c) Show that if f is a closed immersion then so is h (use Lemma 3.3.15(b)). Deduce from this that $\mathcal{L} \otimes \mathcal{M}$ is very ample if $\mathcal{L}$ is very ample.

Solution. The first line of the statement says X'; since f, g are defined on X, we read $X' = X$ (the projectivity of X is not used).

(a) $h^*\mathcal{N} = h^*p^*\mathcal{O}(1) \otimes h^*q^*\mathcal{O}(1) = (ph)^*\mathcal{O}(1) \otimes (qh)^*\mathcal{O}(1) = f^*\mathcal{O}(1) \otimes g^*\mathcal{O}(1) \simeq \mathcal{L} \otimes \mathcal{M}$, since $ph = f$, $qh = g$ and inverse images commute with tensor products.

(b) Let $T_0, \dots, T_n$ and $U_0, \dots, U_m$ be the coordinates of $\mathbb{P}^n_A$ and $\mathbb{P}^m_A$ and S_{kl} ($0 \le k \le n$, $0 \le l \le m$) those of $\mathbb{P}^{nm+n+m}_A$. Recall that the Segre embedding i is glued from the morphisms $Y_{kl} := p^{-1}D_+(T_k) \cap q^{-1}D_+(U_l) = D_+(T_k) \times_A D_+(U_l) \to D_+(S_{kl})$ given by the ring homomorphisms $S_{ab}/S_{kl} \mapsto (T_a/T_k)(U_b/U_l)$. Consider the sections $\sigma_{kl} = p^*T_k \otimes q^*U_l \in \mathcal{N}(Y)$. At $y \in Y$ the stalk $(p^*\mathcal{O}(1))_y = \mathcal{O}(1)_{p(y)} \otimes \mathcal{O}_{Y,y}$ is free of rank 1 and p^*T_k generates it iff $p(y) \in D_+(T_k)$ (see (F2) of the solution of 1.23); similarly for q^*U_l. The tensor product of the two free rank 1 modules is generated by σ_{kl} iff both factors are generated by their sections (a product of two elements of a local ring is a unit iff both are). So $Y_{\sigma_{kl}} = Y_{kl}$, these sets cover Y, and the σ_{kl} generate $\mathcal{N}$. On Y_{kl} we have $\sigma_{ab}/\sigma_{kl} = (T_a/T_k)(U_b/U_l)$. Hence the morphism attached to $(\mathcal{N}, \sigma_{kl})$ in Prop. 1.31(b) is glued from the morphisms $Y_{kl} \to D_+(S_{kl})$ above, i.e. it is i. By Prop. 1.31(b), $\mathcal{N} \simeq i^*\mathcal{O}(1)$.

(c) *Claim.* Let $h : X \to Y$ and $p : Y \to Z$ be morphisms of schemes with p separated. If ph is a closed immersion (resp. an immersion), so is h. Indeed $h = \mathrm{pr}_2 \circ \Gamma_h$ with $\Gamma_h = (\mathrm{id}_X, h) : X \to X \times_Z Y$ and $\mathrm{pr}_2 : X \times_Z Y \to Y$. The graph Γ_h is the base change of the diagonal $\Delta : Y \to Y \times_Z Y$ along $h \times \mathrm{id} : X \times_Z Y \to Y \times_Z Y$, hence a closed immersion because p is separated. The projection pr_2 is the base change of $ph : X \to Z$ along $p : Y \to Z$, hence a closed immersion (resp. immersion) if ph is. A composition of two closed immersions is a closed immersion, and a closed immersion followed by an immersion is an immersion, which proves the claim.

Apply the claim with $Z = \mathbb{P}^n_A$ and $p : Y \to \mathbb{P}^n_A$ the projection, which is separated being the base change of $\mathbb{P}^m_A \to \operatorname{Spec} A$. Since $ph = f$, if f is a closed immersion then h is a closed immersion. In the same way, if f is an immersion then h is an immersion.

Now let $\mathcal{L}$ be very ample, i.e. we may take f an immersion (closed or not) with $\mathcal{L} \simeq f^*\mathcal{O}(1)$ (Def. 1.26); g is the given morphism with $\mathcal{M} \simeq g^*\mathcal{O}(1)$. Then h is an immersion, i is a closed immersion (Lemma 3.3.31), so $i \circ h : X \to \mathbb{P}^{nm+n+m}_A$ is an immersion, and by (a), (b)

$$(ih)^*\mathcal{O}(1) = h^*i^*\mathcal{O}(1) \simeq h^*\mathcal{N} \simeq \mathcal{L} \otimes \mathcal{M}.$$

So $\mathcal{L} \otimes \mathcal{M}$ is very ample. □

Exercise 1.29

> **Problem.** Let $\pi : X' \to X$ be a faithfully flat morphism of schemes. (a) Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Show that there exists a homomorphism $\varphi : \mathcal{O}_X^{(I)} \to \mathcal{F}$ such that $\varphi(X)$ is surjective. Show that $\mathcal{F}$ is generated by its global sections if and only if φ is surjective. (b) Show that $\mathcal{F}$ is generated by its global sections if and only if $\pi^*\mathcal{F}$ is generated by its global sections. (c) Let us suppose X, X' are quasi-compact. Let $\mathcal{L}$ be an invertible sheaf on X. Show that if $\pi^*\mathcal{L}$ is ample, then $\mathcal{L}$ is ample.

Solution. (a) Let $I = \mathcal{F}(X)$ and let $\varphi = \psi : \mathcal{O}_X^{(I)} \to \mathcal{F}$ be the homomorphism of the proof of Lemma 1.3: φ sends the canonical basis element ε_s to s ($s \in I$). On the submodule $\mathcal{O}_X(X)^{(I)} \subseteq \mathcal{O}_X^{(I)}(X)$ the map $\varphi(X)$ sends $\varepsilon_s \mapsto s$, hence $\varphi(X)$ is onto $\mathcal{F}(X)$. On stalks, $\varphi_x(\sum a_s\varepsilon_s) = \sum a_s s_x$, so the image of φ_x is the $\mathcal{O}_{X,x}$-module generated by the germs s_x, $s \in \mathcal{F}(X)$. Since a morphism of sheaves is surjective iff it is surjective on stalks, φ is surjective iff $\mathcal{F}$ is generated by its global sections. (Quasi-coherence is not needed.)

The statements (b) and (c) are false as printed. Let k be a field, $X = \mathbb{P}^1_k$, $U = D_+(T_0)$, $V = D_+(T_1)$, $X' = U \sqcup V$ (a disjoint union of two affine schemes, hence affine, and quasi-compact) and $\pi : X' \to X$ the natural morphism; it is flat (open immersions are flat) and surjective, so faithfully flat. Let $\mathcal{F} = \mathcal{L} = \mathcal{O}_X(-1)$. Since X' is affine and $\pi^*\mathcal{F}$ is quasi-coherent (Prop. 1.14(b)), $\pi^*\mathcal{F}$ is generated by its global sections (Thm. 1.7 and Lemma 1.3), and $\pi^*\mathcal{L}^{\otimes r} \otimes \mathcal{G}$ is globally generated for every quasi-coherent $\mathcal{G}$ for the same reason, so $\pi^*\mathcal{L}$ is ample. However $\mathcal{O}_X(-1)(X) = 0$ by Lemma 1.22 while its stalks are nonzero, so $\mathcal{F}$ is not generated by global sections, and $\mathcal{L}^{\otimes r} = \mathcal{O}_X(-r)$ is not generated by global sections for any $r \geq 1$, so $\mathcal{L}$ is not ample (take the f.g. sheaf $\mathcal{O}_X$ in Def. 1.33). The implication "$\Rightarrow$" in (b) is however true for every morphism π (see below).

Corrected hypothesis. (BC): $A \to B$ is a faithfully flat ring homomorphism, X is an A-scheme which is Noetherian or separated and quasi-compact, $X' = X \times_{\operatorname{Spec} A} \operatorname{Spec} B$ and π is the projection. Then π is affine, surjective and flat (base change of $\operatorname{Spec} B \to \operatorname{Spec} A$), so faithfully flat, and X' is quasi-compact. This is the situation of Ex. 1.30.

Lemma (FB), flat base change of sections. Let $R \to R'$ be a flat ring homomorphism, X an R-scheme that is Noetherian or separated and quasi-compact, $X' = X \times_R R'$, $\pi : X' \to X$, and $\mathcal{G}$ a quasi-coherent sheaf on X. Then the canonical map $c : \mathcal{G}(X) \otimes_R R' \to (\pi^*\mathcal{G})(X')$, $s \otimes r \mapsto r \cdot \pi^* s$, is bijective. *Proof.* For an open $W \subseteq X$ put $W' = \pi^{-1}(W) = W \times_R R'$ and let $c_W : \mathcal{G}(W) \otimes_R R' \to (\pi^*\mathcal{G})(W')$ be the analogous map. Step 1: if $W = \operatorname{Spec} C$ is affine, then $W' = \operatorname{Spec}(C \otimes_R R')$ and $\pi^*\mathcal{G}|_{W'} = (\mathcal{G}(W) \otimes_C (C \otimes_R R'))^\sim$ (Prop. 1.14(b), Thm. 1.7), whose global sections are $\mathcal{G}(W) \otimes_R R'$; so c_W is bijective. Step 2: let $W = \operatorname{Spec} C$ be affine and $V = D(g_1) \cup \cdots \cup D(g_m) \subseteq W$. We show c_V bijective by induction on m; $m = 1$ is Step 1. For $m \geq 2$ let $V_1 = D(g_1) \cup \cdots \cup D(g_{m-1})$ and $D_m = D(g_m)$; then $V = V_1 \cup D_m$ and $V_1 \cap D_m = \bigcup_{j<m} D(g_j g_m)$ is a union of $m-1$ principal open sets. The sheaf axiom gives an exact sequence $0 \to \mathcal{G}(V) \to \mathcal{G}(V_1) \oplus \mathcal{G}(D_m) \to \mathcal{G}(V_1 \cap D_m)$, and the same for $\pi^*\mathcal{G}$ on the primed sets. Tensoring the first with the flat module R' keeps it exact, and c gives a morphism between the two exact sequences whose second and third components are bijective by induction. Hence c_V is bijective. Step 3: let $W = U_1 \cup \cdots \cup U_r$ with U_j affine open in X; we prove c_W bijective by induction on r ($r = 1$ is Step 1). Put $W_1 = U_1 \cup \cdots \cup U_{r-1}$, $U = U_r$. The set $W_1 \cap U$ is quasi-compact (in the Noetherian case every open set is; in the separated case it is the finite union of the affine sets $U_j \cap U$), and open in the affine scheme U, hence a finite union of principal open subsets of U. The Mayer–Vietoris argument of Step 2 applied to $W = W_1 \cup U$ shows that c_W is bijective, using the induction hypothesis for W_1, Step 1 for U and Step 2 for $W_1 \cap U$. Finally X is quasi-compact, so it is such a W. $\square$

Corrected (b). The implication "$\Rightarrow$" holds for any π: if $\mathcal{F}_x$ is generated by the germs of $\mathcal{F}(X)$, then $(\pi^*\mathcal{F})_{x'} = \mathcal{F}_{\pi(x')} \otimes \mathcal{O}_{X',x'}$ is generated by the germs of the sections $\pi^* s$. Under (BC), "$\Leftarrow$" holds. Let $\varphi : \mathcal{O}_X^{(I)} \to \mathcal{F}$ be as in (a) and $\mathcal{C} = \operatorname{Coker} \varphi$, a quasi-coherent sheaf (Prop. 1.12(c)). By (FB), $(\pi^*\mathcal{F})(X') = \mathcal{F}(X) \otimes_A B$, so every global section of $\pi^*\mathcal{F}$ is a finite sum $\sum b_k \pi^* s_k$ with $b_k \in B$ (acting through $B \to \mathcal{O}_{X'}(X')$) and $s_k \in \mathcal{F}(X)$. Since $\pi^*\mathcal{F}$ is generated by global sections, its stalks are generated by the germs of such sections, hence by the germs of the sections $\pi^* s$, $s \in \mathcal{F}(X)$. This means that $\pi^*\varphi : \mathcal{O}_{X'}^{(I)} \to \pi^*\mathcal{F}$ is surjective (stalkwise, $(\pi^*\varphi)_{x'} = \varphi_{\pi(x')} \otimes \mathcal{O}_{X',x'}$ has image generated by the $\pi^* s$). So $\pi^*\mathcal{C} = \operatorname{Coker}(\pi^*\varphi) = 0$ (π^* is right exact). For $x \in X$ choose $x' \in \pi^{-1}(x)$ (π is surjective); $\mathcal{O}_{X,x} \to \mathcal{O}_{X',x'}$ is a flat local homomorphism, hence faithfully flat, and $0 = (\pi^*\mathcal{C})_{x'} = \mathcal{C}_x \otimes \mathcal{O}_{X',x'}$ gives $\mathcal{C}_x = 0$. So $\mathcal{C} = 0$, i.e. φ is onto and $\mathcal{F}$ is generated by global sections by (a).

Corrected (c). Assume (BC) and $\pi^*\mathcal{L}$ ample on X' (which is quasi-compact). X is quasi-compact by hypothesis. Let $\mathcal{F}$ be a finitely generated quasi-coherent sheaf on X. Then $\pi^*\mathcal{F}$ is finitely generated quasi-coherent (Prop. 1.14(b)), so there is n_0 such that $\pi^*\mathcal{F}\otimes(\pi^*\mathcal{L})^{\otimes n} = \pi^*(\mathcal{F}\otimes\mathcal{L}^{\otimes n})$ is generated by global sections for $n \geq n_0$. By corrected (b), $\mathcal{F}\otimes\mathcal{L}^{\otimes n}$ is generated by global sections for $n \geq n_0$. Hence $\mathcal{L}$ is ample. □

Exercise 1.30

> **Problem.** Let X be a proper scheme over a Noetherian ring A, and let $\mathcal{L}$ be an invertible sheaf on X. Let us suppose that there exists a faithfully flat base change $\operatorname{Spec} B \to \operatorname{Spec} A$ such that B is Noetherian and that $\pi^*\mathcal{L}$ is very ample relative to B, where $\pi : X_B \to X$ is the canonical projection. We are going to show that $\mathcal{L}$ is very ample relative to A. (a) Show that $\mathcal{L}$ is ample and generated by some global sections $s_0, \ldots, s_n$. Let $f : X \to Y := \mathbb{P}^n_A$ be the morphism associated to $\mathcal{L}$ and to the s_i. Show that $f_B : X_B \to Y_B$ is a closed immersion. (b) Let $y \in Y$ and let $y' \in Y_B$ be a point lying above y. Show that the fiber of X_B over y' is isomorphic to $X_y \times_{\operatorname{Spec} k(y)} \operatorname{Spec} k(y')$. Deduce from this that $f^{-1}(y)$ consists of at most one point, and that f is a topological closed immersion. (c) Show that $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective and that f is a closed immersion.

Solution. Throughout, X is proper over A, hence separated and quasi-compact over A, and $X_B = X \times_A B$, so (BC) of Ex. 1.29 holds and Lemma (FB) of the solution of 1.29 is available. Write $\mathcal{L}_B = \pi^*\mathcal{L}$, $Y_B = \mathbb{P}^n_B = \mathbb{P}^n_A \times_A B$, $p : Y_B \to Y$ the projection, and $f_B = f \times_A \mathrm{id}_B$, so that $pf_B = f\pi$.

(a) *Ampleness and generation.* By hypothesis there is an immersion $i : X_B \to \mathbb{P}^m_B$ with $\mathcal{L}_B \simeq i^*\mathcal{O}(1)$, and $t_j := i^*T_j$ (viewed in $\mathcal{L}_B(X_B)$) generate $\mathcal{L}_B$ (Prop. 1.31(a)). The scheme X_B is proper over B and $\mathbb{P}^m_B$ is separated, so i is proper, hence has closed image; an immersion with closed image is a closed immersion (it is a closed immersion into an open subscheme $W \subseteq \mathbb{P}^m_B$ with image closed in $\mathbb{P}^m_B$, and the sheaf map $\mathcal{O}_{\mathbb{P}^m_B} \to i_*\mathcal{O}_{X_B}$ is onto at points of W and has zero target stalk outside the closed image). So X_B is projective over B and $\mathcal{L}_B = \mathcal{O}_{X_B}(1)$ is ample by Thm. 1.27. By corrected 1.29(c), $\mathcal{L}$ is ample; since $\mathcal{L}_B$ is globally generated, corrected 1.29(b) shows that $\mathcal{L}$ is generated by global sections.

Choice of the sections. By (FB), $\mathcal{L}_B(X_B) = \mathcal{L}(X)\otimes_A B$, so we can write $t_j = \sum_k b_{jk}\pi^*r_{jk}$ with $b_{jk} \in B$, $r_{jk} \in \mathcal{L}(X)$ (finitely many). Since X is quasi-compact and $\mathcal{L}$ is generated by $\mathcal{L}(X)$, finitely many global sections generate $\mathcal{L}$ (each point has a neighborhood where finitely many sections generate; cover). Let $s_0, \ldots, s_n \in \mathcal{L}(X)$ consist of such a finite generating family together with all the r_{jk}. (Some choice like this is necessary: for $X = \mathbb{P}^1_A$, $\mathcal{L} = \mathcal{O}(2)$, which is very ample, the generating sections T_0^2, T_1^2 give a degree 2 morphism $\mathbb{P}^1 \to \mathbb{P}^1$, not an immersion.) Let $f : X \to Y = \mathbb{P}^n_A$ be the morphism of Prop. 1.31(b) with $f^*\mathcal{O}(1) \simeq \mathcal{L}$, $f^*T_i \leftrightarrow s_i$. Then $f_B^*\mathcal{O}_{Y_B}(1) = f_B^*p^*\mathcal{O}_Y(1) = \pi^*f^*\mathcal{O}_Y(1) \simeq \mathcal{L}_B$, with $f_B^*T_i \leftrightarrow \pi^*s_i$.

Each $t_j = \sum_i c_{ji}\pi^*s_i$ with $c_{ji} \in B$ (put $c_{ji} = 0$ for the s_i not occurring). Let $\ell_j = \sum_i c_{ji}T_i \in H^0(Y_B, \mathcal{O}(1))$ and $W = \bigcup_j D_+(\ell_j) \subseteq Y_B$, open. By (F1) of 1.23, ℓ_j generates $\mathcal{O}(1)$ exactly on $D_+(\ell_j)$, so the $\ell_j|_W$ generate $\mathcal{O}(1)|_W$, and Prop. 1.31(b) gives a B-morphism $c : W \to \mathbb{P}^m_B$ with $c^*\mathcal{O}(1) \simeq \mathcal{O}(1)|_W$ and $c^*T_j \leftrightarrow \ell_j$. We have $f_B(X_B) \subseteq W$: for $x \in X_B$ some t_j generates $\mathcal{L}_{B,x}$; as $f_B^*\ell_j \leftrightarrow \sum_i c_{ji}\pi^*s_i = t_j$, by the argument in the proof of (F2) of 1.23 (for any section ℓ of $\mathcal{O}(1)$, write $\ell_{g(z)} = ue$ with e a generator; $g^*\ell$ generates at z iff $g^\#u$ is a unit iff u is a unit, as $g^\#$ is local) $f_B(x) \in D_+(\ell_j)$. Moreover $cf_B : X_B \to \mathbb{P}^m_B$ satisfies $(cf_B)^*\mathcal{O}(1) \simeq f_B^*\mathcal{O}(1) \simeq \mathcal{L}_B$

with $(cf_B)^*T_j \leftrightarrow f_B^*\ell_j = t_j$, and i has the same property. By (F2), $cf_B = i$. Now W and $\mathbb{P}^m_B$ are separated B-schemes, so c is separated, and $i = cf_B$ is a closed immersion. By the Claim in the solution of 1.28(c), $f_B : X_B \to W$ is a closed immersion. Therefore $f_B : X_B \to Y_B$ is an immersion; it is also proper (X_B proper over B, Y_B separated), so its image is closed, and as in the previous paragraph it is a closed immersion.

(b) Since $X_B = X \times_A B = X \times_Y Y_B$ ($Y_B = Y \times_A B$) and f_B is the second projection, the fiber of f_B over y' is

$$X_B \times_{Y_B} \operatorname{Spec} k(y') = X \times_Y \operatorname{Spec} k(y') = (X \times_Y \operatorname{Spec} k(y)) \times_{\operatorname{Spec} k(y)} \operatorname{Spec} k(y') = X_y \times_{k(y)} \operatorname{Spec} k(y'),$$

because $\operatorname{Spec} k(y') \to Y$ factors through $\operatorname{Spec} k(y)$ (as y' lies above y). Such a y' exists for every y, since $p : Y_B \to Y$ is surjective (base change of the surjection $\operatorname{Spec} B \to \operatorname{Spec} A$).

As f_B is a closed immersion, its fiber over y' is $\operatorname{Spec}(\mathcal{O}_{Y_B,y'}/\mathcal{I} \otimes k(y'))$, a quotient of the field $k(y')$: it is empty or $\operatorname{Spec} k(y')$, a single point. The morphism $X_y \times_{k(y)} \operatorname{Spec} k(y') \to X_y$ is surjective (base change of $\operatorname{Spec} k(y') \to \operatorname{Spec} k(y)$). Hence X_y has at most one point, i.e. $f^{-1}(y)$ has at most one point. Thus f is injective. The morphism f is proper (X proper over A, Y separated over A), hence a closed map. An injective, continuous, closed map is a homeomorphism onto a closed subset: f is a topological closed immersion.

(c) The morphism f is separated and quasi-compact, so $f_*\mathcal{O}_X$ is quasi-coherent (Prop. 1.14(c)). Let $Y' = \operatorname{Spec} C \subseteq Y$ be an affine open set, $X'' = f^{-1}(Y')$ (separated and quasi-compact over C, since f is) and let $C_B = C \otimes_A B = \mathcal{O}(p^{-1}Y')$, so that $f_B^{-1}(p^{-1}Y') = X'' \times_C C_B = X''_B$. The sheaf $f_*\mathcal{O}_X|_{Y'}$ is $\mathcal{O}(X'')^\sim$ (Thm. 1.7), so it suffices to show that $u : C \to \mathcal{O}(X'')$ is surjective. Let $M = \operatorname{Coker} u$. By (FB) (applied to X'' over $R = C$ with $R' = C_B$, flat over C), $\mathcal{O}(X'') \otimes_C C_B = \mathcal{O}(X''_B)$, and under this identification $u \otimes \mathrm{id}$ becomes the structure map $C_B \to \mathcal{O}(X''_B)$ of f_B over the affine scheme $p^{-1}Y' = \operatorname{Spec} C_B$. This map is surjective, because f_B restricted to X''_B is a closed immersion into an affine scheme (a closed subscheme of $\operatorname{Spec} C_B$ is $\operatorname{Spec} C_B/J$). Therefore $M \otimes_C C_B = \operatorname{Coker}(u \otimes \mathrm{id}) = 0$. Since $C \to C_B = C \otimes_A B$ is faithfully flat (base change of $A \to B$), $M = 0$. So u is onto for all affine Y', i.e. $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective (Prop. 1.5(b)). Together with (b), f is a closed immersion. Since $\mathcal{L} \simeq f^*\mathcal{O}_Y(1)$, the sheaf $\mathcal{L}$ is very ample relative to A (Def. 1.26). □

Exercise 1.31

> **Problem.** Let $\mathcal{L}$ be an ample sheaf and $\mathcal{M}$ an invertible sheaf on a Noetherian scheme. Show that $\mathcal{L} \otimes \mathcal{M}$ is ample if $\mathcal{M}$ is ample or if $\mathcal{M}$ is generated by its global sections.

Solution. A Noetherian scheme is quasi-compact. We use two remarks. (R1) If $\mathcal{G}, \mathcal{H}$ are generated by global sections, so is $\mathcal{G} \otimes \mathcal{H}$: for $x \in X$, $(\mathcal{G} \otimes \mathcal{H})_x = \mathcal{G}_x \otimes \mathcal{H}_x$ is generated by the elements $u_x \otimes v_x$ with $u \in \mathcal{G}(X)$, $v \in \mathcal{H}(X)$, which are the germs of the global sections $u \otimes v$. (R2) Consequently if $\mathcal{M}$ is generated by global sections, so is $\mathcal{M}^{\otimes n}$ for all $n \geq 1$.

In both cases there is $n_1 \geq 1$ such that $\mathcal{M}^{\otimes n}$ is generated by global sections for all $n \geq n_1$: if $\mathcal{M}$ is globally generated take $n_1 = 1$ by (R2); if $\mathcal{M}$ is ample apply Def. 1.33 to the finitely generated sheaf $\mathcal{O}_X$.

Let $\mathcal{F}$ be a finitely generated quasi-coherent sheaf. As $\mathcal{L}$ is ample there is n_0 with $\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ generated by global sections for $n \geq n_0$. For $n \geq \max(n_0, n_1)$,

$$\mathcal{F} \otimes (\mathcal{L} \otimes \mathcal{M})^{\otimes n} \simeq (\mathcal{F} \otimes \mathcal{L}^{\otimes n}) \otimes \mathcal{M}^{\otimes n}$$

is generated by global sections by (R1). The sheaf $\mathcal{L} \otimes \mathcal{M}$ is invertible; hence it is ample. □

Exercise 1.32

> **Problem.** Let $\mathcal{L}$ be an invertible sheaf on a quasi-compact scheme. Show that the following properties are equivalent: (i) $\mathcal{L}$ is ample; (ii) $\mathcal{L}^n$ is ample for every $n \geq 1$; (iii) there exists an $n \geq 1$ such that $\mathcal{L}^n$ is ample.

Solution. Write $\mathcal{L}^n = \mathcal{L}^{\otimes n}$.

(i)⇒(ii). Let $n \geq 1$ and let $\mathcal{F}$ be finitely generated quasi-coherent. By (i) there is n_0 with $\mathcal{F} \otimes \mathcal{L}^m$ globally generated for all $m \geq n_0$. For $q \geq n_0$ we have $nq \geq n_0$, so $\mathcal{F} \otimes (\mathcal{L}^n)^q = \mathcal{F} \otimes \mathcal{L}^{nq}$ is globally generated. Hence $\mathcal{L}^n$ is ample.

(ii)⇒(iii). Take $n = 1$.

(iii)⇒(i). Let $\mathcal{L}^n$ be ample, and let $\mathcal{F}$ be finitely generated quasi-coherent. For each $r = 0, \ldots, n-1$ the sheaf $\mathcal{F} \otimes \mathcal{L}^r$ is finitely generated quasi-coherent, so there is q_r with $(\mathcal{F} \otimes \mathcal{L}^r) \otimes (\mathcal{L}^n)^q$ globally generated for all $q \geq q_r$. Let $q_0' = \max_r q_r$ and $N_0 = nq_0'$. If $N \geq N_0$, write $N = nq + r$ with $0 \leq r < n$; then $n(q+1) > N \geq nq_0'$ gives $q \geq q_0'$, and $\mathcal{F} \otimes \mathcal{L}^N = (\mathcal{F} \otimes \mathcal{L}^r) \otimes (\mathcal{L}^n)^q$ is globally generated. So $\mathcal{L}$ is ample. □

Exercise 1.33

> **Problem.** Let X be a Noetherian or separated and quasi-compact scheme, and let $f : X \to Y = \operatorname{Spec} A$ be a morphism. (a) Let $\mathcal{F}$ be a finitely generated quasi-coherent sheaf on X. Let M be a sub-A-module of $\mathcal{F}(X)$, and let $x \in X$ be such that $M \otimes \mathcal{O}_{X,x} \to \mathcal{F}_x$ is surjective. Show that this is then true in an open neighborhood of x. (b) Let $\mathcal{F}$ be as in (a). Show that $\mathcal{F}$ is generated by its global sections if and only if for every $y \in Y$, the sheaf $\rho^*\mathcal{F}$, where ρ is the projection $X \times_Y \operatorname{Spec} \mathcal{O}_{Y,y} \to \operatorname{Spec} \mathcal{O}_{Y,y}$, is generated by its global sections. (c) Let $\mathcal{L}$ be an invertible sheaf on X. Let us suppose that $\rho^*\mathcal{L}$ is ample for every $y \in Y$. Show that $\mathcal{L}$ is ample.

Solution. In (b), (c) the map $\rho = \rho_y$ is read as the projection $X_{(y)} := X \times_Y \operatorname{Spec} \mathcal{O}_{Y,y} = X \times_A \operatorname{Spec} A_{\mathfrak{p}} \to X$, where $\mathfrak{p}$ is the prime of y (as in Prop. 1.37), since $\mathcal{F}$ lives on X. The morphism ρ_y is affine and flat, $X_{(y)}$ is quasi-compact, and $\rho_y^*\mathcal{F}$ is finitely generated quasi-coherent when $\mathcal{F}$ is (Prop. 1.14(b)).

(a) The hypothesis says that the images of the elements of M generate the finitely generated $\mathcal{O}_{X,x}$-module $\mathcal{F}_x$. Each of finitely many generators of $\mathcal{F}_x$ is a finite $\mathcal{O}_{X,x}$-combination of germs of elements of M, so there are $m_1, \ldots, m_r \in M$ whose germs generate $\mathcal{F}_x$. Let $\mathcal{G} = \operatorname{Coker}(\mathcal{O}_X^r \to \mathcal{F})$, the map sending the j-th basis vector to m_j. Then $\mathcal{G}$ is a quotient of the finitely generated sheaf $\mathcal{F}$, so finitely generated, and $\mathcal{G}_x = 0$. Choose an open $V \ni x$ and a surjection $\mathcal{O}_V^k \to \mathcal{G}|_V$; the images $g_1, \ldots, g_k \in \mathcal{G}(V)$ of the basis vectors have zero germ at x, so they vanish on an

open $V' \ni x$, $V' \subseteq V$. Then $\mathcal{G}|_{V'} = 0$, i.e. $\mathcal{O}_{V'}^r \to \mathcal{F}|_{V'}$ is surjective, and so $M \otimes \mathcal{O}_{X,x'} \to \mathcal{F}_{x'}$ is surjective for all $x' \in V'$. (No hypothesis on X is needed.)

(b) *Local rings of $X_{(y)}$.* Let $x \in X$ with $f(x) = y$ (y corresponds to $\mathfrak{p}$). If $U = \operatorname{Spec} C$ is an affine open neighborhood of x, x corresponds to a prime $\mathfrak{q}$ of C with $\mathfrak{q} \cap A = \mathfrak{p}$, and $U \times_A \operatorname{Spec} A_{\mathfrak{p}} = \operatorname{Spec} S^{-1}C$ with S the image of $A \setminus \mathfrak{p}$ in C, $S \cap \mathfrak{q} = \emptyset$. So there is a point $x' \in X_{(y)}$ above x, and $\mathcal{O}_{X_{(y)},x'} = C_{\mathfrak{q}} = \mathcal{O}_{X,x}$.

Claim P. Let $\mathcal{G}$ be a quasi-coherent sheaf on X, $x \in X$, $y = f(x)$ and x' as above. If $(\rho_y^*\mathcal{G})_{x'}$ is generated by the germs of the global sections of $\rho_y^*\mathcal{G}$, then $\mathcal{G}_x$ is generated by the germs of $\mathcal{G}(X)$. *Proof.* By Lemma (FB) of the solution of 1.29 (applied with $R = A$, $R' = A_{\mathfrak{p}}$, flat), $(\rho_y^*\mathcal{G})(X_{(y)}) = \mathcal{G}(X) \otimes_A A_{\mathfrak{p}}$; its elements are the sums $\sum(a_k/s_k)\rho^* u_k$ with $u_k \in \mathcal{G}(X)$, $a_k \in A$, $s_k \in A \setminus \mathfrak{p}$. Moreover $(\rho_y^*\mathcal{G})_{x'} = \mathcal{G}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{X_{(y)},x'} = \mathcal{G}_x$, and the germ at x' of such a section is $\sum(a_k/s_k)(u_k)_x$, where a_k/s_k acts through $A_{\mathfrak{p}} \to \mathcal{O}_{X,x}$ (the s_k are units in $\mathcal{O}_{X,x}$ since $f(x) = y$). Thus the germs of global sections of $\rho_y^*\mathcal{G}$ at x' lie in the $\mathcal{O}_{X,x}$-submodule of $\mathcal{G}_x$ generated by the germs of $\mathcal{G}(X)$. If these generate $\mathcal{G}_x$, so does $\mathcal{G}(X)$. □

Now (b). "⇒": if $\mathcal{F}$ is generated by global sections, then for each $x' \in X_{(y)}$ over x, $(\rho^*\mathcal{F})_{x'} = \mathcal{F}_x \otimes \mathcal{O}_{X_{(y)},x'}$ is generated by the germs of the sections $\rho^* u$, $u \in \mathcal{F}(X)$, which are global sections of $\rho^*\mathcal{F}$. "⇐": let $x \in X$ and $y = f(x)$; $\rho_y^*\mathcal{F}$ is generated by global sections, in particular at x', so by Claim P, $\mathcal{F}_x$ is generated by the germs of $\mathcal{F}(X)$. (Finite generation of $\mathcal{F}$ is not needed here.)

(c) X is quasi-compact (Noetherian, or by hypothesis), so the definition of ample applies. Let $\mathcal{F}$ be a finitely generated quasi-coherent sheaf on X. We show that every $x \in X$ has an open neighborhood W_x and an integer N_x such that $\mathcal{F} \otimes \mathcal{L}^N$ is generated by its global sections on W_x (at all points of W_x) for every $N \geq N_x$. Since X is quasi-compact, finitely many W_x cover X, and then $N_0 = \max N_x$ works for $\mathcal{F}$, proving that $\mathcal{L}$ is ample.

Fix x, $y = f(x)$, $x' \in X_{(y)}$ above x. Since $\rho_y^*\mathcal{L}$ is ample on $X_{(y)}$, applying Def. 1.33 to $\rho_y^*\mathcal{F}$ and to $\mathcal{O}_{X_{(y)}}$ gives an n_y and a $c \geq 1$ such that $\rho_y^*(\mathcal{F} \otimes \mathcal{L}^N) = \rho_y^*\mathcal{F} \otimes (\rho_y^*\mathcal{L})^N$ is globally generated for $N \geq n_y$ and $\rho_y^*(\mathcal{L}^c) = (\rho_y^*\mathcal{L})^c$ is globally generated. By Claim P, $\mathcal{F} \otimes \mathcal{L}^N$ is generated by $\mathcal{G}(X)$ at x for all $N \geq n_y$ (with $\mathcal{G} = \mathcal{F} \otimes \mathcal{L}^N$) and $\mathcal{L}^c$ is generated by its global sections at x. The sheaves $\mathcal{F} \otimes \mathcal{L}^N$ and $\mathcal{L}^c$ are finitely generated quasi-coherent, so by (a) (with M the whole module of global sections) there is an open neighborhood W_x of x on which $\mathcal{L}^c$ and the c sheaves $\mathcal{F} \otimes \mathcal{L}^N$, $N = n_y, \ldots, n_y + c - 1$, are all generated by global sections. Let $N \geq n_y$ and write $N = j + qc$ with $n_y \leq j < n_y + c$, $q \geq 0$. Then $\mathcal{F} \otimes \mathcal{L}^N = (\mathcal{F} \otimes \mathcal{L}^j) \otimes (\mathcal{L}^c)^q$ is generated by global sections on W_x by (R1) of the solution of 1.31 (stalkwise tensor product of globally generated sheaves; the case $q = 0$ is trivial). Take $N_x = n_y$. This proves the claim and (c). □

5.2 Cech cohomology

Cohomology means Cech cohomology as in Def. 2.10, and results of the text are cited by number ("Thm. 2.19", "Prop. 1.6", ...); Chapter 2 and 3 results are cited as "Ex. 2.2.8", "Prop. 3.3.6", and so on. For a sequence $\alpha = (\alpha_0, \ldots, \alpha_{q-1})$ of indices, $U_\alpha = U_{\alpha_0} \cap \cdots \cap U_{\alpha_{q-1}}$, and $\alpha^{(l)}$ denotes α with the entry α_l removed.

Honest notes. In Ex. 2.1(c) a total ordering is not enough for the Zorn argument (one needs

to be able to take the least element not yet treated), so the ordering is taken to be a well-ordering and (H_J) is only considered for initial segments J; also (a) is vacuous as printed. In Ex. 2.4(b) the finiteness statement needs X Noetherian (it can fail for a non-Noetherian one-dimensional domain), so (b) is proved for Noetherian X; (c) is not affected. In Ex. 2.6(b) direct limits commute with finite products only, so (b) is proved for finite coverings (the statement for infinite coverings is not proved); this is enough for (c), because the finite coverings are cofinal on the quasi-compact space X. In Ex. 2.7(c) the gluing direction of Ex. 2.2.8 is as in the book. In Ex. 2.8(b) the complex (2.9) is taken with ordered cochains C'' (Cor. 2.4), which is what makes it zero from rank n on. Ex. 2.9 is proved by replacing the kernel of $\mathcal{O}_X^n \to \mathcal{O}_X$ by an induction involving only ideal sheaves.

Exercise 2.1

Problem. (Cohomology of flasque sheaves) Let X be a topological space. We say that a sheaf $\mathcal{F}$ on X is flasque if for every pair of open subsets $V \subseteq U$, the restriction $\mathcal{F}(U) \to \mathcal{F}(V)$ is surjective. Let $\mathcal{U} = \{U_i\}_{i\in I}$ be a covering of X. We are going to show that $H^{p+1}(\mathcal{U}, \mathcal{F}) = 0$ for every $p \geq 0$. Let us endow I with a total ordering with minimal element i_0. Let us fix an $f \in C^{p+1}(\mathcal{U}, \mathcal{F})$ such that $df = 0$.

(a) For any $i \in I$, let $\mathcal{U}_i$ denote the covering $U_i \cap \mathcal{U}$ of U_i. Note that $C^p(\mathcal{U}, \mathcal{F}) = C^{p-1}(\mathcal{U}_i, \mathcal{F}) \times \prod_{i'\neq i} C^{p-1}(\mathcal{U}_{i'}, \mathcal{F})$. Let J be a subset of I containing i_0. Consider the hypothesis (H_J): there exist $g_i = (g_{i\alpha})_{\alpha\in I^p} \in C^{p-1}(\mathcal{U}_i, \mathcal{F}|_{U_i})$, $i \in J$, such that for every $i, j \in J$ with $i < j$ and every $\alpha \in I^p$ we have $f_{ij\alpha} = g_{j\alpha}|_{U_{ij\alpha}} - (dg_i)_{j\alpha}$. Show that ($\mathrm{H}_J$) is satisfied if $J = \{i_0\}$.

(b) Suppose (H_J) is true. Let $k \in I$ be such that $k > i$ for every $i \in J$. Show that for any $i, j \in J$, $i \neq j$, we have $f_{ik\alpha} + (dg_i)_{k\alpha} = f_{jk\alpha} + (dg_j)_{k\alpha}$ on $U_{ijk\alpha}$ (develop $(df)_{ijk\alpha}$). Show that there exist $h_{k\alpha} \in \mathcal{F}(\bigcup_{i\in J} U_{ik\alpha})$ such that $h_{k\alpha}|_{U_{ik\alpha}} = f_{ik\alpha} + (dg_i)_{k\alpha}$ for every $i \in J$. Deduce that ($\mathrm{H}_{J\cup\{k\}}$) is true (lift $h_{k\alpha}$ to $g_{k\alpha} \in \mathcal{F}(U_{k\alpha})$).

(c) Show that (H_I) is true (consider the set of couples $(J, (g_i)_{i\in J})$ verifying (H_J), ordered in a natural manner, and apply Zorn's lemma). Deduce from this that $H^{p+1}(\mathcal{U}, \mathcal{F}) = 0$ (use Corollary 2.4).

Solution. Conventions. For $i \in I$ the complex $C^\bullet(\mathcal{U}_i, \mathcal{F}|_{U_i})$ is the Cech complex of the covering $\{U_i \cap U_m\}_{m\in I}$ of U_i, so $C^q(\mathcal{U}_i, \mathcal{F}|_{U_i}) = \prod_{\beta\in I^{q+1}} \mathcal{F}(U_{i\beta})$. For $p = 0$ the statement uses $C^{-1}(\mathcal{U}_i, \mathcal{F}|_{U_i})$; we set $C^{-1}(\mathcal{U}_i, \mathcal{F}|_{U_i}) := \mathcal{F}(U_i)$ (one empty index sequence) with $d(g) = (g|_{U_{im}})_{m\in I} \in C^0$. Then $d^2 = 0$ in all degrees ≥ -1 (for $q = -1$: $(ddg)_{jk} = g|_{U_{ijk}} - g|_{U_{ijk}} = 0$). (Thus $C^p(\mathcal{U}, \mathcal{F}) = \prod_{i\in I} C^{p-1}(\mathcal{U}_i, \mathcal{F}|_{U_i})$ by singling out the first index, which is the decomposition of the statement.) A family $(g_i)_i$ with $g_i \in C^{p-1}(\mathcal{U}_i, \mathcal{F}|_{U_i})$ is the same as an element $G \in C^p(\mathcal{U}, \mathcal{F})$ via $G_{i\alpha} = g_{i\alpha}$; this is the identification in the statement. We write $g_{i\alpha} = (g_i)_\alpha$; all identities below are identities of sections over the indicated open set, restrictions being understood. Since I may carry any total ordering for Cor. 2.4, we *choose a well-ordering of* I; i_0 is then its least element.

A formula. For $j \in I$, $k \in I$, $\alpha \in I^p$ the definition of d gives

$$(dg_j)_{k\alpha} = g_{j\alpha} - \sum_{l=0}^{p-1} (-1)^l g_{jk\alpha^{(l)}} \qquad (1)$$

(remove k: sign + and get $g_{j\alpha}$; remove α_l, which sits in position $l+1$: sign $(-1)^{l+1}$). For $p=0$ this reads $(dg_j)_k = g_j|_{U_{jk}}$.

(a). For $J=\{i_0\}$ there is no pair $i<j$ in J, so (H_J) is an empty condition and it holds with $g_{i_0}=0$.

(b). Assume (H_J) and let $k>i$ for all $i\in J$. For $i\in J$ put

$$\psi_i := f_{ik\alpha} + (dg_i)_{k\alpha} \in \mathcal{F}(U_{ik\alpha}).$$

Let $i<j$ be in J. We compute $\psi_j-\psi_i$ on $U_{ijk\alpha}$. Since $df=0$, expanding $(df)_{ijk\alpha}$ (positions $0,1,2$ hold i,j,k; α_l sits in position $l+3$) gives

$$f_{jk\alpha} - f_{ik\alpha} + f_{ij\alpha} - \sum_{l=0}^{p-1}(-1)^l f_{ijk\alpha^{(l)}} = 0.$$

By (H_J) applied to the pair $i<j$ and to the index sequences α and $k\alpha^{(l)}\in I^p$,

$$f_{ij\alpha} = g_{j\alpha} - (dg_i)_{j\alpha}, \qquad f_{ijk\alpha^{(l)}} = g_{jk\alpha^{(l)}} - (dg_i)_{jk\alpha^{(l)}}.$$

Substituting, and using (1) for $(dg_j)_{k\alpha}$,

$$\begin{aligned}\psi_j-\psi_i &= \big(f_{jk\alpha}-f_{ik\alpha}\big) + (dg_j)_{k\alpha} - (dg_i)_{k\alpha}\\ &= -g_{j\alpha} + (dg_i)_{j\alpha} + \sum_l (-1)^l\big(g_{jk\alpha^{(l)}} - (dg_i)_{jk\alpha^{(l)}}\big) + g_{j\alpha} - \sum_l (-1)^l g_{jk\alpha^{(l)}} - (dg_i)_{k\alpha}\\ &= (dg_i)_{j\alpha} - (dg_i)_{k\alpha} - \sum_l (-1)^l (dg_i)_{jk\alpha^{(l)}} \;=\; -(d^2 g_i)_{jk\alpha} = 0.\end{aligned}$$

The last equality is the expansion $(d^2g_i)_{jk\alpha} = (dg_i)_{k\alpha} - (dg_i)_{j\alpha} + \sum_l (-1)^l (dg_i)_{jk\alpha^{(l)}}$ and $d^2=0$ in $C^\bullet(\mathcal{U}_i, \mathcal{F}|_{U_i})$. This proves $\psi_i=\psi_j$ on $U_{ijk\alpha}$ for all $i\neq j$ in J (the case $i>j$ is the same with the roles exchanged).

The open sets $U_{ik\alpha}=U_i\cap U_{k\alpha}$, $i\in J$, therefore carry sections ψ_i that agree on overlaps, so they glue to a section $h_{k\alpha}$ over $W_{k\alpha}=\bigcup_{i\in J} U_{ik\alpha}$ with $h_{k\alpha}|_{U_{ik\alpha}}=\psi_i$. As $W_{k\alpha}\subseteq U_{k\alpha}$ and $\mathcal{F}$ is flasque, there is $g_{k\alpha}\in\mathcal{F}(U_{k\alpha})$ with $g_{k\alpha}|_{W_{k\alpha}}=h_{k\alpha}$. Put $g_k=(g_{k\alpha})_{\alpha\in I^p}\in C^{p-1}(\mathcal{U}_k,\mathcal{F}|_{U_k})$. For $i\in J$ we get $g_{k\alpha}|_{U_{ik\alpha}} = f_{ik\alpha}+(dg_i)_{k\alpha}$, that is $f_{ik\alpha}=g_{k\alpha}-(dg_i)_{k\alpha}$, which is the condition of $(\mathrm{H}_{J\cup\{k\}})$ for the pairs $i<k$; the pairs inside J are unchanged.

(c). Let $\mathfrak{S}$ be the set of couples $(J,(g_i)_{i\in J})$ such that J is an initial segment of I (if $j\in J$ and $i<j$ then $i\in J$) containing i_0 and (H_J) holds, ordered by $(J,g)\le(J',g')$ iff $J\subseteq J'$ and $g'_i=g_i$ for $i\in J$. By (a), $(\{i_0\},0)\in\mathfrak{S}$. If $\{(J_\lambda,g^\lambda)\}$ is a chain, put $J=\bigcup J_\lambda$ with the common g_i; J is an initial segment and (H_J) holds because two indices $i<j$ of J lie in a common J_λ. So Zorn's lemma gives a maximal (J,g). If $J\neq I$, let $k=\min(I\setminus J)$; since J is an initial segment, $k>i$ for all $i\in J$, and (b) extends (J,g) to $J\cup\{k\}$ (again an initial segment), contradicting maximality. So $J=I$: (H_I) holds.

Conclusion. By Prop. 2.3 every class of $H^{p+1}(\mathcal{U},\mathcal{F})$ is represented by an alternating cocycle, so we may assume f alternating. Let $G\in C^p(\mathcal{U},\mathcal{F})$ be the cochain with $G_{i\alpha}=g_{i\alpha}$. The expansion of $(dG)_{ij\alpha}=G_{j\alpha}-G_{i\alpha}+\sum_l(-1)^l G_{ij\alpha^{(l)}}$ and formula (1) (with $j\leftrightarrow i$, $k\leftrightarrow j$) give $(dg_i)_{j\alpha}=G_{i\alpha}-\sum_l(-1)^l G_{ij\alpha^{(l)}}$, so (H_I) says exactly

$$f_{ij\alpha}=(dG)_{ij\alpha} \qquad \text{for all } i<j \text{ in } I,\ \alpha\in I^p.$$

Let $\pi : C \to C''$ be the projection onto the coordinates with strictly increasing indices. It commutes with the differentials, since removing an index from an increasing sequence leaves it increasing, and its restriction to C' is an isomorphism of complexes (proof of Cor. 2.4). The displayed identity says $\pi(f) = d\,\pi(G)$ in C''^{p+1}. Choose $G' \in C'^{p}$ with $\pi(G') = \pi(G)$. Then $\pi(f) = \pi(dG')$, and injectivity of $\pi|_{C'}$ gives $f = dG'$. So f is a coboundary and $H^{p+1}(\mathcal{U}, \mathcal{F}) = 0$ for every $p \geq 0$. Passing to the direct limit (Def. 2.10), $H^q(X, \mathcal{F}) = 0$ for all $q \geq 1$ as well. $\square$

Exercise 2.2

> **Problem.** Let $f : X \to Y$ be an affine morphism (i.e., for every affine open subset V of Y, $f^{-1}(V)$ is affine). Show that for any quasi-coherent sheaf $\mathcal{F}$ on X, and for any $p \geq 1$, we have $R^p f_* \mathcal{F} = 0$.

Solution. First, f is quasi-compact and separated, so that Prop. 2.28 applies. Quasi-compactness: for every affine open $V \subseteq Y$, $f^{-1}(V)$ is affine, hence quasi-compact. Separatedness: for affine $V \subseteq Y$, the composition $f^{-1}(V) \to V \to \operatorname{Spec} \mathbf{Z}$ is a morphism between affine schemes, hence separated (affine schemes are separated), so $f^{-1}(V) \to V$ is separated by Prop. 3.3.9(e). Being separated is local on the target (the diagonal being a closed immersion can be checked over an open covering of Y), so f is separated.

By Prop. 2.28, $R^p f_* \mathcal{F}$ is the quasi-coherent sheaf on Y with $R^p f_* \mathcal{F}(V) = H^p(f^{-1}(V), \mathcal{F}|_{f^{-1}(V)})$ for every affine open $V \subseteq Y$. For such V the scheme $f^{-1}(V)$ is affine and $\mathcal{F}|_{f^{-1}(V)}$ is quasi-coherent, so this group is 0 for $p \geq 1$ by Thm. 2.18. Now let $W \subseteq Y$ be any open set and $s \in R^p f_* \mathcal{F}(W)$. Cover W by affine opens V_λ; then $s|_{V_\lambda} = 0$ for all λ, hence $s = 0$ by the sheaf axiom. So $R^p f_* \mathcal{F} = 0$ for $p \geq 1$. $\square$

Exercise 2.3

> **Problem.** Let $f : X \to Y$ be a morphism of schemes. Let $\mathcal{F}$ be a sheaf on X.
>
> (a) Show that we have a canonical homomorphism $H^p(Y, f_* \mathcal{F}) \to H^p(X, \mathcal{F})$ for every $p \geq 0$.
>
> (b) Let us suppose $\mathcal{F}$ is quasi-coherent. Show that the homomorphism in (a) is an isomorphism in each of the following cases: (1) X is separated and f is affine (use Theorem 2.19); (2) f is a closed immersion.

Solution. *(a).* For a covering $\mathcal{V} = \{V_i\}_{i \in I}$ of Y put $f^{-1}\mathcal{V} = \{f^{-1}(V_i)\}_{i \in I}$, a covering of X. Since $f^{-1}(V_{i_0 \ldots i_p}) = f^{-1}(V_{i_0}) \cap \cdots \cap f^{-1}(V_{i_p})$ and $(f_* \mathcal{F})(W) = \mathcal{F}(f^{-1} W)$,

$$C^p(\mathcal{V}, f_* \mathcal{F}) = \prod_{(i_0, \ldots, i_p)} \mathcal{F}\big(f^{-1} V_{i_0 \ldots i_p}\big) = C^p(f^{-1}\mathcal{V}, \mathcal{F}),$$

and this equality is compatible with d (both are alternating sums of restrictions). Hence $H^p(\mathcal{V}, f_* \mathcal{F}) = H^p(f^{-1}\mathcal{V}, \mathcal{F})$. If $\mathcal{W} = \{W_j\}_{j \in J}$ refines $\mathcal{V}$ via $\sigma : J \to I$, then $f^{-1}\mathcal{W}$ refines $f^{-1}\mathcal{V}$ via the same σ and the refinement maps correspond under the identification. Compose the identification with the canonical map $H^p(f^{-1}\mathcal{V}, \mathcal{F}) \to H^p(X, \mathcal{F})$ to get $\varphi_{\mathcal{V}} : H^p(\mathcal{V}, f_* \mathcal{F}) \to H^p(X, \mathcal{F})$. If $\mathcal{W}$ refines $\mathcal{V}$, then $\varphi_{\mathcal{W}} \circ \sigma^* = \varphi_{\mathcal{V}}$, because the canonical maps to the direct limit are compatible

with refinement. Hence the $\varphi_{\mathcal{V}}$ induce $\varphi : H^p(Y, f_*\mathcal{F}) = \varinjlim_{\mathcal{V}} H^p(\mathcal{V}, f_*\mathcal{F}) \to H^p(X, \mathcal{F})$. It is canonical (no choices), and it needs no hypothesis on $\mathcal{F}$ or f.

(b)(1). Let $\mathcal{V} = \{V_i\}_{i\in I}$ be a covering of Y by affine open sets. We show

$$H^p(\mathcal{V}, f_*\mathcal{F}) \to H^p(Y, f_*\mathcal{F}) \quad \text{and} \quad H^p(f^{-1}\mathcal{V}, \mathcal{F}) \to H^p(X, \mathcal{F})$$

are isomorphisms. Since $\varphi \circ (\text{can}) = (\text{can}) \circ (\text{identification of (a)})$ by construction, this shows that φ is an isomorphism. We do not assume Y separated.

The second map: each $f^{-1}(V_i)$ is affine as f is affine, so $f^{-1}\mathcal{V}$ is an affine covering of the separated scheme X and Thm. 2.19 applies.

The first map: we use Thm. 2.12 (Leray) for the sheaf $\mathcal{G} = f_*\mathcal{F}$ on Y and the covering $\mathcal{V}$, with $(\mathcal{V}^\lambda)$ the family of all coverings of Y by affine open sets (cofinal by Rem. 2.11). Let $\alpha = (i_0, \dots, i_n)$ and $\mathcal{W} = \{W_j\}_{j\in J}$ an affine covering of Y. The covering of V_α is $\mathcal{W}_\alpha = \{V_\alpha \cap W_j\}_j$. By (a) applied over V_α (the morphism $f^{-1}(V_\alpha) \to V_\alpha$),

$$H^q(\mathcal{W}_\alpha, \mathcal{G}|_{V_\alpha}) = H^q\big(f^{-1}\mathcal{W}_\alpha, \mathcal{F}|_{f^{-1}V_\alpha}\big).$$

Now $f^{-1}(V_\alpha) = \bigcap_k f^{-1}(V_{i_k})$ is an intersection of affine opens of the separated scheme X, hence affine (Prop. 3.3.6), and likewise $f^{-1}(V_\alpha \cap W_j) = f^{-1}(V_\alpha) \cap f^{-1}(W_j)$ is affine, since $f^{-1}(W_j)$ is affine. So $f^{-1}\mathcal{W}_\alpha$ is an affine covering of the affine (separated) scheme $f^{-1}(V_\alpha)$, and by Thm. 2.19 and Thm. 2.18, for $q \geq 1$,

$$H^q(f^{-1}\mathcal{W}_\alpha, \mathcal{F}) = H^q(f^{-1}(V_\alpha), \mathcal{F}) = 0.$$

So the hypothesis of Thm. 2.12 holds and $H^p(\mathcal{V}, f_*\mathcal{F}) \to H^p(Y, f_*\mathcal{F})$ is an isomorphism. This proves (1).

(b)(2). Let f be a closed immersion. Then f is a homeomorphism onto the closed set $f(X)$ (the argument below does not use quasi-coherence). Let Φ send the class of a covering $\mathcal{V}$ of Y to the class of $f^{-1}\mathcal{V}$; this is well defined and order preserving (a refinement of $\mathcal{V}$ pulls back to a refinement). By the identification of (a), it suffices to show that

$$\varinjlim_{[\mathcal{V}]} H^p(f^{-1}\mathcal{V}, \mathcal{F}) \longrightarrow \varinjlim_{[\mathcal{U}]} H^p(\mathcal{U}, \mathcal{F}) = H^p(X, \mathcal{F})$$

is an isomorphism (the left limit over classes of coverings of Y, the right one over classes of coverings of X).

Φ *is surjective.* (If $X = \emptyset$ then $f_*\mathcal{F} = 0$ and both sides vanish, so assume $X \neq \emptyset$.) Let $\mathcal{U} = \{U_j\}_{j\in J}$ be a covering of X. Because X has the topology induced from Y, there are open $W_j \subseteq Y$ with $f^{-1}(W_j) = U_j$. Then $\mathcal{V} = \{W_j\}_j \cup \{Y \setminus f(X)\}$ is a covering of Y ($f(X)$ is closed), and $f^{-1}\mathcal{V} = \{U_j\}_j \cup \{\emptyset\}$. This is equivalent to $\mathcal{U}$: $\mathcal{U}$ refines it by the inclusion of indices, and it refines $\mathcal{U}$ by sending the extra index to any $j_0 \in J$. Consequently, by Cor. 2.9 the refinement map is an isomorphism $H^p(\mathcal{U}, \mathcal{F}) \simeq H^p(f^{-1}\mathcal{V}, \mathcal{F})$, compatible with the maps to $H^p(X, \mathcal{F})$ (Lemma 2.8). Since every element of $H^p(X, \mathcal{F})$ comes from some $H^p(\mathcal{U}, \mathcal{F})$, the limit map displayed above is surjective.

Injectivity. Let $\xi \in H^p(f^{-1}\mathcal{V}, \mathcal{F})$ be zero in $H^p(X, \mathcal{F})$. Then ξ becomes zero in $H^p(\mathcal{U}', \mathcal{F})$ for some covering $\mathcal{U}'$ refining $f^{-1}\mathcal{V}$. Choose $\mathcal{V}'$ with $f^{-1}\mathcal{V}'$ equivalent to $\mathcal{U}'$ and put $\mathcal{V}'' = \mathcal{V} \cap \mathcal{V}'$.

Then $\mathcal{V}''$ refines $\mathcal{V}$, and $f^{-1}\mathcal{V}'' = f^{-1}\mathcal{V} \cap f^{-1}\mathcal{V}'$ refines $\mathcal{U}'$. By Lemma 2.8 and compatibility of refinement maps with composition, the image of ξ in $H^p(f^{-1}\mathcal{V}'', \mathcal{F})$ is the image of its (zero) image in $H^p(\mathcal{U}', \mathcal{F})$, hence is 0. Since $\mathcal{V}''$ refines $\mathcal{V}$, ξ is zero in the left-hand direct limit.

Thus the map is bijective and $H^p(Y, f_*\mathcal{F}) \simeq H^p(X, \mathcal{F})$ for closed immersions (indeed for every sheaf $\mathcal{F}$). □

Exercise 2.4

Problem. Let X be an integral scheme. Let $\mathcal{K}_X$ denote the constant sheaf $K(X)$ on X.

(a) Show that $\mathcal{K}_X$ is a quasi-coherent sheaf.

(b) For any point $x \in X$, let $\mathcal{Q}_x$ denote the skyscraper sheaf with support $\{x\}$ and fiber $K(X)/\mathcal{O}_{X,x}$ over x. Let us suppose X is of dimension 1 and quasi-compact. Show that we have an exact sequence of quasi-coherent sheaves on X: $0 \to \mathcal{O}_X \to \mathcal{K}_X \to \bigoplus_{x \in X} \mathcal{Q}_x \to 0$ (show that every section s of $\mathcal{K}_X/\mathcal{O}_X$ over an open subset U verifies $s_x = 0$ for all but finitely many points x).

(c) Let $X = \mathbf{P}^1_k$ be the projective line over a field k. Let $x_1, \ldots, x_n$ be points of X that are pairwise distinct, and let $f_1, \ldots, f_n \in K(X)$. Show that there exists an $f \in K(X)$ such that $f - f_i \in \mathcal{O}_{X,x_i}$ for every i (apply Proposition 2.15 to the exact sequence in (b)).

Solution. Write $K = K(X)$ and η for the generic point. The sheaf $\mathcal{K}_X$ is the sheaf $U \mapsto K$ for $U \neq \emptyset$ (and 0 for $U = \emptyset$), with identity restrictions. It is a sheaf by Ex. 2.2.1, because X is irreducible, so every nonempty open subset is connected. For $U \neq \emptyset$ open, $\mathcal{O}_X(U) \to \mathcal{O}_{X,\eta} = K$ is injective (X integral), so $\mathcal{O}_X \subseteq \mathcal{K}_X$ is a subsheaf of $\mathcal{O}_X$-modules, and $(\mathcal{K}_X/\mathcal{O}_X)_x = K/\mathcal{O}_{X,x}$.

(a). Let $U = \operatorname{Spec} A$ be a nonempty affine open subset; A is a domain with fraction field K. For $0 \neq g \in A$ we have $D(g) \neq \emptyset$ and $\widetilde{K}(D(g)) = K_g = K$, because g is invertible in the field K; and $D(0) = \emptyset$ with $K_0 = 0$. The restriction $K_g \to K_{gh}$ is the identity of K. Hence $\widetilde{K}$ and $\mathcal{K}_X|_U$ have the same sections, with the same restriction maps and the same A-module structure, on the base of principal open sets. A morphism of sheaves defined on a base extends uniquely (Ex. 2.2.7), so $\mathcal{K}_X|_U \simeq \widetilde{K}$. By Prop. 1.5(c) the sheaf $\mathcal{K}_X|_U$ is quasi-coherent; as U ranges over a covering of X, $\mathcal{K}_X$ is quasi-coherent (Thm. 1.7).

(b). **Hypothesis.** I assume that X is moreover Noetherian. The finiteness statement used below (an element of K has finitely many poles on X) is not available for arbitrary quasi-compact one-dimensional integral schemes: in the ring of all algebraic integers A (a domain of dimension 1), the element $1/2$ is non-integral at every one of the infinitely many primes of A containing 2. For Noetherian X everything works; (c) concerns $\mathbf{P}^1_k$, which is Noetherian.

Points of X. Every point $x \neq \eta$ is closed: otherwise $\overline{\{x\}}$ contains a point $y \neq x$ and $\{y\} \subsetneq \overline{\{x\}} \subsetneq X$ would be a chain of irreducible closed sets of length 2. Also $\mathcal{O}_{X,\eta} = K$, so $\mathcal{Q}_\eta = 0$. For closed x, $\mathcal{Q}_x(U) = K/\mathcal{O}_{X,x}$ if $x \in U$ and 0 otherwise (identity restrictions) and $(\mathcal{Q}_x)_y = K/\mathcal{O}_{X,x}$ for $y = x$ and 0 otherwise.

Finiteness of poles. For $a \in K$ let $S_a = \{x \in X : a \notin \mathcal{O}_{X,x}\}$. We claim S_a is finite. Cover X by finitely many affine open subsets $\operatorname{Spec} A_m$ (X is quasi-compact); each A_m is a Noetherian domain with fraction field K. Write $a = b/c$ with $b, c \in A_m$, $c \neq 0$. For $\mathfrak{p} \in D(c)$ we have $a \in (A_m)_\mathfrak{p} = \mathcal{O}_{X,\mathfrak{p}}$, so $S_a \cap \operatorname{Spec} A_m \subseteq V(c)$. Now $V(c)$ is a closed subset of the Noetherian

space $\operatorname{Spec} A_m$ not containing η ($c \neq 0$), so it has finitely many irreducible components $\overline{\{z\}}$ with $z \neq \eta$; each z is closed, so each component is the point z. So $V(c)$ is finite, and S_a is finite.

If $U \subseteq X$ is open and $s \in (\mathcal{K}_X/\mathcal{O}_X)(U)$, choose an open covering of U by sets U_j with $s|_{U_j}$ the class of some $a_j \in K$. As X is a Noetherian space, U is quasi-compact, so finitely many $U_1, \dots, U_r$ suffice. If $x \in U_j$ and $s_x \neq 0$ then $a_j \notin \mathcal{O}_{X,x}$, i.e. $x \in S_{a_j}$. So $\{x \in U : s_x \neq 0\} \subseteq S_{a_1} \cup \cdots \cup S_{a_r}$ is finite.

The sheaf $\bigoplus \mathcal{Q}_x$. For an open U let $\mathcal{Q}(U) \subseteq \prod_x \mathcal{Q}_x(U)$ be the families with only finitely many nonzero components. It is a sheaf: if $t \in \prod_x \mathcal{Q}_x(U)$ glues from finitely supported families on an open covering of U, then finitely many members $U_1, \dots, U_r$ cover U (U is quasi-compact) and $\operatorname{supp} t \subseteq \bigcup \operatorname{supp}(t|_{U_j})$ is finite. Hence $\mathcal{Q} = \bigoplus_x \mathcal{Q}_x$ is the direct sum in the category of sheaves.

The isomorphism. Define $\theta : \mathcal{K}_X/\mathcal{O}_X \to \mathcal{Q}$ by $\theta(s) = (s_x)_{x \in U}$, with $s_x \in K/\mathcal{O}_{X,x} = \mathcal{Q}_x(U)$ (and 0 for $x \notin U$); it lands in $\mathcal{Q}$ by the finiteness shown above and commutes with restrictions. On stalks: at η, both stalks are 0 (a finitely supported family can be shrunk to 0 near η by removing its finitely many closed support points). At a closed point y, θ_y is injective: if $s \in (\mathcal{K}_X/\mathcal{O}_X)(U)$ has $\theta(s) = 0$ near y, then all germs of s vanish near y, so $s = 0$ near y. And θ_y is surjective: a germ in $\mathcal{Q}_y$ is represented by $t \in \mathcal{Q}(U)$, finitely supported; removing from U the finitely many closed points of $\operatorname{supp} t$ other than y we may assume $t = (t_y, 0, \dots)$. Lift t_y to $a \in K$ and let s be the class of a in $(\mathcal{K}_X/\mathcal{O}_X)(U)$. Removing the finite set $S_a \setminus \{y\}$ from U, $\theta(s) = t$ near y. So θ is an isomorphism and we get the exact sequence $0 \to \mathcal{O}_X \to \mathcal{K}_X \to \bigoplus_x \mathcal{Q}_x \to 0$.

Quasi-coherence. $\mathcal{O}_X$ and $\mathcal{K}_X$ are quasi-coherent; $\bigoplus_x \mathcal{Q}_x \simeq \operatorname{Coker}(\mathcal{O}_X \to \mathcal{K}_X)$ is quasi-coherent by Prop. 1.12(c). Each $\mathcal{Q}_x$ is the image of the projection endomorphism of $\bigoplus_{x'} \mathcal{Q}_{x'}$ onto the summand $\mathcal{Q}_x$, hence quasi-coherent by Prop. 1.12(c).

(c). $X = \mathbf{P}^1_k$ is integral, Noetherian, of dimension 1, so (b) applies. Prop. 2.15 applied to $0 \to \mathcal{O}_X \to \mathcal{K}_X \to \bigoplus \mathcal{Q}_x \to 0$ gives an exact sequence

$$K(X) = \mathcal{K}_X(X) \to \Big(\bigoplus_x \mathcal{Q}_x\Big)(X) \to H^1(X, \mathcal{O}_X) = 0,$$

the last equality by Example 2.20 (with $A = k$). So the first map is surjective; it sends f to the family $(f \bmod \mathcal{O}_{X,x})_x$. Let t be the family with $t_{x_i} = f_i \bmod \mathcal{O}_{X,x_i}$ and $t_x = 0$ for $x \notin \{x_1, \dots, x_n\}$; this is well defined since the x_i are pairwise distinct and it has finite support. (If $x_i = \eta$ the condition is empty, as $\mathcal{O}_{X,\eta} = K$.) A preimage $f \in K(X)$ of t satisfies $f - f_i \in \mathcal{O}_{X,x_i}$ for all i. □

Exercise 2.5

Problem. Let X be an open subset of the affine plane $\mathbf{A}^2_k$ over a field k.

(a) Let F be a finite set of closed points $p_1, \dots, p_n$ of $\mathbf{A}^2_k$. Show that p_i corresponds to a maximal ideal of $k[T,S]$ generated by two irreducible polynomials $f_i \in k[T]$, $g_i \in k[T,S]$. Changing the numbering if necessary, we may assume that $f_1, \dots, f_m$ are pairwise prime to each other, and that f_i is equal to one of the $f_1, \dots, f_m$ if $i \geq m+1$. Let us set

$$f = \prod_{1 \leq i \leq m} f_i, \qquad g = \sum_{1 \leq i \leq m} \Big(\prod_{1 \leq j \leq m,\, j \neq i} f_j \Big) \Big(\prod_{f_l = f_i} g_l \Big).$$

Show that $V(f,g) = \{p_1, \dots, p_n\}$.

(b) Show that X is the union of two affine open sets (more generally, one can show that any non-empty open subset of $\mathbf{A}^d_k$ is the union of d affine open subsets, see [53], Corollary V.1.5). Deduce from this that for any quasi-coherent sheaf $\mathcal{F}$ and for any $p \geq 2$, we have $H^p(X, \mathcal{F}) = 0$.

Solution. *(a), first claim.* A closed point of $\mathbf{A}^2_k = \operatorname{Spec} k[T,S]$ is a maximal ideal $\mathfrak{m}$. The field $k[T,S]/\mathfrak{m}$ is a finite extension of k (Zariski's lemma). Hence the subring $k[T]/(\mathfrak{m} \cap k[T])$ is a domain of finite dimension over k, thus a field; and $\mathfrak{m} \cap k[T] \neq 0$ since $k[T]$ is infinite-dimensional. So $\mathfrak{m} \cap k[T] = (f_0)$ with f_0 a monic irreducible polynomial. Put $L = k[T]/(f_0)$ (a field). Then $k[T,S]/(f_0) = L[S]$ and $\mathfrak{m}/(f_0)$ is a maximal ideal of the principal ideal domain $L[S]$, nonzero since $L[S]$ is not a field. So $\mathfrak{m}/(f_0) = (q)$ with q monic irreducible of degree $d \geq 1$. Lift the coefficients of q to polynomials in $k[T]$ and let $g_0 = S^d + \sum_{j<d} c_j(T) S^j \in k[T,S]$ be the lift. Then $\mathfrak{m} = (f_0, g_0)$. The polynomial g_0 is irreducible: if $g_0 = ab$ in $k[T][S]$, then, as g_0 is monic in S, we may take a, b monic in S; if both had positive degree in S, then reducing mod f_0 would factor $q = \bar{a}\bar{b}$ with factors of positive degree, a contradiction. So one factor has degree 0 in S and divides 1, a unit. Thus $\mathfrak{m} = (f_0, g_0)$ with $f_0 \in k[T]$, $g_0 \in k[T,S]$ irreducible. Apply this to each p_i to get $\mathfrak{m}_i = (f_i, g_i)$ with f_i monic.

(a), second claim. After renumbering, $f_1, \ldots, f_m$ are the distinct polynomials among $f_1, \ldots, f_n$; as distinct monic irreducible polynomials they are pairwise coprime in $k[T]$. Put $\hat{f}_i = \prod_{j \leq m,\, j \neq i} f_j$ and $h_i = \prod_{l : f_l = f_i} g_l$ (product over all $l \in \{1, \ldots, n\}$), so $f = \prod_{i \leq m} f_i$ and $g = \sum_{i \leq m} \hat{f}_i h_i$.

$V(f,g) \subseteq \{p_l\}$. Let $\mathfrak{P}$ be a prime containing f and g. Since $\mathfrak{P}$ is prime and contains f, it contains some f_{i_0} with $i_0 \leq m$. For $i \neq i_0$, $\hat{f}_i$ contains the factor $f_{i_0} \in \mathfrak{P}$, so $\hat{f}_i h_i \in \mathfrak{P}$. Therefore $\hat{f}_{i_0} h_{i_0} = g - \sum_{i \neq i_0} \hat{f}_i h_i \in \mathfrak{P}$. Now $\hat{f}_{i_0} \notin \mathfrak{P}$: if $f_j \in \mathfrak{P}$ for some $j \neq i_0$ then $\mathfrak{P} \cap k[T] \supseteq (f_j, f_{i_0}) = k[T]$ (coprime in the PID $k[T]$), which is absurd. Hence $h_{i_0} \in \mathfrak{P}$, so some $g_l \in \mathfrak{P}$ with $f_l = f_{i_0}$. Then $\mathfrak{P} \supseteq (f_l, g_l) = \mathfrak{m}_l$, so $\mathfrak{P} = \mathfrak{m}_l$ by maximality, i.e. the point is p_l.

$\{p_l\} \subseteq V(f,g)$. Let $l \in \{1, \ldots, n\}$ and $f_l = f_{i_0}$ with $i_0 \leq m$. Then $f \in (f_{i_0}) \subseteq \mathfrak{m}_l$. In $g = \hat{f}_{i_0} h_{i_0} + \sum_{i \neq i_0} \hat{f}_i h_i$, the first term contains the factor $g_l \in \mathfrak{m}_l$ and each other term contains $f_{i_0} \in \mathfrak{m}_l$. So $g \in \mathfrak{m}_l$. Thus $V(f,g) = \{p_1, \ldots, p_n\}$ (for $n = 0$ nothing is to be proved).

(b). If $X = \emptyset$ there is nothing to prove, so let $X \neq \emptyset$. Let $Z = \mathbf{A}^2_k \setminus X = V(I)$ with $I \subset k[T,S]$ an ideal; $I \neq 0$ as $X \neq \emptyset$. Since $k[T,S]$ is a UFD, let $h = \gcd(I)$ and $I = hJ$, where J is an ideal with no common nonunit divisor. Then $Z = V(h) \cup V(J)$. We claim $V(J)$ is a finite set of closed points. Indeed, let $\mathfrak{P}$ be a minimal prime over J. If $\operatorname{ht} \mathfrak{P} = 1$ then $\mathfrak{P} = (q)$ is principal in the UFD, with q irreducible, and q divides every element of J, contrary to the choice of J; and $\mathfrak{P} = 0$ is impossible as $J \neq 0$. So $\operatorname{ht} \mathfrak{P} = 2 = \dim k[T,S]$, and $\mathfrak{P}$ is maximal (for a finitely generated domain over a field, $\dim R/\mathfrak{P} = \dim R - \operatorname{ht} \mathfrak{P} = 0$). As $k[T,S]$ is Noetherian there are finitely many minimal primes, so $V(J) = \{p_1, \ldots, p_n\}$ is a finite set of closed points (possibly empty, if $J = (1)$).

Take f, g as in (a) for these points (if $n = 0$ take $f = 1$, $g = 0$ and read $D(f) \cup D(g)$ as $D(f)$). Then $\mathbf{A}^2_k \setminus V(J) = D(f) \cup D(g)$ by (a), so

$$X = \mathbf{A}^2_k \setminus (V(h) \cup V(J)) = D(h) \cap \big(D(f) \cup D(g)\big) = D(hf) \cup D(hg).$$

Both $D(hf)$ and $D(hg)$ are principal open subsets of an affine scheme, hence affine. So X is the union of two affine open sets.

Finally X is a separated scheme (an open subscheme of an affine scheme: Prop. 3.3.9(a),(b))

covered by $m = 2$ affine open subsets. By Rem. 2.21, $H^p(X, \mathcal{F}) = 0$ for every quasi-coherent $\mathcal{F}$ and every $p \geq 2$. □

Exercise 2.6

Problem. Let X be a Noetherian topological space. Let $(\mathcal{F}_\lambda)_{\lambda\in\Lambda}$ be a direct system of sheaves on X.

(a) Show that the correspondence $U \mapsto \varinjlim_\lambda \mathcal{F}_\lambda(U)$, for every open set U, defines a sheaf $\mathcal{F}$ on X.

(b) Let us fix $p \geq 0$. Let $\mathcal{U}$ be an open covering of X. Show that the cohomology groups $H^p(C(\mathcal{U}, \mathcal{F}_\lambda))$, $\lambda \in \Lambda$, naturally form a direct system and that we have a canonical isomorphism $\varinjlim_\lambda H^p(C(\mathcal{U}, \mathcal{F}_\lambda)) \simeq H^p(C(\mathcal{U}, \mathcal{F}))$.

(c) Show property (b) when replacing $H^p(C(\mathcal{U}, *))$ by $H^p(X, *)$.

Solution. As usual a direct system is indexed by a directed set Λ (so that direct limits are exact), and the sheaves are sheaves of Abelian groups (or modules over a fixed ring). Every open subset of the Noetherian space X is quasi-compact.

(a). The presheaf $\mathcal{F}(U) = \varinjlim_\lambda \mathcal{F}_\lambda(U)$ has the obvious restriction maps. Let $U = \bigcup_i U_i$ be an open covering.

Locality. Let $s \in \mathcal{F}(U)$ with $s|_{U_i} = 0$ for all i. Represent s by $s_\lambda \in \mathcal{F}_\lambda(U)$. Since U is quasi-compact, finitely many $U_1, \ldots, U_r$ cover U. For each $i \leq r$, $s|_{U_i} = 0$ means that $s_\lambda|_{U_i}$ becomes 0 in some $\mathcal{F}_{\mu_i}(U_i)$. Choose $\mu \geq \lambda, \mu_1, \ldots, \mu_r$ (so that μ_i may be taken $\geq \lambda$). The image of s_λ in $\mathcal{F}_\mu(U)$ restricts to 0 on each U_i, $i \leq r$, hence is 0 since $\mathcal{F}_\mu$ is a sheaf. So $s = 0$.

Gluing. Let $s_i \in \mathcal{F}(U_i)$ be sections with $s_i|_{U_{ij}} = s_j|_{U_{ij}}$. Take finitely many $U_1, \ldots, U_r$ covering U. Represent s_i $(i \leq r)$ by $s'_i \in \mathcal{F}_{\lambda_i}(U_i)$. For each pair $i, j \leq r$, $s'_i|_{U_{ij}}$ and $s'_j|_{U_{ij}}$ become equal in some $\mathcal{F}_{\nu_{ij}}(U_{ij})$. Choose ν larger than all λ_i and ν_{ij} (finitely many). The images $t_i \in \mathcal{F}_\nu(U_i)$ of s'_i agree on overlaps, so they glue to $t \in \mathcal{F}_\nu(U)$; let $s \in \mathcal{F}(U)$ be its class. Then $s|_{U_i} = s_i$ for $i \leq r$. For an arbitrary index i, the sections $s|_{U_i}$ and s_i agree on each $U_i \cap U_j$, $j \leq r$, which cover U_i; by locality they are equal. So $\mathcal{F}$ is a sheaf (a direct limit of sheaves over a Noetherian space is the presheaf direct limit).

(b). **Scope.** I prove this for *finite* coverings $\mathcal{U}$. For an infinite covering the product $\prod_\alpha \mathcal{F}_\lambda(U_\alpha)$ is an infinite product, which does not commute with direct limits, and the statement is not proved here. For (c) finite coverings are all we need.

Let $\mathcal{U} = \{U_i\}_{i\in I}$ be finite. For $\lambda \leq \mu$ the transition $\mathcal{F}_\lambda \to \mathcal{F}_\mu$ induces $C(\mathcal{U}, \mathcal{F}_\lambda) \to C(\mathcal{U}, \mathcal{F}_\mu)$, which commutes with d (restrictions commute with sheaf morphisms); so the complexes $C^\bullet(\mathcal{U}, \mathcal{F}_\lambda)$ form a direct system and, by functoriality of cohomology, so do the $H^p(C(\mathcal{U}, \mathcal{F}_\lambda))$. For $q \geq 0$, I^{q+1} is finite and direct limits commute with finite direct sums, so

$$\varinjlim_\lambda C^q(\mathcal{U}, \mathcal{F}_\lambda) = \varinjlim_\lambda \prod_{\alpha\in I^{q+1}} \mathcal{F}_\lambda(U_\alpha) = \prod_{\alpha\in I^{q+1}} \varinjlim_\lambda \mathcal{F}_\lambda(U_\alpha) = C^q(\mathcal{U}, \mathcal{F}),$$

compatibly with d. Since $\varinjlim$ over a directed set is exact, it commutes with kernels, images and quotients: $\ker(\varinjlim d) = \varinjlim \ker d$, $\operatorname{im}(\varinjlim d) = \varinjlim \operatorname{im} d$, and so $H^p(\varinjlim C_\lambda) = \varinjlim H^p(C_\lambda)$. The resulting isomorphism is induced by the canonical maps $\mathcal{F}_\lambda \to \mathcal{F}$, hence is canonical.

(c). Since X is Noetherian, it is quasi-compact, so the finite coverings are cofinal among all coverings (Rem. 2.11). Hence $H^p(X,\mathcal{G}) = \varinjlim_{\mathcal{U}} H^p(\mathcal{U},\mathcal{G})$, the limit over (classes of) finite coverings, for any sheaf $\mathcal{G}$. The isomorphisms of (b) are compatible with refinement maps (these are index maps and restrictions, which commute with the transition maps $\mathcal{F}_\lambda \to \mathcal{F}_\mu$ and $\mathcal{F}_\lambda \to \mathcal{F}$). Direct limits commute with direct limits (both sides are the colimit over $\Lambda \times \{\text{finite coverings}\}$), so

$$\varinjlim_\lambda H^p(X,\mathcal{F}_\lambda) = \varinjlim_\lambda \varinjlim_{\mathcal{U}} H^p(\mathcal{U},\mathcal{F}_\lambda) = \varinjlim_{\mathcal{U}} \varinjlim_\lambda H^p(\mathcal{U},\mathcal{F}_\lambda) \overset{(b)}{=} \varinjlim_{\mathcal{U}} H^p(\mathcal{U},\mathcal{F}) = H^p(X,\mathcal{F}).$$

□

Exercise 2.7

Problem. Let X be a scheme. Let $\mathcal{O}_X^*$ denote the sheaf of Abelian groups $U \mapsto \mathcal{O}_X(U)^*$.

(a) Let $\mathcal{L}$ be an invertible sheaf on X. Let $\mathcal{U} = \{U_i\}_{i\in I}$ be a covering of X such that for every $i \in I$, $\mathcal{L}|_{U_i}$ is free, generated by a section $e_i \in \mathcal{L}(U_i)$. Show that there exist $f_{ij} \in \mathcal{O}_X^*(U_{ij})$ such that $e_i|_{U_{ij}} = e_j|_{U_{ij}} f_{ij}$ for every $i,j \in I$. Let $f := (f_{ij})_{i,j\in I} \in C'^1(\mathcal{U},\mathcal{O}_X^*)$. Show that the image of f in $H^1(\mathcal{U},\mathcal{O}_X^*)$ is independent of the choice of the e_i. Show that the image $\varphi(\mathcal{L})$ of f in $H^1(X,\mathcal{O}_X^*)$ is independent of the choice of $\mathcal{U}$.

(b) Show that $\varphi(\mathcal{L}) = 1$ if and only if $\mathcal{L}$ is free.

(c) Show that the correspondence $\mathcal{L} \mapsto \varphi(\mathcal{L})$ induces a canonical group isomorphism $\mathrm{Pic}(X) \simeq H^1(X,\mathcal{O}_X^*)$ (to prove the surjectivity, use Exercise 2.2.8 by taking $\mathcal{F}_i = \mathcal{O}_{U_i}$, and ϕ_{ij} to be the isomorphism defined by multiplication by f_{ij}).

(d) Let $f : X \to Y$ be a morphism of schemes. Then $\mathcal{L} \mapsto f^*\mathcal{L}$ induces a group homomorphism $\mathrm{Pic}(Y) \to \mathrm{Pic}(X)$. In addition, the homomorphism $\mathcal{O}_Y^* \to f_*\mathcal{O}_X^*$ canonically induces a homomorphism $H^1(Y,\mathcal{O}_Y^*) \to H^1(X,\mathcal{O}_X^*)$. Show that these two homomorphisms are identical via the isomorphisms of (c).

Solution. Notation. The group $\mathcal{O}_X^*(U)$ is written multiplicatively, so the differential of $C^\bullet(\mathcal{U},\mathcal{O}_X^*)$ is $(du)_{ij} = u_j u_i^{-1}$ for a 0-cochain, and $(df)_{ijk} = f_{jk} f_{ik}^{-1} f_{ij}$ for a 1-cochain. A 1-cochain f is an alternating cocycle iff $f_{ii} = 1$, $f_{ji} = f_{ij}^{-1}$ and $f_{ik} = f_{ij}f_{jk}$ on U_{ijk}. The neutral element of H^1 is written 1. By Prop. 2.3 every class in $H^1(\mathcal{U},\mathcal{O}_X^*)$ has an alternating representative.

(a), the cochain. On U_{ij} both $e_i|$ and $e_j|$ generate the free rank one module $\mathcal{L}|_{U_{ij}}$. So there are unique $f_{ij}, f_{ji} \in \mathcal{O}_X(U_{ij})$ with $e_i = e_j f_{ij}$ and $e_j = e_i f_{ji}$; then $e_i = e_i f_{ji} f_{ij}$ gives $f_{ij}f_{ji} = 1$ by uniqueness of coordinates in the basis e_i. So $f_{ij} \in \mathcal{O}_X^*(U_{ij})$, $f_{ji} = f_{ij}^{-1}$, and $f_{ii} = 1$. On U_{ijk}, $e_i = e_j f_{ij} = e_k f_{jk} f_{ij}$ and $e_i = e_k f_{ik}$, so $f_{ik} = f_{ij}f_{jk}$ by uniqueness of coordinates with respect to e_k. So $f \in C'^1(\mathcal{U},\mathcal{O}_X^*)$ is an alternating cocycle.

Independence of the e_i. Let e_i' be other generators of $\mathcal{L}|_{U_i}$. Then $e_i' = e_i u_i$ with $u_i \in \mathcal{O}_X^*(U_i)$ (write $e_i = e_i' v_i$ also; then $u_i v_i = 1$). With $e_i' = e_j' f_{ij}'$ we get $e_i u_i = e_j u_j f_{ij}'$ and $e_i u_i = e_j f_{ij} u_i$, so $f_{ij}' = f_{ij} u_i u_j^{-1} = f_{ij} \cdot (du^{-1})_{ij}$ with $u^{-1} = (u_i^{-1})_i$. Thus f' and f differ by a coboundary and have the same class in $H^1(\mathcal{U},\mathcal{O}_X^*)$.

Independence of $\mathcal{U}$. Let $\mathcal{V} = \{V_j\}_{j\in J}$ be another covering with generators $e_j' \in \mathcal{L}(V_j)$, and let $f^{\mathcal{U}}, f^{\mathcal{V}}$ be the corresponding cocycles. Let $\mathcal{W} = \{U_i \cap V_j\}_{(i,j)}$, with refinement maps $\sigma(i,j) = i$

and $\tau(i,j)=j$. On $W_{(i,j)}$ both $e_i|$ and $e'_j|$ are generators of $\mathcal{L}$. Using the generators $e_{\sigma(w)}|_{W_w}$, the cocycle on $\mathcal{W}$ is $\left(f^{\mathcal{U}}_{\sigma(w)\sigma(w')}|_{W_{ww'}}\right)_{w,w'} = \sigma(f^{\mathcal{U}})$; using the generators $e'_{\tau(w)}|$ it is $\tau(f^{\mathcal{V}})$. By the previous paragraph these two cocycles define the same class in $H^1(\mathcal{W},\mathcal{O}_X^*)$, i.e. $\sigma^*[f^{\mathcal{U}}]=\tau^*[f^{\mathcal{V}}]$. Both $[f^{\mathcal{U}}]$ and $[f^{\mathcal{V}}]$ have the same image in $\varinjlim = H^1(X,\mathcal{O}_X^*)$ as their images in $H^1(\mathcal{W},\mathcal{O}_X^*)$. So $\varphi(\mathcal{L})$ is well defined.

Moreover $\varphi(\mathcal{L})$ depends only on the isomorphism class of $\mathcal{L}$: if $\iota:\mathcal{L}\to\mathcal{L}'$ is an isomorphism, then $\iota(e_i)$ are generators of $\mathcal{L}'|_{U_i}$, and $\iota(e_i)=\iota(e_j)f_{ij}$ by $\mathcal{O}_X$-linearity, which gives the same cocycle. Also φ is multiplicative: if $\mathcal{L},\mathcal{M}$ have generators e_i,e'_i on a common covering (refine), with cocycles f,f', then $e_i\otimes e'_i$ generates $(\mathcal{L}\otimes\mathcal{M})|_{U_i}$ and $e_i\otimes e'_i=(e_j\otimes e'_j)f_{ij}f'_{ij}$, so $\varphi(\mathcal{L}\otimes\mathcal{M})=\varphi(\mathcal{L})\varphi(\mathcal{M})$.

(b). If $\mathcal{L}$ is free, take $\mathcal{U}=\{X\}$ and a generator e; the only cochain entry is $f_{11}=1$, so $\varphi(\mathcal{L})=1$. Conversely let $\varphi(\mathcal{L})=1$. By definition of the direct limit there is a refinement $\sigma:\mathcal{V}\to\mathcal{U}$ with $\sigma^*[f]=0$ in $H^1(\mathcal{V},\mathcal{O}_X^*)$. As computed above, $\sigma(f)$ is the cocycle of $\mathcal{L}$ for $\mathcal{V}$ and the generators $e_{\sigma(j)}|_{V_j}$. So replacing $\mathcal{U}$ by $\mathcal{V}$ we may assume $[f]=0$ in $H^1(\mathcal{U},\mathcal{O}_X^*)$: there exist $u_i\in\mathcal{O}_X^*(U_i)$ with $f_{ij}=u_ju_i^{-1}$ on U_{ij}. Then $e'_i:=e_iu_i\in\mathcal{L}(U_i)$ satisfy on U_{ij}: $e'_i=e_jf_{ij}u_i=e_ju_j=e'_j$. So the e'_i glue to a global section $e\in\mathcal{L}(X)$, which generates $\mathcal{L}$ on each U_i. Hence $\mathcal{O}_X\to\mathcal{L}$, $1\mapsto e$ is an isomorphism on every U_i, so it is an isomorphism, and $\mathcal{L}$ is free.

(c). By (a), $\varphi:\mathrm{Pic}(X)\to H^1(X,\mathcal{O}_X^*)$ is a well-defined group homomorphism (Pic is the group of isomorphism classes of invertible sheaves under $\otimes$). By (b) its kernel is trivial, so it is injective. Surjectivity: let $\xi\in H^1(X,\mathcal{O}_X^*)$. It is the image of a class in $H^1(\mathcal{U},\mathcal{O}_X^*)$ for some covering $\mathcal{U}=\{U_i\}$, represented by an alternating cocycle $f=(f_{ij})$ (Prop. 2.3). Let $\mathcal{F}_i=\mathcal{O}_{U_i}$ and let $\phi_{ij}:\mathcal{F}_i|_{U_{ij}}\to\mathcal{F}_j|_{U_{ij}}$ be multiplication by f_{ij}, an isomorphism of $\mathcal{O}_{U_{ij}}$-modules. We have $\phi_{ii}=\mathrm{Id}$ since $f_{ii}=1$, and $\phi_{ik}=\phi_{jk}\circ\phi_{ij}$ on U_{ijk} since $f_{ik}=f_{ij}f_{jk}$. By Ex. 2.2.8 there is a sheaf $\mathcal{L}$ on X with isomorphisms $\psi_i:\mathcal{L}|_{U_i}\to\mathcal{O}_{U_i}$ such that $\psi_j=\phi_{ij}\circ\psi_i$ on U_{ij}. (The construction there, $\mathcal{L}(W)=\{(s_i)_i : s_i\in\mathcal{F}_i(W\cap U_i),\ \phi_{ij}(s_i)=s_j \text{ on } W\cap U_{ij}\}$, gives an $\mathcal{O}_X$-module because the ϕ_{ij} are $\mathcal{O}$-linear, and the ψ_i are then $\mathcal{O}$-linear.) So $\mathcal{L}$ is invertible. Put $e_i=\psi_i^{-1}(1)$, a generator of $\mathcal{L}|_{U_i}$. On U_{ij}, $\psi_j(e_i)=\phi_{ij}(\psi_i(e_i))=\phi_{ij}(1)=f_{ij}=\psi_j(e_jf_{ij})$, so $e_i=e_jf_{ij}$. Hence the cocycle of $\mathcal{L}$ for $\mathcal{U}$ and the e_i is f, and $\varphi(\mathcal{L})=\xi$. So φ is an isomorphism.

(d). Let $\mathcal{L}$ be invertible on Y, $\mathcal{V}=\{V_i\}$ a covering with generators e_i and cocycle $g=(g_{ij})$, $e_i=e_jg_{ij}$. Let $\varepsilon_i\in(f^*\mathcal{L})(f^{-1}V_i)$ be the image of e_i under $\mathcal{L}(V_i)\to(f^*\mathcal{L})(f^{-1}V_i)$. This map is compatible with restrictions and satisfies $a\,s\mapsto f^{\#}(a)\cdot(\text{image of } s)$ for $a\in\mathcal{O}_Y(V_i)$. Moreover ε_i generates $f^*\mathcal{L}$ on $f^{-1}V_i$, since $f^*(\mathcal{L}|_{V_i})\simeq f^*\mathcal{O}_{V_i}=\mathcal{O}_{f^{-1}V_i}$ with $1\leftrightarrow\varepsilon_i$. So $\varepsilon_i=\varepsilon_jf^{\#}(g_{ij})$ on $f^{-1}(V_{ij})$. Therefore $\varphi(f^*\mathcal{L})$ is the class of the cocycle $(f^{\#}(g_{ij}))_{ij}$ of the covering $f^{-1}\mathcal{V}$ of X.

On the other hand the homomorphism $H^1(Y,\mathcal{O}_Y^*)\to H^1(X,\mathcal{O}_X^*)$ is the composition $H^1(Y,\mathcal{O}_Y^*)\to H^1(Y,f_*\mathcal{O}_X^*)\to H^1(X,\mathcal{O}_X^*)$ of the map induced by $f^{\#}:\mathcal{O}_Y^*\to f_*\mathcal{O}_X^*$ (Rem. 2.13) and the map of Ex. 2.3(a). On the class of $g\in C^1(\mathcal{V},\mathcal{O}_Y^*)$ the first map gives the class of $(f^{\#}(g_{ij}))\in C^1(\mathcal{V},f_*\mathcal{O}_X^*)$, and under the identification $C^1(\mathcal{V},f_*\mathcal{O}_X^*)=C^1(f^{-1}\mathcal{V},\mathcal{O}_X^*)$ of Ex. 2.3(a) this is the cochain $(f^{\#}(g_{ij}))_{ij}$, whose class in $H^1(X,\mathcal{O}_X^*)$ is the image under the second map. So the two homomorphisms agree on $\varphi(\mathcal{L})$, i.e. $\varphi_X(f^*\mathcal{L})=f^*\varphi_Y(\mathcal{L})$. □

Exercise 2.8

Problem. Let X be a quasi-compact open subscheme of an affine scheme $\mathrm{Spec}\,A$. We suppose that $H^p(X,\mathcal{O}_X)=0$ for every $p\geq 1$.

(a) Show that there exist $f_1, \dots, f_n \in A$ such that $U_i := D(f_i) \subseteq X$ and that $X = \bigcup_{1 \le i \le n} U_i$.

(b) Let us set $B = \mathcal{O}_X(X)$ and $\mathcal{U} = \{U_i\}_{1 \le i \le n}$. Show that $0 \to B \to C^0(\mathcal{U}, \mathcal{O}_X) \to C^1(\mathcal{U}, \mathcal{O}_X) \to \dots$ (2.9) is an exact sequence of flat B-modules (use Proposition 2.3.12), zero from a finite rank on. Show that for any B-module M, (2.9)$\otimes_B M$ stays exact (use Proposition 1.2.6).

(c) Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Show that $H^p(X, \mathcal{F}) = 0$ for every $p \ge 1$ (use Proposition 1.6 and (b)). Deduce from this that X is affine.

Solution. *(a).* The principal open sets $D(f)$ form a base of the topology of $\operatorname{Spec} A$, so X is the union of the principal open sets $D(f)$ contained in X. As X is quasi-compact, finitely many of them, $D(f_1), \dots, D(f_n)$, suffice.

Preliminaries. X is separated (open subscheme of an affine scheme: Prop. 3.3.9(a),(b)) and quasi-compact, so Prop. 1.6 applies to X. For $g \in B$ write $X_g = \{x \in X : g_x \in \mathcal{O}^*_{X,x}\}$. Let $\bar{f}_i \in B$ be the restriction of f_i to X. At $x = \mathfrak{p} \in X$, $\bar{f}_{i,x}$ is the image of f_i in $A_\mathfrak{p}$, a unit iff $f_i \notin \mathfrak{p}$, so $X_{\bar{f}_i} = D(f_i) = U_i$. For an index sequence $\alpha = (i_0, \dots, i_p)$ put $\bar{f}_\alpha = \bar{f}_{i_0} \cdots \bar{f}_{i_p} \in B$; then $X_{\bar{f}_\alpha} = U_{i_0} \cap \dots \cap U_{i_p} = U_\alpha$. By Prop. 1.6 (for $\mathcal{F}$ quasi-coherent, in particular for $\mathcal{O}_X$)

$$\mathcal{F}(U_\alpha) = \mathcal{F}(X)_{\bar{f}_\alpha} = \mathcal{F}(X) \otimes_B B_{\bar{f}_\alpha}, \qquad \mathcal{O}_X(U_\alpha) = B_{\bar{f}_\alpha}, \qquad (2)$$

and these identifications are compatible with the restriction maps (they are induced by the restrictions $\mathcal{F}(X) \to \mathcal{F}(U_\alpha)$ via the universal property of localization).

(b). Use the complex $C''(\mathcal{U}, \mathcal{O}_X)$ of strictly increasing index sequences (Cor. 2.4) as the complex (2.9); then $C''^p = 0$ for $p \ge n$ (Cor. 2.5), so (2.9) is zero from rank n on. By (2), $C''^p(\mathcal{U}, \mathcal{O}_X) = \bigoplus_{i_0 < \dots < i_p} B_{\bar{f}_{i_0 \dots i_p}}$ is a finite direct sum of localizations of B, hence a flat B-module; B itself is flat. *Exactness.* $B \to C''^0 = \prod_i \mathcal{O}_X(U_i)$ is injective with image the kernel of $d : C''^0 \to C''^1$ (sheaf axiom for the covering $\mathcal{U}$ of X). For $p \ge 1$, $H^p(C''(\mathcal{U}, \mathcal{O}_X)) = H^p(\mathcal{U}, \mathcal{O}_X)$ (Cor. 2.4) $= H^p(X, \mathcal{O}_X)$ (Thm. 2.19: X is separated and $\mathcal{U}$ is an affine covering) $= 0$ by hypothesis. So (2.9) is exact.

Lemma. Let $0 \to E^{-1} \to E^0 \to \dots \to E^{n-1} \to 0$ be an exact complex of flat B-modules. Then it stays exact after $\otimes_B M$, for every B-module M. *Proof.* Let $Z^q = \ker(d^q)$ ($Z^n = 0$, $Z^{-1} = 0$), so we have short exact sequences $0 \to Z^q \to E^q \to Z^{q+1} \to 0$ ($-1 \le q \le n-1$). By descending induction Z^q is flat: $Z^n = 0$ is flat, and if Z^{q+1} and E^q are flat then the long exact Tor sequence gives $\operatorname{Tor}_1^B(Z^q, M)$ between $\operatorname{Tor}_2^B(Z^{q+1}, M) = 0$ and $\operatorname{Tor}_1^B(E^q, M) = 0$, so Z^q is flat. Since Z^{q+1} is flat, $0 \to Z^q \otimes M \to E^q \otimes M \to Z^{q+1} \otimes M \to 0$ is exact (the Tor_1 term vanishes; compare Prop. 1.2.6). The map $d^q \otimes 1$ factors as $E^q \otimes M \twoheadrightarrow Z^{q+1} \otimes M \hookrightarrow E^{q+1} \otimes M$, so $\ker(d^q \otimes 1) = \operatorname{im}(Z^q \otimes M \to E^q \otimes M) = \operatorname{im}(d^{q-1} \otimes 1)$, using that $E^{q-1} \otimes M \to Z^q \otimes M$ is onto. For $q = -1$: $E^{-1} \otimes M \to E^0 \otimes M$ factors through $Z^0 \otimes M \hookrightarrow E^0 \otimes M$ and $E^{-1} \otimes M \to Z^0 \otimes M$ is an isomorphism ($Z^{-1} = 0$ and $Z^{-1} \otimes M = 0$).

Applying the Lemma to $E^{-1} = B$ and $E^q = C''^q(\mathcal{U}, \mathcal{O}_X)$, (2.9)$\otimes_B M$ is exact for all M.

(c). Let $\mathcal{F}$ be quasi-coherent on X and $M = \mathcal{F}(X)$. By (2), $C''^p(\mathcal{U}, \mathcal{F}) = \bigoplus_{i_0 < \dots < i_p} M \otimes_B B_{\bar{f}_{i_0 \dots i_p}} = C''^p(\mathcal{U}, \mathcal{O}_X) \otimes_B M$, compatibly with d (the differential is an alternating sum of the restriction maps, which correspond to the localization maps tensored with M). By (b), $C''(\mathcal{U}, \mathcal{O}_X) \otimes_B M$ is exact in degrees ≥ 1, so $H^p(\mathcal{U}, \mathcal{F}) = H^p(C''(\mathcal{U}, \mathcal{F})) = 0$ for $p \ge 1$ (Cor. 2.4). By Thm. 2.19 (X separated, $\mathcal{U}$ affine), $H^p(X, \mathcal{F}) = H^p(\mathcal{U}, \mathcal{F}) = 0$ for $p \ge 1$.

Thus (ii) of Serre's criterion (Thm. 2.23) holds for X, which is separated and quasi-compact, so X is affine. □

Exercise 2.9

> **Problem.** Show that in Serre's criterion (Theorem 2.23), we can replace condition (iii) by $H^1(X, \mathcal{J}) = 0$ for every quasi-coherent sheaf of ideals $\mathcal{J}$.

Solution. Let X be Noetherian, or separated and quasi-compact. We show: X is affine if and only if (iii′) $H^1(X, \mathcal{J}) = 0$ for every quasi-coherent sheaf of ideals $\mathcal{J}$ of $\mathcal{O}_X$. If X is affine, then (iii′) holds by Thm. 2.18, since an ideal sheaf is a quasi-coherent sheaf. Conversely assume (iii′). We follow the proof of Thm. 2.23; the only places where it used H^1 of a sheaf that is not an ideal sheaf are the last step ($\ker \psi$), which we replace below, and we check that the first step uses an ideal sheaf. Recall $X_f = \{y \in X : f_y \in \mathcal{O}_{X,y}^*\}$ for $f \in A := \mathcal{O}_X(X)$.

Step 1: every point lies in an affine open X_f. Let $\phi : X \to \operatorname{Spec} A$ be the canonical morphism. Let $x \in X$. As in the book, $\overline{\{x\}}$ with its reduced structure is a nonempty quasi-compact scheme, so it has a point closed in it, which is closed in X (Ex. 2.4.8). An open set containing this closed point $y \in \overline{\{x\}}$ contains x. So it is enough to find an affine open of the form X_f containing a given closed point y; rename y as x.

Let $U \ni x$ be an affine open and let $\mathcal{J}$ (resp. $\mathcal{M}$) be the quasi-coherent ideal sheaf of the reduced closed subscheme with underlying set $X \setminus U$ (resp. $\{x\}$). The product $\mathcal{M}\mathcal{J}$ is again a quasi-coherent sheaf of ideals (image of $\mathcal{M} \otimes \mathcal{J} \to \mathcal{O}_X$; Prop. 1.12(b),(c)). In the exact sequence $0 \to \mathcal{M}\mathcal{J} \to \mathcal{J} \to \mathcal{J}/\mathcal{M}\mathcal{J} \to 0$ we have $(\mathcal{J}/\mathcal{M}\mathcal{J})_y = 0$ for $y \neq x$ (because $\mathcal{M}_y = \mathcal{O}_{X,y}$), and $(\mathcal{J}/\mathcal{M}\mathcal{J})_x = \mathcal{O}_{X,x}/\mathfrak{m}_x = k(x)$ since $x \in U$ gives $\mathcal{J}_x = \mathcal{O}_{X,x}$ and $\mathcal{M}_x = \mathfrak{m}_x$. As x is closed, $(\mathcal{J}/\mathcal{M}\mathcal{J})(X) = k(x)$ (a sheaf with zero stalks off the closed point x has its global sections equal to its stalk at x: a germ at x extends by 0 on $X \setminus \{x\}$). By (iii′), $H^1(X, \mathcal{M}\mathcal{J}) = 0$, so Prop. 2.15 shows $\mathcal{J}(X) \to k(x)$ is surjective. Take $f \in \mathcal{J}(X)$ with class 1 in $k(x)$, i.e. $f_x \notin \mathfrak{m}_x$. Then $x \in X_f$. Also $X_f \subseteq U$: for $y \notin U$, $f_y \in \mathcal{J}_y \subseteq \mathfrak{m}_y$. Since U is affine and $X_f = D(f|_U)$ inside U, X_f is affine.

Step 2: reduction to $\sum f_i A = A$. Under our hypotheses, condition (3.2) of Subsection 2.3.1 holds, so $\mathcal{O}_X(X_f) = A_f$ (Prop. 1.6 with $\mathcal{F} = \mathcal{O}_X$) and $\phi^{-1}(D(f)) = X_f$. If X_f is affine, then $\phi|_{X_f} : X_f = \operatorname{Spec} A_f \to D(f) = \operatorname{Spec} A_f$ is an isomorphism. By Step 1 and quasi-compactness of X there are $f_1, \ldots, f_n \in A$ with $X = \bigcup X_{f_i}$ and every X_{f_i} affine. If we show $\sum_i f_i A = A$, then $\operatorname{Spec} A = \bigcup D(f_i)$ and ϕ is an isomorphism over each $D(f_i)$, so ϕ is an isomorphism and X is affine.

Step 3: $\sum f_i A = A$. Let $\mathcal{J}_k \subseteq \mathcal{O}_X$ be the image of $\mathcal{O}_X^k \to \mathcal{O}_X$, $(a_1, \ldots, a_k) \mapsto \sum_{i \le k} f_i a_i$, i.e. the ideal sheaf generated by $f_1, \ldots, f_k$; it is quasi-coherent (Prop. 1.12(c)), and $\mathcal{J}_0 = 0$. At each $x \in X$ some f_i is a unit in $\mathcal{O}_{X,x}$, so $\mathcal{J}_n = \mathcal{O}_X$. We prove by induction on k that $\mathcal{J}_k(X) = \sum_{i \le k} f_i A$. For $k = 0$ both sides are 0. Let $k \ge 1$. The quotient $\mathcal{J}_k/\mathcal{J}_{k-1}$ is quasi-coherent, and the morphism $\mathcal{O}_X \to \mathcal{J}_k/\mathcal{J}_{k-1}$, $a \mapsto a f_k \bmod \mathcal{J}_{k-1}$ is surjective. Its kernel $\mathcal{I}' = (\mathcal{J}_{k-1} : f_k)$ is a quasi-coherent sheaf of *ideals* (Prop. 1.12(c)), so $H^1(X, \mathcal{I}') = 0$ by (iii′). Prop. 2.15 for $0 \to \mathcal{I}' \to \mathcal{O}_X \to \mathcal{J}_k/\mathcal{J}_{k-1} \to 0$ shows that $A \to (\mathcal{J}_k/\mathcal{J}_{k-1})(X)$, $a \mapsto a f_k \bmod \mathcal{J}_{k-1}$, is surjective. Let $s \in \mathcal{J}_k(X)$. Its image in $(\mathcal{J}_k/\mathcal{J}_{k-1})(X)$ equals that of $a f_k$ for some $a \in A$. Hence $s - a f_k$ maps to 0, so lies in $\mathcal{J}_{k-1}(X)$ (left exactness of sections, Ex. 2.2.4), and $\mathcal{J}_{k-1}(X) = \sum_{i<k} f_i A$ by induction. So $s \in \sum_{i \le k} f_i A$; the other inclusion is clear.

For $k = n$ this gives $A = \mathcal{O}_X(X) = \mathcal{J}_n(X) = \sum_i f_i A$. Hence X is affine. So condition (iii) of Thm. 2.23 can be replaced by (iii′). (If X is Noetherian, all the sheaves of ideals used above are coherent, so it is enough to ask $H^1(X, \mathcal{J}) = 0$ for coherent ideal sheaves $\mathcal{J}$.) □

Exercise 2.10

> **Problem.** Let us keep the notation and hypotheses of Exercise 1.16: $f : X \to S$ is a morphism of schemes, $\pi : T \to S$ is a flat morphism, $X_T = X \times_S T$ with projections $p : X_T \to X$ and $f_T : X_T \to T$, $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module, and X is Noetherian or f is separated and quasi-compact. Show that we have a canonical isomorphism $\pi^*(R^n f_*\mathcal{F}) \to R^n f_{T*}(p^*\mathcal{F})$ for every $n \geq 0$.

Solution. Remark on the hypotheses. The sheaves $R^n f_*\mathcal{F}$ are defined in Prop. 2.28 only when f is separated and quasi-compact. So we assume this. If X is merely Noetherian, f need not be separated and Prop. 2.28 does not define $R^n f_*$ for $n \geq 1$; for $n = 0$ the statement is Ex. 1.16(b) and is covered by the case below only if f is separated. The base change f_T is again separated (Prop. 3.3.9(c)) and quasi-compact (stability of quasi-compactness under base change), and $p^*\mathcal{F}$ is quasi-coherent (Prop. 1.14(b)), so $R^n f_{T*}(p^*\mathcal{F})$ is defined.

Lemma 1. Let $\rho : A \to B$ be a ring homomorphism and M a B-module such that, for every prime $\mathfrak{q}$ of B, the module $M_\mathfrak{q}$ is flat over $A_\mathfrak{p}$, where $\mathfrak{p} = \rho^{-1}(\mathfrak{q})$. Then M is flat over A.

Proof. Let $N' \to N$ be an injective homomorphism of A-modules, and let $\kappa : M \otimes_A N' \to M \otimes_A N$ be the induced map; it is B-linear. For a prime $\mathfrak{q}$ of B we have $(M \otimes_A N)_\mathfrak{q} = M_\mathfrak{q} \otimes_A N$. The elements of $A \setminus \mathfrak{p}$ act invertibly on $M_\mathfrak{q}$, so $M_\mathfrak{q} \otimes_A N = M_\mathfrak{q} \otimes_{A_\mathfrak{p}} N_\mathfrak{p}$. Hence $\kappa_\mathfrak{q}$ is $M_\mathfrak{q} \otimes_{A_\mathfrak{p}} (N'_\mathfrak{p} \to N_\mathfrak{p})$, which is injective because $M_\mathfrak{q}$ is flat over $A_\mathfrak{p}$. So $(\operatorname{Ker}\kappa)_\mathfrak{q} = 0$ for all $\mathfrak{q}$, hence $\operatorname{Ker}\kappa = 0$. □

If $V = \operatorname{Spec} A$ and $W = \operatorname{Spec} B$ are affine and $\varphi : W \to V$ is a flat morphism, then B is flat over A: for a prime $\mathfrak{q}$ of B, $B_\mathfrak{q} = \mathcal{O}_{W,w}$ is flat over $A_\mathfrak{p} = \mathcal{O}_{V,\varphi(w)}$ by definition of flatness, and Lemma 1 applies with $M = B$.

Construction of the map. Let $V = \operatorname{Spec} A \subseteq S$ and $W = \operatorname{Spec} B \subseteq T$ be affine open subsets with $\pi(W) \subseteq V$. Then B is flat over A (just shown). Put $X_V = f^{-1}(V)$ and $Z_W = f_T^{-1}(W) = X_V \times_V W$, so that p restricts to $p_W : Z_W \to X_V$. The scheme X_V is separated and quasi-compact. Cor. 2.27 (applied to $X_V \to \operatorname{Spec} A$, the flat A-algebra B and the sheaf $\mathcal{F}|_{X_V}$) gives an isomorphism

$$\varphi_{V,W} : \; H^n(X_V, \mathcal{F}) \otimes_A B \;\xrightarrow{\sim}\; H^n(Z_W, p^*\mathcal{F}).$$

Recall how it arises. Take a finite affine covering $\mathcal{U} = \{U_i\}$ of X_V. Then $\{p^{-1}(U_i)\}$ is an affine covering of Z_W, all intersections $U_{i_0 \ldots i_k}$ are affine (X_V is separated), and $(p^*\mathcal{F})(p^{-1}U_\alpha) = \mathcal{F}(U_\alpha) \otimes_A B$ by Prop. 1.14(b). So the Cech complex of the new covering is $C(\mathcal{U}, \mathcal{F}) \otimes_A B$. Since B is flat, its cohomology is $H^n(\mathcal{U}, \mathcal{F}) \otimes_A B$. One concludes with Thm. 2.19 on X_V and on Z_W. This map is built from restrictions of sections and tensor products with B, so it commutes with refinement of coverings, and it is compatible with shrinking: if $V'' \subseteq V$ and $W'' \subseteq W$ are affine opens with $\pi(W'') \subseteq V''$, the square formed by $\varphi_{V,W}$, $\varphi_{V'',W''}$ and the restriction maps $H^n(X_V, \mathcal{F}) \otimes_A B \to H^n(X_{V''}, \mathcal{F}) \otimes_{A''} B''$ and $H^n(Z_W, p^*\mathcal{F}) \to H^n(Z_{W''}, p^*\mathcal{F})$ commutes (check it on Cech cochains for coverings of X_V and the restricted, then refined, coverings of $X_{V''}$).

Now $R^n f_* \mathcal{F}$ is quasi-coherent with $(R^n f_* \mathcal{F})(V) = H^n(X_V, \mathcal{F})$ (Prop. 2.28), so by Prop. 1.14(b)

$$(\pi^* R^n f_* \mathcal{F})|_W = \big(H^n(X_V, \mathcal{F}) \otimes_A B\big)^\sim.$$

On the other side, $R^n f_{T*}(p^* \mathcal{F})|_W$ is quasi-coherent on the affine scheme W with module of sections $H^n(Z_W, p^* \mathcal{F})$ (Prop. 2.28 and Thm. 1.7), so it equals $H^n(Z_W, p^* \mathcal{F})^\sim$. Let

$$\Phi_W : (\pi^* R^n f_* \mathcal{F})|_W \to R^n f_{T*}(p^* \mathcal{F})|_W$$

be $\varphi^\sim_{V,W}$; it is an isomorphism.

Gluing. Let (V, W) and (V', W') be two admissible pairs as above and $t \in W \cap W'$. Choose an affine open $V'' \ni \pi(t)$ contained in $V \cap V'$ and then an affine open $W'' \ni t$ contained in $W \cap W' \cap \pi^{-1}(V'')$. By compatibility with shrinking, both Φ_W and $\Phi_{W'}$ restrict on W'' to $\Phi_{W''}$ (computed with V''). Hence Φ_W and $\Phi_{W'}$ agree near every point of $W \cap W'$, so they agree on $W \cap W'$. Since the admissible W cover T, the Φ_W glue to a homomorphism $\Phi : \pi^*(R^n f_* \mathcal{F}) \to R^n f_{T*}(p^* \mathcal{F})$. It is canonical (no choices were made) and it is an isomorphism because it is one on each W. For $n = 0$ it is the homomorphism of Ex. 1.16(a), which is thereby shown to be an isomorphism under the stated hypotheses. □

Exercise 2.11

> **Problem.** Let $f : X \to Y$ be a morphism. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module, flat over Y. (a) Show that for any affine open subset V of Y and any affine open subset U of $f^{-1}(V)$, $\mathcal{F}(U)$ is a flat $\mathcal{O}_Y(V)$-module (draw inspiration from the proof of Corollary 1.2.15). (b) Let us suppose that f is separated and quasi-compact, and that $R^p f_* \mathcal{F} = 0$ for every $p \geq 1$. Show that $f_* \mathcal{F}$ is a flat $\mathcal{O}_Y$-module (use Exercise 1.2.15).

Solution. Hypothesis. In (a) the sheaf $\mathcal{F}$ must be quasi-coherent: for a general $\mathcal{O}_X$-module the module $\mathcal{F}(U)$ is not controlled by the stalks, and the argument below (as the argument of Cor. 1.2.15) uses $\mathcal{F}|_U = \mathcal{F}(U)^\sim$. In (b) quasi-coherence is needed anyway for $R^p f_* \mathcal{F}$ to be defined (Prop. 2.28). So $\mathcal{F}$ is quasi-coherent throughout.

(a) Let $V = \operatorname{Spec} A$, $U = \operatorname{Spec} B \subseteq f^{-1}(V)$, and let $\rho : A \to B$ correspond to $f|_U : U \to V$. Put $M = \mathcal{F}(U)$, so $\mathcal{F}|_U = M^\sim$. For a point $x \in U$ corresponding to a prime $\mathfrak{q}$ of B, we have $f(x) \leftrightarrow \mathfrak{p} = \rho^{-1}(\mathfrak{q})$, $\mathcal{F}_x = M_\mathfrak{q}$, $\mathcal{O}_{Y,f(x)} = A_\mathfrak{p}$. By hypothesis $M_\mathfrak{q}$ is flat over $A_\mathfrak{p}$. Lemma 1 of Ex. 2.10 gives that M is flat over A.

(b) Let $V = \operatorname{Spec} A$ be an affine open subset of Y. By Lemma 2.31(a) and the quasi-coherence of $f_* \mathcal{F}$ (Prop. 1.14(c)), it suffices to show that $(f_* \mathcal{F})(V) = \mathcal{F}(X_V)$ is flat over A, where $X_V = f^{-1}(V)$. If $X_V = \emptyset$ there is nothing to prove. Otherwise, X_V is quasi-compact and separated; choose a finite affine covering $\mathcal{U} = \{U_1, \ldots, U_m\}$ of X_V, totally ordered by the index. All $U_{i_0 \ldots i_k}$ are affine (Prop. 3.3.6). Consider the complex of ordered cochains $C^k = \bigoplus_{i_0 < \cdots < i_k} \mathcal{F}(U_{i_0 \ldots i_k})$, $0 \leq k \leq m-1$. By (a), each C^k is a finite direct sum of flat A-modules, hence flat. By Cor. 2.4, Prop. 2.6 and Thm. 2.19, $H^k(C^\bullet) = H^k(X_V, \mathcal{F})$, which is $\mathcal{F}(X_V)$ for $k = 0$ and is $(R^k f_* \mathcal{F})(V) = 0$ for $k \geq 1$ (Prop. 2.28). So

$$0 \to \mathcal{F}(X_V) \to C^0 \to C^1 \to \cdots \to C^{m-1} \to 0$$

is exact.

We use Ex. 1.2.15 in the following form: if $0 \to M' \to M \to M'' \to 0$ is exact with M, M'' flat, then M' is flat. (Indeed, for every module N the Tor sequence contains $\operatorname{Tor}_2^A(M'', N) \to \operatorname{Tor}_1^A(M', N) \to \operatorname{Tor}_1^A(M, N)$ with both ends zero.) Let $Z^k = \operatorname{Ker}(C^k \to C^{k+1})$. Then $Z^{m-1} = C^{m-1}$ is flat, and for $k < m-1$ exactness gives $0 \to Z^k \to C^k \to Z^{k+1} \to 0$. By descending induction Z^k is flat for all k, in particular $\mathcal{F}(X_V) = Z^0$ is flat over A. □

Exercise 2.12

> **Problem.** Let S be a scheme and $f : X \to S$ a morphism with X Noetherian or f separated and quasi-compact. We suppose that there exists a faithfully flat S-scheme T such that $X_T \to T$ is affine. We want to show that f is an affine morphism. (a) Let us suppose S is affine. Let us fix a quasi-coherent sheaf $\mathcal{F}$ on X. Show that for any $s \in S$, we have $H^1(X, \mathcal{F}) \otimes_{\mathcal{O}_S(S)} \mathcal{O}_{S,s} = 0$ (use Corollary 2.27). (b) Deduce from this that X is affine if S is affine (use Lemma 1.2.12). Conclude.

Solution. Here "faithfully flat" means that the structure morphism $\pi : T \to S$ is flat and surjective; $X_T = X \times_S T$ and $f_T : X_T \to T$ is the projection.

Two lemmas. We prove two lemmas, also used in Ex. 2.13 and 2.15.

Lemma A (descent of separatedness). Let $g : X \to Y$ be a morphism of schemes with X Noetherian, let $q : Y' \to Y$ be flat and surjective, and let $g' : X' = X \times_Y Y' \to Y'$ be the base change. If g' is separated, then g is separated (and quasi-compact, as X is Noetherian).

Proof. Separatedness is local on Y (Def. 3.2), and the hypotheses pass to $g^{-1}(V) \to V$ and $q^{-1}(V) \to V$ for $V \subseteq Y$ an affine open. So we may assume Y affine. Put $W = X \times_Y X$, with diagonal $\Delta : X \to W$, and $W' = W \times_Y Y' = X' \times_{Y'} X'$ with the projection $q_W : W' \to W$, which is flat and surjective (base change of q). The diagonal Δ' of g' is the base change of Δ along q_W, so $\Delta'(X') = q_W^{-1}(\Delta(X))$ (points of fibre products, Ex. 3.1.7). By assumption Δ' is a closed immersion, so $q_W^{-1}(\Delta(X))$ is closed. By Cor. 3.3.5 (a morphism whose diagonal has closed image is separated) we must show that $\Delta(X)$ is closed in W.

Cover X by finitely many affine opens $X_1, \dots, X_m$. As Y is affine, the opens $W_{ab} = X_a \times_Y X_b$ are affine and cover W, and $\Delta(X) \cap W_{ab} = \Delta(X_a \cap X_b) =: E$. The scheme $X_a \cap X_b$ is Noetherian, hence a finite union of affine opens $N_c = \operatorname{Spec} C_c$. Write $W_{ab} = \operatorname{Spec} R$. It suffices to show that E is closed in W_{ab}.

Claim: every point z of the closure $\overline{E}$ of E in W_{ab} is a specialization of a point of E. Indeed, $\overline{E} = \bigcup_c \overline{\Delta(N_c)}$ (finite union). The restriction $N_c \to W_{ab}$ corresponds to a ring homomorphism $\theta : R \to C_c$; the closure of its image is $V(\operatorname{Ker}\theta)$. Let $P \in V(\operatorname{Ker}\theta)$, put $R' = R/\operatorname{Ker}\theta \subseteq C_c$, and let P' be a minimal prime of R' contained in the image of P. Then $R'_{P'}$ is local with the unique prime $P'R'_{P'}$, and $R'_{P'} \to (C_c)_{P'}$ is injective with $(C_c)_{P'} \neq 0$. Any prime of $(C_c)_{P'}$ contracts to the unique prime of $R'_{P'}$, so P' lies in the image of $\operatorname{Spec} C_c$. Thus P is a specialization of a point of $\Delta(N_c)$. This proves the claim.

Now let $z \in \overline{E}$ and $e \in E$ a generization of z. Choose $z' \in W'$ over z. The local homomorphism $\mathcal{O}_{W,z} \to \mathcal{O}_{W',z'}$ is flat, hence faithfully flat (Prop. 1.2.17: it is flat and $\mathfrak{m}_z \mathcal{O}_{W',z'} \neq \mathcal{O}_{W',z'}$), so $\operatorname{Spec} \mathcal{O}_{W',z'} \to \operatorname{Spec} \mathcal{O}_{W,z}$ is surjective. The points of these spectra are the generizations of z', z. So there is a generization e' of z' with $q_W(e') = e$. Then $e' \in q_W^{-1}(\Delta(X)) = \Delta'(X')$, which is closed, so $z' \in \overline{\{e'\}} \subseteq q_W^{-1}(\Delta(X))$. Hence $z = q_W(z') \in \Delta(X) \cap W_{ab} = E$. So E is closed. □

Lemma B. Let $\operatorname{Spec} C = Y$ be affine, $q : Y' \to Y$ flat, $y \in Y$, $t \in Y'$ a point over y, and $W = \operatorname{Spec} B \ni t$ an affine open subset of Y' (so $W \to Y$ gives $C \to B$, which is flat). If N is a C-module with $N \otimes_C B = 0$, then $N \otimes_C \mathcal{O}_{Y,y} = 0$.

Proof. $\mathcal{O}_{Y',t} = B_{\mathfrak{q}}$ is a localization of B, so $N \otimes_C \mathcal{O}_{Y',t} = (N \otimes_C B) \otimes_B B_{\mathfrak{q}} = 0$. Now $\mathcal{O}_{Y,y} \to \mathcal{O}_{Y',t}$ is a flat local homomorphism, hence faithfully flat (Prop. 1.2.17), and $N \otimes_C \mathcal{O}_{Y',t} = (N \otimes_C \mathcal{O}_{Y,y}) \otimes_{\mathcal{O}_{Y,y}} \mathcal{O}_{Y',t}$. Faithful flatness gives $N \otimes_C \mathcal{O}_{Y,y} = 0$. □

Reduction to f separated and quasi-compact. The morphism f_T is affine, hence separated (Prop. 3.3.4), and it is the base change of f along the flat surjective π. If X is Noetherian, Lemma A shows that f is separated and quasi-compact. So in all cases f is separated and quasi-compact.

(a) Let $S = \operatorname{Spec} A$, $s \in S$, $M = H^1(X, \mathcal{F})$. Since π is surjective, choose $t \in T$ with $\pi(t) = s$ and an affine open $W = \operatorname{Spec} B \ni t$ of T. Then $W \to S$ is flat, so B is flat over A (Ex. 2.10). The scheme $X_W = f_T^{-1}(W) = X \times_S W$ is affine, as f_T is affine. Cor. 2.27 (for $f : X \to \operatorname{Spec} A$ and the flat A-algebra B) and Thm. 2.18 give

$$M \otimes_A B \simeq H^1(X_W, p^*\mathcal{F}) = 0.$$

Lemma B gives $M \otimes_A \mathcal{O}_{S,s} = 0$.

(b) Let $S = \operatorname{Spec} A$. By (a) all localizations of the A-module $H^1(X, \mathcal{F})$ at primes of A vanish, so $H^1(X, \mathcal{F}) = 0$ (Lemma 1.2.12). This holds for every quasi-coherent $\mathcal{F}$, and X is Noetherian or separated and quasi-compact (see above), so X is affine by Serre's criterion (Thm. 2.23).

Conclusion. Let S be arbitrary and $V \subseteq S$ an affine open subset. Then $\pi^{-1}(V) \to V$ is flat and surjective, $(f^{-1}(V))_{\pi^{-1}(V)} \to \pi^{-1}(V)$ is the restriction of f_T over $\pi^{-1}(V)$, hence affine, and $f^{-1}(V)$ is Noetherian (open in X) or $f^{-1}(V) \to V$ is separated and quasi-compact. By (b), $f^{-1}(V)$ is affine. Since V was an arbitrary affine open, f is affine. □

Exercise 2.13

> **Problem.** Let S be a scheme and $g : X \to Y$ a morphism of S-schemes such that X is Noetherian or g is separated and quasi-compact. We suppose that there exists a faithfully flat S-scheme T such that g_T is an isomorphism. We want to show that g is an isomorphism. (a) Show that we can reduce to S and Y affine. (b) Show that under the hypotheses of (a), we have X affine and g is an isomorphism (use Lemma 1.2.12).

Solution. We use Lemma A and Lemma B from Ex. 2.12 and Ex. 2.10 (Lemma 1 there for flatness of B over A).

(a) Being an isomorphism is local on Y. Let $V \subseteq Y$ be an affine open subset mapping into an affine open subset S_0 of S. Then $g^{-1}(V) \to V$ is a morphism of S_0-schemes, $T_0 = T \times_S S_0 \to S_0$ is faithfully flat, and the base change of $g^{-1}(V) \to V$ to T_0 is the restriction of the isomorphism g_T to g_T^{-1} of $V \times_S T_0 = V_T$, hence an isomorphism. The hypothesis on X or g passes to $g^{-1}(V) \to V$ ($g^{-1}(V)$ is open in X; separatedness and quasi-compactness restrict). So it suffices to prove the claim for $g^{-1}(V) \to V$, i.e. we may assume S and Y affine.

(b) Let $Y = \operatorname{Spec} C$. Put $Y' = Y_T = Y \times_S T$, with $q : Y' \to Y$ flat and surjective (base change of $T \to S$), and note $X \times_Y Y' = X_T$, so that g_T is the base change of g along q. By assumption

g_T is an isomorphism, hence separated. By Lemma A (if X is Noetherian) g is separated and quasi-compact; in the other case this is the hypothesis.

X is affine. Let $y \in Y$. Since q is surjective choose $t \in Y'$ over y and an affine open $W = \operatorname{Spec} B \ni t$ of Y'; $C \to B$ is flat. Then $X \times_Y W = g_T^{-1}(W) \to W$ is an isomorphism, so $X_W := X \times_Y W$ is affine. For a quasi-coherent $\mathcal{F}$ on X, Cor. 2.27 and Thm. 2.18 give $H^1(X, \mathcal{F}) \otimes_C B \simeq H^1(X_W, \mathcal{F}_W) = 0$, hence $H^1(X, \mathcal{F}) \otimes_C \mathcal{O}_{Y,y} = 0$ by Lemma B. As y was arbitrary, $H^1(X, \mathcal{F}) = 0$ (Lemma 1.2.12). By Serre's criterion (Thm. 2.23; X is Noetherian, or separated and quasi-compact), X is affine, say $X = \operatorname{Spec} D$, and g corresponds to a ring homomorphism $\varphi : C \to D$.

φ is an isomorphism. With y, t, W as above, $X_W = \operatorname{Spec}(D \otimes_C B)$ and g_W is the isomorphism $\operatorname{Spec}(D \otimes_C B) \to \operatorname{Spec} B$, i.e. $B \to D \otimes_C B$, $b \mapsto 1 \otimes b$, is an isomorphism. Let $K = \operatorname{Ker} \varphi$, $Q = \operatorname{Coker} \varphi$. Tensoring $0 \to K \to C \to D \to Q \to 0$ with the flat C-module B gives the exact sequence $0 \to K \otimes_C B \to B \to D \otimes_C B \to Q \otimes_C B \to 0$ whose middle map is an isomorphism. So $K \otimes_C B = 0 = Q \otimes_C B$, and Lemma B gives $K_y = 0 = Q_y$ for every $y \in Y$. Hence $K = Q = 0$ (Lemma 1.2.12), so φ is an isomorphism and g is an isomorphism. □

Exercise 2.14

> **Problem.** Let $f : X \dashrightarrow Y$ be a birational map as in Exercise 4.3.15 but without the Y affine hypothesis. Using Exercises 2.13 and 3.3.14, show that f is defined everywhere.
>
> (Exercise 4.3.15: X, Y are integral schemes, separated and of finite type over a locally Noetherian scheme S; $f : X \dashrightarrow Y$ is a birational map; $T \to S$ is a faithfully flat morphism such that X_T and Y_T are integral; $f_T : X_T \dashrightarrow Y_T$ is defined everywhere.)

Solution. Set-up. The map f is given by an isomorphism of S-schemes $u : U \to V$ between non-empty (dense) open subschemes of X and Y. Then f_T is given by the isomorphism $u_T : U_T \to V_T$. Here U_T is non-empty ($U_T \to U$ is surjective) and open in the integral scheme X_T, hence integral and dense. That f_T is defined everywhere means: there is a T-morphism $F : X_T \to Y_T$ agreeing with u_T on a dense open subset of U_T. As U_T is reduced and Y_T is separated over T, $F|_{U_T} = u_T$ (Prop. 3.3.11).

Let $W = X \times_S Y$, $p : W \to X$, $q_Y : W \to Y$ the projections. As Y/S is separated, $i = (\mathrm{Id}_U, u) : U \to W$ is a closed immersion into the open subscheme $p^{-1}(U) = U \times_S Y$. Its image is the graph Γ_U. By definition (Ex. 3.3.13(c)), Γ_f is the closure of Γ_U in W with the reduced structure, and $g = p|_{\Gamma_f} : \Gamma_f \to X$. By Ex. 3.3.14 (the proof of which uses only that Y/S is separated), f is defined everywhere iff g is an isomorphism. For completeness we give the direction we need: if g is an isomorphism then $\hat{f} = q_Y \circ g^{-1} : X \to Y$ is an S-morphism; and $\Gamma_f \cap p^{-1}(U) = \Gamma_U$ (it is closed in $p^{-1}(U)$), so $g^{-1}|_U = i$ and $\hat{f}|_U = q_Y \circ i = u$. Thus $\hat{f}$ extends u and f is defined everywhere.

So it remains to show that g is an isomorphism. We use Ex. 2.13 for the S-morphism $g : \Gamma_f \to X$ and the faithfully flat S-scheme T.

Hypotheses of Ex. 2.13. g is the composite of the closed immersion $\Gamma_f \to W$ and p. The projection p is separated (base change of Y/S) and of finite type (base change of Y/S of finite type). Hence g is separated and of finite type, so quasi-compact.

The base change of g. The schemes X, Y, W are of finite type over the locally Noetherian S, hence locally Noetherian, so every open immersion into them is quasi-compact. Therefore i is a

quasi-compact and separated morphism. Put $\mathcal{I} = \mathrm{Ker}(\mathcal{O}_W \to i_*\mathcal{O}_U)$. A local section a of $\mathcal{O}_W$ lies in $\mathcal{I}$ iff $i^\#(a) = 0$ in $\mathcal{O}_U$. Since U is reduced, this holds iff a vanishes (in the residue fields) at all points of $i(U)$, i.e. iff a vanishes at all points of the closure of $i(U)$, since the vanishing locus of a section is closed. So $\mathcal{I}$ is the ideal sheaf of the reduced closed subscheme Γ_f.

Let $q_W : W_T = X_T \times_T Y_T \to W$ be the projection; it is flat. Then $(\Gamma_f)_T = \Gamma_f \times_W W_T$ is the closed subscheme of W_T with ideal sheaf $q_W^*\mathcal{I} \subseteq \mathcal{O}_{W_T}$ (flatness). Moreover $q_W^*(i_*\mathcal{O}_U) = (i_T)_*\mathcal{O}_{U_T}$, where $i_T = (\mathrm{Id}, u_T) : U_T \to W_T$ is the base change of i: this is Ex. 1.16(b) for the quasi-compact separated morphism i and the quasi-coherent sheaf $\mathcal{O}_U$, which on an affine open $\mathrm{Spec}\, R \subseteq W$ and an affine open $\mathrm{Spec}\, R'$ over it amounts to $\mathcal{O}(i^{-1}\,\mathrm{Spec}\, R) \otimes_R R' = \mathcal{O}(i_T^{-1}\,\mathrm{Spec}\, R')$ (Cor. 2.27, case $p = 0$). By flatness of q_W the kernel is preserved, so

$$q_W^*\mathcal{I} = \mathrm{Ker}\big(\mathcal{O}_{W_T} \to (i_T)_*\mathcal{O}_{U_T}\big).$$

Since U_T is reduced, the same argument as above shows that this is the ideal of the reduced closed subscheme of W_T supported on the closure of $i_T(U_T)$. Now $(\mathrm{Id}, F) : X_T \to W_T$ is a closed immersion (Y_T/T separated) with image Γ_F, the first projection restricts to an inverse isomorphism $\Gamma_F \to X_T$, and Γ_F is reduced ($\simeq X_T$). It contains $i_T(U_T)$ (as $F|_{U_T} = u_T$) as a dense subset (U_T is dense in X_T). Hence Γ_F is the reduced subscheme on the closure of $i_T(U_T)$, and so $(\Gamma_f)_T = \Gamma_F$. Under this identification $g_T : (\Gamma_f)_T \to X_T$ is the first projection $\Gamma_F \to X_T$, an isomorphism.

Conclusion. By Ex. 2.13, g is an isomorphism, hence f is defined everywhere by the above. □

Exercise 2.15

> **Problem.** Let S be a locally Noetherian scheme, $f : X \to S$ a proper morphism. (a) Let us suppose that $S = \mathrm{Spec}\, A$, with A a complete local ring. Let s be the closed point of S. Show that if X_s is finite, then X is finite over S (use Lemma 4.4.1). (b) Show that (a) is true without the A complete hypothesis (use Exercise 2.13 and Lemma 3.3.17). (c) Let us suppose that X_s is finite for every $s \in S$. Show that f is a finite morphism (compare with Exercise 4.4.2).

Solution. "X_s is finite" means that $X_s \to \mathrm{Spec}\, k(s)$ is a finite morphism, i.e. $X_s =$ Spec of a finite $k(s)$-algebra; its underlying space is finite and discrete. (Since X_s is of finite type over $k(s)$, this is equivalent to X_s having finitely many points.) Recall that a proper morphism is closed and universally so, and that a morphism which is finite over an affine open covering of the base is finite (Ex. 3.3.15).

(a) Here A is a Noetherian complete local ring. Let $X_s = \{x_1, \ldots, x_n\}$ (possibly $n = 0$). Because X_s is discrete, there is an open subset $X_i \subseteq X$ with $X_i \cap X_s = \{x_i\}$, and X_i is of finite type over S. By Lemma 4.4.1 there is an affine open neighbourhood $U_i \subseteq X_i$ of x_i finite over S. Then $U_i \to S$ is proper (Ex. 3.3.17(d)), so $U_i \to X$ is proper (Prop. 3.3.16(e); $X \to S$ is separated), in particular closed. As U_i is also open, U_i is open and closed in X. Put $U = \bigcup_i U_i$, open and closed. The set $f(X \setminus U)$ is closed (f is proper). If $X \setminus U \neq \emptyset$, this closed non-empty subset of the spectrum of a local ring contains the closed point s, so $X \setminus U$ meets $X_s \subseteq U$, absurd. So $X = U$.

Put $V_1 = U_1$ and $V_k = U_k \setminus (U_1 \cup \cdots \cup U_{k-1})$. These are open and closed in X, pairwise disjoint, and cover X. V_k is a clopen subscheme of the affine scheme $U_k = \mathrm{Spec}\, C_k$, so $V_k = \mathrm{Spec}\, C_k'$

with $C_k = C'_k \times C''_k$ (idempotent); C'_k is a quotient of C_k, hence finite over A. Therefore $X = \bigsqcup_k V_k = \operatorname{Spec}(\prod_k C'_k)$ is affine and finite over A. By Ex. 3.3.15, f is finite.

(b) Now A is a Noetherian local ring with closed point s. Let $\widehat{A}$ be its completion, $T = \operatorname{Spec} \widehat{A}$, $\pi : T \to S$. By Thm. 1.3.16, $\widehat{A}$ is flat over A, and $\widehat{A}/\mathfrak{m}\widehat{A} = A/\mathfrak{m} \neq 0$, so $A \to \widehat{A}$ is faithfully flat (Prop. 1.2.17) and π is flat and surjective. The morphism $f_T : X_T \to T$ is proper, T is the spectrum of a complete local ring, and its closed fibre is $X_s \otimes_{k(s)} k(s') = X_s$ (the residue field of $\widehat{A}$ is $k(s)$), which is finite. By (a), f_T is finite; in particular affine. By Ex. 2.12 (here f is separated and quasi-compact) f is affine, so X is affine, and it is proper over $\operatorname{Spec} A$. Lemma 3.3.17 shows that $\mathcal{O}_X(X)$ is finite over A. So f is finite. (One may use Ex. 2.13 instead: it applies to $g : X \to \operatorname{Spec} \mathcal{O}_X(X)$, using that X_T is affine and $\mathcal{O}(X_T) = \mathcal{O}_X(X) \otimes_A \widehat{A}$ by Cor. 2.27 with $p = 0$; the route via Ex. 2.12 is shorter.)

(c) Let $T = \bigsqcup_{s \in S} \operatorname{Spec} \mathcal{O}_{S,s}$, with $\pi : T \to S$ the canonical morphism. Each $\operatorname{Spec} \mathcal{O}_{S,s} \to S$ is flat, and π is surjective, so T is a faithfully flat S-scheme. For $s \in S$, the component of f_T over $\operatorname{Spec} \mathcal{O}_{S,s}$ is the base change $X \times_S \operatorname{Spec} \mathcal{O}_{S,s} \to \operatorname{Spec} \mathcal{O}_{S,s}$. It is proper, $\mathcal{O}_{S,s}$ is a Noetherian local ring, and its closed fibre is X_s, finite. By (b) it is finite, in particular affine. A morphism whose restriction over each member of a disjoint open covering of the base is affine is affine, so f_T is affine. By Ex. 2.12 f is affine. For an affine open $V = \operatorname{Spec} A \subseteq S$, $f^{-1}(V) = \operatorname{Spec} B$ is proper over $\operatorname{Spec} A$, so B is finite over A (Lemma 3.3.17). Hence f is finite. □

Exercise 2.16

> **Problem.** Let $f : X \to Y$ be an affine morphism. (a) Let $\mathcal{F}$ be a quasi-coherent sheaf on X. Show that the canonical homomorphism $f^* f_* \mathcal{F} \to \mathcal{F}$ (Exercise 1.1) is surjective. (b) Let us suppose Y is quasi-compact and f finite. Let $\mathcal{L}$ be an ample sheaf on Y. Show that $f^*\mathcal{L}$ is ample on X (use Propositions 1.14(d) and 2.32).

Solution. (a) The sheaf $f_*\mathcal{F}$ is quasi-coherent (Prop. 1.14(c); f is affine, hence separated and quasi-compact). The canonical homomorphism $f^* f_* \mathcal{F} = f^{-1} f_* \mathcal{F} \otimes_{f^{-1}\mathcal{O}_Y} \mathcal{O}_X \to \mathcal{F}$ is induced by $s \otimes a \mapsto as$. The question is local on Y, so let $V = \operatorname{Spec} A$ be an affine open subset of Y. Then $f^{-1}(V) = \operatorname{Spec} B$ is affine. Put $M = \mathcal{F}(f^{-1}V)$, a B-module with $\mathcal{F}|_{f^{-1}V} = M^\sim$; as an A-module it gives $(f_*\mathcal{F})|_V = M^\sim$ (Thm. 1.7). By Prop. 1.14(b), $(f^* f_* \mathcal{F})|_{f^{-1}V} = (M \otimes_A B)^\sim$, and the canonical homomorphism is $(M \otimes_A B \to M,\ m \otimes b \mapsto bm)^\sim$. This map of modules is surjective, since $m \otimes 1 \mapsto m$. Hence the sheaf homomorphism is surjective over each $f^{-1}(V)$, so everywhere.

(b) As Y is quasi-compact, it is a finite union of affine opens Y_a, and $f^{-1}(Y_a)$ is affine; so X is quasi-compact. The sheaf $f^*\mathcal{L}$ is invertible. Let $\mathcal{F}$ be a finitely generated quasi-coherent sheaf on X. By Prop. 1.14(d), $f_*\mathcal{F}$ is quasi-coherent and finitely generated. Since $\mathcal{L}$ is ample, there is n_0 with $f_*\mathcal{F} \otimes \mathcal{L}^{\otimes n}$ generated by global sections for all $n \geq n_0$. Fix such an n and put $\mathcal{G} = \mathcal{F} \otimes (f^*\mathcal{L})^{\otimes n} = \mathcal{F} \otimes f^*(\mathcal{L}^{\otimes n})$, a quasi-coherent sheaf. The projection formula (Prop. 2.32 with $p = 0$; $\mathcal{L}^{\otimes n}$ is locally free, hence flat, and f is separated and quasi-compact) gives

$$f_*\mathcal{G} \simeq f_*\mathcal{F} \otimes \mathcal{L}^{\otimes n}.$$

So $f_*\mathcal{G}$ is generated by global sections: there are $s_i \in (f_*\mathcal{G})(Y) = \mathcal{G}(X)$, $i \in I$, such that $\mathcal{O}_Y^{(I)} \to f_*\mathcal{G}$, $e_i \mapsto s_i$, is surjective. Applying f^*, which is right exact, $\mathcal{O}_X^{(I)} = f^*\mathcal{O}_Y^{(I)} \to f^* f_* \mathcal{G}$ is

surjective. Composing with the surjection $f^*f_*\mathcal{G} \to \mathcal{G}$ of (a), we get a surjection $\mathcal{O}_X^{(I)} \to \mathcal{G}$ that sends the ith basis vector to $s_i \in \mathcal{G}(X)$. So $\mathcal{G} = \mathcal{F} \otimes (f^*\mathcal{L})^{\otimes n}$ is generated by global sections for all $n \geq n_0$. Hence $f^*\mathcal{L}$ is ample (Def. 1.33). □

Exercise 2.17

Problem. Let $(X, \mathcal{O}_X)$ be a ringed topological space, and let $\mathcal{F}$, $\mathcal{G}$ be two $\mathcal{O}_X$-modules. We are going to define an operation called the cup product $H^p(X, \mathcal{F}) \times H^q(X, \mathcal{G}) \to H^{p+q}(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$ for any $p, q \geq 0$. Let $\mathcal{U} = \{U_i\}_i$, $\mathcal{V} = \{V_j\}_j$ be two coverings of X. Let $\mathcal{W} = \{U_i \cap V_j\}_{(i,j)}$ be the covering $\mathcal{U} \cap \mathcal{V}$ of X. We will denote the index (i, j) by α. (a) Let $f \in C^p(\mathcal{U}, \mathcal{F})$, $g \in C^q(\mathcal{V}, \mathcal{G})$, and set $f \smile g \in C^{p+q}(\mathcal{W}, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$ equal to the cochain defined by $(f \smile g)_{\alpha_0 \dots \alpha_{p+q}} = f_{i_0 \dots i_p}|_{W_\alpha} \otimes g_{j_p \dots j_{p+q}}|_{W_\alpha}$, $W_\alpha = U_i \cap V_j$. Show that $d(f \smile g) = df \smile g + (-1)^p (f \smile dg)$. (b) Show that the operation $\smile$ in (a) induces a homomorphism $H^p(\mathcal{U}, \mathcal{F}) \times H^q(\mathcal{V}, \mathcal{G}) \to H^{p+q}(\mathcal{W}, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$. By taking the direct limit, we thus obtain a homomorphism $\smile$: $H^p(X, \mathcal{F}) \times H^q(X, \mathcal{G}) \to H^{p+q}(X, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$ called the cup product. (c) Show that the cup product is associative in an evident sense. We can show that it is anti-commutative, that is to say that $\xi \smile \eta = (-1)^{pq} \eta \smile \xi$ if $\xi \in H^p(X, \mathcal{F})$, $\eta \in H^q(X, \mathcal{G})$, and if we identify $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ with $\mathcal{G} \otimes_{\mathcal{O}_X} \mathcal{F}$ (see [40], Section II.6.6).

Solution. Conventions. Cochains are the (not necessarily alternating) cochains $C^p(\mathcal{U}, \mathcal{F}) = \prod_{(i_0, \dots, i_p)} \mathcal{F}(U_{i_0 \dots i_p})$ with the differential of the text. For $\alpha_k = (i_k, j_k)$ and $n = p + q$ one has $W_{\alpha_0 \dots \alpha_n} = W_{\alpha_0} \cap \dots \cap W_{\alpha_n} = U_{i_0 \dots i_n} \cap V_{j_0 \dots j_n}$, and both $f_{i_0 \dots i_p}$ and $g_{j_p \dots j_n}$ are restricted to this open set. Here $a \otimes b \in (\mathcal{F} \otimes \mathcal{G})(W)$ for $a \in \mathcal{F}(W)$, $b \in \mathcal{G}(W)$ denotes the image of $a \otimes b$ under the canonical map $\mathcal{F}(W) \otimes_{\mathcal{O}_X(W)} \mathcal{G}(W) \to (\mathcal{F} \otimes \mathcal{G})(W)$. This construction is bilinear and compatible with restriction to smaller open sets. Observe also that $\smile$ is biadditive.

(a) Write $n = p + q$ and $h = f \smile g$. Fix $\alpha_0, \dots, \alpha_{n+1}$ with $\alpha_k = (i_k, j_k)$; all sections below are restricted to $W_{\alpha_0 \dots \alpha_{n+1}}$. By definition $(dh)_{\alpha_0 \dots \alpha_{n+1}} = \sum_{k=0}^{n+1} (-1)^k h_{\alpha_0 \dots \hat{\alpha}_k \dots \alpha_{n+1}}$. Consider the omitted index k.

If $k \leq p$, the first $p+1$ entries of the sequence $(\alpha_0, \dots, \hat{\alpha}_k, \dots, \alpha_{n+1})$ are $\alpha_0, \dots, \hat{\alpha}_k, \dots, \alpha_{p+1}$ and the entries from the $(p+1)$-th on are $\alpha_{p+1}, \dots, \alpha_{n+1}$. So the term is $A_k = f_{i_0 \dots \hat{i}_k \dots i_{p+1}} \otimes g_{j_{p+1} \dots j_{n+1}}$.

If $k \geq p + 1$, the first $p + 1$ entries are $\alpha_0, \dots, \alpha_p$ and the entries from the $(p + 1)$-th on are $\alpha_p, \dots, \hat{\alpha}_k, \dots, \alpha_{n+1}$. So the term is $B_k = f_{i_0 \dots i_p} \otimes g_{j_p \dots \hat{j}_k \dots j_{n+1}}$.

Thus $dh = \sum_{k=0}^{p} (-1)^k A_k + \sum_{k=p+1}^{n+1} (-1)^k B_k$.

On the other hand,

$$(df \smile g)_{\alpha_0 \dots \alpha_{n+1}} = \sum_{k=0}^{p+1} (-1)^k f_{i_0 \dots \hat{i}_k \dots i_{p+1}} \otimes g_{j_{p+1} \dots j_{n+1}} = \sum_{k=0}^{p} (-1)^k A_k + (-1)^{p+1} f_{i_0 \dots i_p} \otimes g_{j_{p+1} \dots j_{n+1}},$$

and, with $k = p + m$,

$$(-1)^p (f \smile dg)_{\alpha_0 \dots \alpha_{n+1}} = (-1)^p \sum_{m=0}^{q+1} (-1)^m f_{i_0 \dots i_p} \otimes g_{j_p \dots \hat{j}_{p+m} \dots j_{n+1}} = (-1)^p f_{i_0 \dots i_p} \otimes g_{j_{p+1} \dots j_{n+1}} + \sum_{k=p+1}^{n+1} (-1)^k B_k.$$

Adding these two expressions, the two terms $\pm f_{i_0 \dots i_p} \otimes g_{j_{p+1} \dots j_{n+1}}$ cancel and we obtain dh. So $d(f \smile g) = df \smile g + (-1)^p (f \smile dg)$.

(b) Let $f \in C^p(\mathcal{U},\mathcal{F})$ and $g \in C^q(\mathcal{V},\mathcal{G})$ be cocycles, $df = 0 = dg$. By (a), $d(f \smile g) = 0$. If $f = d\varphi$ with $\varphi \in C^{p-1}(\mathcal{U},\mathcal{F})$ and $dg = 0$, then (a) gives $d(\varphi \smile g) = d\varphi \smile g + (-1)^{p-1}\varphi \smile dg = f \smile g$. If $g = d\psi$ with $\psi \in C^{q-1}(\mathcal{V},\mathcal{G})$ and $df = 0$, then $d(f \smile \psi) = df \smile \psi + (-1)^p f \smile d\psi = (-1)^p f \smile g$. So the cup product of a cocycle with a coboundary is a coboundary. Since $\smile$ is biadditive, it induces a biadditive (indeed $\mathcal{O}_X(X)$-bilinear) map $H^p(\mathcal{U},\mathcal{F}) \times H^q(\mathcal{V},\mathcal{G}) \to H^{p+q}(\mathcal{W},\mathcal{F}\otimes\mathcal{G})$.

Passage to the limit. Let $\mathcal{U}' = \{U'_a\}_{a\in I'}$ refine $\mathcal{U}$ via $\sigma : I' \to I$ and $\mathcal{V}' = \{V'_b\}_{b\in J'}$ refine $\mathcal{V}$ via $\tau : J' \to J$. Then $\mathcal{W}' = \mathcal{U}' \cap \mathcal{V}'$ refines $\mathcal{W}$ via $(\sigma,\tau) : (a,b) \mapsto (\sigma(a),\tau(b))$, because $U'_a \cap V'_b \subseteq U_{\sigma(a)} \cap V_{\tau(b)}$. For cochains one checks directly from the definitions that

$$\sigma(f) \smile \tau(g) = (\sigma,\tau)(f \smile g) :$$

the component at $((a_0,b_0),\dots,(a_n,b_n))$ of the left side is $f_{\sigma(a_0)\dots\sigma(a_p)|} \otimes g_{\tau(b_p)\dots\tau(b_n)|}$, and that of the right side is $(f \smile g)_{(\sigma a_0,\tau b_0)\dots(\sigma a_n,\tau b_n)|}$, which is the same element. Hence the cup product of classes commutes with the refinement maps (which on cohomology do not depend on the choice of the refinement map, Lemma 2.8). Two coverings always have a common refinement, so the pairs $(\mathcal{U},\mathcal{V})$ form a directed set, and

$$\varinjlim_{(\mathcal{U},\mathcal{V})} \big(H^p(\mathcal{U},\mathcal{F}) \times H^q(\mathcal{V},\mathcal{G})\big) = H^p(X,\mathcal{F}) \times H^q(X,\mathcal{G}).$$

(Indeed, the directed set of coverings is cofinal in the product directed set via the diagonal $\mathcal{U} \mapsto (\mathcal{U},\mathcal{U})$ up to refinement, and a limit of products over the product directed set is the product of the limits.) The maps $H^p(\mathcal{U},\mathcal{F})\times H^q(\mathcal{V},\mathcal{G}) \to H^{p+q}(\mathcal{U}\cap\mathcal{V},\mathcal{F}\otimes\mathcal{G}) \to H^{p+q}(X,\mathcal{F}\otimes\mathcal{G})$ are compatible with the transition maps by what we have just shown. So they define in the limit the cup product $H^p(X,\mathcal{F}) \times H^q(X,\mathcal{G}) \to H^{p+q}(X,\mathcal{F}\otimes\mathcal{G})$: for $\xi = [f]$, $\eta = [g]$, we have $\xi \smile \eta = [f \smile g]$.

(c) *Associativity.* Let $\mathcal{H}$ be a third $\mathcal{O}_X$-module, $\mathcal{U},\mathcal{V},\mathcal{W}$ coverings indexed by I,J,K, and $f \in C^p(\mathcal{U},\mathcal{F})$, $g \in C^q(\mathcal{V},\mathcal{G})$, $h \in C^r(\mathcal{W},\mathcal{H})$ cochains; put $n = p+q+r$. The coverings $(\mathcal{U}\cap\mathcal{V})\cap\mathcal{W}$ and $\mathcal{U}\cap(\mathcal{V}\cap\mathcal{W})$ are both the family $\{U_i \cap V_j \cap W_k\}$ indexed, respectively, by $(I\times J)\times K$ and $I\times(J\times K)$; we identify both index sets with $I\times J\times K$ (a relabelling of the index set, under which the Cech complexes correspond to each other). For indices $\gamma_m = (i_m,j_m,k_m)$, $0 \le m \le n$, and with all sections restricted to the open set $U_{i_0\dots i_n} \cap V_{j_0\dots j_n} \cap W_{k_0\dots k_n}$,

$$\big((f \smile g) \smile h\big)_{\gamma_0\dots\gamma_n} = (f_{i_0\dots i_p} \otimes g_{j_p\dots j_{p+q}}) \otimes h_{k_{p+q}\dots k_n},$$

$$\big(f \smile (g \smile h)\big)_{\gamma_0\dots\gamma_n} = f_{i_0\dots i_p} \otimes (g_{j_p\dots j_{p+q}} \otimes h_{k_{p+q}\dots k_n}).$$

Let $\mathrm{as} : (\mathcal{F}\otimes\mathcal{G})\otimes\mathcal{H} \to \mathcal{F}\otimes(\mathcal{G}\otimes\mathcal{H})$ be the canonical associativity isomorphism of sheaves, $(a\otimes b)\otimes c \mapsto a\otimes(b\otimes c)$. It commutes with restrictions, hence acts on cochains, and the displayed formulas say $\mathrm{as}\big((f \smile g) \smile h\big) = f \smile (g \smile h)$. Since as commutes with the differentials, on cohomology of $(\mathcal{U}\cap\mathcal{V})\cap\mathcal{W}$ we get $\mathrm{as}_*\big(([f] \smile [g]) \smile [h]\big) = [f] \smile ([g] \smile [h])$. The two sides are compatible with the refinement maps of the three coverings (the same computation as in (b)). Passing to the limit, for $\xi \in H^p(X,\mathcal{F})$, $\eta \in H^q(X,\mathcal{G})$, $\zeta \in H^r(X,\mathcal{H})$,

$$\mathrm{as}_*\big((\xi \smile \eta) \smile \zeta\big) = \xi \smile (\eta \smile \zeta) \quad \text{in } H^{p+q+r}\big(X,\mathcal{F}\otimes(\mathcal{G}\otimes\mathcal{H})\big).$$

This is associativity of the cup product.

Anti-commutativity. The exercise says "we can show" this, and asks to prove only associativity. I prove only the easy cases $p = 0$ or $q = 0$ and do not prove the general sign rule here. (A complete proof needs an explicit homotopy between $f \smile g$ and $\pm t^*(g \smile f)$, where t is the swap of the index pairs, and a sign calculation that I have not carried out.)

Case $p = 0$: here $\xi = s \in H^0(X, \mathcal{F}) = \mathcal{F}(X)$ (Prop. 2.6) is represented by the one-element covering $\mathcal{U} = \{X\}$ and $f = s \in C^0(\mathcal{U}, \mathcal{F})$. Let $\eta = [g]$ with $g \in C^q(\mathcal{V}, \mathcal{G})$. Then $\mathcal{U} \cap \mathcal{V}$ is $\mathcal{V}$ with index set $\{*\} \times J$, and $(s \smile g)_{j_0 \dots j_q} = s|_{V_{j_0 \dots j_q}} \otimes g_{j_0 \dots j_q}$, while $(g \smile s)_{j_0 \dots j_q} = g_{j_0 \dots j_q} \otimes s|_{V_{j_0 \dots j_q}}$. Under the identification $\mathcal{F} \otimes \mathcal{G} = \mathcal{G} \otimes \mathcal{F}$ they coincide. So $\xi \smile \eta = \eta \smile \xi$, which is the claim since $(-1)^{pq} = 1$. The case $q = 0$ is the same with the roles exchanged. □

5.3 Cohomology of projective schemes

Cohomology is Cech cohomology as in Def. 2.10, computed on a finite affine covering of a separated scheme (Thm. 2.19) with ordered cochains (Cor. 2.4). Results of the text are cited by number ("Thm. 3.2", "Cor. 3.22", ...) and results of other chapters as "Ex. 2.2.8", "Prop. 3.3.6", ... Exercises of Chapter 5 are cited as "Ex. 2.3" and so on. For a ring A we write $\mathbf{P}^d_A = \operatorname{Proj} A[T_0, \dots, T_d]$, and $\mathcal{O}(q)$ for $\mathcal{O}_{\mathbf{P}^d}(q)$. An algebraic variety is a separated scheme of finite type over a field (not necessarily reduced), so a closed subscheme $V_+(F_1, \dots, F_c)$ of $\mathbf{P}^d_k$ is a variety.

Honest notes. Several statements of this section are not correct as printed or need a hypothesis; the solutions say so where it happens. Ex. 3.2: the formula of Lemma 3.1(c) needs $d \geq 1$. Ex. 3.3: "defined as a variety by $F_1, \dots, F_{d-r}$" is read as "equal to the closed subscheme $V_+(F_1, \dots, F_{d-r})$" (varieties need not be reduced), and standard dimension theory (Krull's theorem, Cohen–Macaulayness of regular local rings) is used without proof. Ex. 3.5(a),(b) are false when the special fibre has dimension 0 (a counterexample with three points in $\mathbf{P}^2$ is given), and in (a) the ideal I has to be the saturated ideal of Z; both are proved for $r \geq 1$ (using Rem. 3.21 of the text and the upper semicontinuity of the dimension of the fibres of a proper morphism, Chevalley, EGA IV 13.1.5, which is not proved in the text). Part (c) is proved as printed. Ex. 3.6: the relative Frobenius is the morphism $X \to X^{(p)}$ and not the other way round; both $F_{X/S}$ and the projection $X^{(p)} \to X$ are treated. Ex. 3.9: f has to be dominant. Ex. 3.10 is false if f is not surjective (counterexample: a closed immersion of two points into the affine line), so it is proved for surjective f; this affects Ex. 3.9(b) only through the (automatic) surjectivity of f. Ex. 3.12: the geometrically connected case is proved as printed. The geometrically integral case is false for non-reduced Y (counterexample given); it is proved here only for Y integral and f flat. The case of a reduced Y with f not flat is *not proved*. Ex. 3.13 and 3.14 rest on Thm. 3.20 and Lemma 3.14 of the text; in Ex. 3.14(e) the phrase "X is a complete intersection in some $\mathbf{P}^d_Y$" is read as "$X = V_+(F_1, \dots, F_c)$ and all fibres have dimension $d - c$". Ex. 3.4 and 3.15 are proved completely.

Exercise 3.1

Problem. Let k be a field, and let X be the open subvariety $D(x) \cup D(y)$ of $\operatorname{Spec} k[x, y]$. We want to show that $H^1(X, \mathcal{O}_X)$ is of infinite dimension over k.

(a) Show that the $x^i y^j$, $i, j \in \mathbf{Z}$, form a basis of $k[x, 1/x, y, 1/y]$ as a k-vector space.

(b) Show that $k[x,1/x,y]+k[x,y,1/y]$ is generated by the vectors x^iy^j with $\max\{i,j\}\geq 0$, and that

$$H^1(X,\mathcal{O}_X)\simeq k[x,1/x,y,1/y]/\big(k[x,1/x,y]+k[x,y,1/y]\big)$$

has infinite dimension over k.

Solution. (a) The ring $R=k[x,1/x,y,1/y]=k[x,y]_{xy}$ is spanned by the elements $f/(xy)^n$ with $f\in k[x,y]$, that is by the monomials $x^iy^j=x^{i+n}y^{j+n}/(xy)^n$ with $i,j\in\mathbf{Z}$. Suppose a finite combination $\sum c_{ij}x^iy^j$ vanishes in R. Choose n with $i,j\geq -n$ for all (i,j) occurring. Multiplying by $(xy)^n$ we get $\sum c_{ij}x^{i+n}y^{j+n}=0$ in R, hence in $k[x,y]$, because $k[x,y]\to R$ is injective ($k[x,y]$ is a domain). Distinct monomials of $k[x,y]$ are linearly independent, so all $c_{ij}=0$.

(b) We have $\mathcal{O}(D(x))=k[x,1/x,y]$, which by (a) is the span of the x^iy^j with $i\in\mathbf{Z}$, $j\geq 0$ (it is the localization of $k[x,y]$ at x), and $\mathcal{O}(D(y))=k[x,y,1/y]$ is the span of the x^iy^j with $i\geq 0$, $j\in\mathbf{Z}$. The sum of two subspaces spanned by subsets of a basis is spanned by the union of these subsets: the monomials with $j\geq 0$ or $i\geq 0$, that is with $\max\{i,j\}\geq 0$.

The scheme X is separated (open in an affine scheme) and $\mathcal{U}=\{D(x),D(y)\}$ is an affine covering with $D(x)\cap D(y)=D(xy)=\operatorname{Spec}R$. By Thm. 2.19 and Cor. 2.4 (order $D(x)<D(y)$), $H^1(X,\mathcal{O}_X)=H^1(\mathcal{U},\mathcal{O}_X)$ is the cohomology in degree 1 of the complex $C^0\to C^1\to C^2=0$ with $C^0=k[x,1/x,y]\oplus k[x,y,1/y]$, $C^1=R$, $d(a,b)=b-a$. Hence

$$H^1(X,\mathcal{O}_X)=R/\big(k[x,1/x,y]+k[x,y,1/y]\big).$$

By (a) and the description of the sum, the classes of the monomials x^iy^j with $i\leq -1$ and $j\leq -1$ form a basis of this quotient (the sum is spanned by a subset of a basis, the quotient has as basis the remaining basis vectors). There are infinitely many such monomials, so $\dim_k H^1(X,\mathcal{O}_X)=\infty$. □

Exercise 3.2

Problem. We are going to show Lemma 3.1(c). Let A be a ring, and let $X=\operatorname{Proj}A[T_0,\dots,T_d]$.

(a) Let $\mathcal{U}=\{U_i\}_{0\leq i\leq d}$ denote the covering of X with $U_i=D_+(T_i)$. Let $q\in\mathbf{Z}$. Show that $\mathcal{O}_X(q)(U_{01\dots d})$ admits as a basis over A the elements of the form D/N, where D,N are monomials without common variable and such that $\deg D=\deg N+q$. Show that if a T_i does not divide N, then D/N belongs to $\mathcal{O}_X(q)(U_{01\dots\hat{\imath}\dots d})|_{U_{01\dots d}}$.

(b) Show that the images under $\mathcal{O}_X(q)(U_{01\dots d})\to H^d(X,\mathcal{O}_X(q))$ of the elements of the form $T_0^{r_0}\cdots T_d^{r_d}$ with $r_i\leq -1$ and $\sum_i r_i=q$ form a basis of $H^d(X,\mathcal{O}_X(q))$ over A. In particular, the module $H^d(X,\mathcal{O}_X(-d-1))$ is free of rank 1 over A, with $e:=[(T_0\cdots T_d)^{-1}]$ as a basis.

(c) Let $n\geq 0$. Show that the cup product (Ex. 2.17)

$$\smile:\ H^0(X,\mathcal{O}_X(n))\times H^d(X,\mathcal{O}_X(-d-1-n))\to H^d(X,\mathcal{O}_X(-d-1))$$

induces a non-degenerate bilinear map (let M,N be monomials of degree n, show that $[M]\smile[(T_0\cdots T_dN)^{-1}]=e$ if $M=N$ and 0 otherwise).

Solution. Hypothesis $d \geq 1$**.** The proof (and Lemma 3.1(c)) needs $d \geq 1$: for $d = 0$ we have $X = \operatorname{Spec} A$, and $H^0(X, \mathcal{O}_X) = A$ while $H^0(X, \mathcal{O}_X(-1))^\vee = 0$, so the formula of Lemma 3.1(c) fails. From now on $d \geq 1$.

Set-up. Put $B = A[T_0, \dots, T_d]$ and $B^\pm = B[T_0^{-1}, \dots, T_d^{-1}]$. This is a free A-module with basis the Laurent monomials $T^r = T_0^{r_0} \cdots T_d^{r_d}$, $r \in \mathbf{Z}^{d+1}$ (the argument of Ex. 3.1(a) applies), graded by $|r| = \sum r_i$. For $I \subseteq \{0, \dots, d\}$ let $U_I = \bigcap_{i \in I} U_i = D_+(T_I)$, $T_I = \prod_{i \in I} T_i$. Then $\mathcal{O}_X(q)(U_I) = (B_{T_I})_q$, the degree q part of B_{T_I}, and all these modules and the restriction maps are inclusions of A-submodules of $B^\pm$ (as in the proof of Lemma 1.22). For $I = \{0, \dots, d\} \setminus \{i\}$ write $\hat{i}$ for the index set.

(a) The module $\mathcal{O}_X(q)(U_{01\dots d}) = (B^\pm)_q$ is free with basis the T^r with $|r| = q$. Write $r = \rho - \nu$ with $\rho_i = \max(r_i, 0)$ and $\nu_i = \max(-r_i, 0)$. Then $T^r = D/N$ with $D = T^\rho$, $N = T^\nu$ monomials without common variable and $\deg D - \deg N = |r| = q$. Conversely a pair (D, N) of monomials without common variable determines $r = \rho - \nu$ uniquely. So the D/N of the statement are exactly the T^r, $|r| = q$, and form a basis. A variable T_i does not divide N iff $\nu_i = 0$ iff $r_i \geq 0$. The module $(B_{T_{\hat{i}}})_q$ is spanned by the T^r with $|r| = q$ and $r_i \geq 0$ (the other exponents are arbitrary). So if $T_i \nmid N$, then $D/N = T^r \in \mathcal{O}_X(q)(U_{\hat{i}})$ restricted to $U_{01\dots d}$. Conversely every element of $\mathcal{O}_X(q)(U_{\hat{i}})$ is a combination of such T^r with $r_i \geq 0$.

(b) By Thm. 2.19 and Cor. 2.4, $H^d(X, \mathcal{O}_X(q))$ is the cokernel of $d : C^{d-1} \to C^d$ in the complex of ordered cochains for $0 < 1 < \cdots < d$ (here $d \geq 1$ and $C^{d+1} = 0$). We have $C^d = \mathcal{O}_X(q)(U_{01\dots d})$, $C^{d-1} = \bigoplus_{i=0}^d \mathcal{O}_X(q)(U_{\hat{i}})$ and $d(g) = \sum_i (-1)^i g_{\hat{i}}|_{U_{01\dots d}}$. By (a) the image of d is the span of the T^r ($|r| = q$) having some $r_i \geq 0$ (each T^r with $r_i \geq 0$ is the image of $\pm T^r$ placed in the component $\hat{i}$). Hence the cokernel is the quotient of the free module with basis $\{T^r\}_{|r|=q}$ by the span of a subset of this basis, so it is free with basis the classes of the T^r with all $r_i \leq -1$ and $|r| = q$. For $q = -d-1$ the only such r is $(-1, \dots, -1)$, so $H^d(X, \mathcal{O}_X(-d-1))$ is free of rank 1 with basis e. (If $q > -d-1$ there is no such r, so $H^d(X, \mathcal{O}_X(q)) = 0$, which is the last assertion of Lemma 3.1(c).)

(c) *Cup product with a global section.* Let $s \in H^0(X, \mathcal{F})$ and $\xi = [g] \in H^d(\mathcal{U}, \mathcal{G})$ with $g \in C^d(\mathcal{U}, \mathcal{G})$ a cocycle. In the definition of Ex. 2.17 take both coverings equal to $\mathcal{U}$ and $f = (s|_{U_i})_i \in C^0(\mathcal{U}, \mathcal{F})$. The covering $\mathcal{W} = \{U_i \cap U_j\}_{(i,j)}$ refines $\mathcal{U}$ through $\rho : (i,j) \mapsto j$, and $(f \smile g)_\alpha = s|_{W_\alpha} \otimes g_{j_0 \dots j_d}|_{W_\alpha}$ for $\alpha = ((i_0, j_0), \dots, (i_d, j_d))$. This is $\rho^*(sg)$ where $(sg)_{j_0 \dots j_d} = s| \otimes g_{j_0 \dots j_d}$ is the image of g under the sheaf homomorphism $\mathcal{G} \to \mathcal{F} \otimes \mathcal{G}$, $u \mapsto s \otimes u$. Since refinement maps induce the canonical maps into $H^d(X, -)$, we get

$$s \smile [g] = [s\, g] \in H^d(X, \mathcal{F} \otimes \mathcal{G}).$$

Take $\mathcal{F} = \mathcal{O}_X(n)$, $\mathcal{G} = \mathcal{O}_X(-d-1-n)$ and identify $\mathcal{F} \otimes \mathcal{G} = \mathcal{O}_X(-d-1)$ canonically. Then $s\, g$ is the product of Laurent polynomials.

Let M, N be monomials of degree n with exponents $\mu, \nu \in \mathbf{N}^{d+1}$, $|\mu| = |\nu| = n$. By Lemma 1.22, the monomials M form a basis of $H^0(X, \mathcal{O}_X(n)) = B_n$. By (b) with $q = -d-1-n$, the classes $[(T_0 \cdots T_d N)^{-1}] = [T^{-\nu-\mathbf{1}}]$ form a basis of $H^d(X, \mathcal{O}_X(-d-1-n))$ (the exponents $r = -\nu - \mathbf{1}$ are exactly the r with $r_i \leq -1$ and $|r| = -d-1-n$). Now

$$M \cdot (T_0 \cdots T_d N)^{-1} = T^{\mu-\nu-\mathbf{1}}.$$

If $M = N$ this is $(T_0 \cdots T_d)^{-1}$ and the class is e. If $M \neq N$, then $\mu \neq \nu$ and $|\mu| = |\nu|$, so there is i with $\mu_i > \nu_i$; then the exponent $\mu_i - \nu_i - 1 \geq 0$, and by (b) the class of $T^{\mu-\nu-\mathbf{1}}$ is

0. Thus, after identifying $H^d(X, \mathcal{O}_X(-d-1))$ with A through the basis e, the pairing has the monomials M and the classes $[(T_0 \cdots T_d N)^{-1}]$ as dual bases: the matrix of the pairing is the identity. Both modules are free of the same finite rank, and the induced maps $H^d(X, \mathcal{O}_X(-d-1-n)) \to \mathrm{Hom}_A(H^0(X, \mathcal{O}_X(n)), A)$ and $H^0(X, \mathcal{O}_X(n)) \to \mathrm{Hom}_A(H^d(X, \mathcal{O}_X(-d-1-n)), A)$ are isomorphisms: the pairing is non-degenerate (perfect). With $m = -d-1-n$ this gives Lemma 3.1(c): $H^d(X, \mathcal{O}_X(m)) \simeq H^0(X, \mathcal{O}_X(-m-d-1))^\vee$ for $m \leq -d-1$, while for $m > -d-1$ both sides are 0 by (b) and Lemma 1.22. □

Exercise 3.3

Problem. Let $P = \mathbf{P}_k^d$ be a projective space over a field k. Let X be a closed subvariety of P of dimension r. We say that X is a complete intersection if it is defined (as a variety) by $d-r$ homogeneous polynomials $F_1, \ldots, F_{d-r}$. This notion depends on the embedding $X \to P$.

(a) Let $Z = V_+(F_1, \ldots, F_{d-r-1})$ ($Z = P$ if $d - r = 1$). Show that Z is a complete intersection. Let $\mathcal{J}$ be the sheaf of ideals of $\mathcal{O}_Z$ that defines the subscheme X of Z. Show that $\mathcal{J} \simeq \mathcal{O}_Z(-\deg F_{d-r})$.

(b) Let $q \in \mathbf{Z}$. Show the following properties by induction on $d-r$: (1) $H^p(X, \mathcal{O}_X(q)) = 0$ if $1 \leq p \leq r-1$. (2) $H^0(P, \mathcal{O}_P(q)) \to H^0(X, \mathcal{O}_X(q))$ is surjective if $r \geq 1$. (3) $\dim_k H^p(X, \mathcal{O}_X(q))$ is an integer that only depends on p, q, d, and the $\deg F_i$.

(c) Show that X is geometrically connected if $r \geq 1$.

(d) Let $d = 2$, and let $X = V_+(F)$ with $\deg F = n$. Show that $\dim_k H^1(X, \mathcal{O}_X) = (n-1)(n-2)/2$.

Solution. Interpretation. "Defined as a variety by $F_1, \ldots, F_c$" ($c = d - r$) means that X is the closed subscheme $V_+(F_1, \ldots, F_c)$ of P (the scheme whose homogeneous ideal is generated by the F_i), and $\dim X = r$. We put $e_i = \deg F_i$, $Z_i = V_+(F_1, \ldots, F_i)$ ($Z_0 = P$, $Z_c = X$) and $S = k[T_0, \ldots, T_d]$. For a point $x \in P$ and $x \in D_+(T_l)$ let $f_i = F_i / T_l^{e_i} \in \mathcal{O}_{P,x}$; then $\mathcal{O}_{Z_i,x} = \mathcal{O}_{P,x}/(f_1, \ldots, f_i)$.

We use the following standard facts of dimension theory. (K1) (Krull) If $m \leq d$, every irreducible component of $V_+(F_1, \ldots, F_m) \subseteq \mathbf{P}_k^d$ is non-empty of dimension $\geq d - m$. (K2) If $C \subseteq \mathbf{P}_k^d$ is an integral closed subscheme of dimension $n \geq 1$ and F is a form of positive degree not vanishing identically on C, then $C \cap V_+(F)$ is non-empty and all its irreducible components have dimension $n-1$. (K3) For an integral scheme C of finite type over a field and $x \in C$, $\dim \mathcal{O}_{C,x} = \dim C - \dim \overline{\{x\}}$. (K4) A regular local ring is Cohen–Macaulay; hence in it, if an ideal of height i is generated by i elements, these elements form a regular sequence (Matsumura, Thm. 17.4), and so does every initial segment of them.

Lemma A. Let $F_1, \ldots, F_c \in S$ be homogeneous, $c \leq d$, with $\dim V_+(F_1, \ldots, F_c) = d - c$. Then for every $0 \leq i \leq c$, every irreducible component of Z_i has dimension exactly $d-i$, and for every $x \in Z_i$ the elements $f_1, \ldots, f_i$ form a regular sequence in $\mathcal{O}_{P,x}$. Consequently, for $1 \leq i \leq c$ the multiplication by F_i,

$$0 \to \mathcal{O}_{Z_{i-1}}(-e_i) \xrightarrow{F_i} \mathcal{O}_{Z_{i-1}} \to \mathcal{O}_{Z_i} \to 0,$$

is an exact sequence of sheaves on Z_{i-1}.

Proof. (The forms F_i have positive degree: a non-zero constant would give $X = \emptyset$, which is excluded by $\dim X = r \geq 0$, and the zero form is excluded by the argument below, since it vanishes on every C.) We show by descending induction on i that $\dim Z_i = d - i$ and that Z_i is pure of dimension $d - i$. For $i = c$ the dimension is the hypothesis. Suppose $\dim Z_i = d - i$ for some $i \geq 1$. Let C be an irreducible component of Z_{i-1} (with reduced structure). By (K1) $\dim C \geq d - i + 1 \geq 1$. If F_i vanished identically on C, then $C \subseteq Z_i$, contradicting $\dim Z_i = d - i < \dim C$. So by (K2) the components of $C \cap V_+(F_i)$ have dimension $\dim C - 1$. As $Z_i = \bigcup_C (C \cap V_+(F_i))$, we get $d - i = \dim Z_i = \max_C \dim C - 1$. Hence $\dim C = d - i + 1$ for every component C of Z_{i-1} (all are $\geq d - i + 1$ by (K1)), so Z_{i-1} is pure of dimension $d - (i - 1)$, and the same argument shows that Z_i (whose components are among those of the $C \cap V_+(F_i)$) is pure of dimension $d - i$. This proves the first claim.

Now let $x \in Z_i$ and $R = \mathcal{O}_{P,x}$, a regular local ring of dimension $d - \dim \overline{\{x\}}$. The minimal primes of $R/(f_1, \ldots, f_i) = \mathcal{O}_{Z_i,x}$ correspond to the components C of Z_i through x, and by (K3) $\dim \mathcal{O}_{C,x} = d - i - \dim \overline{\{x\}} = \dim R - i$. So $\dim R/(f_1, \ldots, f_i) = \dim R - i$, i.e. $(f_1, \ldots, f_i)$ has height i (R is Cohen–Macaulay). By (K4), $f_1, \ldots, f_i$ is a regular sequence.

For the exact sequence, check stalks at $x \in Z_{i-1}$. If $x \in Z_i$, f_i is a non-zero-divisor on $\mathcal{O}_{P,x}/(f_1, \ldots, f_{i-1}) = \mathcal{O}_{Z_{i-1},x}$ by what we have shown, so multiplication by F_i is injective. If $x \notin Z_i$, then $f_i(x) \neq 0$, f_i is a unit and the multiplication is bijective and $\mathcal{O}_{Z_i,x} = 0$. The cokernel is $\mathcal{O}_{Z_{i-1}}/F_i\mathcal{O}_{Z_{i-1}}(-e_i) = \mathcal{O}_{Z_i}$. □

(a) Let $c = d - r$. The hypotheses of Lemma A hold for $F_1, \ldots, F_c$ ($c \leq d$, $\dim X = r = d - c$). Put $Z = Z_{c-1}$. By Lemma A, $\dim Z = d - (c - 1) = r + 1$, and $Z = V_+(F_1, \ldots, F_{c-1})$ is defined by $d - (r + 1)$ forms, so Z is a complete intersection of dimension $r + 1$ (for $c = 1$, $Z = P$ of dimension $d = r + 1$). The ideal sheaf of X in Z is the image $\mathcal{J}$ of $F_c : \mathcal{O}_Z(-e_c) \to \mathcal{O}_Z$, which is injective by Lemma A. So $\mathcal{J} \simeq \mathcal{O}_Z(-e_c)$, $e_c = \deg F_{d-r}$, and

$$0 \to \mathcal{O}_Z(-e_c) \to \mathcal{O}_Z \to \mathcal{O}_X \to 0 \tag{1}$$

is exact (as a sequence of sheaves on Z, $\mathcal{O}_X$ meaning $i_*\mathcal{O}_X$ for $i : X \to Z$).

(b) We prove (1), (2), (3) by induction on $c = d - r \geq 0$, for all d, k, q and all complete intersections. If $c = 0$, then $X = P$ and (1) holds by Lemma 3.1(b) (there $r = d$, so p ranges over $1 \leq p \leq d - 1$), (2) is the identity, (3) holds by Lemma 3.1 because $\dim_k H^p(P, \mathcal{O}(q))$ is 0 for $p \neq 0, d$ and equals $\dim B_q$, resp. $\dim B_{-q-d-1}$, for $p = 0$, resp. $p = d$.

Let $c \geq 1$, and let Z, $e = e_c$ be as in (a); Z is a complete intersection of dimension $r + 1$ defined by $c - 1$ forms, so the properties hold for Z (all q): (1_Z) $H^p(Z, \mathcal{O}_Z(m)) = 0$ for $1 \leq p \leq r$; (2_Z) $H^0(P, \mathcal{O}_P(m)) \to H^0(Z, \mathcal{O}_Z(m))$ is surjective ($r + 1 \geq 1$); (3_Z) $h^p(Z, \mathcal{O}_Z(m)) := \dim_k H^p(Z, \mathcal{O}_Z(m))$ depends only on $p, m, d, e_1, \ldots, e_{c-1}$. Twisting (1) by the invertible, hence flat, sheaf $\mathcal{O}(q)$ gives the exact sequence $0 \to \mathcal{O}_Z(q - e) \to \mathcal{O}_Z(q) \to \mathcal{O}_X(q) \to 0$ and, by Cor. 2.22 and Ex. 2.3(b), the long exact sequence

$$\cdots \to H^p(Z, \mathcal{O}_Z(q-e)) \to H^p(Z, \mathcal{O}_Z(q)) \to H^p(X, \mathcal{O}_X(q)) \to H^{p+1}(Z, \mathcal{O}_Z(q-e)) \to H^{p+1}(Z, \mathcal{O}_Z(q)) \to \cdots \tag{2}$$

(Z is projective, hence separated). Moreover $H^p(X, -) = 0$ for $p > r$ by Prop. 2.24.

(1) Let $1 \leq p \leq r - 1$. In (2), $H^p(Z, \mathcal{O}_Z(q)) = 0$ and $H^{p+1}(Z, \mathcal{O}_Z(q - e)) = 0$ by (1_Z) because $1 \leq p < p + 1 \leq r$. So $H^p(X, \mathcal{O}_X(q)) = 0$.

(2) Let $r \geq 1$. In (2) with $p = 0$, $H^1(Z, \mathcal{O}_Z(q - e)) = 0$ by (1_Z) $(1 \leq r)$, so $H^0(Z, \mathcal{O}_Z(q)) \to$

$H^0(X, \mathcal{O}_X(q))$ is surjective. The map $H^0(P, \mathcal{O}_P(q)) \to H^0(X, \mathcal{O}_X(q))$ is the composition of this map with $H^0(P, \mathcal{O}_P(q)) \to H^0(Z, \mathcal{O}_Z(q))$, which is surjective by (2_Z). So (2) holds.

(3) By (1) and the vanishing for $p > r$ it remains to treat $p = 0$ and $p = r$. If $r \geq 1$: for $p = 0$ (2) gives $0 \to H^0(Z, \mathcal{O}_Z(q-e)) \to H^0(Z, \mathcal{O}_Z(q)) \to H^0(X, \mathcal{O}_X(q)) \to H^1(Z, \mathcal{O}_Z(q-e)) = 0$ (the first zero because multiplication by F_c is injective on sections), so $h^0(X, \mathcal{O}_X(q)) = h^0(Z, q) - h^0(Z, q-e)$. For $p = r \geq 1$: since $H^r(Z, \mathcal{O}_Z(q)) = 0$ (by (1_Z)) and $H^{r+1}(X, \mathcal{O}_X(q)) = 0$, (2) gives $0 \to H^r(X, \mathcal{O}_X(q)) \to H^{r+1}(Z, \mathcal{O}_Z(q-e)) \to H^{r+1}(Z, \mathcal{O}_Z(q)) \to 0$, so $h^r(X, q) = h^{r+1}(Z, q-e) - h^{r+1}(Z, q)$. If $r = 0$: $p = 0 = r$ and (2) gives the exact sequence $0 \to H^0(Z, q-e) \to H^0(Z, q) \to H^0(X, q) \to H^1(Z, q-e) \to H^1(Z, q) \to H^1(X, q) = 0$, so $h^0(X, q) = h^0(Z, q) - h^0(Z, q-e) + h^1(Z, q-e) - h^1(Z, q)$. In all cases $\dim_k H^p(X, \mathcal{O}_X(q))$ is an integer combination of the numbers $h^*(Z, q)$, $h^*(Z, q-e)$, which by (3_Z) depend only on the indicated data. This proves (3).

(c) Let $r \geq 1$ and let k'/k be any field extension. The scheme $X_{k'} = V_+(F_1, \ldots, F_c) \subseteq \mathbf{P}^d_{k'}$ is non-empty, has dimension r (dimension does not change under extension of the base field) and is defined by $d-r$ forms, so it is a complete intersection of dimension r over k', and the whole of (b) applies to it. By (b)(2), $k' = H^0(\mathbf{P}^d_{k'}, \mathcal{O}) \to H^0(X_{k'}, \mathcal{O}_{X_{k'}})$ is surjective; it is also injective (it is the structure map, $X_{k'} \neq \emptyset$), so $H^0(X_{k'}, \mathcal{O}_{X_{k'}}) = k'$. If $X_{k'} = U_1 \sqcup U_2$ with U_i open and closed and non-empty, the characteristic functions give a k'-subalgebra $k' \times k' \subseteq H^0(X_{k'}, \mathcal{O})$ of dimension 2, a contradiction. So $X_{k'}$ is connected for every k', in particular for $k' = \bar{k}$: X is geometrically connected.

(d) Here $d = 2$, $c = 1$, $r = 1$ and $X = V_+(F)$, $\deg F = n \geq 1$. We apply sequence (1) with $Z = P = \mathbf{P}^2_k$ and cohomology: $H^1(P, \mathcal{O}_P) \to H^1(X, \mathcal{O}_X) \to H^2(P, \mathcal{O}_P(-n)) \to H^2(P, \mathcal{O}_P)$. By Lemma 3.1 (with $d = 2$), $H^1(P, \mathcal{O}_P) = 0$ and $H^2(P, \mathcal{O}_P) = 0$ (as $0 \geq -d$). Therefore $H^1(X, \mathcal{O}_X) \simeq H^2(P, \mathcal{O}_P(-n))$, and by Lemma 3.1(c) (applied with Ex. 3.2),

$$\dim_k H^1(X, \mathcal{O}_X) = \dim_k H^0(P, \mathcal{O}_P(n-3)) = \dim_k B_{n-3} = \binom{n-1}{2} = \frac{(n-1)(n-2)}{2},$$

where $B = k[T_0, T_1, T_2]$; for $n = 1, 2$ the module B_{n-3} is 0 and the binomial coefficient $\binom{n-1}{2}$ is 0 as well. □

Exercise 3.4

> **Problem.** Let Y be a Noetherian local scheme with closed point y_0, and $X = V_+(F_1, \ldots, F_{d-r})$ be a closed subscheme of $\mathbf{P}^d_Y$. Let us suppose that X_{y_0} is a complete intersection in $\mathbf{P}^d_{k(y_0)}$, of dimension r. Show that for any $y \in Y$, X_y is a complete intersection in $\mathbf{P}^d_{k(y)}$, of dimension r (use induction on $d-r$). Likewise, show that X is flat over Y (use Lemma 4.3.16).

Solution. Let $Y = \operatorname{Spec} A$ with A Noetherian local, $\mathfrak{m}$ its maximal ideal, $k_0 = k(y_0) = A/\mathfrak{m}$. Here $F_j \in A[T_0, \ldots, T_d]$ are homogeneous of degree e_j, $c = d-r$, and $Z_i = V_+(F_1, \ldots, F_i) \subseteq \mathbf{P}^d_Y$ ($Z_0 = \mathbf{P}^d_Y$, $Z_c = X$). For $y \in Y$ the fibre $(Z_i)_y$ is $V_+(F_1(y), \ldots, F_i(y)) \subseteq \mathbf{P}^d_{k(y)}$, where $F_j(y)$ is the image of F_j in $k(y)[T]$. In particular $X_y = V_+(F_1(y), \ldots, F_c(y))$, so to be a complete intersection of dimension r it only remains to see that $\dim X_y = r$. By (K1) of Ex. 3.3 every non-empty V_+ of $m \leq d$ forms in $\mathbf{P}^d_{k(y)}$ has all components of dimension $\geq d - m$; so $(Z_i)_y \neq \emptyset$ and $\dim (Z_i)_y \geq d - i$, and what must be shown is $\dim (Z_i)_y \leq d - i$.

We prove by induction on $i = 0, \ldots, c$ the statement

$S(i)$: Z_i is flat over Y, and $\dim(Z_i)_y \leq d - i$ for every $y \in Y$.

$S(0)$ is clear. Assume $S(i-1)$, $i \geq 1$. The hypothesis says that $\dim V_+(F_1(y_0), \ldots, F_c(y_0)) = d - c$; so Lemma A of Ex. 3.3 applies over the field k_0 to the forms $F_j(y_0)$ and shows: for every $x \in (Z_i)_{y_0}$ the element f_i (local equation of F_i) is a non-zero-divisor on $\mathcal{O}_{(Z_{i-1})_{y_0},x}$.

Flatness at the points over y_0. Let $x \in Z_i$ lie over y_0. Put $B = \mathcal{O}_{Z_{i-1},x}$. It is a Noetherian local ring, flat over A by $S(i-1)$, and $A \to B$ is a local homomorphism. We have $B/\mathfrak{m}B = \mathcal{O}_{(Z_{i-1})_{y_0},x}$, and the image of f_i in it is a non-zero-divisor. By Lemma 4.3.16, $B/f_iB = \mathcal{O}_{Z_i,x}$ is flat over A. Moreover f_i is a non-zero-divisor on B (this is Ex. 4.3.14; here is the argument). Put $b = f_i$ and $J = \mathrm{Ann}_B(b)$. The module $bB = \mathrm{Ker}(B \to B/bB)$ is flat, since B and B/bB are ($\mathrm{Tor}_j^A(bB, N) = \mathrm{Tor}_{j+1}^A(B/bB, N) = 0$ for $j \geq 1$). So both sequences $0 \to J \to B \to bB \to 0$ and $0 \to bB \to B \to B/bB \to 0$ remain exact after $\otimes_A A/\mathfrak{m}$. The composite $B/\mathfrak{m}B \to bB \otimes A/\mathfrak{m} \hookrightarrow B/\mathfrak{m}B$ is multiplication by b, which is injective; hence the first map is injective, so $J \otimes A/\mathfrak{m} = J/\mathfrak{m}J = 0$, and $J = 0$ by Nakayama's lemma.

Flatness everywhere. Let $x' \in Z_i$ be arbitrary and y' its image. The closure T of $\{x'\}$ in Z_i is closed, and $Z_i \to Y$ is projective, hence closed (proper). So $f(T)$ is a closed set containing y', hence contains the closure of y', which contains the closed point y_0 (Y is local). Thus there is $x \in T$ over y_0, and x' is a generization of x. So $\mathcal{O}_{Z_i,x'}$ is a localization of $\mathcal{O}_{Z_i,x}$, which is flat over A, hence $\mathcal{O}_{Z_i,x'}$ is flat over A. As $\mathcal{O}_{Z_i,x'}$ is a module over the localization $A_{\mathfrak{p}'} = \mathcal{O}_{Y,y'}$ of A, it is flat over $\mathcal{O}_{Y,y'}$ (for an $A_{\mathfrak{p}'}$-module N, $M \otimes_{A_{\mathfrak{p}'}} N = M \otimes_A N$). So Z_i is flat over Y.

Dimension of the fibres. Let $y \in Y$ and let x' be a closed point of $(Z_i)_y$. Put $B' = \mathcal{O}_{Z_{i-1},x'}$, flat over A and $B'/f_iB' = \mathcal{O}_{Z_i,x'}$ flat over A by what we just proved. Tensoring $0 \to B' \xrightarrow{f_i} B' \to B'/f_iB' \to 0$ (exact since f_i is a non-zero-divisor on B': with $x \in Z_i$ over y_0 such that x' is a generization of x, as in the previous step, B' is a localization of $\mathcal{O}_{Z_{i-1},x}$ on which f_i is regular) with $k(y)$ over A, we obtain an exact sequence, because $\mathrm{Tor}_1^A(B'/f_iB', k(y)) = 0$:

$$0 \to B' \otimes_A k(y) \xrightarrow{f_i} B' \otimes_A k(y) \to (B'/f_iB') \otimes_A k(y) \to 0.$$

Here $B' \otimes_A k(y) = \mathcal{O}_{(Z_{i-1})_y,x'}$. So f_i is a non-zero-divisor in the Noetherian local ring $\mathcal{O}_{(Z_{i-1})_y,x'}$, and therefore $\dim \mathcal{O}_{(Z_i)_y,x'} = \dim \mathcal{O}_{(Z_{i-1})_y,x'} - 1 \leq \dim(Z_{i-1})_y - 1 \leq d - i$ by $S(i-1)$. A scheme of finite type over a field has dimension equal to the supremum of the dimensions of its local rings at closed points, so $\dim(Z_i)_y \leq d - i$. This proves $S(i)$.

For $i = c$, $S(c)$ says that X is flat over Y and $\dim X_y \leq r$; with $\dim X_y \geq r$ we get $\dim X_y = r$ for all $y \in Y$. Hence every X_y is a complete intersection of dimension r in $\mathbf{P}^d_{k(y)}$. □

Exercise 3.5

Problem. Let Y be a locally Noetherian scheme. Let Z be a closed subscheme of $\mathbf{P}^d_Y$, flat over Y. Let $y_0 \in Y$ be a point such that Z_{y_0} is a complete intersection in $\mathbf{P}^d_{k(y_0)}$. Let us set $r = \dim Z_{y_0}$.

(a) Let us suppose Y is affine. Let us write $Z = V_+(I)$. Show that there exist $F_1, \ldots, F_{d-r} \in I$ such that if we set $X = V_+(F_1, \ldots, F_{d-r})$, then $X_{y_0} = Z_{y_0}$ as a subscheme of $\mathbf{P}^d_{k(y_0)}$.

(b) Show that there exists an open neighborhood V of y_0 such that $Z \times_Y V = X \times_Y V$ (Ex. 4.3.8). Show that for V small enough, Z_y is a complete intersection in $\mathbf{P}^d_{k(y)}$ of dimension r for every $y \in V$.

(c) Let us suppose that Y is connected and that Z_y is a complete intersection in $\mathbf{P}^d_{k(y)}$ for all $y \in Y$. Let $p \geq 0$, $q \in \mathbf{Z}$. Show that the function $y \mapsto \dim_{k(y)} H^p(Z_y, \mathcal{O}_{Z_y}(q))$ is constant on Y.

Solution. Summary of what is true. (a) and the second assertion of (b) are *false as printed* when $r = 0$, and in (a) the ideal I must be the saturated ideal of Z. We prove (a), (b) for $r \geq 1$ (with $I = \bigoplus_n H^0(\mathbf{P}^d_A, \mathcal{I}_Z(n))$), and (c) as printed.

Counterexample for $r = 0$. Let $A = k[t]_{(t)}$ (a discrete valuation ring), $Y = \operatorname{Spec} A$, $d = 2$, and let $Z \subset \mathbf{P}^2_Y$ be the union of the three sections $P_1 = (1 : 0 : 0)$, $P_2 = (0 : 1 : 0)$, $P_3 = (1 : 1 : t)$; $Z \simeq \bigsqcup_3 \operatorname{Spec} A$ is flat over Y. Their reductions modulo t are three distinct points of the line $z = 0$, so $Z_{y_0} = V_+(z, xy(x - y))$ is a complete intersection with $r = 0$, $d - r = 2$. If $F_1, F_2 \in I$ (any homogeneous ideal I with $Z = V_+(I)$ is contained in the saturated ideal) satisfied $V_+(F_1, F_2)_{y_0} = Z_{y_0}$, then $V_+(F_1(y_0), F_2(y_0))$ is a zero-dimensional complete intersection of length 3 in $\mathbf{P}^2_k$, so by Bezout $\{\deg F_1, \deg F_2\} = \{1, 3\}$. A linear form $ax + by + cz$ in the saturated ideal vanishes on the three sections: $a = 0$ (at P_1), $b = 0$ (at P_2), $ct = 0$ (at P_3), so $c = 0$ as A is a domain, and the form is 0; but $V_+(0, F_2)$ is a curve. So (a) fails. For (b): the generic fibre Z_η consists of the three points $(1 : 0 : 0), (0 : 1 : 0), (1 : 1 : t)$ of $\mathbf{P}^2_K$, $K = k(t)$; these are not collinear, while a zero-dimensional complete intersection of length 3 in $\mathbf{P}^2_K$ has degrees $(1, 3)$ and lies on a line. So Z_η is not a complete intersection and the second assertion of (b) fails for every neighborhood V of y_0.

Preliminaries. (i) *The ideal sheaf commutes with base change.* Let $\mathcal{I}$ be the ideal sheaf of Z in $\mathcal{O}_{\mathbf{P}^d_Y}$. As $\mathcal{O}_{\mathbf{P}^d_Y}$ and $\mathcal{O}_Z$ are flat over Y, so is $\mathcal{I}$ (the kernel of a surjection of flat modules with flat target is flat), and for every $y \in Y$ the sequence $0 \to \mathcal{I} \otimes k(y) \to \mathcal{O}_{\mathbf{P}^d_{k(y)}} \to \mathcal{O}_{Z_y} \to 0$ is exact because $\operatorname{Tor}_1(\mathcal{O}_Z, k(y)) = 0$. So $\mathcal{I}|_{\mathbf{P}^d_{k(y)}}$ is the ideal sheaf $\mathcal{I}_y$ of Z_y. (ii) By definition, $Z_{y_0} = V_+(G_1, \ldots, G_c)$ with $c = d - r$ and $G_j \in k[T_0, \ldots, T_d]$ forms of degree e_j, $k = k(y_0)$. The G_j are sections of $\mathcal{I}_{y_0}(e_j)$ on $\mathbf{P}^d_k$. (iii) *Fibre cohomology of $\mathcal{I}_{y_0}$ if $r \geq 1$ and $c \geq 1$.* Suppose $r \geq 1$ and $c = d - r \geq 1$ (so $d \geq 2$) and let $X_0 = Z_{y_0}$. By Ex. 3.3(b)(2), $H^0(\mathbf{P}^d_k, \mathcal{O}(q)) \to H^0(X_0, \mathcal{O}_{X_0}(q))$ is surjective for all q. The exact sequence $0 \to \mathcal{I}_{y_0}(q) \to \mathcal{O}(q) \to \mathcal{O}_{X_0}(q) \to 0$ and Lemma 3.1(b) ($H^1(\mathbf{P}^d_k, \mathcal{O}(q)) = 0$, since $d \geq 2$) give $H^1(\mathbf{P}^d_k, \mathcal{I}_{y_0}(q)) = 0$ for every q. (iv) *Dimension of fibres.* We use the upper semicontinuity of the dimension of the fibres of a proper morphism (Chevalley; EGA IV 13.1.5), which is not proved in the text: for $g : W \to Y$ proper, $\{y \mid \dim W_y \geq n\}$ is closed.

(a) Let $r \geq 1$. If $c = 0$ then $Z_{y_0} = \mathbf{P}^d_k$, the list $F_1, \ldots, F_c$ is empty, $X = \mathbf{P}^d_Y$, and (a) is vacuous; so let $c \geq 1$ in the rest of (a). Let $Y = \operatorname{Spec} A$, y_0 corresponds to the prime $\mathfrak{p}$, $R = A_\mathfrak{p}$ (local with residue field k), and $I = \bigoplus_{n \geq 0} I_n$, $I_n = H^0(\mathbf{P}^d_A, \mathcal{I}(n)) \subseteq H^0(\mathbf{P}^d_A, \mathcal{O}(n)) = A[T]_n$ (Lemma 1.22).

Claim. For $e \in \mathbf{Z}$ the map $H^0(\mathbf{P}^d_R, \mathcal{I}_R(e)) \to H^0(\mathbf{P}^d_k, \mathcal{I}_{y_0}(e))$ is surjective. Let $\mathcal{F} = \mathcal{I}_R(e)$; it is coherent, flat over R, and $\mathcal{F} \otimes_R k = \mathcal{I}_{y_0}(e)$ by (i). Let $f : \mathbf{P}^d_R \to \operatorname{Spec} R$. By (iii) $H^1(\mathbf{P}^d_k, \mathcal{F}_{y_0}) = 0$. By Rem. 3.21(a) the function $y \mapsto \dim H^1(\mathbf{P}^d_{k(y)}, \mathcal{F}_y)$ is upper semicontinuous on $\operatorname{Spec} R$; the set where it is ≥ 1 is closed and does not contain the closed point y_0, but a closed subset containing some point contains its closure, hence y_0; so this set is empty: $H^1(\mathbf{P}^d_{k(y)}, \mathcal{F}_y) = 0$

for all $y \in \operatorname{Spec} R$. Therefore the map (3.15) for $p = 1$ is surjective at every point, and by Rem. 3.21(c) (first part: bijective at y_0, so $H^1(\mathbf{P}^d_R, \mathcal{F}) \otimes k \simeq H^1(\mathbf{P}^d_k, \mathcal{F}_{y_0}) = 0$; this module is finitely generated by Thm. 3.2(a), hence $H^1(\mathbf{P}^d_R, \mathcal{F}) = 0$ by Nakayama; second part with $\mathcal{G} = \widetilde{\mathfrak{m}}$) we get $H^1(\mathbf{P}^d_R, \mathcal{F} \otimes_R \mathfrak{m}) \simeq H^1(\mathbf{P}^d_R, \mathcal{F}) \otimes_R \mathfrak{m} = 0$. Now $\mathcal{F}$ is flat, so $0 \to \mathcal{F} \otimes_R \mathfrak{m} \to \mathcal{F} \to \mathcal{F} \otimes_R k \to 0$ is exact, and Cor. 2.22 and Ex. 2.3(b) (closed immersion $\mathbf{P}^d_k \to \mathbf{P}^d_R$) give an exact sequence $H^0(\mathbf{P}^d_R, \mathcal{F}) \to H^0(\mathbf{P}^d_k, \mathcal{I}_{y_0}(e)) \to H^1(\mathbf{P}^d_R, \mathcal{F} \otimes \mathfrak{m}) = 0$. This proves the claim.

By the claim there are $F'_j \in H^0(\mathbf{P}^d_R, \mathcal{I}_R(e_j)) = I_{e_j} \otimes_A R$ (flat base change, Cor. 2.27) whose image in $k[T]$ is G_j. Write $F'_j = F_j/s_j$ with $F_j \in I_{e_j}$, $s_j \in A \setminus \mathfrak{p}$. The reduction of F_j in $k[T]$ is $\bar{s}_j G_j$ with $\bar{s}_j \in k^\times$. So $F_1, \dots, F_c \in I$ are forms of degrees e_j with $V_+(F_1(y_0), \dots, F_c(y_0)) = V_+(G_1, \dots, G_c)$, i.e. $X = V_+(F_1, \dots, F_c)$ satisfies $X_{y_0} = Z_{y_0}$ as subschemes of $\mathbf{P}^d_k$. Since $F_j \in I$, $Z \subseteq X$ is a closed subscheme.

(b) Take Y affine and F_j, X as in (a) (if $c = 0$ take $X = \mathbf{P}^d_Y$); Z is a closed subscheme of X with $Z_{y_0} = X_{y_0}$. Let $\mathcal{J} \subseteq \mathcal{O}_X$ be the (coherent) ideal sheaf of Z in X. Since $\mathcal{O}_Z$ is flat over Y, tensoring $0 \to \mathcal{J} \to \mathcal{O}_X \to \mathcal{O}_Z \to 0$ with $k(y_0)$ stays exact, and $\mathcal{O}_{X_{y_0}} \to \mathcal{O}_{Z_{y_0}}$ is an isomorphism, so $\mathcal{J} \otimes k(y_0) = 0$. For $x \in X_{y_0}$ this says $\mathcal{J}_x = \mathfrak{m}_{y_0}\mathcal{J}_x \subseteq \mathfrak{m}_x\mathcal{J}_x$, hence $\mathcal{J}_x = 0$ by Nakayama. So $\operatorname{Supp} \mathcal{J}$ is a closed subset of X disjoint from X_{y_0}. The morphism $X \to Y$ is projective, hence closed, so $V = Y \setminus f(\operatorname{Supp} \mathcal{J})$ is an open neighborhood of y_0, and $\mathcal{J} = 0$ over V, i.e. $Z \times_Y V = X \times_Y V$. (This is Ex. 4.3.8(b).)

Now let $y \in V$. Then $Z_y = X_y = V_+(F_1(y), \dots, F_c(y))$ is non-empty of dimension $\geq d - c = r$ by (K1) of Ex. 3.3. By (iv) applied to $X \to Y$ the set $\{y \mid \dim X_y \geq r + 1\}$ is closed and does not contain y_0; replacing V by the complement in V of this closed set, we get $\dim X_y = r$ for all $y \in V$ (also $Z \times_Y V = X \times_Y V$ still holds on the smaller V). So Z_y is a complete intersection of dimension r in $\mathbf{P}^d_{k(y)}$, defined by forms of the same degrees $e_1, \dots, e_c$ for all $y \in V$.

(c) Let $\varphi_{p,q}(y) = \dim_{k(y)} H^p(Z_y, \mathcal{O}_{Z_y}(q))$. As Y is connected, it suffices to show that $\varphi_{p,q}$ is locally constant. Let $y_0 \in Y$ and $r = \dim Z_{y_0}$; we may replace Y by an affine neighborhood of y_0.

Case $r \geq 1$. By (b) there is a neighborhood V of y_0 and forms $F_1, \dots, F_c$ of degrees $e_1, \dots, e_c$ such that $Z_y = V_+(F_1(y), \dots, F_c(y))$ is a complete intersection of dimension r for every $y \in V$. By Ex. 3.3(b)(3), $\varphi_{p,q}(y)$ depends only on $p, q, d, e_1, \dots, e_c$, hence is constant on V.

Case $r = 0$. By (iv) there is a neighborhood V of y_0 on which $\dim Z_y \leq 0$; the fibres Z_y are non-empty complete intersections, hence 0-dimensional. Then $Z_V \to V$ is proper and quasi-finite, hence finite, and flat, so $\mathcal{E} = f_*\mathcal{O}_{Z_V}$ is a coherent flat, hence locally free, $\mathcal{O}_V$-module of finite rank, and the rank at y is $\dim_{k(y)} H^0(Z_y, \mathcal{O}_{Z_y})$ (formation of f_* commutes with base change for an affine morphism); the rank is locally constant. For $p \geq 1$ we have $H^p(Z_y, -) = 0$ since Z_y is affine. For $p = 0$, $\mathcal{O}_{Z_y}(q) \simeq \mathcal{O}_{Z_y}$ because Z_y is the spectrum of an Artinian ring, so every invertible sheaf on it is trivial; so $\varphi_{0,q}(y) = \operatorname{rank} \mathcal{E}_y$, locally constant.

Thus $\varphi_{p,q}$ is locally constant, hence constant on the connected scheme Y. □

Exercise 3.6

Problem. Let S be a scheme over a finite field $\mathbb{F}_p$, and let $X \to S$ be a scheme of finite type. Show that the relative Frobenius $X^{(p)} \to X$ (Definition 3.2.23) is a purely inseparable morphism.

Solution. Which morphism. Recall (Def. 3.2.23): F_S is the absolute Frobenius of S, $X^{(p)} = X \times_{S,F_S} S$ with projections $\phi : X^{(p)} \to X$, $\pi : X^{(p)} \to S$, $q : X \to S$ the structure morphism, and the *relative Frobenius* is the morphism $F_{X/S} : X \to X^{(p)}$ determined by $\phi \circ F_{X/S} = F_X$ (absolute Frobenius of X) and $\pi \circ F_{X/S} = q$ (it exists since $q \circ F_X = F_S \circ q$). So the relative Frobenius goes $X \to X^{(p)}$ and not $X^{(p)} \to X$ as printed. We prove that $F_{X/S}$ is purely inseparable, and also (Remark at the end) the projection $\phi : X^{(p)} \to X$. Finite type is not needed.

Recall that F_X is the identity on the underlying space and induces $a \mapsto a^p$ on $\mathcal{O}_X$; hence on residue fields it induces the map $k(x) \to k(x)$, $a \mapsto a^p$.

Injectivity. If $F_{X/S}(x_1) = F_{X/S}(x_2)$, then applying ϕ gives $F_X(x_1) = F_X(x_2)$, i.e. $x_1 = x_2$.

Residue fields. Let $x \in X$ and $x' = F_{X/S}(x)$; then $\phi(x') = F_X(x) = x$. The identity $\phi \circ F_{X/S} = F_X$ gives on residue fields that the composite

$$k(x) \xrightarrow{\phi^\#} k(x') \xrightarrow{F^\#_{X/S}} k(x)$$

is $a \mapsto a^p$. So for $\beta \in k(x)$ we have $\beta^p = F^\#_{X/S}(\phi^\#(\beta)) \in F^\#_{X/S}(k(x'))$. Thus every element of $k(x)$ has its p-th power in the subfield $F^\#_{X/S}(k(x'))$, i.e. $k(x') \to k(x)$ is an algebraic purely inseparable extension. Hence $F_{X/S}$ is purely inseparable.

Remark. The absolute Frobenius F_S is purely inseparable by the same reasoning (identity on points, $a \mapsto a^p$ on residue fields), hence universally injective by Ex. 3.8; this property is stable under base change, so ϕ, the base change of F_S along q, is universally injective, hence purely inseparable by Ex. 3.8 again. □

Exercise 3.7

Problem. Let k be an algebraically closed field. Let $f : X \to Y$ be a finite surjective morphism of integral algebraic varieties over k.

(a) Let us suppose that $K(X)/K(Y)$ is separable of degree n. Show that there exists a non-empty open subset V of Y such that $f^{-1}(V) \to V$ decomposes into a finite birational morphism $f^{-1}(V) \to W$ followed by a finite etale morphism $W \to V$. Show that for any $y \in V(k)$, $f^{-1}(y)$ consists of at least n points.

(b) Use (a) to show that if f induces an injective map $X(k) \to Y(k)$, then it is purely inseparable.

Solution. Write $K = K(Y)$, $L = K(X)$. We first prove a lemma which contains (a) and which also serves for (b).

Lemma. Let $K \subseteq M \subseteq L$ be an intermediate field with M/K separable of degree m. Then there are a non-empty affine open $V = \operatorname{Spec} A' \subseteq Y$ with $f^{-1}(V) = \operatorname{Spec} B'$, a domain C which is a free A'-module of rank m with $\operatorname{Frac} C = M$ and $\operatorname{Spec} C \to V$ finite etale, and an injective A'-algebra map $C \to B'$ making B' a finite C-module, such that for every $y \in V(k)$ the fibre $f^{-1}(y)$ contains at least m points.

Proof. Take a non-empty affine open $\operatorname{Spec} A \subseteq Y$, with $f^{-1}(\operatorname{Spec} A) = \operatorname{Spec} B$ (f is finite, hence affine); B is a domain, finite over the domain A, and $\operatorname{Frac} A = K$, $\operatorname{Frac} B = L$. Then

$L = B \otimes_A K$, so every element of L has the form b/a with $b \in B$, $0 \neq a \in A$. Let θ_0 be a primitive element of the finite separable extension M/K, and choose $0 \neq a \in A$ with $\theta := a\theta_0 \in B$; still $M = K(\theta)$. Let $P \in K[T]$ be the minimal polynomial of θ: monic of degree m, separable, so its discriminant $\mathrm{disc}(P) \in K^\times$. Choose $0 \neq a_1 \in A$ such that the coefficients of P and $\mathrm{disc}(P)^{-1}$ lie in $A' := A_{a_1}$. Then $P \in A'[T]$ is monic with $\mathrm{disc}(P) \in A'^\times$, so $C := A'[T]/(P)$ is a free A'-module of rank m and etale over A'. Put $V = D(a_1)$ and $B' = B_{a_1}$.

Since $P(\theta) = 0$ in L, the map $T \mapsto \theta$ defines an A'-algebra map $C \to B'$. Since C is free over A', $C \hookrightarrow C \otimes_{A'} K = K[T]/(P) \simeq M$ (as P is irreducible), so C is a domain with fraction field M, and the composite $C \to B' \subseteq L$ is this inclusion, so $C \to B'$ is injective. B' is finite over A', hence over C, and $C \subseteq B'$ are domains, so $\operatorname{Spec} B' \to \operatorname{Spec} C$ is surjective (lying over). Let $y \in V(k)$, corresponding to a maximal ideal $\mathfrak{n}$ of A' with $A'/\mathfrak{n} = k$. The fibre of $\operatorname{Spec} C$ over y is $\operatorname{Spec} k[T]/(\bar{P})$ with $\bar{P}$ monic of degree m and non-zero discriminant, so it has m distinct points; each is the image of a point of $\operatorname{Spec} B'$ lying over y, and distinct points of $\operatorname{Spec} C$ have distinct preimages. So $f^{-1}(y)$ has at least m points. □

(a) Apply the lemma with $M = L$, $m = n$. Put $W = \operatorname{Spec} C$. Then $f^{-1}(V) = \operatorname{Spec} B' \to W$ is finite, surjective, and birational since $\operatorname{Frac} B' = L = M = \operatorname{Frac} C$; and $W \to V$ is finite etale. So $f^{-1}(V) \to V$ is the composition of these two, and each $f^{-1}(y)$, $y \in V(k)$, has at least n points.

(b) Suppose $X(k) \to Y(k)$ is injective.

Step 1: if f is as in the statement, then L/K is purely inseparable. The extension L/K is finite. Let K_s be the separable closure of K in L and $n_s = [K_s : K]$; then L/K_s is purely inseparable. Apply the lemma with $M = K_s$: for $y \in V(k)$ (a non-empty open subset of a variety over an algebraically closed field has a k-point) the fibre $f^{-1}(y)$ has at least n_s points, and these points are closed points over a closed point, i.e. elements of $X(k)$. By injectivity, $n_s \leq 1$. So $K_s = K$ and L/K is purely inseparable.

Step 2: residue fields. Let $x \in X$, $T = \overline{\{x\}}$ and $T' = f(T) = \overline{\{f(x)\}}$ (closed as f is finite), with reduced structures. Then $T \to T'$ is a finite surjective morphism of integral varieties, $T(k) \subseteq X(k)$, $T'(k) \subseteq Y(k)$, and $T(k) \to T'(k)$ is injective. By Step 1, $K(T)/K(T')$, that is $k(x)/k(f(x))$, is purely inseparable.

Step 3: injectivity on all points. Let $x_1, x_2 \in X$ with $f(x_1) = f(x_2) = y$, $T_i = \overline{\{x_i\}}$, $T' = \overline{\{y\}} = f(T_1) = f(T_2)$. Let y_0 be any closed point of T'. Since $T_i \to T'$ is surjective, T_i contains a closed point over y_0; since f is injective on k-points, the closed points of T_1 over y_0 and the closed points of T_2 over y_0 are the same unique point. Every closed point of T_i lies over a closed point of T', so T_1 and T_2 have the same closed points. In a scheme of finite type over a field a closed subset is the closure of the set of its closed points, so $T_1 = T_2$, whence $x_1 = x_2$ (generic points).

By Steps 2 and 3, f is injective with purely inseparable residue extensions, i.e. purely inseparable. □

Exercise 3.8

Problem. Let $f : X \to Y$ be a morphism of schemes. Show that the following properties are equivalent:

(i) f is purely inseparable.

(ii) For every Y-scheme Z, the projection morphism $X \times_Y Z \to Z$ is injective (in other words, f is universally injective).

(iii) For every field K, the canonical map $X(K) \to Y(K)$ is injective.

Solution. Field lemma. For a field extension E/F the following are equivalent: (1) E/F is algebraic and purely inseparable (if $\operatorname{char} F = p > 0$: every $e \in E$ has $e^{p^n} \in F$ for some n; if $\operatorname{char} F = 0$: $E = F$); (2) for every algebraically closed field $\Omega \supseteq F$, there is at most one F-embedding $E \to \Omega$ (we then say "at most one F-point of E in Ω"). *Proof.* (1)⇒(2): in characteristic 0, $E = F$. In characteristic p, if σ is an F-embedding and $e \in E$ with $e^{p^n} = a \in F$, then $\sigma(e)^{p^n} = a$, and $T^{p^n} - a = (T-\alpha)^{p^n}$ has a unique root in Ω; so $\sigma(e)$ is determined. (2)⇒(1): if some $e \in E$ is transcendental over F, choose a transcendence basis $B \ni e$ of E/F; the F-automorphism of $F(B)$ with $e \mapsto e+1$ fixing the other elements of B extends to an embedding $E \to \overline{E}$ (as $E/F(B)$ is algebraic and $\overline{E}$ is algebraically closed), distinct from the inclusion. If all elements are algebraic but some e has an irreducible polynomial with at least two distinct roots in $\Omega = \overline{E}$ (this happens exactly when e is not purely inseparable over F), we get two embeddings of $F(e)$, which extend to E ($E/F(e)$ algebraic). So (2) fails in both cases. □

(ii)⇒(iii). Let $a, b : \operatorname{Spec} K \to X$ with $f \circ a = f \circ b =: c$. Regard $Z = \operatorname{Spec} K$ as a Y-scheme through c. The pairs (a, id) and (b, id) define two sections $\sigma_a, \sigma_b : \operatorname{Spec} K \to X_Z = X \times_Y Z$ of the projection $\pi : X_Z \to Z = \operatorname{Spec} K$. The one-point space Z has one point, so π injective implies that σ_a and σ_b have the same image point w. A section with image w corresponds to a ring homomorphism $\sigma^\# : k(w) \to K$ with $\sigma^\# \circ \pi^\# = \mathrm{id}_K$, where $\pi^\# : K \to k(w)$. Then $\sigma^\#$ is injective (field homomorphism) and surjective ($\sigma^\#\pi^\#(K) = K$), so $\sigma^\# = (\pi^\#)^{-1}$ does not depend on the section. A morphism from $\operatorname{Spec} K$ is determined by its image point and the map on residue fields, so $\sigma_a = \sigma_b$ and $a = \operatorname{pr}_1 \sigma_a = \operatorname{pr}_1 \sigma_b = b$.

(iii)⇒(i). *Injectivity.* Let $x_1, x_2 \in X$ with $f(x_1) = f(x_2) = y$. The ring $k(x_1) \otimes_{k(y)} k(x_2)$ is non-zero; let K be its quotient by a maximal ideal. We get $a_i : \operatorname{Spec} K \to \operatorname{Spec} k(x_i) \to X$ with $f \circ a_1 = f \circ a_2$ (both factor through $\operatorname{Spec} k(y) \to Y$ and the two maps $k(y) \to K$ agree by construction). By (iii), $a_1 = a_2$, so $x_1 = x_2$. *Residue fields.* Let $x \in X$, $y = f(x)$, $F = k(y)$, $E = k(x)$, and let Ω be an algebraically closed field containing F and let $\sigma_1, \sigma_2 : E \to \Omega$ be two F-embeddings. They define two morphisms $\operatorname{Spec} \Omega \to \operatorname{Spec} k(x) \to X$ with the same composition with f (both factor through $\operatorname{Spec} F \to Y$ with the inclusion $F \subseteq \Omega$). By (iii) they are equal, so $\sigma_1 = \sigma_2$. By the field lemma E/F is purely inseparable.

(i)⇒(ii). Let $Z \to Y$ be a morphism and $z \in Z$ a point with image $y \in Y$. The fibre of $X_Z \to Z$ over z is $X \times_Y \operatorname{Spec} k(z) = X_y \times_{k(y)} \operatorname{Spec} k(z)$, whose points are the points of X_Z above z. By (i), $f^{-1}(y)$ is empty or a single point x. If it is empty there is nothing to prove. Otherwise the underlying space of X_y is the single point x, so $X_y = \operatorname{Spec} D$ with D a local ring whose maximal ideal $\mathfrak{m}_D$ is the nilradical (the only prime) and $D/\mathfrak{m}_D = k(x)$. Then $X_y \times_{k(y)} \operatorname{Spec} k(z) = \operatorname{Spec}(D \otimes_{k(y)} k(z))$, in which the ideal generated by $\mathfrak{m}_D$ is nil; so its spectrum is that of $G := k(x) \otimes_{k(y)} k(z)$. We claim G has exactly one prime ideal. It is non-zero. If $\operatorname{char} k(y) = 0$, then $k(x) = k(y)$ and $G = k(z)$. If $\operatorname{char} k(y) = p > 0$: every element $u = \sum e_i \otimes g_i$ of G satisfies $u^{p^n} = \sum e_i^{p^n} \otimes g_i^{p^n} \in 1 \otimes k(z)$ for n large (the $e_i^{p^n}$ lie in $k(y)$). If $\mathfrak{P}$ is a prime of G and $u \in \mathfrak{P}$, then $u^{p^n} \in \mathfrak{P} \cap (1 \otimes k(z)) = 0$ (a proper ideal of the field $1 \otimes k(z)$), so u is nilpotent. Hence every prime consists of nilpotents, i.e. equals the nilradical: G has exactly one prime. So each fibre of $X_Z \to Z$ has at most one point: the map is injective. □

Exercise 3.9

Problem. Let Y be a normal, locally Noetherian, integral scheme, and let $f : X \to Y$ be a projective morphism with X integral.

(a) Let us suppose f is finite. Show that f is purely inseparable if and only if the extension $K(Y) \to K(X)$ is purely inseparable. Moreover, if f is purely inseparable, it will be a homeomorphism.

(b) Let us suppose that the algebraic closure of $K(Y)$ in $K(X)$ is purely inseparable over $K(Y)$. Show that the fibers of $X \to Y$ are geometrically connected.

Solution. In both parts f is assumed *dominant*, as is implicit in the statement (the map $K(Y) \to K(X)$ must exist). A dominant projective (hence closed) morphism is surjective. We write $K = K(Y)$, $L = K(X)$.

Lemma. Let A be a normal domain with fraction field K, and B a domain containing A and integral over A, with $L' = \operatorname{Frac} B$ purely inseparable over K (so $L' = K$ if $\operatorname{char} K = 0$). Then $\operatorname{Spec} B \to \operatorname{Spec} A$ is purely inseparable; if B is finite over A, it is a homeomorphism. *Proof.* Let $b \in B$. There is $n \geq 0$ with $b^{p^n} \in K$ (take $p^n = 1$ in characteristic 0). But b^{p^n} is integral over A and A is integrally closed, so $b^{p^n} \in A$. Let $\mathfrak{q}_1, \mathfrak{q}_2$ be primes of B with $\mathfrak{q}_1 \cap A = \mathfrak{q}_2 \cap A = \mathfrak{p}$. For $b \in \mathfrak{q}_1$, $b^{p^n} \in \mathfrak{q}_1 \cap A = \mathfrak{p} \subseteq \mathfrak{q}_2$, so $b \in \mathfrak{q}_2$. Thus $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$ and by symmetry they are equal: $\operatorname{Spec} B \to \operatorname{Spec} A$ is injective. For the residue fields, $k(\mathfrak{q}) = \operatorname{Frac}(B/\mathfrak{q})$, and an element $\bar{u}/\bar{v}$ $(u, v \in B)$ satisfies $(\bar{u}/\bar{v})^{p^n} = \overline{u^{p^n}}/\overline{v^{p^n}} \in k(\mathfrak{p})$ for n large, since $u^{p^n}, v^{p^n} \in A$. So $k(\mathfrak{q})/k(\mathfrak{p})$ is purely inseparable. If B is finite over A, the map is closed (integral extension), surjective (lying over) and bijective; a continuous closed bijection is a homeomorphism. □

(a) If f is purely inseparable, the generic point ξ_X of X maps to the generic point ξ_Y (f is dominant), and the residue field extension there is $K \to L$, which is purely inseparable. Conversely suppose L/K is purely inseparable. Being purely inseparable is local on Y: for an affine open $V = \operatorname{Spec} A \subseteq Y$ we have $f^{-1}(V) = \operatorname{Spec} B$, with A a normal domain, B a domain finite over A and $\operatorname{Frac} B = L$ ($f^{-1}(V) \neq \emptyset$ as f is dominant). The Lemma shows $f^{-1}(V) \to V$ is purely inseparable and a homeomorphism; these properties glue, so f is purely inseparable. If f is purely inseparable, it is injective, and surjective (dominant and closed), and closed (finite), hence a homeomorphism.

(b) By Cor. 3.5, $f_*\mathcal{O}_X$ is a coherent $\mathcal{O}_Y$-algebra, and $g : Y' = \operatorname{Spec} f_*\mathcal{O}_X \to Y$ (Ex. 1.17) is finite. For an affine open $V = \operatorname{Spec} A$ of Y we have $g^{-1}(V) = \operatorname{Spec} B$ with $B = \mathcal{O}_X(f^{-1}V)$, a domain ($X$ integral) containing A, finite over A and contained in L. Every element of B is algebraic over K, so it lies in the algebraic closure K' of K in L, which is purely inseparable over K by hypothesis; so $\operatorname{Frac} B \subseteq K'$ is purely inseparable over K. By the Lemma, g is purely inseparable; this says that $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is purely inseparable (Def. 3.13). As f is surjective, Ex. 3.10 (proved below) applies: for any connected $Z \to Y$, the scheme $X \times_Y Z$ is connected. Let $y \in Y$ and let Ω be an algebraic closure of $k(y)$, $Z = \operatorname{Spec} \Omega \to Y$. Then Z is connected and $X \times_Y Z = X_y \times_{k(y)} \Omega$, so X_y is geometrically connected. □

Exercise 3.10

Problem. Let Y be a locally Noetherian scheme, and $f : X \to Y$ be a projective morphism such that $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is purely inseparable. Show that for any morphism $Z \to Y$ with Z

connected, the fibered product $X \times_Y Z$ is connected. Also show that $f^{-1}(F)$ is connected for any connected subset F of Y.

Solution. A needed hypothesis. The statement is false if f is not surjective: let $Y = \mathbf{A}^1_k$ and $X = \{0,1\} \hookrightarrow Y$ the closed immersion of two rational points (finite, hence projective). Then $\operatorname{Spec} f_*\mathcal{O}_X \to Y$ is this immersion, which is purely inseparable (injective, trivial residue extensions), but $X \times_Y Y = X$ is not connected. So we assume f *surjective*. (Here a space is connected if it is not the union of two disjoint non-empty open subsets; surjectivity makes everything non-empty when $Z \neq \emptyset$.)

Step 0 (Stein factorization; used also in Ex. 3.11 and 3.12). Let $f : X \to Y$ be projective, Y locally Noetherian. By Cor. 3.5, $\mathcal{A} = f_*\mathcal{O}_X$ is a coherent $\mathcal{O}_Y$-algebra; let $h : Y' = \operatorname{Spec} \mathcal{A} \to Y$ (Ex. 1.17), a finite morphism with $h_*\mathcal{O}_{Y'} = \mathcal{A}$. Over an affine open $V \subseteq Y$, a morphism $f^{-1}(V) \to h^{-1}(V) = \operatorname{Spec} \mathcal{A}(V)$ over V is the same as an algebra homomorphism $\mathcal{A}(V) \to \mathcal{O}_X(f^{-1}V)$; taking the identity we obtain (compatibly) a morphism $g : X \to Y'$ with $f = h \circ g$. Then: (S1) $\mathcal{O}_{Y'} \to g_*\mathcal{O}_X$ is an isomorphism: for V affine, $W = h^{-1}(V) = \operatorname{Spec} \mathcal{O}_X(f^{-1}V)$ is affine, $g^{-1}(W) = f^{-1}(V)$, and $\mathcal{O}_{Y'}(W) = \mathcal{O}_X(f^{-1}V) = (g_*\mathcal{O}_X)(W)$. Both sheaves are quasi-coherent on W, so they agree. (S2) g is projective: choose a closed immersion $i : X \to \mathbf{P}^n_Y$ over Y. The graph $(\mathrm{id}, g) : X \to X \times_Y Y'$ is a closed immersion, being a base change of the diagonal of the separated morphism h; and $i \times \mathrm{id} : X \times_Y Y' \to \mathbf{P}^n_Y \times_Y Y' = \mathbf{P}^n_{Y'}$ is a closed immersion. The composition is a closed immersion $X \to \mathbf{P}^n_{Y'}$ over Y'. (S3) g is surjective: g is proper, so $g(X)$ is closed, and $\operatorname{Supp} g_*\mathcal{O}_X \subseteq g(X)$ (if $y' \notin g(X)$ then $g_*\mathcal{O}_X$ vanishes near y'); by (S1) $\operatorname{Supp} g_*\mathcal{O}_X = \operatorname{Supp} \mathcal{O}_{Y'} = Y'$. (S4) *For every $y' \in Y'$ and every field extension $E \supseteq k(y')$, the scheme $X_{y'} \otimes_{k(y')} E$ is non-empty and connected.* Non-emptiness is (S3). By (S1), (S2) and Thm. 3.15 with Cor. 3.17 applied to g (Y' is locally Noetherian since h is finite), $X_\kappa := X_{y'} \otimes_{k(y')} \kappa$ is connected, $\kappa = \overline{k(y')}$. It is projective over κ, so $\Gamma = H^0(X_\kappa, \mathcal{O})$ is a finite-dimensional κ-algebra (Thm. 3.2(a)) without idempotents other than $0, 1$ (Ex. 2.4.6), hence a local Artinian ring with residue field κ. For an algebraically closed $\Omega \supseteq \kappa$, $H^0(X_\Omega, \mathcal{O}) = \Gamma \otimes_\kappa \Omega$ (Cor. 2.27) is local (its nilpotent maximal ideal is $\mathfrak{m}_\Gamma \otimes \Omega$, with quotient Ω), so again has no non-trivial idempotents, and X_Ω is connected (Ex. 2.4.6). Given E, take Ω an algebraic closure of E; it contains a copy of κ, and $X_\Omega \to X_{y'} \otimes E$ is a surjective continuous map, so $X_{y'} \otimes E$ is connected.

A topological lemma (TL). Let $q : T \to W$ be a closed, surjective, continuous map with connected non-empty fibres, W connected. Then T is connected. Indeed, if $T = T_1 \sqcup T_2$ with T_i open and closed, each fibre lies in T_1 or T_2; so $W = q(T_1) \sqcup q(T_2)$ with both closed and disjoint; W connected forces one of them to be empty, so one T_i is empty.

Solution. Let f be surjective with $h : Y' \to Y$ purely inseparable, in the notation of Step 0; h is finite and surjective ($f = h \circ g$), and universally injective by Ex. 3.8(ii). Let $Z \to Y$ be a morphism, Z connected, and $Y'_Z = Y' \times_Y Z$. Then $h_Z : Y'_Z \to Z$ is closed (finite), injective (universal injectivity), and surjective (base change of a surjection), hence a homeomorphism, so Y'_Z is connected. Now $X_Z = X \times_Y Z = X \times_{Y'} Y'_Z$ and the projection $g_Z : X_Z \to Y'_Z$ is the base change of g along $Y'_Z \to Y'$. It is proper (hence closed) and surjective, and its fibre over $w \in Y'_Z$ with image $y' \in Y'$ is $X_{y'} \otimes_{k(y')} k(w)$, which is non-empty and connected by (S4). By TL, X_Z is connected.

For the second assertion, let $F \subseteq Y$ be a connected subset. The map $f : X \to Y$ is closed and surjective; its fibres are connected: h is bijective (injective since purely inseparable, surjective),

so $f^{-1}(y) = g^{-1}(y')$ where y' is the unique point over y, i.e. the topological space of $X_{y'}$, which is connected by (S4) with $E = k(y')$. The restriction $f^{-1}(F) \to F$ is closed (for C closed in $f^{-1}(F)$, $C = C' \cap f^{-1}(F)$ with C' closed in X, and $f(C) = f(C') \cap F$), surjective, with connected fibres. By TL, $f^{-1}(F)$ is connected. □

Exercise 3.11

Problem. (Stein factorization). Let $f : X \to Y$ be a projective morphism to a locally Noetherian scheme Y.

(a) Show that f factors into $g : X \to Z$ and $h : Z \to Y$ such that g has only geometrically connected fibres, and h is finite.

(b) Show that we can factor f as in (a) with moreover $\mathcal{O}_Z \simeq g_*\mathcal{O}_X$. Show that this condition implies that Z is unique up to isomorphism (Ex. 1.17).

Solution. (b) (and thereby (a)). Take $Z = Y' = \operatorname{Spec} f_*\mathcal{O}_X$, $h : Z \to Y$ and $g : X \to Z$ as in Step 0 of Ex. 3.10. Then h is finite, $f = h \circ g$ and $\mathcal{O}_Z \simeq g_*\mathcal{O}_X$ (S1). Moreover g is projective (S2) and Z is locally Noetherian, so Thm. 3.15 and Cor. 3.17 applied to g show that every fibre X_z, $z \in Z$, is (non-empty and) geometrically connected. This gives (b) existence, and (a), because the factorization of (b) satisfies the requirements of (a).

Uniqueness. Let $f = h_1 \circ g_1$ with $g_1 : X \to Z_1$, $h_1 : Z_1 \to Y$ finite and $\mathcal{O}_{Z_1} \simeq g_{1*}\mathcal{O}_X$. Then $h_{1*}\mathcal{O}_{Z_1} \simeq h_{1*}g_{1*}\mathcal{O}_X = f_*\mathcal{O}_X$ as $\mathcal{O}_Y$-algebras (the isomorphism being induced by $g_1^{\#}$). As h_1 is affine, $Z_1 \simeq \operatorname{Spec} h_{1*}\mathcal{O}_{Z_1}$ over Y (Ex. 1.17(c)), so we get an isomorphism of Y-schemes $\varphi : Z_1 \to \operatorname{Spec} f_*\mathcal{O}_X = Z$. By the description of morphisms $X \to \operatorname{Spec}\mathcal{A}$ as algebra homomorphisms $\mathcal{A} \to f_*\mathcal{O}_X$ (Step 0), g_1 corresponds to the isomorphism above, composed with which the identity of $f_*\mathcal{O}_X$ is the morphism g; hence $\varphi \circ g_1 = g$. So Z is unique up to a unique isomorphism compatible with g and h. □

Exercise 3.12

Problem. Let $f : X \to Y$ be a projective morphism. Let us suppose that for every $y \in Y$, X_y is geometrically connected (resp. geometrically integral). Show that $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is purely inseparable (resp. an isomorphism).

Solution. Throughout, Y is locally Noetherian (as in the whole section) and "geometrically connected/integral" includes non-empty, so f is surjective. Let $f = h \circ g$, $Z = \operatorname{Spec} f_*\mathcal{O}_X$, be the factorization of Step 0 (Ex. 3.10). Recall h is finite, g is projective and surjective, $\mathcal{O}_Z \simeq g_*\mathcal{O}_X$.

Geometrically connected fibres. Let $y \in Y$, $k = k(y)$ and $\kappa = \bar{k}$. The fibre $Z_y = h^{-1}(y)$ is $\operatorname{Spec} C_y$ with C_y a finite k-algebra (as $f_*\mathcal{O}_X$ is coherent), so Z_y is a finite discrete set $\{z_1, \dots, z_n\}$. We have $X_y = X \times_Z Z_y$, so $X_y \to Z_y$ is surjective (g is) and $X_y = \bigsqcup_i g^{-1}(z_i)$ is a decomposition into n disjoint non-empty open and closed subsets. As X_y is connected ($X_\kappa \to X_y$ is surjective and X_κ is connected), $n = 1$. So h is injective. Let z be the point of Z_y. The map $X_\kappa \to Z_y \otimes_k \kappa = \operatorname{Spec}(C_y \otimes_k \kappa)$ is surjective and X_κ is connected, so $\operatorname{Spec}(C_y \otimes_k \kappa)$ is connected; it is a finite discrete set, hence a point. Its underlying space is that of $\operatorname{Spec}(k(z) \otimes_k \kappa)$ (C_y

is local Artinian with residue field $k(z)$, finite over k, and the kernel of $C_y \otimes \kappa \to k(z) \otimes \kappa$ is nilpotent). The prime ideals of the Artinian ring $k(z) \otimes_k \kappa$ correspond to the κ-points, i.e. to the k-embeddings $k(z) \to \kappa$ (the residue field at each prime is finite over κ, so equals κ). So there is exactly one k-embedding of $k(z)$ in κ, and by the field lemma of Ex. 3.8 $k(z)/k$ is purely inseparable. Hence h is injective with purely inseparable residue extensions, i.e. purely inseparable: $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is purely inseparable (Def. 3.13). This proves the first assertion, as printed.

Geometrically integral fibres. As printed, the second assertion is false for non-reduced Y: let $Y = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ and $X = \operatorname{Spec} k \to Y$ the closed immersion (projective). The only fibre is $\operatorname{Spec} k$, geometrically integral, but $\mathcal{O}_Y = k[\varepsilon] \to f_*\mathcal{O}_X = k$ is not an isomorphism. We prove the assertion when Y is *integral* and f is *flat*; the case of a reduced Y with a non-flat f is not proved here.

For every $y \in Y$ the scheme X_y is projective over $k = k(y)$ and geometrically integral, so X_κ is integral and projective over $\kappa = \bar{k}$, and $H^0(X_\kappa, \mathcal{O})$ is a finite-dimensional κ-algebra (Thm. 3.2(a)) which is a domain (it embeds in the function field), hence a field finite over κ, i.e. $= \kappa$. By flat base change (Cor. 2.27), $H^0(X_y, \mathcal{O}_{X_y}) \otimes_k \kappa = \kappa$, so $\dim_{k(y)} H^0(X_y, \mathcal{O}_{X_y}) = 1$ for every $y \in Y$. Now $\mathcal{F} = \mathcal{O}_X$ is coherent and flat over Y, and Y is integral; the function $y \mapsto \dim H^0(X_y, \mathcal{F}_y)$ is constant, so by Rem. 3.21(b) ($p = 0$) the sheaf $f_*\mathcal{O}_X = R^0 f_*\mathcal{O}_X$ is locally free and $f_*\mathcal{O}_X \otimes k(y) \to H^0(X_y, \mathcal{O}_{X_y})$ is an isomorphism for every y; so $f_*\mathcal{O}_X$ is locally free of rank 1. The unit section $\mathcal{O}_Y \to f_*\mathcal{O}_X$ induces on $k(y)$-fibres a map $k(y) \to H^0(X_y, \mathcal{O}_{X_y}) = k(y)$ sending 1 to $1 \neq 0$ ($X_y \neq \emptyset$), which is an isomorphism. By Nakayama's lemma $\mathcal{O}_{Y,y} \to (f_*\mathcal{O}_X)_y$ is surjective, hence an isomorphism (a surjection between free modules of rank 1 over a local ring is an isomorphism). Thus $\mathcal{O}_Y \simeq f_*\mathcal{O}_X$. □

Exercise 3.13

Problem. Let $X \to S$ be a projective morphism to a locally Noetherian scheme. Let us suppose that the fibers of $X \to S$ are of dimension $\leq d$.

(a) Show that if $\mathcal{F} \to \mathcal{G}$ is a surjective homomorphism of quasi-coherent sheaves on X, then $H^d(X, \mathcal{F}) \to H^d(X, \mathcal{G})$ is surjective (use Prop. 2.34). Show that the canonical homomorphism
$$R^d f_*\mathcal{F} \otimes_{\mathcal{O}_S} k(s) \to H^d(X_s, \mathcal{F}_s) \tag{3.16}$$
is surjective for all $s \in S$.

(b) Let us suppose that S is a Dedekind scheme, and that $\mathcal{F}$ is coherent and flat over S. Show that (3.16) is bijective.

Solution. Let $f : X \to S$ be the morphism. In the first assertion of (a), S is affine (for general S the statement is that $R^d f_*\mathcal{F} \to R^d f_*\mathcal{G}$ is surjective, which is local on S by Prop. 2.28); then $H^p(X, -) = R^p f_*(-)(S)$.

(a) Let $\mathcal{K} = \operatorname{Ker}(\mathcal{F} \to \mathcal{G})$; it is quasi-coherent. X is separated, so Cor. 2.22 gives an exact sequence $H^d(X, \mathcal{F}) \to H^d(X, \mathcal{G}) \to H^{d+1}(X, \mathcal{K})$, and $H^{d+1}(X, \mathcal{K}) = 0$ by Prop. 2.34 (the fibres have dimension $\leq d$, and f is quasi-projective). So $H^d(X, \mathcal{F}) \to H^d(X, \mathcal{G})$ is surjective.

Now let $s \in S$; choose an affine open $\operatorname{Spec} A \ni s$ and replace S by it; s corresponds to a prime $\mathfrak{p}$ and $k(s) = \kappa(\mathfrak{p})$. The source of (3.16) is $H^d(X, \mathcal{F}) \otimes_A k(s)$, and the target is $H^d(X_s, \mathcal{F}_s) =$

$H^d(X, \mathcal{F} \otimes_A k(s))$ (Ex. 2.3(b)(1): $X_s \to X$ is affine and X is separated; $\mathcal{F} \otimes_A k(s)$ is the direct image of $\mathcal{F}_s$). The map (3.16) is the map ρ_M of Lemma 2.26 for $M = k(s)$. This map is natural in M, and it is bijective if M is free (if $M = A^{(I)}$, both sides are $H^d(X, \mathcal{F})^{(I)}$: cohomology of the quasi-compact scheme X commutes with direct sums, as one sees on a finite affine covering). Choose a surjection $A^{(I)} \to k(s)$; it induces a surjection of quasi-coherent sheaves $\mathcal{F} \otimes A^{(I)} \to \mathcal{F} \otimes k(s)$, hence a surjection on H^d by the first part. By naturality of ρ in M we have a commutative square, in which the left vertical arrow is $\mathrm{id} \otimes (A^{(I)} \to k(s))$, the right one is induced by the surjection of sheaves, and the horizontal arrows are $\rho_{A^{(I)}}$ and $\rho_{k(s)}$. Here the top arrow is an isomorphism, the right arrow is surjective, so $\rho_{k(s)}$ is surjective. This is (3.16).

(b) The formation of $R^d f_*$ commutes with the flat base change $\operatorname{Spec} \mathcal{O}_{S,s} \to S$ (Cor. 2.27) and X_s, $\mathcal{F}_s$ do not change; so we may assume $S = \operatorname{Spec} \mathcal{O}_{S,s}$. The ring $\mathcal{O}_{S,s}$ of a Dedekind scheme is a field (then s is the only point, $X_s = X$, and (3.16) is the identity) or a discrete valuation ring with closed point s or generic point s. If s is the generic point, $k(s) = K$ is flat over $\mathcal{O}_K$, so (3.16) is bijective by Lemma 2.26 and Ex. 2.3. Let s be the closed point. Then (3.16) is the homomorphism (3.14) of Thm. 3.20 for $p = d$; by Thm. 3.20(d) it is an isomorphism if and only if $H^{d+1}(X, \mathcal{F})$ is free over $\mathcal{O}_K$. But $H^{d+1}(X, \mathcal{F}) = 0$ by Prop. 2.34. So (3.16) is bijective. □

Exercise 3.14

Problem. Let Y be a locally Noetherian scheme, and let $f : X \to Y$ be a flat projective morphism. We say that f is cohomologically flat (in dimension 0) if the construction of $f_*\mathcal{O}_X$ commutes with base change (Cor. 3.22(i)).

(a) Show that f is cohomologically flat if and only if for every $y \in Y$, the morphism $X \times_Y \operatorname{Spec} \mathcal{O}_{Y,y} \to \operatorname{Spec} \mathcal{O}_{Y,y}$ is cohomologically flat.

(b) Let $Z \to Y$ be a faithfully flat morphism. Show that f is cohomologically flat if and only if $X \times_Y Z \to Z$ is cohomologically flat.

(c) Let us suppose that Y is local with closed point y, and that $H^0(X_y, \mathcal{O}_{X_y})$ is a direct sum of fields isomorphic to $k(y)$. Show that

$$H^0(X, \mathcal{O}_X) \otimes_{\mathcal{O}_{Y,y}} k(y) \to H^0(X_y, \mathcal{O}_{X_y}) \tag{3.17}$$

is surjective (use Lemma 3.14 with $A = H^0(X, \mathcal{O}_X)$).

(d) Let Y be local with closed point y. Let R be a finite etale algebra over $k(y)$. Show that there exists a finite free (hence faithfully flat) $\mathcal{O}_{Y,y}$-algebra A such that $A \otimes_{\mathcal{O}_{Y,y}} k(y) = R$. Deduce from this that if X_y is geometrically reduced, then (3.17) is surjective (consider $R = H^0(X_y, \mathcal{O}_{X_y})$ and $X_A \to \operatorname{Spec} A$).

(e) Let us suppose Y is regular of dimension 1 and that either X_y is geometrically reduced for every closed point $y \in Y$, or X is a complete intersection in some $\mathbf{P}^d_Y$. Show that f is cohomologically flat.

Solution. Reformulation. Cohomological flatness of f means that for every morphism $\pi : T \to Y$ the canonical map $\pi^* f_* \mathcal{O}_X \to f_{T*} \mathcal{O}_{X_T}$ is an isomorphism. This is local on T, so (taking T affine and mapping into an affine open) it is equivalent to:

(CF) for every affine open $V = \operatorname{Spec} A \subseteq Y$, with $B = H^0(f^{-1}V, \mathcal{O}_X)$, and every A-algebra A', the canonical map $\varphi_{A'} : B \otimes_A A' \to H^0(X_{A'}, \mathcal{O}_{X_{A'}})$ is bijective ($X_{A'} = X_V \times_V \operatorname{Spec} A'$).

(a) If f is cohomologically flat and $y \in Y$, put $Y_y = \operatorname{Spec} \mathcal{O}_{Y,y}$, $X^y = X \times_Y Y_y$. For an affine open $V = \operatorname{Spec} A \ni y$, $H^0(X^y, \mathcal{O}) = B \otimes_A A_{\mathfrak{p}}$ (Cor. 2.27), and for an $A_{\mathfrak{p}}$-algebra A' the map $\varphi_{A'}$ for f^y is $B \otimes_A A_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} A' = B \otimes_A A' \to H^0(X_{A'}, \mathcal{O})$, i.e. the one for f with A' regarded as an A-algebra. So f^y satisfies (CF). Conversely, let $V = \operatorname{Spec} A \subseteq Y$ be affine and A' an A-algebra. The map $\varphi_{A'}$ is a homomorphism of A-modules, bijective iff $\varphi_{A'} \otimes_A A_{\mathfrak{p}}$ is bijective for all primes $\mathfrak{p}$. By flat base change (Cor. 2.27) $\varphi_{A'} \otimes_A A_{\mathfrak{p}}$ is the map $\varphi_{A'_{\mathfrak{p}}}$ of f^y ($y \leftrightarrow \mathfrak{p}$) for the $A_{\mathfrak{p}}$-algebra $A'_{\mathfrak{p}} = A' \otimes_A A_{\mathfrak{p}}$, which is bijective by hypothesis.

(b) Let $\rho : Z \to Y$ be faithfully flat and $f_Z : X_Z \to Z$. Let $W = \operatorname{Spec} C \subseteq Z$ be an affine open mapping into an affine open $V = \operatorname{Spec} A$ of Y; C is flat over A, $H^0(X_W, \mathcal{O}) = B \otimes_A C$ (Cor. 2.27), and for a C-algebra C' the map $\varphi^Z_{C'} : (B \otimes_A C) \otimes_C C' = B \otimes_A C' \to H^0(X_{C'}, \mathcal{O})$ is $\varphi_{C'}$, with C' regarded as an A-algebra. ($\Rightarrow$) If f satisfies (CF), then so does f_Z (for all such W, hence on Z). ($\Leftarrow$) Let $V = \operatorname{Spec} A \subseteq Y$ be affine, A' an A-algebra, and $\mathfrak{p}$ a prime of A. Let $z \in Z$ lie over $\mathfrak{p}$ (ρ is surjective), $W = \operatorname{Spec} C \ni z$ an affine open mapping into V, and $C' = A' \otimes_A C$. Since $A' \to C'$ is flat, $H^0(X_{C'}, \mathcal{O}) = H^0(X_{A'}, \mathcal{O}) \otimes_{A'} C' = H^0(X_{A'}, \mathcal{O}) \otimes_A C$ (Cor. 2.27), and $\varphi^Z_{C'} = \varphi_{A'} \otimes_A C$. By (CF) for f_Z this is bijective. Localizing at z: $\varphi_{A'} \otimes_A \mathcal{O}_{Z,z}$ is bijective. The local homomorphism $A_{\mathfrak{p}} \to \mathcal{O}_{Z,z}$ is flat, hence faithfully flat, so $\varphi_{A'} \otimes_A A_{\mathfrak{p}}$ is bijective (its kernel and cokernel become 0 after the faithfully flat extension, so are 0). As $\mathfrak{p}$ was arbitrary, $\varphi_{A'}$ is bijective.

(c) Let $Y = \operatorname{Spec} A$ with A local Noetherian, $\mathfrak{m}$ its maximal ideal, $k = k(y) = A/\mathfrak{m}$, $B = H^0(X, \mathcal{O}_X)$. By Step 0 of Ex. 3.10, $B = \mathcal{A}(Y)$ is a finite A-algebra, $Y' = \operatorname{Spec} B$, and $g : X \to \operatorname{Spec} B$ is projective with $g_* \mathcal{O}_X = \mathcal{O}_{\operatorname{Spec} B}$; so B is a Noetherian ring, X is projective over B, and $B \to H^0(X, \mathcal{O}_X)$ is the identity, in particular purely inseparable. Apply Lemma 3.14 to the ring B and the ideal $I = \mathfrak{m}B$: there is an ideal J of B with $J \subseteq \mathfrak{m}B \subseteq \sqrt{J}$ such that $B/J \to R_J := H^0(X, \mathcal{O}_X/J\mathcal{O}_X)$ is purely inseparable. Let X_J be the closed subscheme of X defined by $J\mathcal{O}_X$. As $\sqrt{J} = \sqrt{\mathfrak{m}B}$, X_J and $X_y = X \otimes_A k$ (defined by $\mathfrak{m}\mathcal{O}_X = \mathfrak{m}B\,\mathcal{O}_X$) have the same underlying space, and $X_y \subseteq X_J$ is a closed subscheme. Also B/J is Artinian ($J \supseteq (\mathfrak{m}B)^n$ for some n and $B/\mathfrak{m}B$ is a finite k-algebra), and R_J is a finite B/J-module (Thm. 3.2), hence Artinian.

Components. The space $|X_y|$ is Noetherian, with finitely many connected components $X_1, \ldots, X_N$, all open and closed. So $H^0(X_y, \mathcal{O}) = \prod_i H^0(X_i, \mathcal{O})$, each factor non-zero without idempotents other than $0, 1$ (Ex. 2.4.6). By hypothesis this ring is $\simeq k^m$, a product of m fields k; a decomposition of k^m into N non-zero rings without non-trivial idempotents has $N = m$ factors, each equal to k. So $H^0(X_i, \mathcal{O}) = k$ for all i. The same components X_i^J are open and closed in X_J, so $R_J = \prod_i R_i$ with $R_i = H^0(X_i^J, \mathcal{O})$, Artinian without non-trivial idempotents, hence local, with residue field κ_i.

Residue fields. The restriction to the closed subscheme $X_i \subseteq X_i^J$ gives a ring homomorphism $R_i \to H^0(X_i, \mathcal{O}) = k$ over A; it factors through a homomorphism $\sigma_i : \kappa_i \to k$ (its kernel is the maximal ideal). The elements of $\mathfrak{m}$ are nilpotent in R_i ($\mathfrak{m} \subseteq \sqrt{J}$ and $JR_i = 0$), so $A \to R_i \to \kappa_i$ factors through k: we get $\tau_i : k \to \kappa_i$, and $\sigma_i \tau_i = \mathrm{id}_k$ (both are A-algebra maps from $k = A/\mathfrak{m}$). A field homomorphism σ_i with a section is bijective, so $\kappa_i \simeq k$ via σ_i, with inverse τ_i.

Points of $\operatorname{Spec} B$ *over* y. Since $B/J \to R_J$ is purely inseparable, $\operatorname{Spec} R_J = \bigsqcup \operatorname{Spec} R_i \to \operatorname{Spec} B/J$ is injective, so its N points map to N distinct points $z_1, \ldots, z_N$ of $\operatorname{Spec} B/J$, with $\kappa_i/k(z_i)$ purely inseparable. The composition $k \to k(z_i) \to \kappa_i \simeq k$ is the identity, so $k(z_i) = k$. Every point z of $\operatorname{Spec} B/J = h^{-1}(y)$ is of the form $g(x)$ with $x \in X$ (g is surjective), and

$x \in X_y = |X_J|$, say $x \in X_i$; the morphism $g|_{X_J}$ factors as $X_J \to \operatorname{Spec} R_J \to \operatorname{Spec} B/J$, so $z = z_i$. Thus $\operatorname{Spec} B/J = \{z_1, \dots, z_N\}$, and $B/\mathfrak{m}B$ is an Artinian ring with exactly N maximal ideals, all with residue field k. By the Chinese remainder theorem $B/\mathfrak{m}B \to \prod_i k(z_i) = k^N$ is surjective.

Conclusion. The map $B \to H^0(X_y, \mathcal{O}) = \prod_i H^0(X_i, \mathcal{O}) = k^N$ (restriction of sections) factors through $B/\mathfrak{m}B$ and, in the i-th component, is $B \to R_i \to \kappa_i \xrightarrow{\sigma_i} k$, which is $B \to k(z_i) = k$. So it is the surjection above, and (3.17) is surjective.

(d) Write $R = \prod_{j=1}^n k_j$ with k_j finite separable field extensions of $k = k(y)$, $k_j = k[T]/(\tilde{P}_j)$ with $\tilde{P}_j$ monic irreducible (primitive element theorem). Let $P_j \in \mathcal{O}_{Y,y}[T]$ be monic lifts of $\tilde{P}_j$ and $A = \prod_j \mathcal{O}_{Y,y}[T]/(P_j)$. It is a free $\mathcal{O}_{Y,y}$-module of rank $\sum \deg P_j$, hence faithfully flat, and $A \otimes k = \prod_j k[T]/(\tilde{P}_j) = R$.

Now let X_y be geometrically reduced and $R_0 = H^0(X_y, \mathcal{O}_{X_y})$, a finite k-algebra (Thm. 3.2). With $\kappa = \bar{k}$, $R_0 \otimes_k \kappa = H^0(X_\kappa, \mathcal{O})$ (Cor. 2.27), a reduced ring because X_κ is reduced. So R_0 is reduced and Artinian, $R_0 = \prod_j k_j$ with fields k_j, and each $k_j \otimes_k \kappa$ is reduced, so k_j/k is separable: R_0 is finite etale. Choose a finite Galois extension E/k containing a copy of every k_j; then $k_j \otimes_k E \simeq E^{[k_j:k]}$, hence $R_0 \otimes_k E \simeq E^{\dim_k R_0}$. Apply the construction above to the single field $E = k[T]/(\tilde{P})$: $A_0 = \mathcal{O}_{Y,y}[T]/(P)$ is finite free over $\mathcal{O}_{Y,y}$ with $A_0 \otimes k = E$; it is a local ring (it is finite over a local ring and $A_0/\mathfrak{m}A_0 = E$ is a field), with closed point y_0 and $k(y_0) = E$. Let $X_0 = X \times_Y Y_0$, $Y_0 = \operatorname{Spec} A_0$. It is projective over the local Noetherian scheme Y_0 and its closed fibre is $X_y \otimes_k E$ with $H^0 = R_0 \otimes_k E \simeq E^{\dim R_0}$, a direct sum of fields isomorphic to $k(y_0)$. By (c), $H^0(X_0, \mathcal{O}) \otimes_{A_0} E \to H^0(X_y \otimes E, \mathcal{O})$ is surjective. By Cor. 2.27 (A_0 is flat over $\mathcal{O}_{Y,y}$), $H^0(X_0, \mathcal{O}) = B \otimes A_0$, so this map is $\big(B \otimes_{\mathcal{O}_{Y,y}} k \to H^0(X_y, \mathcal{O})\big) \otimes_k E$, i.e. (3.17) tensored with E. As E is faithfully flat over k, (3.17) is surjective.

(e) *Reduction.* By (a) it suffices to treat $Y_y = \operatorname{Spec} \mathcal{O}_{Y,y}$ for each y ($X^y \to Y_y$ is again flat, projective, and satisfies the hypothesis). As Y is regular of dimension 1, $\mathcal{O}_{Y,y}$ is a discrete valuation ring (closed point) or a field (generic point). If $\mathcal{O}_{Y,y}$ is a field K, every K-algebra A' is flat over K, so $\varphi_{A'}$ is bijective by Lemma 2.26, Cor. 2.27; (CF) holds. So let $S = \operatorname{Spec} \mathcal{O}_K$, $\mathcal{O}_K = \mathcal{O}_{Y,y}$ a discrete valuation ring with closed point s, $f : X \to S$ flat projective. The sheaf $\mathcal{O}_X$ is coherent and flat over S. For an $\mathcal{O}_K$-module M, $H^0(X, \mathcal{O}_X \otimes M)$ is $H^0(X_M, \mathcal{O})$ if M is an algebra ($X_M \to X$ affine, Ex. 2.3), so (CF) for f says exactly that the map ρ_M of Thm. 3.20(c) is bijective for $p = 0$, $\mathcal{F} = \mathcal{O}_X$, and all algebras $M = A'$. By Thm. 3.20(c) (and Cor. 3.22), it is enough to show that

$$H^0(X, \mathcal{O}_X) \otimes k(s) \to H^0(X_s, \mathcal{O}_{X_s}) \qquad (*)$$

is surjective ((3.14) for $p = 0$).

First case: X_s is geometrically reduced. Then $(*)$ is surjective by (d) (applied to the local scheme S).

Second case: $X = V_+(F_1, \dots, F_c) \subseteq \mathbf{P}^d_S$ is a complete intersection, which we understand (as in Ex. 3.4 and 3.5) as: every fibre X_y has dimension $r = d - c$. If $r = 0$, then f is proper with finite fibres, hence finite, hence affine; then $H^0(X_{A'}, \mathcal{O}) = B \otimes_A A'$ trivially and (CF) holds. If $r \geq 1$, then by Ex. 3.3(b)(2) (and the argument of Ex. 3.3(c)) $H^0(X_s, \mathcal{O}_{X_s}) = k(s)$, spanned by the image of 1, and $(*)$ is surjective because $1 \in H^0(X, \mathcal{O}_X)$ maps to 1.

In both cases f is cohomologically flat. □

Exercise 3.15

Problem. (A non-cohomologically flat scheme). Let $S = \operatorname{Spec} A$ be a Noetherian integral affine scheme, and let $t \in A$ be a non-zero non-invertible element. Let us consider the S-scheme

$$X = \operatorname{Proj} A[w,x,y,z]/(x^2 - t^2wy,\ xz - ty^2,\ xy - twz,\ z^2w - y^3),$$

and let $f : X \to S$ denote the structural morphism.

(a) Show that over the open subset $T := D(t)$ of S, there exists a morphism $\mathbf{P}^1_T \to \mathbf{P}^3_T$ that induces an isomorphism $\mathbf{P}^1_T \simeq X \times_S T$ (consider the 3-uple embedding $\mathbf{P}^1_T \to \mathbf{P}^3_T$ (Ex. 1.27) followed by the automorphism $(w,x,y,z) \to (w,tx,y,z)$).

(b) Show that $D_+(z) \cap X \simeq \mathbf{A}^1_S$ and that $\mathcal{O}_X(D_+(w) \cap X)$ is free over A as an A-module. Conclude that X is flat over S.

(c) Let $s \in V(t)$. Show that X_s is an irreducible curve, reduced except at the point $(1,0,0,0)$. Show that we have $\dim_{k(s)} H^0(X_s, \mathcal{O}_{X_s}) = 2$ and that $A \to H^0(X, \mathcal{O}_X)$ is an isomorphism. Deduce from this that $X \to S$ is not cohomologically flat.

(d) Show that X is not normal, and that its normalization is isomorphic to $\mathbf{P}^1_S$ if S is normal (Ex. 4.1.5).

Solution. Let I be the homogeneous ideal generated by the four relations, $B_0 = A[w,x,y,z]/I$ so $X = \operatorname{Proj} B_0$. Write $A[u,v]$ for the homogeneous coordinate ring of $\mathbf{P}^1_A$ and $s = v/u$ for the affine coordinate on $D_+(u)$. Define

$$\nu : \mathbf{P}^1_S \to \mathbf{P}^3_S, \qquad (u:v) \mapsto (w:x:y:z) = (u^3 : t\,u^2v : uv^2 : v^3).$$

The four sections u^3, tu^2v, uv^2, v^3 of $\mathcal{O}(3)$ have no common zero (already u^3, v^3 have none), so ν is a morphism. The four relations of I vanish when we substitute these four elements:

$$x^2 - t^2wy \mapsto t^2u^4v^2 - t^2u^3 \cdot uv^2 = 0, \qquad xz - ty^2 \mapsto tu^2v \cdot v^3 - t\,u^2v^4 = 0,$$
$$xy - twz \mapsto t\,u^2v \cdot uv^2 - t\,u^3v^3 = 0, \qquad z^2w - y^3 \mapsto v^6u^3 - u^3v^6 = 0.$$

Hence ν factors through X: $\nu : \mathbf{P}^1_S \to X$.

Charts of X. The sets $D_+(w) \cap X$ and $D_+(z) \cap X$ cover X: if a homogeneous prime $\mathfrak{P}$ of B_0 contains w and z, then it contains $x^2 = t^2wy + (x^2 - t^2wy)$ and $y^3 = z^2w - (z^2w - y^3)$, hence x, y, hence the whole irrelevant ideal; so it is not a point of X.

$D_+(z)$. Setting $z = 1$, the relations become $x^2 - t^2wy$, $x - ty^2$, $xy - tw$, $w - y^3$. The last two eliminate $w = y^3$ and $x = ty^2$, and the other two relations then read $t^2y^4 - t^2y^4 = 0$ and $ty^3 - ty^3 = 0$. Hence the coordinate ring of $D_+(z) \cap X$ is $A[y]$, and $D_+(z) \cap X \simeq \mathbf{A}^1_S$; $\nu^{-1}(D_+(z)) = D_+(v) = \operatorname{Spec} A[Y]$, $Y = u/v = s^{-1}$, and ν restricts to an isomorphism $D_+(v) \to D_+(z) \cap X$ $(y/z = Y)$.

$D_+(w)$. Setting $w = 1$ we get $C := A[x,y,z]/(x^2 - t^2y,\ xz - ty^2,\ xy - tz,\ z^2 - y^3)$. The A-algebra map $\theta : C \to A[s]$, $x \mapsto ts$, $y \mapsto s^2$, $z \mapsto s^3$, is well defined (the four relations are satisfied: $t^2s^2 = t^2s^2$, $ts \cdot s^3 = t\,s^4$, $ts \cdot s^2 = t\,s^3$, $s^6 = s^6$); it is the restriction of ν to $\nu^{-1}(D_+(w)) = D_+(u) = \operatorname{Spec} A[s]$. We claim that C is spanned over A by $\mathcal{B} = \{1,\ x\} \cup \{y^b \mid b \geq 1\} \cup \{y^bz \mid b \geq 0\}$. Using $x^2 = t^2y$, $xy = tz$, $xz = ty^2$, every monomial $x^ay^bz^c$ is an

A-combination of monomials whose x-degree is smaller, unless it is x or has $a = 0$ already; and for $a = 0$, $z^2 = y^3$ reduces c to 0 or 1. Now θ sends $\mathcal{B}$ to 1, ts, s^{2b} $(b \geq 1)$, s^{2b+3} $(b \geq 0)$, i.e. to 1, ts and each s^k, $k \geq 2$, exactly once. These elements are A-linearly independent in $A[s]$: a relation $a_0 + a_1\, ts + \sum_{k\geq 2} a_k s^k = 0$ in $A[s]$ gives $a_0 = 0$, $a_1 t = 0$ and $a_k = 0$ for $k \geq 2$, and then $a_1 = 0$ because A is a domain and $t \neq 0$. Hence $\mathcal{B}$ is an A-basis of C, θ is injective, and

$$C \simeq A[ts, s^2, s^3] \subseteq A[s], \qquad C \text{ is a free } A\text{-module with basis } \mathcal{B}.$$

(b) We have shown $D_+(z) \cap X \simeq \mathbf{A}^1_S$ and that $\mathcal{O}_X(D_+(w) \cap X) = C$ (the coordinate ring of this affine open) is free over A with basis $\mathcal{B}$. Both affine opens have flat, even free, coordinate rings over A, and they cover X, so X is flat over S.

(a) Over $T = D(t)$, t is invertible. Then $C \otimes_A A_t = A_t[ts, s^2, s^3] = A_t[s]$ (as $s = t^{-1} \cdot ts$) so θ becomes an isomorphism $C_t \to A_t[s]$, i.e. $\nu : D_+(u)_T \to (D_+(w) \cap X)_T$ is an isomorphism; and $\nu : D_+(v)_T \to (D_+(z) \cap X)_T$ is an isomorphism (already over S). Since $\nu^{-1}(D_+(w)) = D_+(u)$, $\nu^{-1}(D_+(z)) = D_+(v)$, and these two cover $\mathbf{P}^1$ and X respectively, $\nu_T : \mathbf{P}^1_T \to X_T = X \times_S T$ is an isomorphism. Finally ν is the composition of the 3-uple embedding $(u : v) \mapsto (u^3 : u^2v : uv^2 : v^3)$ with the automorphism $(w, x, y, z) \mapsto (w, tx, y, z)$ of $\mathbf{P}^3_T$ (t is invertible on T), followed by the inclusion $X_T \subseteq \mathbf{P}^3_T$. This is the morphism asked for.

(c) Let $s \in V(t)$ (non-empty as t is not invertible), $k = k(s)$; the image of t in k is 0. Then $X_s = \operatorname{Proj} k[w, x, y, z]/(x^2, xz, xy, z^2w - y^3)$, covered by

$U_1 = D_+(z) \cap X_s = \operatorname{Spec} k[y]$ and $U_2 = D_+(w) \cap X_s = \operatorname{Spec} C_s$, $C_s = C \otimes_A k = k[x, y, z]/(x^2, xy, xz, z^2 - y^3)$.

By the basis $\mathcal{B}$ of C, C_s has k-basis $\mathcal{B}$; we have $x^2 = xy = xz = 0$, so kx is an ideal and $C_s/kx = \Gamma := k[y, z]/(z^2 - y^3) \simeq k[\sigma^2, \sigma^3]$ ($y \mapsto \sigma^2$, $z \mapsto \sigma^3$, σ an indeterminate), a domain. Thus $C_s = \Gamma \oplus kx$ as a vector space, and the underlying space of U_2 is $\operatorname{Spec} \Gamma$, irreducible of dimension 1. So U_1 and U_2 are irreducible curves with non-empty intersection $D(z)$ (a non-empty open subset of each), hence X_s is an irreducible curve.

Reducedness. U_1 is reduced. In C_s, the element x is nilpotent and non-zero, and the prime ideal $\mathfrak{o} = (x, y, z)$ (the point $(1, 0, 0, 0)$) is the only prime of C_s containing y and z ($\Gamma/(y, z) = k$). At any other point, y or z is invertible in the local ring, and then $x = 0$ (from $xy = 0$, $xz = 0$); the local ring is a localization of Γ, reduced. At the point $\mathfrak{o}$ the local ring has the non-zero nilpotent x and is not reduced. So X_s is reduced except at $(1, 0, 0, 0)$.

$H^0(X_s, \mathcal{O})$. $U_1 \cap U_2 = D_+(wz) = \operatorname{Spec}(C_s)_z$; in $(C_s)_z$ we have $x = 0$ (as $xz = 0$) and y is a unit (as $y^3 = z^2$), so $(C_s)_z = \Gamma_z = k[\sigma, \sigma^{-1}]$. The restriction maps are: $k[Y] \to k[\sigma, \sigma^{-1}]$, $Y \mapsto y/z = \sigma^{-1}$ (on U_1, $y/z = Y$ is the coordinate), and $C_s \to k[\sigma, \sigma^{-1}]$, $x \mapsto 0$, $y \mapsto \sigma^2$, $z \mapsto \sigma^3$. A global section is a pair $(f(Y),\, p(y, z) + cx)$ ($f \in k[Y]$, $p \in \Gamma$, $c \in k$) with $f(\sigma^{-1}) = p(\sigma^2, \sigma^3)$ in $k[\sigma, \sigma^{-1}]$. The left side lies in $k[\sigma^{-1}]$, the right side in $k[\sigma]$, and $k[\sigma^{-1}] \cap k[\sigma] = k$; so $f = \lambda$ is constant, and $p = \lambda$ because $\Gamma \to k[\sigma]$ is injective. Thus the global sections are exactly $\lambda \cdot 1 + c \cdot \xi$ with $\xi = (0, x)$, so $\dim_k H^0(X_s, \mathcal{O}_{X_s}) = 2$.

$H^0(X, \mathcal{O}_X) = A$. Similarly, $U_1^A = D_+(z) \cap X = \operatorname{Spec} A[Y]$, $U_2^A = \operatorname{Spec} C$ and $U_1^A \cap U_2^A = \operatorname{Spec} C_z$. Now $C \subseteq A[s]$ and $C_z = A[ts, s^2, s^3, s^{-3}] \subseteq A[s, s^{-1}]$; it contains $s^{-1} = y\, z^{-1}$ and $s = y^2 z^{-1}$ (as $y = s^2$, $z = s^3$), thus $C_z = A[s, s^{-1}]$. All rings are domains inside $A[s, s^{-1}]$ ($A[Y] = A[s^{-1}]$ because $Y \mapsto y/z = s^{-1}$), so the restriction maps are injective and a global section is an element of $A[s^{-1}] \cap C$. This lies in $A[s^{-1}] \cap A[s] = A$, and $A \subseteq C$. So $H^0(X, \mathcal{O}_X) = A$

and $A \to H^0(X, \mathcal{O}_X)$ is an isomorphism.

Not cohomologically flat. If f were cohomologically flat, then by (CF) of Ex. 3.14 (the base change $\operatorname{Spec} k(s) \to S$) the map $H^0(X, \mathcal{O}_X) \otimes_A k(s) \to H^0(X_s, \mathcal{O}_{X_s})$ would be bijective. Its source is $A \otimes_A k(s) = k(s)$, of dimension 1, and its target has dimension 2. So f is not cohomologically flat (although f is flat and projective by (b)).

(d) The scheme X is integral: it is covered by the integral opens $D_+(z) \cap X \simeq \mathbf{A}^1_S$ (coordinate ring $A[y]$) and $D_+(w) \cap X = \operatorname{Spec} C$ ($C \subseteq A[s]$), whose intersection is non-empty (it contains X_T). Its function field is $K(X) = \operatorname{Frac}(A)(s)$.

Not normal. The element $s = z/y \in \operatorname{Frac} C$ satisfies $s^2 = y \in C$, so it is integral over C. But $s \notin C$: an element of C, written in the basis $\mathcal{B}$, has coefficient of s (in $A[s]$) in tA, and $1 \notin tA$ since t is not invertible. So C is not integrally closed in its fraction field, hence some localization of C at a maximal ideal is not integrally closed, and X is not normal.

Normalization. Assume S normal, i.e. A is a normal domain. Then $\mathbf{P}^1_S$ is normal (it is covered by two copies of $\operatorname{Spec} A[s]$, and a polynomial ring over a normal domain is normal) and integral, with function field $\operatorname{Frac}(A)(s) = K(X)$. The morphism $\nu : \mathbf{P}^1_S \to X$ is birational (isomorphism over the dense open X_T by (a)) and finite: over $D_+(v)$ it is an isomorphism, and over $D_+(w) \cap X = \operatorname{Spec} C$ it is $C \subseteq A[s]$, where $A[s] = C + Cs$ is a finite C-module: $A[s] = C + Cs$, because A and the As^k ($k \geq 2$) lie in C and $As \subseteq Cs$. So $\nu_* \mathcal{O}_{\mathbf{P}^1_S}$ is a finite $\mathcal{O}_X$-algebra, contained in $K(X)$ and integrally closed in it (it is the sheaf of rings of a normal scheme with function field $K(X)$); hence it equals the integral closure of $\mathcal{O}_X$ in $K(X)$, and ν is the normalization of X (Ex. 4.1.5). Thus the normalization of X is $\mathbf{P}^1_S$. □